

SMITHIAN FOLD THEORY V3

Smithian Fold Theory V3 Scientific Audit Layer

Complete claim, evidence, chronology, correction and machine-identity audit

Ernos Labs

Open Source Science Platform and Knowledge Tree

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Version 0.1.0 - first standalone V3 preliminary publication

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New standalone V3 concept record 21717583 - Paper: CC BY 4.0 - Code: Apache-2.0

FULL-SCALE PRELIMINARY VERSION - COMPUTATIONAL PROGRAMMES CONTINUE

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Smithian Fold Theory of Everything V3 preliminary synthesis

Author and publication authority: Maria Smith

Organisation: Ernos Labs

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Status: Scientific audit for the authorised first standalone V3 preliminary publication. Final computational proofs, downstream full-field censuses, canonical current-graph reconciliation and final Grand Lock verification remain open.

This layer preserves complete claim identities and current evidence classes. It does not promote implementation tests to empirical confirmation, historical programme results to current V3 proof authority, or foundational closure to full-field closure.

1. Global direct-source verification

Quantity	Verified value
Live claims	2,751
Model-admitted claims	2,751
Generated candidates	892,246
Unique survivors	2,751
Adverse controls	11,004
Passed adverse controls	11,004
Receipt identities matched	2,751/2,751
Candidate censuses reconstructed	2,751/2,751
Claims with a path to the root theorem	2,751/2,751
Missing declared dependencies	0
Unclassified obligations in the live claim ledger	0

The candidate total is reconstructed directly from every current `candidate_census.json`, including the larger Foundation and Physics grammars. Receipt validation compares the registered claim row with the receipt's internal identity and status fields; the file SHA-256 is retained separately.

2. Branch status and exact totals

Branch	Current boundary	Claims	Candidates	Survivors	Controls	Root-traced
Astronomy and Cosmology	published foundation and prior-return family complete; full-field census not frozen	77	19,712	77	308	77/77
Biology and Life Sciences	82/424 full-field obligations closed; continuation active	82	20,992	82	328	82/82
Chemistry	272/272 registered obligations plus nine separately retained returns; 281 live claims	281	71,936	281	1,124	281/281
Classical Computation	369/369 dated full-field boundary complete	369	94,464	369	1,476	369/369
Consciousness and Cognitive Science	published foundation and prior-return family complete; full-field census not frozen	77	19,712	77	308	77/77
Cross-Branch Synthesis	12/12 inherited-return claims admitted; final global current-graph merge remains open	12	3,072	12	48	12/12
Earth and Environmental Sciences	published foundation and prior-return family complete; full-field census not frozen	75	19,200	75	300	75/75
Engineering Translation	published foundation and prior-return family complete; full-field census not frozen	80	20,480	80	320	80/80
Foundation	16/16 current Foundation boundary complete; extension-open	16	5,222	16	64	16/16
Information Science	262/262 dated full-field boundary complete	262	75,776	262	1,048	262/262
Materials Science	289/289 dated full-field boundary complete	289	73,984	289	1,156	289/289

Branch	Current boundary	Claims	Candidates	Survivors	Controls	Root-traced
Mathematics	323/323 dated full-field boundary complete	323	97,280	323	1,292	323/323
Medicine and Health Sciences	published foundation and prior-return family complete; full-field census not frozen	76	19,456	76	304	76/76
Physics	368/368 current categorical boundary complete	368	257,776	368	1,472	368/368
Reversible and Quantum Computation	288/288 dated full-field boundary complete	288	73,728	288	1,152	288/288
Social prior-return continuation	four admitted prior-return claims retained separately	4	1,024	4	16	4/4
Social and Collective Systems	published 72-law foundation	72	18,432	72	288	72/72

3. Active paper registry

Corpus role	Active source	SHA-256	Lines	Headings	DOI tokens
astronomy_cosmology	output/release/astronomy-cosmology-1.0.0/02_From-One-Sky-to-Cosmos_Astronomy-and-Cosmology-Foundation-Paper-001-v1.0.md	135b2ada93caf788571b9e7c0d384d60ec0851d9c47a85fe432d6ad2618ef643	3,705	461	10.5281/zenodo.21640812
biology	output/release/biology-1.0.0/02_From-Fold-to-Life_Biology-and-Life-Sciences-Foundation-Paper-001-v1.0.md	af18711f0520f03a6d2e7fe5004166856d2203ad370a4932178c59eb00abddb5	8,203	108	10.5281/zenodo.21515629, 10.5281/zenodo.21516146, 10.5281/zenodo.21516916, 10.5281/zenodo.21520881, 10.5281/zenodo.21531455, 10.5281/zenodo.21630203
chemistry	output/release/chemistry-1.3.0/01_From-Fold-to-Chemistry_Chemistry-Branch-Paper-001-v1.3.md	d1880beab7d6eec028167d8f353a86936231dc7a2cfd7c2b318cf5512f121dc1	59,729	409	10.5281/zenodo.21531454, 10.5281/zenodo.21627782
computation	output/release/classical-computation-1.4.0/01_After-Turing-The-Fold-Machine_Classical-Computation-Branch-Paper-001-v1.4.md	e1406b1c91db69b88d92107c4fc96d017fef968299583ef97eec217f60b1e54f	19,146	1,470	10.5281/zenodo.21518310, 10.5281/zenodo.21627721

Corpus role	Active source	SHA-256	Lines	Headings	DOI tokens
consciousness_cognitive_science	output/release/consciousness-cognitive-science-1.0.0/02_From-Fold-to-Consciousness-Consciousness-and-Cognitive-Science-Foundation-Paper-001-v1.0.md	62f473b326aa9cf0ecfab2dcaaf763aa752df468210aef52b0d12c40f3aebaf9	8,725	108	10.5281/zenodo.21515629, 10.5281/zenodo.21516146, 10.5281/zenodo.21516916, 10.5281/zenodo.21520881, 10.5281/zenodo.21630203, 10.5281/zenodo.21630785, 10.5281/zenodo.21636397
earth_environment	output/release/earth-environment-1.0.0/02_From-One-World-to-Earth_Earth-and-Environmental-Sciences-Foundation-Paper-001-v1.0.md	044d381d988635cb69a368ff9dbfe2fad8d87b31cec7f4edd76faee959c60354	9,556	101	10.5281/zenodo.21515629, 10.5281/zenodo.21516146, 10.5281/zenodo.21516916, 10.5281/zenodo.21520881, 10.5281/zenodo.21630203, 10.5281/zenodo.21640810
engineering_translation	output/release/engineering-translation-1.0.0/02_From-One-Law-to-a-Working-World_Engineering-Translation-Foundation-Paper-001-v1.0.md	e232febfaidd820c6ac4557b42bf9b9fcf0b894c316c272e87e19b34a6bc44d0	5,161	533	10.5281/zenodo.21640816
foundation	publications/successors/foundation/FROM_NOTHING_TO_FOLD_FOUNDATION_PAPER_001_V1_3.md	f3387b75f402150e3d88f9216a0cb92c26449ae26d4761a028c0e7872c3c5d48	1,695	137	10.5281/zenodo.21627656
information_science	output/release/information-science-1.4.0/01_From-Distinction-to-Information_Information-Science-Branch-Paper-001-v1.4.md	ee0d40407b908dcd2fc15e89b809a727525cd2448dc53440ea0f1b021215f838	18,755	440	10.5281/zenodo.21515629, 10.5281/zenodo.21516146, 10.5281/zenodo.21516915, 10.5281/zenodo.21627717
materials	output/release/materials-1.3.0/02_From-Fold-to-Materials_Materials-Science-Branch-Paper-001-v1.3.md	79fe7890cf31017536cfc5d76bf816783ab805f484de22b675e311e88d12c1da	30,104	463	10.5281/zenodo.21515629, 10.5281/zenodo.21516146, 10.5281/zenodo.21516916, 10.5281/zenodo.21518311, 10.5281/zenodo.21518313, 10.5281/zenodo.21520881, 10.5281/zenodo.21531455, 10.5281/zenodo.21532481, 10.5281/zenodo.21629306
mathematics	output/release/mathematics-1.5.0/02_From-Fold-to-Mathematics_Mathematics-Branch-Paper-001-v1.5.md	95dba6282d60c6b99b7298498f92352c758fafb080ed723b62b30227f66d6202	6,894	617	10.5281/zenodo.21516145, 10.5281/zenodo.21535636, 10.5281/zenodo.21627708
medicine	output/release/medicine-1.0.0/02_From-Fold-to-Medicine_Medicine-and-Health-Sciences-Foundation-Paper-001-v1.0.md	1f3acb2f7b0100da9e2530779f0a8e556570ae2bab46302554af8ef3fb15fd22	8,785	105	10.5281/zenodo.21515629, 10.5281/zenodo.21516146, 10.5281/zenodo.21516916, 10.5281/zenodo.21520881, 10.5281/zenodo.21531455, 10.5281/zenodo.21630203, 10.5281/zenodo.21630785
methods	publications/successors/methods/THERE_IS_NO_NOTHING_METHODS_PAPER_001_V0_3.md	d427d9803b04001ec6b83fd0612ea7ee7c894d172d8970bd3dc46c17bb4ea19f	1,347	71	10.5281/zenodo.21028523, 10.5281/zenodo.21028591, 10.5281/zenodo.21035460, 10.5281/zenodo.21512799, 10.5281/zenodo.21627646

Corpus role	Active source	SHA-256	Lines	Headings	DOI tokens
physics	output/release/physics-1.3.0/01_From-Fold-to-Physics_Physics-Branch-Paper-001-v1.3.md	3e0c2136752e3683ee6f599f234072b12a6ec01e08a4aa287d267717150c4883	32,411	552	10.5281/zenodo.21520880, 10.5281/zenodo.21591160, 10.5281/zenodo.21591169, 10.5281/zenodo.21591170, 10.5281/zenodo.21591171, 10.5281/zenodo.21591174, 10.5281/zenodo.21591175, 10.5281/zenodo.21627765
protein_fold_preliminary	applications/frontier/v3_computational_proofs/protein_folding/paper/SMITHIAN_FOLD_THEORY_V3_PROTEIN_FOLD_COMPUTATIONAL_PROOF.md	5e15a91525acbb85f5fedbd1b35e40e1e21874bf6c4114acb8f5049284ff27e2	1,026	41	10.5281/zenodo.21630203, 10.5281/zenodo.21717581
quantum_computation	output/release/quantum-computation-1.4.0/01_The-Quantum-Fold-Machine_Quantum-Computation-Branch-Paper-001-v1.4.md	1ef0a0e6908d486d8573092f1378ae99f5dfce493093c859b456aa0a00825ca7	6,722	538	10.5281/zenodo.21518312, 10.5281/zenodo.21627748
social_collective_systems	output/release/social-collective-sciences-1.0.0/02_From-One-Relation-to-Society_Social-Collective-Sciences-Foundation-Paper-001-v1.0.md	83105339ba4dd4f419f06935ff7926264510e62588f78bc58bdd79d1788c9ef8	5,162	533	10.5281/zenodo.21640814

4. Computational proof and programme chronology

4.1 Historical versioned programme evidence

Historical programme manifests are evidence for their producing versions. They are not imported as V3 proof authority.

Programme	Frozen version	Principal retained result	Current use
Chess Fold	2.2	1700: 54.2%; 1900: 62.5%; latest completed 2100 batch: 41.7%; legality/endgame verification retained	Read-only evidence and recreation target
Go Fold	2.2	Exact small-board census; two replayed 2-0 GNU Go 9x9 batches; one recovered 19x19 harness-reported 2-0 point-at-cutoff record with disclosed Round-1 synchronisation defect	Read-only evidence and recreation target
Protein Fold	historical release dated 2026-07-19	Sealed 76-residue run: whole-chain TM 0.0269927379497572; three strong local windows preserved	Read-only evidence and V3 provenance target
Unison AI	6.3	326 suites, 2,002 checks, zero main-corpus failures; standalone 21/21; weak generation benchmarks preserved	Read-only evidence and clean-rebuild target

4.2 Current V3 recreation status

Programme	Current V3 status	Publication or empirical boundary
Protein Fold	Preliminary v0.9.4 published; 42,173,082-candidate parent and 583-state high-attraction slice independently verified; complete 21-record claim audit added; lower bands, global frontier and whole-chain recurrence remain open	Frozen 100-target AlphaFold parity campaign has not started
Chess Fold	Clean V3 workspace not yet created in the current repository	Historical results remain historical until independently recreated
Go Fold	Clean V3 workspace not yet created in the current repository	Historical results remain historical until independently recreated
Unison Fold AI	V3 clean-rebuild scaffold present; mathematical closure open; empirical validation not run	Discord deployment and sealed generalisation remain unauthorised and unexecuted

5. Current corrections, adverse results and unresolved boundaries

- Biology retains corrections to rigid-one-shape protein language, inevitable beneficial fixation, universal three-quarter allometry, strong scale-free-network universality, parity-selected homochirality, and denominator-only accounts of ageing, cancer and ecosystems.
- Earth retains the adverse first mixed-magnitude earthquake catalogue and the later compatible homogeneous holdout as different registered records.
- Astronomy retains the adverse unweighted SPARC rank-four comparison and the separate uncertainty-aware source result without conflating them.
- Protein Fold retains adverse historical 4APD, 4B19 and 8HJC selector outcomes beside the stronger six-axis successor machinery; the current full relation has not yet received its primary blind empirical test.
- Chess and Go retain disclosed benchmark losses, cut-offs and synchronisation defects; historical victories are not relabelled as V3 results.
- Unison retains weak or null generation benchmarks; passing architectural checks are not conversational validation.
- The complete final ToE remains downstream of the open full-field programme, canonical current-graph reconciliation, final computational proofs and a new heavy global verification run.

6. Complete current claim inventory

The following table is generated from all 2,751 live claim packages. Full statements, dependency arrays and file identities remain in `AUTHORITATIVE_CORPUS_INVENTORY.json`.

#	Branch	Claim ID	Title	Formal closure	External status	Candidates	Controls	Receipt ID
1	Foundation	SFT-ROOT-THERE-IS-NO-NOTHING	Premise-free operational root theorem	depth_independent	independently_replicated	2	4	sha256:711864171e4d3a2f2734f0c2890965bcd81a0228349538751a3c80699c27d669
2	Foundation	SFT-FOUNDATION-ONE-001	The structural One	depth_independent	independently_replicated	6	4	sha256:68332624c276dc5d0ddfd568a26b95e20d7d5665fcd7e825ff1b7b19bf1d51c
3	Foundation	SFT-FOUNDATION-COUNT-001	Exact positive finite count	depth_independent	independently_replicated	10	4	sha256:4b2411716cbef2fb9fa7aac58b24ce5eb3c0c59fe7230fd703fd89b5260f6e23
4	Foundation	SFT-FOUNDATION-PART-001	Exact positive part coordinate	depth_independent	independently_replicated	192	4	sha256:87b40affae50458ba5492af91778fbca79a8de0bc3b951e520535093c2b7b728

#	Branch	Claim ID	Title	Formal closure	External status	Candidates	Controls	Receipt ID
5	Foundation	SFT-FOUNDATION-FOLD-001	The minimal structural Fold	depth_independent	independently_replicated	288	4	sha256:9e99f91d15e2b68a5c24f825093686863c168595319c2083e6a5b52b0314e003
6	Foundation	SFT-FOUNDATION-PART-EQUIVALENCE-001	Cross-partition exact part equivalence	depth_independent	independently_replicated	1,024	4	sha256:4f7c913f240527ec64d7688bf57fe95ae508f5445786dcbbdc2e6a7094476680
7	Foundation	SFT-FOUNDATION-FOLD-ASSEMBLY-001	Complete finite Fold-word assembly	depth_independent	independently_replicated	192	4	sha256:8fa632be9e4048a5f2cf6749da4110e3a7b8b4d937051610dcce0583d7ea610d
8	Foundation	SFT-FOUNDATION-FORM-GRAMMAR-001	Complete foundational form grammar	depth_independent	independently_replicated	288	4	sha256:b768689e7b061fdafa3a620befe79ddcee8bf5750cbbaf19f7b2a3156942be71
9	Foundation	SFT-FOUNDATION-FORM-ENFORCEMENT-001	Canonical foundational form enforcement	depth_independent	independently_replicated	192	4	sha256:84d18de18b1345bf506138493aebd7808bb0bd00cc3b318040e00751929c7c8a
10	Foundation	SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001	One-way derivation-to-measurement boundary	depth_independent	independently_replicated	256	4	sha256:a4e409f6bb8b203de746bd0c21d3f2566e8052003219490c657249e11f6f2086
11	Mathematics	SFT-MATH-EXACT-ARITHMETIC-001	Exact arithmetic and generated number structure	depth_independent	independently_replicated	256	4	sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbaf71609cb05c4f40196418
12	Mathematics	SFT-MATH-DISCRETE-001	Generated discrete objects, relations and induction	depth_independent	independently_replicated	256	4	sha256:632de46c995329ed86f8397e8cf2cc493d5074027941a50069df66b4ac9e29bb
13	Mathematics	SFT-MATH-COMBINATORICS-001	Exact finite combinatorial generation	depth_independent	independently_replicated	256	4	sha256:52e1db94bb7865270bb3387c47c609c438de3338b542cc31d910b981e4250968
14	Mathematics	SFT-MATH-GRAPH-NETWORK-001	Exact finite graph and network structure	depth_independent	independently_replicated	512	4	sha256:b4072f93147c8cd9b7233fc96cfeadeb791210239ab4e4ff7dec0ed7462e6d2a
15	Mathematics	SFT-MATH-ALGEBRA-001	Generated finite algebraic structures	depth_independent	independently_replicated	512	4	sha256:1f08bfaf9f602a3825dc75979070e05f9d51a7d01cdf3c34bab31c06849855bd
16	Mathematics	SFT-MATH-ORDER-LATTICE-001	Exact finite order and conditional lattice structure	depth_independent	independently_replicated	512	4	sha256:d0e20f40782f51cc85c1f572b624570e65b51bbb80135a5e8a6b3cf69f71a5f6
17	Mathematics	SFT-MATH-GEOMETRY-TOPOLOGY-001	Exact finite geometry and topology for computation	depth_independent	independently_replicated	1,024	4	sha256:a04f7de0ede6e2348896cf16e59981eebb5391198e498a1ec546b68a4a8d3a6e
18	Mathematics	SFT-MATH-PROBABILITY-STATISTICS-001	Exact finite probability and statistics from Fold observation	depth_independent	independently_replicated	512	4	sha256:2a86d644881cdacb757327d5aac01de41c3458c043980c66a16f5084cdfb45b
19	Mathematics	SFT-MATH-OPTIMIZATION-001	Exact finite optimization and retained optima	depth_independent	independently_replicated	512	4	sha256:3385414ceea0fd4212111e42bd6ce7d6d35556b7c97d66563813d5b8d1531849

#	Branch	Claim ID	Title	Formal closure	External status	Candidates	Controls	Receipt ID
20	Mathematics	SFT-MATH-DYNAMICAL-SYSTEMS-001	Exact generated finite dynamical systems	depth_independent	independently_replicated	1,024	4	sha256:3e0e44b65f6e60fe2a29b9927b1870a40da25c4b0c786e2adaa3df9a2f7cabe
21	Mathematics	SFT-MATH-LOGIC-PROOF-001	Exact finite logic, inference and proof boundaries	depth_independent	independently_replicated	1,024	4	sha256:3d366d57b26d002cb27a16e74b022a0856e133c616b7da6a83233c84bdd50813
22	Mathematics	SFT-MATH-CATEGORY-TYPE-COMPOSITION-001	Forced category, type and compositional structure	depth_independent	independently_replicated	1,024	4	sha256:47873a83a90e29fa22d065ff13aa0152edb6212414e35cc0435e8eff2c9ea5a8
23	Information Science	SFT-INFO-SYMBOL-DISTINCTION-001	Exact symbols, observation classes and distinguishability	depth_independent	independently_replicated	512	4	sha256:e387d64c93f5241d90ecf2a447eb6a3518c1e9f473e7ec627d38b7f03ac9d941
24	Information Science	SFT-INFO-ENCODING-DECODING-001	Exact encoding, decoding and declared representational loss	depth_independent	independently_replicated	1,024	4	sha256:dc3cbc708f5b88b3d9bbae5a646a644dc8feb7fe4a5d6f8367052f5062846d0f
25	Information Science	SFT-INFO-QUANTITY-001	Exact information quantity as retained distinction structure	depth_independent	independently_replicated	512	4	sha256:a06381112b197ea1762d8dd06197c87f03055a4554429f10ead4a93bbb2362d
26	Information Science	SFT-INFO-ENTROPY-UNCERTAINTY-001	Exact entropy and uncertainty from deterministic observation classes	depth_independent	independently_replicated	512	4	sha256:d604932214499ea1e1fc2df7b71f4ea0b67014e3001023720b325d6203d2b72
27	Information Science	SFT-INFO-COMPRESSION-REDUNDANCY-001	Exact lossless compression and redundancy accounting	depth_independent	independently_replicated	1,024	4	sha256:4d46ed451a5cd3c3a19fcb9d8b8cc3c32ffc8f92a9cc1d55cfeb93f331023ee07
28	Information Science	SFT-INFO-CHANNEL-CAPACITY-001	Exact finite channels, confusability and capacity	depth_independent	independently_replicated	1,024	4	sha256:fb47c7f7be3e832cb67658042b2c7c3224113d9262cb4c45df7924a1f3a1cc01
29	Information Science	SFT-INFO-NOISE-ERROR-001	Exact noise, error, detection and correction boundaries	depth_independent	independently_replicated	1,024	4	sha256:bc42a3ab0291a59b278654bed369bb1d64b6abe1f9278af9f89eac78211f48b9
30	Information Science	SFT-INFO-CODING-001	Exact finite coding, detection and correction	depth_independent	independently_replicated	2,048	4	sha256:7c7648bdba686025c8b1f647b3e63a242da24234aa953cac184021f878b07711
31	Information Science	SFT-INFO-MUTUAL-CONDITIONAL-001	Exact mutual and conditional information from joint observation incidence	depth_independent	independently_replicated	1,024	4	sha256:c311f27de12d70ea2307acb2d5b23f83cb065b6b28fe5c08ac4742f6f1aadcd26
32	Information Science	SFT-INFO-CONSERVATION-LOSS-001	Exact information conservation, loss and transformation	depth_independent	independently_replicated	1,024	4	sha256:4e2892e85c1624a2234163fdd51377a7086219ce0945cbf6acfaaa41664303f4
33	Information Science	SFT-INFO-CLASSICAL-PROBABILISTIC-001	Exact classical and probabilistic information correspondence	depth_independent	independently_replicated	1,024	4	sha256:ce0515fdf69d368bddb313faad3a88043550e90e3df21c4049267e1a5e54fe7c
34	Information Science	SFT-INFO-QUANTUM-CORRESPONDENCE-001	Exact quantum-information support correspondence	depth_independent	independently_replicated	1,024	4	sha256:01279dbd48775d7f5fa92b0c9abe07dcafeb2ba670d10374d9f3bb355582cc81

#	Branch	Claim ID	Title	Formal closure	External status	Candidates	Controls	Receipt ID
35	Classical Computation	SFT-COMP-FORM-STATE-TRANSITION-001	Exact state and transition law	depth_independent	independently_replicated	256	4	sha256:6ff1fdc0b1d041805add6ef9350dee1fe38a41082072c9f7888d4a22975228f
36	Classical Computation	SFT-COMP-FORM-LANGUAGE-GRAMMAR-001	Fold formal-language and grammar law	depth_independent	independently_replicated	256	4	sha256:9247a863bf580966b3db1091ef6024a0f4562506076136dce33acda2a6039d14
37	Classical Computation	SFT-COMP-FORM-AUTOMATON-001	Fold finite automaton law	depth_independent	independently_replicated	256	4	sha256:a28d36b12b07be8aa4f32af257b75d6da331b0cebfea99fda3d6240e9ccaf109
38	Classical Computation	SFT-COMP-FORM-REWRITING-001	Fold rewriting-system law	depth_independent	independently_replicated	256	4	sha256:b1f6c54c1f8be568e0dbb809dc19f1f882840e53c693aa208db0c2410860defe
39	Classical Computation	SFT-COMP-FORM-RECURSIVE-FUNCTION-001	Fold recursion law	depth_independent	independently_replicated	256	4	sha256:291230ba67c75760c319e58fe0ae120c6c300a83e39d5f51c70de936c73e580c
40	Classical Computation	SFT-COMP-FORM-LAMBDA-CALCULUS-001	Fold binding and self-application calculus law	depth_independent	independently_replicated	256	4	sha256:c5df90aeea7c0f8459bb98cdfa9343167723cc7b320af2c9fec60f7f5f5bee63
41	Classical Computation	SFT-COMP-FORM-ABSTRACT-MACHINE-001	Fold abstract-machine law	depth_independent	independently_replicated	256	4	sha256:2b2471771a06a549143d109bd88b40339b476c36ab65b278fd86a7b54adbdd1b
42	Classical Computation	SFT-COMP-FORM-CIRCUIT-001	Fold circuit syntax and evaluation law	depth_independent	independently_replicated	256	4	sha256:0a974615e0f53938a3c8551caab041da5b7c241dfdc059febcb5d38269a0c94
43	Classical Computation	SFT-COMP-FORM-OPERATIONAL-PROCESS-001	Fold operational-process law	depth_independent	independently_replicated	256	4	sha256:d3349d79836c6e8edf5199206fe0dc818a6210e9d3dc7f8000eb83f0b55ef282
44	Classical Computation	SFT-COMP-FORM-COMPOSITION-001	Fold process composition and decomposition law	depth_independent	independently_replicated	256	4	sha256:b0fea8f79a1c2c934c5b2dfaa37c9ba09e8b0f1416377175a645afe9f58738ad
45	Classical Computation	SFT-COMP-FORM-MODEL-EQUIVALENCE-001	Fold computational-model equivalence law	depth_independent	independently_replicated	256	4	sha256:911ff906b2d8407111a076f3abb06abe79334098a0e87faa54460f8453338139
46	Classical Computation	SFT-COMP-FORM-UNIVERSALITY-001	Fold universal-computation law	depth_independent	independently_replicated	256	4	sha256:83a401709b86991de5f38a49e7794379ea0c041fb2a406a245f900bf11bcb598
47	Classical Computation	SFT-COMP-CBL-RECOGNITION-DECISION-001	Fold recognition and decision law	depth_independent	independently_replicated	256	4	sha256:c9c992b18e4408ecfd5bb004622b3d62d836752b38773eecbfdfc0e1daed9e8
48	Classical Computation	SFT-COMP-CBL-HALTING-001	Fold halting boundary law	depth_independent	independently_replicated	256	4	sha256:a33ae0f08db036bcde17a25eb3d6605dcfbcd3fa0492176faaf553110d2368e2
49	Classical Computation	SFT-COMP-CBL-ENUMERATION-001	Fold effective-enumeration law	depth_independent	independently_replicated	256	4	sha256:b4a3d735406f9ab57023c47ed0a12a7a221b278c75727cb0a7170dbdfec940f2

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50	Classical Computation	SFT-COMP-CBL-REDUCTION-001	Fold computability-reduction law	depth_independent	independently_replicated	256	4	sha256:7d03f54baf0fc6d4bbe16f2ea73fac601ede53dceacbd55e2b032545196aaff2
51	Classical Computation	SFT-COMP-CBL-DEGREES-001	Fold degrees-of-computability law	depth_independent	independently_replicated	256	4	sha256:40189bc4fb7d083c3ffc7a7610e29e77cc5232bbde8514860c3afa328fe11a3e
52	Classical Computation	SFT-COMP-CBL-UNIVERSAL-MACHINE-001	Fold universal-machine execution law	depth_independent	independently_replicated	256	4	sha256:63466fb169111c50266901bf7ac48255ced18ef727cd66d1f3b376b6beaa7633
53	Classical Computation	SFT-COMP-CBL-UNDECIDABILITY-001	Fold undecidable-problem law	depth_independent	independently_replicated	256	4	sha256:9de0276cb48ddbcea65e0a35f6f2deeac179a225ac85a8f93121e5e7ed97e692
54	Classical Computation	SFT-COMP-CBL-INCOMPLETENESS-001	Fold incompleteness-boundary law	depth_independent	independently_replicated	256	4	sha256:f7fb969bc4fc339ff473e5ec2620e703695e6a12c3ab0b704594e9664e9ea277
55	Classical Computation	SFT-COMP-CBL-RELATIVE-ORACLE-001	Fold relative and oracle-computation law	depth_independent	independently_replicated	256	4	sha256:2a09b83377a56d06136df6922d9e9830c3b336e3487d3654d349c7f6f6746d01
56	Classical Computation	SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001	Fold hypercomputation admissibility-limit law	depth_independent	independently_replicated	256	4	sha256:3d6f7b8fdd8b1ccaf30622dad922140dd6b441b19df9dd4b76ff17ee191439cc
57	Classical Computation	SFT-COMP-CPLX-INPUT-SIZE-001	Fold input and problem-size law	depth_independent	independently_replicated	256	4	sha256:3f002d11d6315fc6f32536022cef2a3fad2bf857ea1761637481a827a8a2fc4
58	Classical Computation	SFT-COMP-CPLX-TIME-SPACE-001	Fold time and space resource law	depth_independent	independently_replicated	256	4	sha256:d6187e34cd91181461fa6614ebb82d710719cf6bebf4644d1bc08c81641df62b
59	Classical Computation	SFT-COMP-CPLX-CIRCUIT-RESOURCE-001	Fold circuit size and depth law	depth_independent	independently_replicated	256	4	sha256:9f3e28733488ad63f954689caad30204963adbe8e2fa60f985be1140722116ae
60	Classical Computation	SFT-COMP-CPLX-COMMUNICATION-QUERY-001	Fold communication and query complexity law	depth_independent	independently_replicated	256	4	sha256:c352ad5637226cc1d41ecbc8fb375f9593e7928d4157a97e8b7050ac18d19f3e
61	Classical Computation	SFT-COMP-CPLX-RANDOMNESS-001	Fold randomness-resource law	depth_independent	independently_replicated	256	4	sha256:e4311dc1af067c40ad88072efc8c155d1647ad6b071c1003c48b0e597be64f8b
62	Classical Computation	SFT-COMP-CPLX-REVERSIBILITY-COST-001	Fold reversibility-cost law	depth_independent	independently_replicated	256	4	sha256:0a8552676c105fc6b3fdcd6aac90a7a68c428962c209460caacea2b81c496ed2
63	Classical Computation	SFT-COMP-CPLX-PARALLEL-001	Fold parallel-complexity law	depth_independent	independently_replicated	256	4	sha256:d451551a66fdaf127bfd2e9738af1c7818753ea3dd2862a46db3108e32354424
64	Classical Computation	SFT-COMP-CPLX-BOUNDS-001	Fold lower- and upper-bound law	depth_independent	independently_replicated	256	4	sha256:7ba9b6310c1f0dd170c3c9b6abc0db4c257241df7af511faf3fb232b2d39db02

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65	Classical Computation	SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001	Fold reduction and completeness law	depth_independent	independently_replicated	256	4	sha256:401b5e76d63dd3d31c2a38156dd3fab752e10b289cb7b0e38c9a71f0cf6f673a
66	Classical Computation	SFT-COMP-CPLX-AVERAGE-WORST-001	Fold average- and worst-case law	depth_independent	independently_replicated	256	4	sha256:231135bfa6605c585af24c7a6f89d21b099f42462dc4b7a6c0820d761bce6bfb
67	Classical Computation	SFT-COMP-CPLX-APPROXIMATION-001	Fold approximation-complexity law	depth_independent	independently_replicated	256	4	sha256:e8062d9de51f472dc26ca19cd2e838de843c9b1449d16c71962ad2c06e9f8357
68	Classical Computation	SFT-COMP-CPLX-PARAMETERIZED-001	Fold parameterized-complexity law	depth_independent	independently_replicated	256	4	sha256:1afcbcd7d33437e51f057b4a63715fc3c5e41890025b3d093edb4fbeb15b370ac
69	Classical Computation	SFT-COMP-CPLX-DESCRIPTIVE-001	Fold information and descriptive-complexity law	depth_independent	independently_replicated	256	4	sha256:bdbb14bfbdd53b05dd80b9b4fd0ea77e3f9eb303ab637ec6714029665a1f6eef
70	Classical Computation	SFT-COMP-ALG-SEARCH-ORDER-001	Fold searching and ordering law	depth_independent	independently_replicated	256	4	sha256:915d9a65bf080d49519f15583d8ffab68375f99e1af014be611cdc25c6626c78
71	Classical Computation	SFT-COMP-ALG-ARITHMETIC-001	Fold exact arithmetic-algorithm law	depth_independent	independently_replicated	256	4	sha256:af254db83c3bc4c67b1c2daa5195c8f74a1a9e4fbcd08b159c677579b9c1ab52
72	Classical Computation	SFT-COMP-ALG-STRINGS-SEQUENCES-001	Fold string and sequence algorithm law	depth_independent	independently_replicated	256	4	sha256:5c2fb101d63e8d37ba076927e1de853261936a53c33400eba91c9adbd135caaf
73	Classical Computation	SFT-COMP-ALG-TREES-GRAPHS-001	Fold tree and graph algorithm law	depth_independent	independently_replicated	256	4	sha256:0c1d76188752d9f584b9bec6db855143f954a2f65b9ad57375bcd3e68809f604
74	Classical Computation	SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001	Fold algebraic and geometric algorithm law	depth_independent	independently_replicated	256	4	sha256:b2ba1fc787d153cd8673cf1166703cd608291c7136d7412bff9c9dc66f3de430
75	Classical Computation	SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001	Fold dynamic-programming law	depth_independent	independently_replicated	256	4	sha256:f6b1c405c189bb4f86db10ec31ad2dc8adc8041e0833baeb6184ac1832ed2e17
76	Classical Computation	SFT-COMP-ALG-OPTIMIZATION-001	Fold optimization-algorithm law	depth_independent	independently_replicated	256	4	sha256:74aac9551388975891fa29de8f8a7843eb1afcbcd2a3df316784f34caf7ebd110
77	Classical Computation	SFT-COMP-ALG-RANDOMIZED-001	Fold randomized-algorithm law	depth_independent	independently_replicated	256	4	sha256:1c9f175ed883e3267b8722b8481cc424959ae5980bab9a6cf81e691d0785e0ae
78	Classical Computation	SFT-COMP-ALG-PARALLEL-001	Fold parallel-algorithm law	depth_independent	independently_replicated	256	4	sha256:c16c0747c8994451d421ddb5bc14c4715d7cd4ea6c78588c3da0d318a66ca285
79	Classical Computation	SFT-COMP-ALG-DISTRIBUTED-001	Fold distributed-algorithm law	depth_independent	independently_replicated	256	4	sha256:5dfc71b4941c7e3f629b7ee0ab6cd3ca1851222fbfaf68690fa67875ed1f7e60

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80	Classical Computation	SFT-COMP-ALG-ONLINE-STREAMING-001	Fold online and streaming algorithm law	depth_independent	independently_replicated	256	4	sha256:1978c4757d70afe4f9f64a74ddad4b9df88db850c160cb3840dbe18f2ada7a15
81	Classical Computation	SFT-COMP-ALG-NUMERICAL-001	Fold numerical-algorithm law	depth_independent	independently_replicated	256	4	sha256:3f1c19281f4ac6c86c5bb13ae391ede42fa858c66e0571a4d6d66f27eeec098f
82	Classical Computation	SFT-COMP-ALG-SYMBOLIC-001	Fold symbolic-computation law	depth_independent	independently_replicated	256	4	sha256:3a84f78267c8d51b1e8c1ff0981ba89929b1c2b5231b5fedcb0d522b87f9fa8d
83	Classical Computation	SFT-COMP-ALG-APPROXIMATE-001	Fold approximation-algorithm law	depth_independent	independently_replicated	256	4	sha256:52dfa606dc9c680d10c2c990b5df3ed29fe4f91be95690eb0d4880df5fd7dffc
84	Classical Computation	SFT-COMP-SEM-SYNTAX-001	Fold program-syntax law	depth_independent	independently_replicated	256	4	sha256:0da324b3d1cd63a836a8494235259bffe143a912f6aace6745f55c1bcae10246
85	Classical Computation	SFT-COMP-SEM-BINDING-SUBSTITUTION-001	Fold binding and substitution law	depth_independent	independently_replicated	256	4	sha256:1200dc420f7ff8fcd37e5c3fd7e862fa3aa0d1466963d23d9fe3ce0a22ac103d
86	Classical Computation	SFT-COMP-SEM-EVALUATION-001	Fold evaluation law	depth_independent	independently_replicated	256	4	sha256:dfb202d42339a8c83ffbd8d1939ec3d6dd29819fb621b60f6dd559175315089a
87	Classical Computation	SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001	Fold operational and denotational correspondence law	depth_independent	independently_replicated	256	4	sha256:6db365c1b3c22883c276eec14485e6fb962528278f521654cba633fe6536e425
88	Classical Computation	SFT-COMP-SEM-TYPE-001	Fold type-theory law	depth_independent	independently_replicated	256	4	sha256:f62ebd67f072c2e126b16d47d16cf1982426bc5c333be0c79d9148e43a505620
89	Classical Computation	SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001	Fold program-equivalence law	depth_independent	independently_replicated	256	4	sha256:ad7d0dba57a7c30f17d67ef1e94ba22b9fe9e126565b894d97f435d1f83e4a17
90	Classical Computation	SFT-COMP-SEM-TERMINATION-001	Fold termination law	depth_independent	independently_replicated	256	4	sha256:dc2378dbe15b3cfe2521a462c2bae6a7cef270b027719a9345618a9f1971c752
91	Classical Computation	SFT-COMP-SEM-CORRECTNESS-001	Fold program-correctness law	depth_independent	independently_replicated	256	4	sha256:95af5c4c8a0cf03d91f3d1ceff4decef32ba66537923e85eebe51ca3335e78bf
92	Classical Computation	SFT-COMP-SEM-SPECIFICATION-001	Fold formal-specification law	depth_independent	independently_replicated	256	4	sha256:4aeb9f85a830e501877b8cc19d41095d2bbf9c96656ff92be8fe5a6655d351a51
93	Classical Computation	SFT-COMP-SEM-TRANSFORMATION-001	Fold program-transformation law	depth_independent	independently_replicated	256	4	sha256:cdc9731adba6f15b42e9eeba2e0c93d29bfeadf4ae281fbfed527d54918a7226
94	Classical Computation	SFT-COMP-SEM-COMPILATION-001	Fold semantics-preserving compilation law	depth_independent	independently_replicated	256	4	sha256:1d1314650abf2f20bb952ce22c2d83de00e3bb90fa515fb5f25560c3e3bb96ae

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95	Classical Computation	SFT-COMP-SEM-VERIFICATION-001	Fold verification and proof-system law	depth_independent	independently_replicated	256	4	sha256:e3312b01e5f796d522ced745f449997e595aade4f15b3b82b216840116f09f97
96	Classical Computation	SFT-COMP-DIST-CONCURRENCY-001	Fold concurrency law	depth_independent	independently_replicated	256	4	sha256:a3a2d3ef12a8166cc2117b17c3a470314146123a2c474c73ab363c2d5cf2e1c7
97	Classical Computation	SFT-COMP-DIST-CAUSALITY-001	Fold computational-causality law	depth_independent	independently_replicated	256	4	sha256:902e87f46e187a03a813011d318a637eaaab7c9fc3dedfc0fc2c59efcb0dffccf
98	Classical Computation	SFT-COMP-DIST-SYNCHRONIZATION-001	Fold synchronization law	depth_independent	independently_replicated	256	4	sha256:e1b5a01e71410f40f763377dcfc336f2d75bc664db00a510dbba8a075fa5aa0f
99	Classical Computation	SFT-COMP-DIST-COMMUNICATION-001	Fold communication law	depth_independent	independently_replicated	256	4	sha256:59defe90f59d3eeef926714873a97cc9e2a4d9d14391e5d6d7458f51687b23423
100	Classical Computation	SFT-COMP-DIST-PARTIAL-ORDER-001	Fold distributed partial-order law	depth_independent	independently_replicated	256	4	sha256:d04af273dac3a8a8cd3d64b1b7c72e3130c1e674320c6632b0c2b9e2a8225a0f
101	Classical Computation	SFT-COMP-DIST-CONSENSUS-001	Fold consensus law	depth_independent	independently_replicated	256	4	sha256:48ae9b96daea76ffeeffb0c3b2b9869b3beea88dc03cfd1b5a90e6f250a4c4d1e
102	Classical Computation	SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001	Fold agreement-impossibility law	depth_independent	independently_replicated	256	4	sha256:30b83eb403fb428502002c7b9d96f569a91195720f36d8e1675385e92919f48d
103	Classical Computation	SFT-COMP-DIST-REPLICATION-CONSISTENCY-001	Fold replication and consistency law	depth_independent	independently_replicated	256	4	sha256:79243bdec0bda3d90c090785f5a2b43444ff4a323b5d7b48f0a36f17c11dcf34
104	Classical Computation	SFT-COMP-DIST-FAULT-MODEL-001	Fold computational fault-model law	depth_independent	independently_replicated	256	4	sha256:812aad3547d57947e1e18963470f883725de08c4e299351306e308a1923bdf46
105	Classical Computation	SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001	Fold distributed-knowledge law	depth_independent	independently_replicated	256	4	sha256:2e47ee8465ef950f9ad2d30eb331453e764abf2567a6fc83ed0ec99b9fceda41
106	Classical Computation	SFT-COMP-DIST-LOCALITY-001	Fold locality law	depth_independent	independently_replicated	256	4	sha256:d75dcc0dc6937266743f829559bfe4b59dfe31fc0d2b186ccbd7979b13cba9fa
107	Classical Computation	SFT-COMP-DIST-NETWORK-COMPUTATION-001	Fold network-computation law	depth_independent	independently_replicated	256	4	sha256:504861c17b0bc318c7dc6cfbf43377ac42819e316735ab1b066b4dd11bb1951
108	Classical Computation	SFT-COMP-SEC-ONE-WAYNESS-001	Fold one-wayness law	depth_independent	independently_replicated	256	4	sha256:a4de8e0ec3e624835896831a4545adc63b5aa37cb7b2cecb26404bda0175eb62
109	Classical Computation	SFT-COMP-SEC-SECREC-001	Fold secrecy law	depth_independent	independently_replicated	256	4	sha256:73a1685ef3e59a0b4cf8b8b67a1590f8a3b94e1dd5ace2d7eff0a6bdd5183b74

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110	Classical Computation	SFT-COMP-SEC-INTEGRITY-001	Fold integrity law	depth_independent	independently_replicated	256	4	sha256:54e959203614f553ff7ac956f549af5e057a240855b4175f237fb7363e080943
111	Classical Computation	SFT-COMP-SEC-AUTHENTICATION-001	Fold authentication law	depth_independent	independently_replicated	256	4	sha256:2c39303d9b2dd557650fe04f8acb2f132510ca3af8258dc10792a276796cb29c
112	Classical Computation	SFT-COMP-SEC-HASHING-001	Fold hashing law	depth_independent	independently_replicated	256	4	sha256:30adbff79b53883ef1c3266d37eabc5389dc245bd879ff5be7c3c81df62767984
113	Classical Computation	SFT-COMP-SEC-COMMITMENT-001	Fold commitment law	depth_independent	independently_replicated	256	4	sha256:402789a36ddf90549527f52c193fbb0d0065ed548efae8b178b629a367c59e5f
114	Classical Computation	SFT-COMP-SEC-SIGNATURE-001	Fold signature law	depth_independent	independently_replicated	256	4	sha256:1e80fcc3c81615ba2f131e83a91759abecf9a02f7a23b0a19b191a612d820bce
115	Classical Computation	SFT-COMP-SEC-PROOF-KNOWLEDGE-001	Fold proof-of-knowledge law	depth_independent	independently_replicated	256	4	sha256:30375e18292a970fd81f642586376049976d61928fff9bfff79bb0ee0692f0eb6
116	Classical Computation	SFT-COMP-SEC-ZERO-KNOWLEDGE-001	Fold zero-knowledge law	depth_independent	independently_replicated	256	4	sha256:060ee3e0322b39395749298b995a9f4d67c148ade8eeba389c05906219d720ed
117	Classical Computation	SFT-COMP-SEC-MULTIPARTY-001	Fold multiparty-computation law	depth_independent	independently_replicated	256	4	sha256:211b27cbbbc88f0e82c4441ed0d47afe01703b5dd89892a3e35759025c5f29eb
118	Classical Computation	SFT-COMP-SEC-ADVERSARIAL-001	Fold adversarial-computation law	depth_independent	independently_replicated	256	4	sha256:feaa06a2fbc1430f8930075dad167efdfcb60df06ed5e21424557ce89b81b78
119	Classical Computation	SFT-COMP-SEC-SECURITY-DEFINITION-001	Fold information-theoretic and computational-security law	depth_independent	independently_replicated	256	4	sha256:e4af2a902c19997a8a81d6df7fe8ebd8246a2900aac642a6003b287d4bde7007
120	Classical Computation	SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001	Fold post-quantum security-boundary law	depth_independent	independently_replicated	256	4	sha256:4e450e475eb74f6968064e4ba58530226f4a58f8a6c3fff12176306a4d028c4d
121	Classical Computation	SFT-COMP-LEARN-INFERENCE-001	Fold inference law	depth_independent	independently_replicated	256	4	sha256:483c9c1cb618e3c57985fbbcd9d4c3450b6dc50874c3f86a5a09fd4a8e54160
122	Classical Computation	SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001	Fold classification and prediction law	depth_independent	independently_replicated	256	4	sha256:2902c0a1035fc27073a7a1be9ad39806838e05a1ef7fb2cbf8e045f1705be080
123	Classical Computation	SFT-COMP-LEARN-REPRESENTATION-001	Fold representation-learning boundary law	depth_independent	independently_replicated	256	4	sha256:aaf1bb06300eddd353f33d549cc9ee8ace48e5f131d530ff9b9b7976c1cd0061
124	Classical Computation	SFT-COMP-LEARN-GENERALIZATION-001	Fold generalization law	depth_independent	independently_replicated	256	4	sha256:c99db1b2646ffad5f54e060cf1ae1b32a9d8ebdc719d36a6e3aec10abf446d

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125	Classical Computation	SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001	Fold sample-complexity law	depth_independent	independently_replicated	256	4	sha256:d4b13522a727a382b31bddb83eda909f6706c4f7b9bc48633b8e1fd28c133092
126	Classical Computation	SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001	Fold optimization-in-learning law	depth_independent	independently_replicated	256	4	sha256:543256016991fee945bc9d01d128355d0a9777a814dc7428ff9146ac93a8a398
127	Classical Computation	SFT-COMP-LEARN-INDUCTION-001	Fold computational-induction law	depth_independent	independently_replicated	256	4	sha256:919a71359ffd05922865e02e03ef9013abb551bc0ccb5b0b95363aee5cd998ed
128	Classical Computation	SFT-COMP-LEARN-SEARCH-PLANNING-001	Fold search and planning law	depth_independent	independently_replicated	256	4	sha256:fae483c37bc6bdfa9b29553a47c1620a9038fb736fd8cf2af77fe96535e780c7
129	Classical Computation	SFT-COMP-LEARN-REINFORCEMENT-001	Fold reinforcement-computation law	depth_independent	independently_replicated	256	4	sha256:0cb38ecfea78976c1452c6d9974fff417f27c4cb05a6721698d9d8c1c4328a1a
130	Classical Computation	SFT-COMP-LEARN-MULTIAGENT-001	Fold multi-agent computation law	depth_independent	independently_replicated	256	4	sha256:9d95cd6493bea3f2e926a8be970becf6593a0aad4c08b4ea2f999e6760202a2
131	Classical Computation	SFT-COMP-LEARN-ADAPTATION-001	Fold adaptation law	depth_independent	independently_replicated	256	4	sha256:64ec3907b0e627e9e06f675e15a8ee10818781609c850b20948008691aa9d1cb
132	Classical Computation	SFT-COMP-LEARN-LEARNING-LIMITS-001	Fold computational limits-of-learning law	depth_independent	independently_replicated	256	4	sha256:969046a8a85fc0802d0269d6a4a46e706d278b71450e8b7c7dbd1188fe21bc90
133	Classical Computation	SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001	Fold interpretability and verification law	depth_independent	independently_replicated	256	4	sha256:c0246290a461042d4797f427dfd27adaebf0954e5372091899b896b3418a1424
134	Classical Computation	SFT-COMP-LEARN-CLASSICAL-LEARNING-001	Fold classical-learning model law	depth_independent	independently_replicated	256	4	sha256:86cdfa5fb4a647883b4b1300ea87c83dddd01b8f34f96c3738fec0baa8f8a8ce
135	Classical Computation	SFT-COMP-SCI-EXACT-APPROXIMATE-001	Fold exact and approximate calculation law	depth_independent	independently_replicated	256	4	sha256:d04dba138ab4ed09a8fba0c3751fc8588a7ad3b34790ac1f536b44b013b1a905
136	Classical Computation	SFT-COMP-SCI-STABILITY-001	Fold numerical-stability law	depth_independent	independently_replicated	256	4	sha256:f50a038a131659cc3156af54451f66c24d7562bb4284ab1ad6a71b6e97cc6980
137	Classical Computation	SFT-COMP-SCI-ERROR-PROPAGATION-001	Fold error-propagation law	depth_independent	independently_replicated	256	4	sha256:083dc87700741205332c48c4a74aa5658b9f205141bcb39a13bb88288354230b
138	Classical Computation	SFT-COMP-SCI-DISCRETIZATION-001	Fold discretization law	depth_independent	independently_replicated	256	4	sha256:bcbf98ba0535b49f76382780ec0f1e22b9f184249bb3ccb596463f48d0532080c
139	Classical Computation	SFT-COMP-SCI-CONVERGENCE-001	Fold convergence-certificate law	depth_independent	independently_replicated	256	4	sha256:a3ae509175198d515e8ae636ad4b669d942524409b96f97c4ce26a94a98c03b5

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140	Classical Computation	SFT-COMP-SCI-SYMBOLIC-001	Fold scientific symbolic-computation law	depth_independent	independently_replicated	256	4	sha256:484049210c99c3cf945baa6cf8f64f14beb56db6f50fe8ede3d322dee49c41b8
141	Classical Computation	SFT-COMP-SCI-SIMULATION-001	Fold simulation law	depth_independent	independently_replicated	256	4	sha256:b06dd069154f18ea84aadfe5c47b070e0ec017113e01945e55690303f274e293
142	Classical Computation	SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001	Fold computational-dynamics law	depth_independent	independently_replicated	256	4	sha256:9f40f7a193ca6e09bbc51fd61d7a526dc81c9224a6797bac0c0c57e8005ae0de
143	Classical Computation	SFT-COMP-SCI-INVERSE-PROBLEM-001	Fold inverse-problem law	depth_independent	independently_replicated	256	4	sha256:41ca83901e2ba93484c943df39462f4196c046678f4d9cb17afafb5277cacdde
144	Classical Computation	SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001	Fold computational-statistics law	depth_independent	independently_replicated	256	4	sha256:af3c12e592f489613af6f00e0bed93b948f0ccc0a7f975ec89a55fce888ceffa
145	Classical Computation	SFT-COMP-SCI-HIGH-DIMENSIONAL-001	Fold high-dimensional computation law	depth_independent	independently_replicated	256	4	sha256:08b4870cfa97a44e956720ebf6ccfef2216790ad6ffcc0261c127f46625f84d
146	Classical Computation	SFT-COMP-SCI-MANY-BODY-001	Fold many-body computation law	depth_independent	independently_replicated	256	4	sha256:4e4db714accde19f571063b6d1c1c64c9f07650378a797107cb424bea54de4ca
147	Classical Computation	SFT-COMP-SCI-MATHEMATICAL-MODELLING-001	Fold mathematical-modelling law	depth_independent	independently_replicated	256	4	sha256:465c067c8c543effc5ba03cf8677a5caa2386f4695d1f3905b6e652060ba7ed4
148	Reversible and Quantum Computation	SFT-QUANTUM-REVERSIBLE-MODEL-001	Complete Fold reversible-computation model	depth_independent	independently_replicated	256	4	sha256:068b389a38d47c06118600324bc263c951e6e19c6f591f257c601f6125525298
149	Reversible and Quantum Computation	SFT-QUANTUM-INFORMATION-UNIT-001	Fold quantum information-unit law	depth_independent	independently_replicated	256	4	sha256:3cae970ebb8336d464bcb64e17d0f50049da6f018b95f98b53b04a46873bf72
150	Reversible and Quantum Computation	SFT-QUANTUM-STATE-COMPOSITION-001	Fold quantum state-composition law	depth_independent	independently_replicated	256	4	sha256:a358610587ee2c7c6dfcba8a782e46f89a1668e2c86739eac0e8c88d6af719c2
151	Reversible and Quantum Computation	SFT-QUANTUM-SUPERPOSITION-001	Superposition-equivalent Fold support law	depth_independent	independently_replicated	256	4	sha256:54c09712a4a53c16054fc2c8de1e247007b718e1f6f0cd9cd38a14b1a5cbe7d1
152	Reversible and Quantum Computation	SFT-QUANTUM-PHASE-INTERFERENCE-001	Fold phase and interference operation law	depth_independent	independently_replicated	256	4	sha256:c1c7e3f2c71ea7de6d801c397c ff01ae8a005651deddccc7acd83bddc857f12
153	Reversible and Quantum Computation	SFT-QUANTUM-ENTANGLEMENT-001	Fold entangling-composition law	depth_independent	independently_replicated	256	4	sha256:6bfa64c6f896c5f7eaa3a55872f31cd487923853353a3069ec7a839afa1d3d0d
154	Reversible and Quantum Computation	SFT-QUANTUM-MEASUREMENT-001	Fold quantum measurement-semantics law	depth_independent	independently_replicated	256	4	sha256:e29a71f7dff97835e28850ee8d352ac80aa9ceabb4399e07105d892175776a06

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155	Reversible and Quantum Computation	SFT-QUANTUM-GATE-001	Fold reversible transformation and gate law	depth_independent	independently_replicated	256	4	sha256:495704206fd6c701e1baf8d1ec0b4bcfd430a002aea0bb76453a091e01ccfdaf
156	Reversible and Quantum Computation	SFT-QUANTUM-CIRCUIT-001	Fold quantum circuit syntax and semantics law	depth_independent	independently_replicated	256	4	sha256:63e751cf9ab382cdc5c4b5dfd0c6c2f01ad90659bf2c9189357f141475d9e8aa
157	Reversible and Quantum Computation	SFT-QUANTUM-UNIVERSALITY-001	Fold quantum universality law	depth_independent	independently_replicated	256	4	sha256:28b6a73e02450c70bf3ee49f3fcdcf2e28b30c463e05218b99cdc8cda723e86a
158	Reversible and Quantum Computation	SFT-QUANTUM-ALGORITHMS-001	Fold quantum algorithm-family law	depth_independent	independently_replicated	256	4	sha256:a33b13375d76ebf039fcc5711c0810ab99ac0b9456295bc7881d76c94558a86b
159	Reversible and Quantum Computation	SFT-QUANTUM-COMPLEXITY-001	Fold quantum complexity law	depth_independent	independently_replicated	256	4	sha256:d6307d961273c2a383fec3b919c56319b910bccb58dbbcf5402a2a76a9ca8b6
160	Reversible and Quantum Computation	SFT-QUANTUM-COMMUNICATION-001	Fold quantum communication law	depth_independent	independently_replicated	256	4	sha256:01059b1cbe86b894ae6137de5625e1dfcd1cd640ac5634c0ca2260977993b97a
161	Reversible and Quantum Computation	SFT-QUANTUM-CODING-001	Fold quantum coding law	depth_independent	independently_replicated	256	4	sha256:b79afc51951dfbebb30cd70fba329965b700fa14a16fb45b5754568ec7407a3e
162	Reversible and Quantum Computation	SFT-QUANTUM-ERROR-CORRECTION-001	Fold quantum error-correction law	depth_independent	independently_replicated	256	4	sha256:a4791dcfbf8753c32e8304c5d5e80f9bf7f21a7356b6adb9422ea60d5eaf59656
163	Reversible and Quantum Computation	SFT-QUANTUM-FAULT-TOLERANCE-001	Fold quantum fault-tolerance law	depth_independent	independently_replicated	256	4	sha256:9f2526c5bd9f1625a3cee65b0cba4e9a600ecdc9a8031ac51367e35c37308b16
164	Reversible and Quantum Computation	SFT-QUANTUM-SIMULATION-001	Fold quantum simulation law	depth_independent	independently_replicated	256	4	sha256:daf61c89738fb9f6370bc244c3a7538a3c4037ab408dfd5d15e3301c57e817a6
165	Reversible and Quantum Computation	SFT-QUANTUM-VERIFICATION-001	Fold quantum verification law	depth_independent	independently_replicated	256	4	sha256:6ca70e1a9487b428bc6e04e5e3b1a11907b79d1ed92f7d03a013a49380ecce54
166	Reversible and Quantum Computation	SFT-QUANTUM-LEARNING-001	Fold quantum learning law	depth_independent	independently_replicated	256	4	sha256:c4388e69ff7f412fe09e53b7e4b2a4eda12168ff6665ca351758c185ac7e018b7
167	Reversible and Quantum Computation	SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001	Full operational classical-quantum correspondence law	depth_independent	independently_replicated	256	4	sha256:e1712589e1aba3ce6536a3ba33d599e06bdf793a335079d1a775975bc234b614
168	Reversible and Quantum Computation	SFT-QUANTUM-LIMITS-001	Limits of Fold quantum computation law	depth_independent	independently_replicated	256	4	sha256:332e3b1f6df5725ae3e5cf97d9080daa26fd501e7138f1b9d1c7b4837469c545
169	Physics	SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001	Capability-closed exact Fold prediction form	depth_independent	independently_replicated	256	4	sha256:bf675d47eb8731760292d542c65fb0af31349d313360d018ee0199a71e23f9a4

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170	Physics	SFT-PHYS-MEAS-TARGET-CUSTODY-001	Portable target commitment and post-seal custody	depth_independent	independently_replicated	256	4	sha256:288826223673d5472e818fb0c53d837710adcfd7abbabf88285100650e3c872
171	Physics	SFT-PHYS-MEAS-HOSTILE-PACKAGE-001	Hostile empirical-package and authority protection	depth_independent	independently_replicated	256	4	sha256:40d8e9ab7ac203495ac06fdf51cc708c5a4e1d528f8701212c9e562da6eaa16
172	Physics	SFT-PHYS-MEAS-OBSERVATION-CARRIER-001	Physical observation and retained distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8d7daef993773a0721b9bf6022aef3b22e9ea445608b80084ce496a35917d955
173	Physics	SFT-PHYS-MEAS-QUANTITY-CARRIER-001	Physical quantity as reference-equivalence carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c98453b691e68a1b5340a5c14d757c54c19f736fc89f9dbfb54c0ce7c7117033
174	Physics	SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001	Dimension composition and independence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a09e3abd01f98710265eecd2397d2cf3df06618ecb18a06c77ea92f703d3ac72b
175	Physics	SFT-PHYS-MEAS-UNIT-COMPARISON-001	Unit comparison and exact scale transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e9b8d7ccb1cd52ca2ab5600a51a684827e6cd36db3f79cd1a01301732bd90827
176	Physics	SFT-PHYS-MEAS-REFERENCE-REALIZATION-001	Reference realization and traceability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fe8d21bca9c2c1721cd7f14ebacc535705a185af0313af00a17f89d5877836f0
177	Physics	SFT-PHYS-MEAS-VALUE-RECORD-001	Measured-value record separated from proof value	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:461c2062551cbfd5aa884249eaea2ee9ae5dbabbe672885eb764deda2e0fa01d
178	Physics	SFT-PHYS-MEAS-UNCERTAINTY-001	Measurement uncertainty as retained admissible alternatives	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9aef08a488c0a15fc4dd2132a1afd9c32d45bb0da6e6ed5c76f90c541f5ffc01
179	Physics	SFT-PHYS-MEAS-CALIBRATION-001	Calibration chain and comparison closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:60c5b750220a91b600f9752669227be225c67e5ec8abb7d760f023557ace561a
180	Physics	SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001	Dimensional consistency and lawful conversion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2e23d3f162cf66416ed05f77336bb9a95254004d09fc44e51e65d2f3b5102984
181	Physics	SFT-PHYS-MECH-EVENT-CHANGE-001	Physical event and generated change	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:71e815aed3d0ceea7f5d5df4ee3d2037e5e3f6fcf5803e7ac66492ef724f9b9c
182	Physics	SFT-PHYS-MECH-DURATION-001	Duration from ordered recurrence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:90b0ed7f1d52601eef77629cc3c1502f2bc9d9c169a5421c06d33d0c6334b649
183	Physics	SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001	Location, separation and held displacement orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fc78abed974ceb7306725b8ac1fa140871e568dcf9d8f37f119dd76138f084ca
184	Physics	SFT-PHYS-MECH-SPEED-VELOCITY-001	Speed and oriented velocity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:032489b9c3a0c26462074860c3419af02d49fb9fcc4ba4384b7f0807881d655e

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185	Physics	SFT-PHYS-MECH-ACCELERATION-001	Change of velocity and acceleration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3726d5e4e02d4b6ab8e04a7dbc2ed1c36a07fccdc05fb44a508a41ddfa27ea6
186	Physics	SFT-PHYS-MECH-INERTIA-001	Inertial persistence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b87863a7c28e6cf7ed9716bac77c53072bb2941651c0368fb542e1b8bb153bc8
187	Physics	SFT-PHYS-MECH-MOMENTUM-001	Momentum carrier and transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9c67363fe5ef6783145dea27a49321411c8b0f3190ec67723878d5a5b277181c
188	Physics	SFT-PHYS-MECH-FORCE-001	Force as constrained momentum transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:55c476d418d54f6bce48a366b936a40c26c77c0ca04ae64e7d2706f1ecc3169d
189	Physics	SFT-PHYS-MECH-WORK-ENERGY-001	Work, kinetic change and energy carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2907d13e9cf208298ac2de4c0b22d6e17db2ca0df3aaf420d4dc9f27fa72d8c0
190	Physics	SFT-PHYS-MECH-POWER-001	Power as ordered energy transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:986113fa723ac5827d698eab494abe9cddfb4054e4629d442d91a0a91f486ecd
191	Physics	SFT-PHYS-MECH-ANGULAR-MOTION-001	Angular motion and cyclic orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f38ac92829c84d31410cbe3fe0d03a2ac5ff997ebdad4e41473498324f322db7
192	Physics	SFT-PHYS-MECH-ANGULAR-MOMENTUM-001	Angular momentum and torque	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b82a7f81b1904737363b439e1d85e573a0dd55fee628475ff139e764c767e28d
193	Physics	SFT-PHYS-MECH-COMPOSITE-CENTRE-001	Composite motion and centre relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ac98f7c19cb0a90e85cb8c728d0728482868cf79c54d6b8a861c8085a1ccf403
194	Physics	SFT-PHYS-MECH-CONSERVATION-001	Mechanical conservation under closed transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7d02feaf423abbf05b62b69c7364f256f4a576c53437611ea6d22b3b06261b5b
195	Physics	SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001	Constrained motion, restoration and oscillation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ad7387e41663cd1f57d403b58863dea2dbd6aa0f0f890a5e1807acaade0b9d46
196	Physics	SFT-PHYS-FIELD-SOURCE-RESPONSE-001	Source, response and field carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f4564473fe6fa51d1f5a4a777b37fb53f9ccf839fff4a2a098e2c25ce4bd5fd7
197	Physics	SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001	Generated support and geometric dilution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9f1c50dbea066ec0e34bc50213ac0f9e7b4e56d35609cb4518927d48b235992b
198	Physics	SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001	Gravitational interaction at the weak relational boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:af25bd6d7cd4af55c92fc4fb93ed4fa2ba2d8dbcdc13a9d22d478a5868a58bed
199	Physics	SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001	Electric distinction and held charge orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3b3e69fb0fa71108a908f8b7299da47df8aa635f40938ae2bddc89bdc72071d8

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200	Physics	SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001	Electric field, potential and work correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c64eacc598d02d73e0d9994f6525b45f4c12a7334e28d7f98351d6c4dfcda6de
201	Physics	SFT-PHYS-FIELD-MAGNETIC-001	Magnetic response to oriented change	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:74b71a149a6537d5819031f72cc561a9e392c7ae5b0fc0985b2d560e0f96d146
202	Physics	SFT-PHYS-FIELD-INDUCTION-001	Induction from changing linked support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b76cd4b60c0f7e29b7a215c36dcd0a400d260df284b2735aaf32617730818f4f
203	Physics	SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001	Electric-magnetic compositional closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4f454fc66603a589ecd4a1cd581d9b51038b33b6f79880f07a168db613027245
204	Physics	SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001	Gauge-equivalent descriptions and retained observables	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c9be792f2d288ae0f370d83dba86c1f4ff503e004ee556fd9363174f6f360412
205	Physics	SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001	Local propagation and causal accessibility	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ddcd093c9d9e17b12469f8992d6cc7bb1458c867ce1a68cffcf6ccc944dd2a77
206	Physics	SFT-PHYS-FIELD-RADIATION-001	Radiative transport and source detachment	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9faled9f5907e60d3c439eebfce928a8a04bec7abf067419b58d0a16a3544361
207	Physics	SFT-PHYS-FIELD-INTERACTION-CLASSES-001	Empirical closure of physical interaction classes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1cc5065d8cae35106aa07ac1b2c6e5fab9685899dff87886320753352e99d0d
208	Physics	SFT-PHYS-FIELD-CONSERVED-SOURCE-001	Source continuity and field conservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:901fcb7327185f1311f179ed4064294e60bf388259a2432f1b3021cbbecba89
209	Physics	SFT-PHYS-FIELD-ACTION-REACTION-001	Reciprocal interaction and transferred distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94eb13578b043cc3a4608bca8bcfb91f9d02f9bec1a12168b3b10d42fc6444f5
210	Physics	SFT-PHYS-WAVE-PERIOD-FREQUENCY-001	Period, recurrence and frequency	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1685681514cfcc033abdaa8cf880a85f252fb69f74f4c1a0e87d8b6581c9c8c5
211	Physics	SFT-PHYS-WAVE-PROPAGATION-001	Generated wave propagation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c5d9e3e5665cc460856932760cb46eb6a7e97867c7487b1f8caca6e2abb47588
212	Physics	SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001	Propagation speed, wavelength and frequency relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d37f2a40ad5f1900c80c049450bbcac88566cd8e1141a4c9b8412a12da209f
213	Physics	SFT-PHYS-WAVE-SUPERPOSITION-001	Wave superposition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1dea738715f2417b0bdc973dc0b2bb0869319d76ec7d426bfc4852710aeca08c
214	Physics	SFT-PHYS-WAVE-INTERFERENCE-001	Wave interference and predecessor merging	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:383616e13353085d904ad29e5ec14ef4012ac635457e5a86d1067266c7c9f586

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215	Physics	SFT-PHYS-WAVE-DIFFRACTION-001	Diffraction at constrained support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1f3027481808a8defa7f394581a65b27fd8bd1d46733f600da60e4c00cf17fd2
216	Physics	SFT-PHYS-WAVE-POLARIZATION-001	Polarization as held transverse orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:611f1b82de00ee945db78924f2951654cefbf5cdd65ac085e8899521b1601461
217	Physics	SFT-PHYS-WAVE-DISPERSION-001	Dispersion and support-dependent propagation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bb15242d9bc421a4ca99ced8941720d9eff92d99c5346e9d2a8336b9ed64450e
218	Physics	SFT-PHYS-WAVE-RESONANCE-001	Resonance and recurrence matching	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:60ae8de521a339460b1d6dc562d275ecba316910f45098981bd3e3b658ceb97a
219	Physics	SFT-PHYS-WAVE-ENERGY-MOMENTUM-001	Wave energy and momentum transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e5984b8f7868064d5ceaa61b26a713dc9d128729fdaae80cafcdeb7986c16876
220	Physics	SFT-PHYS-THERMO-MICRO-MACRO-001	Microstate-to-macrostate observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8b653cd127559fdfc1c7e47ab71c2f17d32f4de088c045a305e44a00abe4bc27
221	Physics	SFT-PHYS-THERMO-TEMPERATURE-001	Temperature carrier and thermal ordering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:120247c26273364ae2733af59411df4ff71663c9a06bf477f3902e0b13f3f304
222	Physics	SFT-PHYS-THERMO-EQUILIBRIUM-001	Thermal equilibrium equivalence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cbd86e5cf8d1493d04fba245ae333d5933fd6afbe11800e515f6651a1a86d7c4
223	Physics	SFT-PHYS-THERMO-HEAT-WORK-001	Heat and work as distinct energy-transfer traces	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:de7c30a2d5ab28ca55a30d35e4973d6197a3db788ad47d8604dca2b49d98e57c
224	Physics	SFT-PHYS-THERMO-FIRST-LAW-001	Closed energy accounting	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7362f8e65ebb689e893823f347b2fdedc17fd5e01e858dff505997936acb79a8
225	Physics	SFT-PHYS-THERMO-ENTROPY-001	Thermodynamic entropy from unresolved support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3ff19937b6ea84cc3aa3220af340ad3e651be1e7fa1bf3928af4f009634769ee
226	Physics	SFT-PHYS-THERMO-SECOND-LAW-001	Irreversible observation and entropy orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:65d57af5d78d0fd2f014c0ed3bc1184cf05bf226566e2b0c951f2fe51b7b0a44
227	Physics	SFT-PHYS-THERMO-THIRD-LAW-001	Finite unattainability boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:35ac14f08f2fe755dadcbaed41b325c257db5500c2e2a8eb796367d632ad9e74
228	Physics	SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001	Exact finite statistical weight	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a1bccdb47bbc54ac9969cabcd193283b550a1258a2bafd47e87a2524a34b28
229	Physics	SFT-PHYS-THERMO-STATE-RELATION-001	Thermodynamic state relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a99aa79962f17d2a4e55a903307a83ca2be54356de263afe8825803f217bf2f6

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230	Physics	SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001	Phase coexistence and transition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4f21021472d3fa831e68e72a049b7a6396e2088b1111327351f3cda0715fd2b8
231	Physics	SFT-PHYS-THERMO-KINETIC-TRANSPORT-001	Kinetic transport and collision mixing	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5beda8e6f719d7fbad9dca80e0b7a8672b5aecef46a1c8df834d6536a8c32550
232	Physics	SFT-PHYS-THERMO-FLUCTUATION-001	Finite fluctuation and recurrence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:30e0865edeacf61bbbac6fdb5cbe11c593b7f41a9366f6cfff95312272e448fc1
233	Physics	SFT-PHYS-THERMO-IRREVERSIBILITY-001	Macroscopic irreversibility with retained microtrace	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2097ca97413226b19cbca02df51a72850c4bdf5b0e4222e45dc2f3bb777a1d2
234	Physics	SFT-PHYS-THERMO-RESPONSE-001	Susceptibility, response and transport correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9152359409657ab66e862af0dc4846b962d19105452c85e184694a16a71c8f5b
235	Physics	SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001	Exact molar Planck measured-value correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:da7b1c59b205af44cd642a3e6c357ddf45fcccec0684e88493a7c6e659c8f3b9f
236	Physics	SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001	Exact momentum measured-value correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6712e0c2581044040f8eaac911bd382d500c1a2d839f2b5c78b1e409436ba5e4
237	Physics	SFT-PHYS-VALIDATION-MECHANICS-FORCE-001	Exact force measured-value correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d1a6cb9ef380156b36f27835ef0c0f23424005c6dbdb19962fc8d0e5bd2deef3
238	Physics	SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001	Exact electric-potential measured-value correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a4c49655afebcc07b564e29e47e2248d648ef15b7a8868b7fbc96dae7cbb962
239	Physics	SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001	Exact electric-field measured-value correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7cbdd805803e22814f2ac485a7d14615c126ad9f32fd60663e5f0148fc0d3465
240	Physics	SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001	Exact wave frequency measured-value correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:647ababd101acfa9e533bfd8d8acfe6e34fdfe0524bdf55b42cd846fca2ea6a
241	Physics	SFT-PHYS-VALIDATION-WAVE-ENERGY-001	Exact wave energy measured-value correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7398c538ac392b7387169fe5c92eedb34897b9342b5769f6f9fb1d7e8898f915
242	Physics	SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001	Exact molar thermal measured-value correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78920ce87ea1f42ad25bed4bb60a3421f288929a14bb97aa5f6eb23241c00349
243	Physics	SFT-PHYS-QUANTUM-PHYSICAL-STATE-001	Physical quantum state correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7206fb62f63acf9c8d3de0b3d35306d94a27b9862aa8a5d548423d931b041e86
244	Physics	SFT-PHYS-QUANTUM-EVOLUTION-001	Reversible physical quantum evolution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e864f02036c1482034ff96e9bd9e01862d3ac6966f1585e24c3218b13d07707b

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245	Physics	SFT-PHYS-QUANTUM-OBSERVABLE-001	Observable and measurement record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:26b996d38a3cac3ea54ec0cc2ac86ba1a852905eb68e6513c46c8986de4f2bbb
246	Physics	SFT-PHYS-QUANTUM-WEIGHT-001	Measurement weights from exact branch support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:17b61d7d856d18d23cf71a7ef468a6c54d9634ed7e8e3245bd60eed81bd8748
247	Physics	SFT-PHYS-QUANTUM-INCOMPATIBILITY-001	Incompatible observations and uncertainty	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cde76117a3c79b8e0893b271dd1de908e4d4840ae15c777ead3e44d21cc9086
248	Physics	SFT-PHYS-QUANTUM-SPIN-001	Spin and finite cyclic label action	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8060070b239eef6926c1eee59b02ffbdbaafe27f7be7f3805c4efc0dd43faaf2
249	Physics	SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001	Identical physical constituents	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c7eb90e57ad8b8deec7bccc6e1ad19e46a854bf51521a54872d749c21d484340
250	Physics	SFT-PHYS-QUANTUM-EXCLUSION-001	Exclusion from antisymmetric composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:713d30ffe5d569ac3ae6a23e68c5dbf10c695577a6639fe42b0662b902e07711
251	Physics	SFT-PHYS-QUANTUM-TUNNELLING-001	Barrier traversal on complete support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b10e5888bd7e44086333276921da133b7559bdc1b7ae27cb43a6164f6c6ed0a9
252	Physics	SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001	Discrete bound-state spectra	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d7b2556daa686f53c3dadec5faff1298af5905404e0ca93a86f5eaf05bceff5c
253	Physics	SFT-PHYS-QUANTUM-ENTANGLEMENT-001	Physical entanglement correlations	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:98925c421294a67280352c088bbc577f428b8651b6d8ca2d0c7414f0f87f7204
254	Physics	SFT-PHYS-QUANTUM-BELL-001	Bell correlation and local-hidden-record boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:51b9838623191c303bc78d402232a72be3ceaac140985c226f12b1550b525ae6
255	Physics	SFT-PHYS-QUANTUM-CONTEXTUALITY-001	Context-dependent joint observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:411350197d503c4bfc3d5ad9812008fdb51646ab99cb4db8d4384ddea5aa32b3
256	Physics	SFT-PHYS-QUANTUM-DECOHERENCE-001	Decoherence through environmental record distribution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1cf4435af3c9ce1069fc175984730c64e333d4be70656537e39e0c7b9d53092f
257	Physics	SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001	Operational quantum-to-classical physical limit	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:31ef665cab0e95b3461a6a67e41fd3d423047fa821e4d6fa5192ffef3cfaa08e
258	Physics	SFT-PHYS-QUANTUM-NO-SIGNALLING-001	Entanglement and no-signalling boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d1fac775f55470f3ddf961e678135bedf89f8be2b80e00aaf162180fe911429cb
259	Physics	SFT-PHYS-MATTER-MASS-ENERGY-001	Mass-energy correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7240300a36d14ae2c841f720ea162b2b67b310dcc912d0d3c1168333603fe955

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260	Physics	SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001	Particle-antiparticle held orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d463c501f9fb01a893d13ccafe10ae7ca8f21c8eab50d501211bd5df99075fd5
261	Physics	SFT-PHYS-MATTER-FERMION-BOSON-001	Fermion and boson composition classes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e9bcfbfe74e5afc64cfa2ff029be021e8f916c3237260b004975ae9136ebcbad6
262	Physics	SFT-PHYS-MATTER-CONSERVED-LABELS-001	Conserved physical labels	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ef9b0dfe3f7d73d541bf398f377a956c483c18fe58492231027b919a02ba694b
263	Physics	SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001	Observed elementary-particle spectrum	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d659b40f4f63f0d453c27364167ab6ea624db37a047ced6792c389e6fc207e14
264	Physics	SFT-PHYS-MATTER-COMPOSITE-HADRONS-001	Composite hadron organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8fe71140f14938a52b5dda186b660635863e5d40109194b299c584bee09ba9a2
265	Physics	SFT-PHYS-MATTER-SCATTERING-001	Scattering and cross-section measure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b2ec3e3f4bf9c77672d9055cd1656562c9972399483eeb9fb81e20c97ebc0393
266	Physics	SFT-PHYS-MATTER-DECAY-001	Particle transition and decay	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b9a36548925456b6db03d9031c9b926b8617d42c0c4c2a536bd39cf814a17627
267	Physics	SFT-PHYS-MATTER-MIXING-001	State mixing and oscillation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:242e8c60ebbd112b1eebde86299605cabae3c8e40e92d0630ca3714b9d954c3
268	Physics	SFT-PHYS-NUCLEAR-BINDING-001	Nuclear binding and mass relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b7d9018630e2f7bc82b25b4f0814dcf216773f2ae4b07242ce80b099ac5546d2
269	Physics	SFT-PHYS-NUCLEAR-LEVELS-001	Nuclear levels and transitions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c44425e09076186d367c47513afb11ccf0f5685fd00df98d7a8ed7beb14b2916
270	Physics	SFT-PHYS-NUCLEAR-RADIOACTIVITY-001	Radioactive transition support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4e90565b4e6d54f5247bec21bd83c2deb7f3748fe72c100d3c4a5473811266d7
271	Physics	SFT-PHYS-NUCLEAR-REACTIONS-001	Nuclear reaction and channel accounting	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a66b3922831670f991c59f828fefce5d5bf36e157f2f16af7ac56886ad38be24
272	Physics	SFT-PHYS-NUCLEAR-FISSION-001	Fission decomposition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3686041dfb9a9bfd34d127884e1642a257f331162f92295faefa8bea1584dc60
273	Physics	SFT-PHYS-NUCLEAR-FUSION-001	Fusion composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:30416752fa5273c24001c4536bab90b3e2eeb804086a18b3c3858485492c91d7
274	Physics	SFT-PHYS-SPACETIME-EVENT-RELATION-001	Relational spacetime event	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c4ac9143c7a478a7432bd5dd1ffca1ff505b170bef1e91f51457b02bdf887002

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275	Physics	SFT-PHYS-SPACETIME-INTERVAL-001	Invariant event interval	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ff799557cb3f67eb01e19774155c51c2be6cd39efd8ef827d6764790e6ffd35b
276	Physics	SFT-PHYS-SPACETIME-CAUSAL-ORDER-001	Causal order and accessibility cone	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b4989c3a9812b65a2902f0c81fe585c555c7f3de0e8060c9b63fc5402fa172f5
277	Physics	SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001	Inertial frame correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:00540e0aabda81441fef87902d4ee27007641146a290d4f50729b05eec1a4c64
278	Physics	SFT-PHYS-SPACETIME-LIMIT-SPEED-001	Invariant limiting propagation speed	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9dd1cbeef40d76297b1a96f762ad084770998e4744f4d437e83d4bbb3e620ec7
279	Physics	SFT-PHYS-SPACETIME-CLOCK-RATE-001	Relative clock-rate relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3fb742859b688cf0a9fcacf65f92b15d654f33f46c2bb193d577117ba6d0ad2d8
280	Physics	SFT-PHYS-SPACETIME-LENGTH-RELATION-001	Relative spatial-extent relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7c3c508f453b0e55987abd30fa084844ea3904397ca514024fffd34b660a5e05e
281	Physics	SFT-PHYS-GRAVITY-EQUIVALENCE-001	Inertial-gravitational equivalence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c49d9f4c5550a7448d5a882431585823d85c51b2829b2becd99d5cc2af0cab52
282	Physics	SFT-PHYS-GRAVITY-CURVATURE-001	Source-linked relational curvature	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:20c01653cd3fea0fca9ac38af60f26ea532e08d7b2797a12004fc79c2f87c200
283	Physics	SFT-PHYS-GRAVITY-GEODESIC-001	Free propagation on curved relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1817eff144b27f906aaa4b4add71f5a101bbd594cfbc57a9f82eda3ffa7be7d9
284	Physics	SFT-PHYS-GRAVITY-FIELD-SOURCE-001	Gravitational field-source closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ca8327c8264470eda6d345068c3565835c80243f2ea069429a672977bec22c79
285	Physics	SFT-PHYS-GRAVITY-WAVE-001	Gravitational-wave propagation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0749b38eecb66d4117113353706603a5344e93996220b1a359f78d667369218d
286	Physics	SFT-PHYS-GRAVITY-HORIZON-001	Causal horizon boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3b1a314d1646e9db9d8213fc14565e9b9dac7cf9cd024251f89cb1cbf6420266
287	Physics	SFT-PHYS-CONTINUUM-COARSE-GRAIN-001	Finite coarse-grained continuum correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:50129d20eaabd2b42eb571667be8ccb5bbfa01aec3e7e39282dd28d74fcadefdf
288	Physics	SFT-PHYS-FLUID-DENSITY-001	Density as exact content-to-support relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5de8d82f4c7c74a083bf5b10cb8f369dc2edb4106e56887704b0fdd3c6d9b86e
289	Physics	SFT-PHYS-FLUID-PRESSURE-STRESS-001	Pressure and stress transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:86c5ce303dc984aca71e28aeecc89beadb8d727c08a50f18f06a369aad7860d4

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290	Physics	SFT-PHYS-FLUID-CONSERVATION-001	Flow conservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:712b58981a69e077c8fb50ddaf ea70ddb5b390d3a2df0dd03b9ca71fa09 794b1
291	Physics	SFT-PHYS-FLUID-INVISCID-001	Inviscid flow correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:952a9eb1da2d46708380c44032 8bb5f77838ecc74808a5d510911f370e2 adddb
292	Physics	SFT-PHYS-FLUID-VISCOSITY-001	Viscous momentum transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:070b3ca4b4303c214c11ef666a aa9cb44b57bdb09cd81a752e519867872 a814d
293	Physics	SFT-PHYS-FLUID-TURBULENCE-001	Finite turbulence and cascade boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fabb7a737b98b1cd23f1820219 14f63de08b58d715aef9ff3798f9fee2 d9eff
294	Physics	SFT-PHYS-PLASMA-COLLECTIVE-001	Plasma collective response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c10a601659f4bc15de4150c9e5 99c2f50d4a8bb30118724f44aa56bbd19 0dd8e
295	Physics	SFT-PHYS-PLASMA-OSCILLATION-001	Plasma oscillation and screening	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2888b7efaca20d0c7f08be5901 1d936c2345b2bb8c459ef7f59cd04ec24 f6498
296	Physics	SFT-PHYS-PLASMA-MHD-001	Magnetofluid composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6cca90cf9ab0c0a753f0aac312 57648c0164daf92d57347119bf0940643 1f404
297	Physics	SFT-PHYS-CONDENSED-LATTICE-001	Lattice order and excitation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9e62f34c6fca48262068bcb2b9 37689e1059d336414ba6df6a4a946dd20 6251c
298	Physics	SFT-PHYS-CONDENSED-BAND-001	Band support and transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2c38da1478ea6ad1a9edfcad1b b1727d7511f937fdea85cd18ab7a7ed6e 38c76
299	Physics	SFT-PHYS-CONDENSED-PHASE-ORDER-001	Collective order and phase transition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3b4572a30962211d40ab1e4947 10304df352743d3478bab102f73c92a43 e4663
300	Physics	SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001	Superconducting coherent transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:28d4be47dd3e3ea7e064dccf a81b4c165deb39fdc07ce034b507d8b39 cf632
301	Physics	SFT-PHYS-CONDENSED-TOPOLOGICAL-001	Topological collective-state correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5d769f8e72acb923996a2284b 2c552f0a25dbc4a175576a9cb27f2e2bf b121f
302	Physics	SFT-PHYS-COSMO-REDSHIFT-001	Propagation redshift relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:011c8f49c26e4ff32c5202fcf3 98b1e5e7934b9b20b9a6a8de3eb31baaa dfd6e
303	Physics	SFT-PHYS-COSMO-EXPANSION-001	Relational expansion observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f3294516adbbba51c7a38681bd1 e4df12fd2b9818b85b73472ea9e2fec76 ed5c6
304	Physics	SFT-PHYS-COSMO-BACKGROUND-001	Background-radiation field relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:65950d7051f8a1cc5261cb6a10 51b1888b4e4b9dff9d52960798ac40e7d f7c02

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305	Physics	SFT-PHYS-COSMO-LENSING-001	Gravitational lensing relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c7e4807f21bea4889c3f78cd9c3e02b73375658afd99cc46da93556c3d4e0189
306	Physics	SFT-PHYS-COSMO-DISTANCE-001	Causal distance and observation-time relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8c54ba35f01cd4082993baabb8e8725cb47d9c42ccc8a2c1ebfb1b33a6e13cc9
307	Physics	SFT-PHYS-COSMO-STRUCTURE-GROWTH-001	Finite structure-growth relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4b216f643356092cdc0906d603b2a3f726e42227e89d2783bd4504aa5d8763dd
308	Physics	SFT-PHYS-COSMO-BRANCH-BOUNDARY-001	Physics-to-astronomy/cosmology handoff boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:23d1f1fc5dfb84012ce63e1c1a63da2c73e3cea268aab818089d15eb42070b4c
309	Chemistry	SFT-CHEM-MEAS-CHEMICAL-ENTITY-001	Chemical entity and retained identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:83fa734a2488422429d604cc55e330a76c5aa8de96029007640440ef5162223d
310	Chemistry	SFT-CHEM-MEAS-CHEMICAL-SPECIES-001	Chemical species and observation-equivalence class	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:865ccab10650d67fedcb6a6f3c31d20887a08fff0ec3ce58de51d4e02a0e3ec4
311	Chemistry	SFT-CHEM-MEAS-SUBSTANCE-001	Chemical substance and composition identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c24408e53e46d5873b5c2f77d638045df99bf6250d5cbae3895cc9a092df15e7
312	Chemistry	SFT-CHEM-MEAS-AMOUNT-001	Amount-of-substance count and reference carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4624037342fc423836b2ca0e1ffa15976d21b67b02a71cbf9e6202063bfd990
313	Chemistry	SFT-CHEM-MEAS-FORMULA-001	Chemical formula as exact composition encoding	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:785d16f1c8f61f25624bfff347d919829dae332431e5a90d27446cc25db6dcb9c
314	Chemistry	SFT-CHEM-MEAS-NOMENCLATURE-001	Reversible chemical nomenclature and identity boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:12634338b801beb64d48817a76f405efc7a77593e5e74c62507fb0a458c63870
315	Chemistry	SFT-CHEM-MEAS-UNCERTAINTY-001	Chemical measurement uncertainty and retained alternatives	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:50c9951b09bb54919aafde3a431ce4b294cf2f6c18f93e8abf52f5d5b78ba37a
316	Chemistry	SFT-CHEM-MEAS-TRACEABILITY-001	Chemical reference traceability and complete record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:74157ee587b7b0d43847e4e753c69637e4b0eececc84a443c2098d40078492269
317	Chemistry	SFT-CHEM-ELEM-ELEMENT-001	Chemical element identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be03c5e39699037ada9127372d6c8c02d713538c36de31904a5015100751f385
318	Chemistry	SFT-CHEM-ELEM-ATOMIC-NUMBER-001	Atomic-number ordering and element distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c63df67479dff616130c735b7e2f407dba06d242ebdbe45067be655337a9cd9
319	Chemistry	SFT-CHEM-ELEM-ISOTOPE-001	Isotope identity within an element	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7cad83d03dcae4993df715fc0b9b92aecbd991c847b48e20d5825a5b12e70d31

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320	Chemistry	SFT-CHEM-ELEM-ATOMIC-WEIGHT-001	Atomic-weight record and isotopic composition boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:56247bdb02421bb72d76605e00a82ac71708b08fb09f0cf34d242408e08ecfa9
321	Chemistry	SFT-CHEM-ELEM-PERIODIC-ORDER-001	Periodic ordering by retained nuclear identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b7092fdb3d12a0149fcf2f857230bb022a7e0badda51d2b8fe530f864433d15
322	Chemistry	SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001	Periodic recurrence of outer chemical organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cc2d14cb33a302f8d6f329c59bfd6c013096e2824e9ebee89284ccfc01cd4042
323	Chemistry	SFT-CHEM-ELEM-GROUP-PERIOD-001	Group and period compositional coordinates	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b99160e0648fadca9668ba6f0f4d6c241115a7d5264c17e878175442e8fb8b62
324	Chemistry	SFT-CHEM-ELEM-VALENCE-001	Valence availability and chemical combination boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:033b82ac5d7e97cafdf88c84ddf9204eb7ab6d4a4fe07d93f47ca6a7a2f2ee4
325	Chemistry	SFT-CHEM-ELEM-ION-001	Ion formation with retained elemental identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:abc77e9274b595b56650212d3eb4f83323c92fffee85569fae93eab34491ca5a7
326	Chemistry	SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001	Known periodic-table observation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:80fd05e50706039e8d8c1a0cca9dcd264c31b7ca0a393199df1de58d7ec320eb
327	Chemistry	SFT-CHEM-STOICH-COMPOSITION-001	Exact chemical composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:074fbc50af44548032b31d2b871481f33477f8a1b1830164c34c9dd4a6a08dae
328	Chemistry	SFT-CHEM-STOICH-CONSERVATION-001	Conservation of elemental identity through reaction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4bbc2d9ff0b0af55f6764854012febfa99ee46bdea5ebde1fd66e7577692f6b9
329	Chemistry	SFT-CHEM-STOICH-COEFFICIENT-001	Positive stoichiometric coefficient and reaction balance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:26cbd66986ff38ff66640b5a85290c9aa4fee32c7b728360e8a45057b89d109b
330	Chemistry	SFT-CHEM-STOICH-LIMITING-001	Limiting-component and complete-consumption boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cc930de298924d974a42c9fd214aa2ee0f156f5dc5e2d2c5bbeed884292284e6
331	Chemistry	SFT-CHEM-STOICH-YIELD-001	Reaction yield as retained product share	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:86bc873a5b4b7126e96d289477d8e3b77de9ccb7f69749da0d1d22c14455c8db
332	Chemistry	SFT-CHEM-STOICH-MIXTURE-001	Mixture components and composition support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:85775961abba9e37de249d62dd2a7653ff2fbf0f92949b00f105cbfbf6b8226
333	Chemistry	SFT-CHEM-STOICH-SOLUTION-001	Solution composition and solute-solvent distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:53c34d8473dac1d8e2567063e83f2d8ad1eab604fa4563b513b4084fa8bf934e
334	Chemistry	SFT-CHEM-BOND-CHEMICAL-BOND-001	Chemical bond as retained interaction closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7c6c37891bbf8d54a3c8e41b6bafae24c29b35c7c380f1266cf62f8db2e28313e

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335	Chemistry	SFT-CHEM-BOND-COVALENT-001	Covalent shared-support bond	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fd341d4ed9119d0e13bada61ffec798c5779cf4a8488bb8ea056f491732d0ef5
336	Chemistry	SFT-CHEM-BOND-IONIC-001	Ionic transferred-label bond	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a77c3443d2e937680cf731cca982bb71b567d83f591baef101f04f8824be8b58
337	Chemistry	SFT-CHEM-BOND-METALLIC-001	Metallic collective-support bond	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:afd35cdaad243c1fa9f9988c0ce7d5a6ee199d081d687f2115d59329041399c7
338	Chemistry	SFT-CHEM-BOND-ORDER-001	Bond order as retained joining multiplicity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:171120df1e712d3def9a761bca06f86050c596c0db773ff0d0f01b6265c84546
339	Chemistry	SFT-CHEM-BOND-LENGTH-STRENGTH-001	Bond length, energy and identity correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:12bf183bd0effffa233f2b3cffc4c423789a93eb997587d9e584c2bfff96a6c09
340	Chemistry	SFT-CHEM-MOL-MOLECULE-001	Molecular identity and complete bonded carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a075ff988b4bd9d4c16a199fa6402277bb945aabfa38f32f9eeebcdbc784f1fb1
341	Chemistry	SFT-CHEM-MOL-GEOMETRY-001	Molecular geometry from held adjacency and orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ac60f99b83d623409f27d5997eae114ad5384e6fb2fdcbcf3c71b50d9a0aa054
342	Chemistry	SFT-CHEM-MOL-ISOMER-001	Isomer distinction under equal composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:93b049013df4a30b2fdd1be11c3b8fd05ade4c1126f22aa7527ae34bf54e335d
343	Chemistry	SFT-CHEM-MOL-INTERMOLECULAR-001	Intermolecular interaction and residual joining	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1ec50ca6dddec341fbce800beedbb12d473460da34391c9047b93c4c684750b40
344	Chemistry	SFT-CHEM-MOL-SUPRAMOLECULAR-001	Supramolecular organization by reversible recognition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a38e3bdbb55e2cfe79d4491665da0f991ab463f874273e8e7b3c13adfa7538d1
345	Chemistry	SFT-CHEM-MOL-NETWORK-001	Molecular network and connected chemical whole	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7a59431e2c40b65e566213a7730b55112b2f1e80edaa9c23f9459bee06b4570c
346	Chemistry	SFT-CHEM-AB-ACID-BASE-001	Conjugate acid-base partition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:318ea7b4fdc414c30a64a83df0ca5a0dfdcac348a4fc207743da2b9d6496827f
347	Chemistry	SFT-CHEM-AB-PROTON-TRANSFER-001	Proton-transfer acid-base relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ddcb09522d19aa8840e8e57745b546ee62be1116bfdf03a850643fe1d24ca6df3
348	Chemistry	SFT-CHEM-AB-LEWIS-001	Electron-pair donor-acceptor relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ef68becb2c22b1c5f781958280bd22f8abfad5a4cbb3c0dfe86cb56cf7d16f598
349	Chemistry	SFT-CHEM-AB-AMPHOTERIC-001	Amphoteric dual-role relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94649424e1a4c49ab8939c2f4eab025306b9262e766ec388c2d1fc97e99eb4ad

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350	Chemistry	SFT-CHEM-AB-BUFFER-001	Buffer response and conjugate-pair retention	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b76553aa9e22f85717dd7cbda96a1e8ba880fb5396756d8049e79fac0f885ca1
351	Chemistry	SFT-CHEM-ELECTRONEGATIVITY-001	Electronegativity ordering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a22665b562002fc2eae1f487748405511ca02f6fb00cd071e660731a5dc0d10d
352	Chemistry	SFT-CHEM-BOND-POLARITY-001	Bond polarity from unequal held affinity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0510531a9b1aa21d40a92effc5592d673d5d631fd052c4f1b6e3e434c66016b7
353	Chemistry	SFT-CHEM-REDOX-OXIDATION-STATE-001	Oxidation-state accounting	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dc3f55fe127993303d0bec15a2f6a53ee4c3ab349377f681bf8b596fa044d2d1
354	Chemistry	SFT-CHEM-REDOX-COUPPING-001	Coupled oxidation and reduction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8e1f6048b5dc8e6fb29718ca5b5075dd0fcdea5bbc4eed6e1d7d1e30b1ac8fb7
355	Chemistry	SFT-CHEM-ELECTROCHEM-CELL-001	Electrochemical cell and separated redox closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bbb6c005a3ccd1c37e538fe895aace47663809fa4ccdaea21caa8183a652eaf4
356	Chemistry	SFT-CHEM-RXN-IDENTITY-001	Chemical reaction as source-bound identity transformation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ec1b06b42e323744b8166f5498671868cd4f9ce5ef68c1f43d718a894712e42d
357	Chemistry	SFT-CHEM-RXN-MECHANISM-001	Reaction mechanism as complete elementary-step trace	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4045d3ae4048a774bb4c5bde502120261bccd2f991683cf8c8415b20ce32dc02
358	Chemistry	SFT-CHEM-RXN-INTERMEDIATE-001	Reaction intermediate and retained path identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d53d9a103971aa8eee09162a1763b71e9572dbc40c1c546c5042d7f571dcf4af
359	Chemistry	SFT-CHEM-KIN-ACTIVATION-001	Activation barrier and transition boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bd0c43a9f211959e83b423962038626e7bc1190a33e82ca1baec27a89d492d8b
360	Chemistry	SFT-CHEM-KIN-RATE-001	Reaction rate from counted transitions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c4c4e2acabcf3b44e5da03ed097bb132c729e017549fbc79dc93d8e64c0a2d1
361	Chemistry	SFT-CHEM-KIN-ORDER-001	Kinetic order and dependency multiplicity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:37abccdd1d68e0f50d2df51585e9b700e106c363b97ff6bc36ef150b3dc6d3a1a
362	Chemistry	SFT-CHEM-EQ-CHEMICAL-001	Chemical equilibrium as balanced reversible support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8f7da55dbccf2e835b129adc43335e84c1ce3336ad696a5b5d7e58b4851c32b7
363	Chemistry	SFT-CHEM-THERMO-REACTION-001	Reaction thermochemical accounting	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ceebccca68fd6438210ac1ee70f21b40819f37b683772597f563902cd17a6c44
364	Chemistry	SFT-CHEM-THERMO-DIRECTION-001	Reaction direction under complete energy and distinction accounting	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78ce22d5ad1ff53bc945906184614c11901a1504bedb5950d5b7bf0ebad5e09a

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365	Chemistry	SFT-CHEM-PHASE-CHEMICAL-001	Chemical phase and coexistence identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:80720a90afc5c84c8c5aa4a6524da264e88628c125b004d9fb370fd81d9baeac
366	Chemistry	SFT-CHEM-SOLUTION-EQUILIBRIUM-001	Solution and solubility equilibrium	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7e7c23b23d756d5bb9ec8d828b0920427adec44e83aefcce24fcba1335b446d3
367	Chemistry	SFT-CHEM-PHOTOCHEM-001	Photochemical excitation and reaction channel	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0ec198ef64ee4470b66083ece9cce7ef8911bffa02b330fd13cbb1a7e9bc6b0e9
368	Chemistry	SFT-CHEM-CAT-CATALYST-001	Catalyst identity conserved through a reaction cycle	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:937aa5f579196ccd541a1fe09dfff3eb5cd095a64ccd378f29c566b97df3793e
369	Chemistry	SFT-CHEM-CAT-PATHWAY-001	Catalytic alternative-path relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:088dfaddc5b0a7b94907fd8bd92a967db8a27ad7c97fef56358dede752f45207
370	Chemistry	SFT-CHEM-CAT-SELECTIVITY-001	Catalytic selectivity among generated products	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0210a8962416f13c5e82b3c11bd91c562fc98a8ad7f0e8ecf7fd8cbd87edf3cf
371	Chemistry	SFT-CHEM-NET-REACTION-001	Chemical reaction-network composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e6af94d430f01d2e10ec7053d39b13705ed69dfb6510c8c0c4504d97721bbe2a
372	Chemistry	SFT-CHEM-NET-AUTOCATALYSIS-001	Autocatalytic closure and self-amplifying path	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8005e646115c8bab3c72478669c99e70f85419d388f4344bb0e059eb8d175f2
373	Chemistry	SFT-CHEM-SURFACE-ADSORPTION-001	Adsorption at a retained interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5d30fb2ced993666788efb16c24431b275bb5847ffdfaf187282ab4b12f97249
374	Chemistry	SFT-CHEM-COLLOID-DISPERSION-001	Colloidal dispersion and phase-interface support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1de99723c818a575de1f05a5c8a230b16ebdc834102d39f9600fe6b89b0e5141
375	Chemistry	SFT-CHEM-INTERFACE-TRANSFER-001	Chemical transfer across an interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ccc945354866ff49d1cb40dc69622afadce5be4e706bac640477352f6c4292a7
376	Chemistry	SFT-CHEM-STEREO-CHIRALITY-001	Chirality and non-superposable held orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f29569bdc7dd166c19b48a69d3dfd95c3299fe9db0fd32f9b00449bf51bfc91
377	Chemistry	SFT-CHEM-STEREO-ENANTIOMER-001	Enantiomer pair and chiral observation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d9a1134bbb9af58c62e6fbbcb65f72c47071183fa1ed0274c2d796392d210077d
378	Chemistry	SFT-CHEM-STEREO-DIASTEREOMER-001	Diastereomer distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:54222fa195d500b1528f8c93492bedd144c895243dba02c49ba8b189f8629bc9
379	Chemistry	SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001	Functional-group recurrence and reaction role	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:84d6b1302565580828881323d45c81917e74c150b85ede407160bc90e33989b8

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380	Chemistry	SFT-CHEM-ORGANIC-REACTION-FAMILY-001	Organic reaction-family compositional law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:89120ae1d10844a8d907ab00b6db94bcb7dfccb75e9ccc93b072f84eef8f8526
381	Chemistry	SFT-CHEM-POLYMER-CHAIN-001	Polymer chain from repeated monomer identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:03f536704d27b6e91a52c136572c1d7596bc5c5d95681b0d56292cbc8e472201
382	Chemistry	SFT-CHEM-POLYMER-DISTRIBUTION-001	Polymer population and retained chain-length distribution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ad81bf1f69ee772b22896d80d25bcb312522f69a90c52418f6f8362a31b29a84
383	Chemistry	SFT-CHEM-BIOMOLECULAR-BOUNDARY-001	Chemistry-to-biology molecular handoff boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dfed7cd5d124ff390399d5adf8b06a01a7edcc102a9e6d0ff59beb201bffd547e
384	Chemistry	SFT-CHEM-ANALYTICAL-SAMPLE-001	Analytical sample, analyte and matrix distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1fd5c56d3970342f09bb0c871b8f2fb12f84d562e0f8e580299e89e91c624b00
385	Chemistry	SFT-CHEM-ANALYTICAL-CALIBRATION-001	Analytical calibration and traceable comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94e195edc3e188f525232747b5c5ca4f893b490ecd46ef9d8bb377ef1d2d2a54
386	Chemistry	SFT-CHEM-ANALYTICAL-SELECTIVITY-001	Analytical selectivity and interference boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6e86b8d37deaba116872b23c046a6cabd8cc701713d7e1d36ffbc0d7478f6bca
387	Chemistry	SFT-CHEM-SPEC-MASS-001	Mass-spectral composition correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dd28b321a35b48d8cb382cfa7b64b2c9ac917bce959c4b52ceeddb9fdcab7a1c
388	Chemistry	SFT-CHEM-SPEC-INFRARED-001	Infrared recurrence and molecular distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:399a21477f3aedee80bdce487e17bbd4ff81213bcc79d35326f359dc5b2949df
389	Chemistry	SFT-CHEM-SPEC-UVVIS-001	Electronic absorption and molecular-state distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:38f612b5b2fcd62dfa158d3c592c930c8611a00d0b1447d9ba59af313bc173f9
390	Chemistry	SFT-CHEM-SPEC-ROT-VIB-001	Rotational-vibrational molecular spectra	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:20367bb1fe68cb5795f63f6d62da369e7cfc5295098f621cc6736c7fd3075e3d
391	Chemistry	SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001	Complete analytical result and falsification record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7787001c4d7b90235d1bf7f3e19482cf5f05cb5f37f98a980be1efc4de413669
392	Physics	SFT-PHYS-MEAS-BOUNDARY-GROWTH-001	Exact source-boundary growth measurement discriminator	depth_independent	independently_replicated	1,024	4	sha256:b2d8360be10ba4be5664b59ee4624d91d909962f83927b3bd98d4caf5b02e43b
393	Physics	SFT-PHYS-STRUCT-GENERATOR-THREE-001	The second structural Fold generator is three	depth_independent	independently_replicated	1,024	4	sha256:51fb37f85fc2774d3118fd08636204778a68aa8c908453ae788179e06de90b0a
394	Physics	SFT-PHYS-SPACE-DIMENSION-THREE-001	Three-dimensional Fold stability	depth_independent	independently_replicated	1,024	4	sha256:1bcc9dbab55d17203193ff3cdc99f0013ba60768a4c42b4fbb98cd000bee6a

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395	Physics	SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001	Rank-two boundary of three-space	depth_independent	independently_replicated	2,048	4	sha256:10f70a4a463a28ed545c101dcf7528b6d5a7b04355e3c89f3911f8b95514717d
396	Physics	SFT-PHYS-FIELD-INVERSE-SQUARE-001	Inverse-square dilution from rank-two boundary support	depth_independent	independently_replicated	4,096	4	sha256:fd630f1572c8e211f410ddefbb26f84133b0c5d91de383d9461861cd44dd195
397	Physics	SFT-PHYS-VALIDATION-INVERSE-SQUARE-001	Post-seal experimental agreement of the inverse-square exponent	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e70905b63729a3bf5619b26115f7fe8b78eb04aba53eba78f6697405d78a220b
398	Physics	SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001	Terminal exact inverse fine-structure Fold ratio	depth_independent	independently_replicated	1,024	4	sha256:b7b1663a8ed78da756a2624d0bca690ac2ade089a9396060c96f141a6f2896a2
399	Physics	SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001	Atomic orbit-cell capacity from three-space boundary orientation	depth_independent	independently_replicated	1,024	4	sha256:1f3ba9a0a82d5ea4f7b36b71f2bc3b135be8f92d2f5f1c3e77182f7b8979e02c
400	Physics	SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001	Exact colour-sector Fold coupling	depth_independent	independently_replicated	128	4	sha256:5ac1c6ce8b4094ba57fd03f96cb30be011ea724f3856903a9ed0b0ab5ca3ddfd
401	Physics	SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001	Complete Fold nuclear-closure sequence	depth_independent	independently_replicated	1,024	4	sha256:eb48cd3f1935f2539cc128ad8259b18deb47ba9338585ceaa9f2ec6b6a4d7295
402	Physics	SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001	Exact atomic existence boundary	depth_independent	independently_replicated	512	4	sha256:7ef00e2c6474de6640916036a369c6bb49c988e45332eb04529113f58b45e2b7
403	Physics	SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001	Post-seal CODATA check of the terminal inverse fine-structure ratio	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:13cf55d69e81157f6a00ec4968703133953f817c337147c7d3f160a86d52210a
404	Physics	SFT-PHYS-VALIDATION-NUCLEAR-CLOSURE-S-001	Post-seal IAEA check of the nuclear-closure sequence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0b4634bdeef77436218ece89bbabd86819a7828d1dc9566b0fbba31970a99b30
405	Chemistry	SFT-CHEM-PRED-G-BLOCK-001	Generated g-block structural prediction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5e21d5cb1ad9758257dfa1fb732792adf4e9b7aa7ae0300fdaf21cb2426772bb
406	Chemistry	SFT-CHEM-PRED-SMITHIUM-001	Smithium element-126 standing prediction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8028271d2d1e27a0c1351974b39d09131916ed17cd349b36268eaa6b8b2b2613
407	Chemistry	SFT-CHEM-PRED-PERIODIC-ENDPOINT-001	Generated periodic endpoint standing prediction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:df9ea26853f6ac0678893d0e005f2a25a4d86ab3084ab3147353432050d1bf16
408	Materials Science	SFT-MAT-MEAS-MATERIAL-001	Material system and retained identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a603da455a0d64fb7ad615c083026bf63365752ae60f5ab151010705873f7a44
409	Materials Science	SFT-MAT-MEAS-SPECIMEN-001	Specimen and sampling relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be89f08f9db698b344dbd6a7bd319ff4302aac70e120cc1531d07aba9e67cd7c

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410	Materials Science	SFT-MAT-MEAS-COMPOSITION-001	Material composition record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:12f436cc0234aa79c73e3e13c80607ee40d2765efedab4566e0e1d3af6e789e3
411	Materials Science	SFT-MAT-MEAS-PHASE-001	Material phase identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:89333be757443d46a610ebec1d82daecb409df0bd61811ebdfa9868f3fb0fc49
412	Materials Science	SFT-MAT-MEAS-MICROSTRUCTURE-001	Microstructure as mesoscale organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:41d5b521ac071b8c9388cc3ffff563dc6c0b8e686e3a062f0657520cfd9d86352
413	Materials Science	SFT-MAT-MEAS-PROPERTY-001	Material property relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bf49f2ad1f00be8833e35eae3aad141cd5213b11094b232ae4fb632a3bad6535
414	Materials Science	SFT-MAT-MEAS-TRACEABILITY-001	Materials uncertainty and metrological traceability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:da14ebc7c6205220ec0993a22cbb9b0fb41777bc23e4a17ffcd978c4b8f8ce53
415	Materials Science	SFT-MAT-CRYST-LATTICE-001	Generated lattice organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:060664b8cdcb7d7ab34749ac5f04012cc025b05d2fcec3ed47cf5e011b39a4da1
416	Materials Science	SFT-MAT-CRYST-UNIT-CELL-001	Primitive unit-cell identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7b95e56aaa1c2c51c79a84cfea8693aa695be4eb0f8fa0c9b948113547ecd68a
417	Materials Science	SFT-MAT-CRYST-TRANSLATION-001	Crystalline translational order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be6c8d29c1753a134eb988fa34e523060c1e197246c7c9ca2dd6f835d2659c7f
418	Materials Science	SFT-MAT-CRYST-CUBIC-COORDINATION-001	Simple-cubic nearest-neighbour coordination	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b49f3d9cf57768f98f6fe15e0133c4df0cadcf3e7ccfc2a0c315ae31453b5ec0
419	Materials Science	SFT-MAT-CRYST-ROTATION-RESTRICTION-001	Crystallographic rotation restriction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a0e401100d13252d5f02d5be754a462b01b737b9b066ec9a1809671a6ac1a252
420	Materials Science	SFT-MAT-CRYST-SYSTEMS-001	Three-dimensional crystal-system classification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8eea622f2a962a3e3b995693f3ba38acf8c0b79242bd30c1804eff03952d7637
421	Materials Science	SFT-MAT-CRYST-BRAVAIS-001	Three-dimensional Bravais classification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c04968088f5382d095822785a3b505081f9a9027b7a8d206438deafdb6f93a52
422	Materials Science	SFT-MAT-CRYST-RECIPROCAL-001	Reciprocal support and diffraction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:257f4aeb601ec2c9cd2681faed0643db4d63f85dedc6ebdeba8bb87ec8c37f72
423	Materials Science	SFT-MAT-CRYST-QUASICRYSTAL-001	Quasicrystalline aperiodic order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c397a36d1bb97f5bbab246e07e290b327c773ea18ec5c2b074d22794d7d1d094
424	Materials Science	SFT-MAT-CRYST-PHONON-001	Acoustic lattice excitation branches	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:574d1650178ebf71cddebdd5128c2d2e9837d02ab67443c66d787cfc246e5b6c

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425	Materials Science	SFT-MAT-DEFECT-POINT-001	Point-defect identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7fac1bb37f212cd0ea2c2fbff94a0c0e1d48392c90c532ef816fe3fee5de5
426	Materials Science	SFT-MAT-DEFECT-VACANCY-001	Vacancy as structural absence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ec38624192bccb935013d71151437c8c90b43a6fae881c3ba54527cd41a85132
427	Materials Science	SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001	Interstitial and substitutional defects	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a0adb50fd640b1941a39bed7dc5e823c0b5a07a979f033e459c25160af154ba0
428	Materials Science	SFT-MAT-DEFECT-DISLOCATION-001	Dislocation line and slip	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5b63a94c1368233f356123a5c3cf72d1e68f8cc92cbd72c5c1758642a4737821
429	Materials Science	SFT-MAT-MICRO-GRAIN-BOUNDARY-001	Grain and grain-boundary identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:97779d2048b92fda5a2968a339e3b2359a2bb5389563cda49f5cdda6aac404e
430	Materials Science	SFT-MAT-MICRO-INTERFACE-001	Material interface law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f8e65b2c9530d410a0a1b2dee9a225b829d130c02ccbd7d31dc52be63ad4e738
431	Materials Science	SFT-MAT-MICRO-DIFFUSION-001	Diffusive constituent transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:07a61b3ac976a385ccdfb2e48ca22dc8d09c675145423d0501d8b4dea7765500
432	Materials Science	SFT-MAT-MICRO-NUCLEATION-GROWTH-001	Nucleation and phase growth	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b314993b37eafa71cd6f643712590798c093ae7113db07b72d5fae73b9e6262c
433	Materials Science	SFT-MAT-ELEC-BAND-GAP-001	Electronic bands and structural gaps	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5fbc05f52167cabe089f6ac5ccb37bc8995972adce2580767b91009ff0d51404
434	Materials Science	SFT-MAT-ELEC-CONDUCTOR-CLASS-001	Conductor, semiconductor and insulator distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0851d0318085f1b7bdf0d9871bf74598b959240df24c4b8714e34f004cad981a
435	Materials Science	SFT-MAT-ELEC-CARRIER-DUALITY-001	Electron and hole carrier duality	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6f848016b57e0d09eae5c587efe10f4da04c6191319e870a6937958303a14727
436	Materials Science	SFT-MAT-ELEC-OCCUPATION-001	Exclusion-bounded state occupation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a86b53942d003089e455938be92486df3b9ea73991cc106a4ab8c3ab2d446f3a
437	Materials Science	SFT-MAT-SEMI-DOPING-001	Doping as controlled constituent substitution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c78add3668c0e8d5e00346b4de0be72933ce45fcb86adc5a3d18e2b1320b6f7
438	Materials Science	SFT-MAT-SEMI-PN-TYPE-001	p-type and n-type organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3ada2c3c62563f43e20c3857ca3ec0c9e8a79bfaf11b2387d938eb380900a96a
439	Materials Science	SFT-MAT-SEMI-JUNCTION-001	p-n junction and depletion organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ab3ac6d4d72771243c560c5ba63f56c21ecf46425253345f6c8e9a067b33a859

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440	Materials Science	SFT-MAT-SEMI-TRANSPORT-001	Semiconductor transport relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b53d46b71c00244ede38457846ac2592276922367ced522052f68e98c03c3e18
441	Materials Science	SFT-MAT-SEMI-OPTICAL-001	Interband optical transition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c4341f21a28539567752ba56fae9c96581b507d5a41289a58c0996693d67d31c
442	Materials Science	SFT-MAT-SC-PAIR-001	Coherent paired-carrier state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:37c8eddac13dfd4d0f10117c6e85695a40f14bdbeb94ddfb21ccbc0c16d7b4
443	Materials Science	SFT-MAT-SC-ZERO-RESISTANCE-001	Dissipation-closed superconducting transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eabc8076794eaf93d8139866bb2cb1b68f83fb9eff249cd9d3f785dab6c7f790
444	Materials Science	SFT-MAT-SC-MEISSNER-001	Magnetic flux exclusion in the superconducting phase	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:958f05ce5f5a809ccd23e229b67fee3c18dc732d43178f85843de9fc89ad4b01
445	Materials Science	SFT-MAT-SC-FLUX-QUANTIZATION-001	Superconducting flux quantization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0ed97a6d6c7de89689c4aed43a9408b27fd7bdd9398b7efd9f9a0525499de4e
446	Materials Science	SFT-MAT-SC-JOSEPHSON-001	Weak-link coherent transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa8fb0a93ee97c5e9aa696f455108f5770b1882a050c6927f250d7a30b144426
447	Materials Science	SFT-MAT-SF-SUPERFLUID-001	Dissipation-closed superfluid transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:13bca5d4d8eefdaaae89f16793de97f2da0956b15f3c909b7dd19a9864f7c4f
448	Materials Science	SFT-MAT-SF-CIRCULATION-001	Quantized superfluid circulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ea513957e5e43cc1e01e7a2187a08a7f2bfb4d9a0d6561c8902174b3eec6b9
449	Materials Science	SFT-MAT-TOPO-INVARIANT-001	Topological material invariant	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fc5c012cceb34acee20825a1baf86f0c979fe88be20c0b06dae65074dca33361
450	Materials Science	SFT-MAT-TOPO-BULK-BOUNDARY-001	Bulk-boundary protected transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d2a5beae9e015469d9a6ba517078466f3918892548d064025b696a2df8421dbc
451	Materials Science	SFT-MAT-MECH-STRESS-STRAIN-001	Stress and strain records	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0445851d4b39830df5906e9cd54ca1b9c33557970627302ef3b5f1261d64e3e4
452	Materials Science	SFT-MAT-MECH-ELASTICITY-001	Elastic recovery	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b763df84f89c26c7f0250b07a2b29a053ece891b3635c136a6a66373fc9936ff
453	Materials Science	SFT-MAT-MECH-PLASTICITY-001	Plastic deformation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:acd1392cca2311cdf999b68957c53230344b1c9b8e31591ebc36802a08cecf2c
454	Materials Science	SFT-MAT-MECH-SLIP-001	Dislocation-mediated slip	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b0bc6c6a3d4d3c3c8f16233b9d41b940ef1302f02c00ff2bb6f4c515e76a092d

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455	Materials Science	SFT-MAT-MECH-MODULUS-001	Elastic modulus as response ratio	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5f7a55c339d12573cad658697b86320f8d447eaa850d2c134ad20b95ef66a30b
456	Materials Science	SFT-MAT-MECH-STRENGTH-HARDNESS-001	Strength and hardness boundaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4de60010c08d9fed3acf8c4728f2783ce2aeb1ef777f628f85195134380de79a
457	Materials Science	SFT-MAT-MECH-FRACTURE-001	Fracture and toughness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d47271030cd2fb14b7c385fec0736f231ee8a1dc4bed34ad8afd00a92bc262d5
458	Materials Science	SFT-MAT-MECH-FATIGUE-CREEP-001	Fatigue and creep	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b94bc1254ab3d3025628c9435c5bc2cd1afa979be5035aebfb0b17dc1436f4d3
459	Materials Science	SFT-MAT-THERM-HEAT-CAPACITY-001	Heat capacity and excitation storage	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6674f76f1ca6440a25798bd45bedccd1ea7e7b5df1bffe9674312f9fd0f409b5
460	Materials Science	SFT-MAT-THERM-CONDUCTION-001	Thermal conduction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:902681fe9ffce6915387d8985f4b5cf95bc767a56094aa0b0b7f44a341150562
461	Materials Science	SFT-MAT-THERM-EXPANSION-001	Thermal expansion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5e9ca77f6756053d098868e4e031d724f44e4c42256357d74fa33a4d4567ac7d
462	Materials Science	SFT-MAT-MAG-FERROMAGNETISM-001	Ferromagnetic order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3860d6dec3e3056aa3eeb5268587b83a21373587f1e399dea61d3b9e840636cb
463	Materials Science	SFT-MAT-MAG-ANTIFERROMAGNETISM-001	Antiferromagnetic order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:042d5e6093d1ae270db43da2ccb72f5299318e53632719ca3714ba648561dbbe
464	Materials Science	SFT-MAT-DIEL-POLARIZATION-001	Dielectric polarization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:46b063c128b7f353fb40d4b67a9fbf26f0266fd2cf441b9a103a8663ec260459
465	Materials Science	SFT-MAT-OPT-REFRACTIVE-001	Optical propagation and refractive response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3210b165450c76b959c9d5abf99c9b2487440238426780d044e5a6e810b2507e
466	Materials Science	SFT-MAT-PHASE-TRANSITION-001	Material phase transition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7657d0f08ff975856f32bc1c00de5bd76507c0880b2ee959818a773e92a45208
467	Materials Science	SFT-MAT-CLASS-METAL-001	Metallic material organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d319f9d2f4dc8869f881d394860212b924efe345b36bd8a11615a789ab14b81a
468	Materials Science	SFT-MAT-CLASS-CERAMIC-001	Ceramic material organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e985201b890a9c99983f9c3286047ccd3bfe488da34cad3615f166d585a97cc
469	Materials Science	SFT-MAT-CLASS-POLYMER-001	Polymeric material organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:afbbe9b832351be2b9532eb36f6b8d9b41a5423cf9d1bdfc3b32d9d94d910175

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470	Materials Science	SFT-MAT-CLASS-COMPOSITE-001	Composite material organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e741199f029ca83879c34656b13c006bc8e54a12637e59a04a32d8f5c738b8d4
471	Materials Science	SFT-MAT-CLASS-GLASS-001	Glass and amorphous organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7a91b06f6cb50235c28be8058235baaa5b839f577f67b30a1ae3cd2b1e8042c2
472	Materials Science	SFT-MAT-CLASS-POROUS-001	Porous material organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7e6e7e7efc51d3e417adb6974dd16f8ec17d9d747971cd2456f92cc019ce0054
473	Materials Science	SFT-MAT-BULK-CORRESPONDENCE-001	Molecular-to-bulk correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d451b17d73ba7d892d5e79c947ca8b0979ee3e7806a0635afaa3285427ef29a5
474	Materials Science	SFT-MAT-BULK-ANISOTROPY-001	Material anisotropy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4118df713bc0e9e8e34ac4bac952f7e9f774f5207bd40efcc4ed5adcb6524981
475	Materials Science	SFT-MAT-BULK-SIZE-SURFACE-001	Size and surface effects	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d3e923caf1c9a615cd149a43345002c9f5695b0b1573e4c82d834400a43fb09a
476	Materials Science	SFT-MAT-PROC-PATH-001	Processing-structure-property path	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:84b881964289b6369ff0a175b1415e110962c47669982eece1ebc77d7cbde690
477	Materials Science	SFT-MAT-PROC-PHASE-DIAGRAM-001	Phase stability and coexistence organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:10b42f4ba45befaa3cc9a5d8c854432e729109908f9feb9e1a3ef541f0a390bd
478	Materials Science	SFT-MAT-PROC-SOLIDIFICATION-001	Solidification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:47e7cf392984fba049ecd86ab40b7df92994589f140f31e58e4dfaebdbb19ab7
479	Materials Science	SFT-MAT-PROC-SINTERING-001	Sintering and densification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:579b77a0e4d157c4027f3b362e50b0947b8ed266da098d1a86f1a7dfa3185c0b
480	Materials Science	SFT-MAT-PROC-HEAT-TREATMENT-001	Heat treatment	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:173f913f1c66a6f72e38a5a5255fa572de13c997acd87bbda1ec5dc1495eb889
481	Materials Science	SFT-MAT-DEGR-CORROSION-001	Corrosion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4542786b0ea6848bf34524495a0a40b678800eecd53efb77249282203f747945
482	Materials Science	SFT-MAT-DEGR-WEAR-001	Wear and tribological material transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c59b96e47896844ac1b5ac4a3b204133efbb551e57bc8b3a48d26793af6577fb
483	Materials Science	SFT-MAT-DEGR-RADIATION-001	Radiation damage	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dbf66ce45a26ae89f38fdc829f0ecaee959d30aff5c58d2d62890771ef2c84c7
484	Materials Science	SFT-MAT-FUNC-PIEZOELECTRIC-001	Piezoelectric coupling	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e334945fb5c5da1183b9dc19935a472a15ea0095fcfabf8cccd2c319375c3bf9

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485	Materials Science	SFT-MAT-FUNC-FERROELECTRIC-001	Ferroelectric order and switching	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:668f7c77951470dd63ae4753fd69bd1d9806e623bbd8cd663f847c86ce7fe460
486	Materials Science	SFT-MAT-FUNC-THERMOELECTRIC-001	Thermoelectric coupling	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a8cd3f6b52008aa2c5e86e4810d8312b4cd9071411cbb3015ce3240076d5405c
487	Materials Science	SFT-MAT-FUNC-PHOTONIC-001	Photonic material organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d90e52c8e5b4c5e0a92c05574a8fa2068d8b1bc5da6caa6ba76957c45c675a3a
488	Materials Science	SFT-MAT-FUNC-MAGNETIC-001	Functional magnetic material response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c01cba9d83be4efcc037953fc82cc5162a40536f07d881413ad9ca9ada247e38
489	Materials Science	SFT-MAT-FUNC-NANOMATERIAL-001	Nanoscale material boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:55317cf042b6531a6fe30ee1532ec73c758b2679f7aab742de64148ac5969b86
490	Materials Science	SFT-MAT-FUNC-BIOMATERIAL-001	Biomaterial interface boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:99ee0945ba72d497738307051dbcf5f9aa26a1f305b8f589e3dce5ccce07b2501
491	Materials Science	SFT-MAT-SUST-LIFECYCLE-001	Material lifecycle and recyclability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be4a9a15df6225f1099d4f7d6c00291b3b6fd3df881617d9b565c3b34d16b32
492	Physics	SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001	Exact charged-lepton cubic invariants	depth_independent	independently_replicated	2,304	4	sha256:9c7202057f7735d46e9b355f8611e63e9043d99b1a3ced16cadb6a468d09e598
493	Physics	SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001	Terminal self-coupling refinement of the charged-lepton cubic	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:720b443c7267c7dc09cf7ecb5ffa226b9a0d6a745136689ef2e42f17f13b2d2f
494	Physics	SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001	Post-seal exact Koide comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5d3bb335ef3740f3b1fd8660a4ab95b3d7fc3dfe8534dd90e7dd2b9b69deb89
495	Physics	SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001	Generation-cover dark-to-baryon fraction	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:d8a5088d10154314b6c018e2daf4ab64bf2c4c94e6f4ee77a814bbf56048f3f3
496	Physics	SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001	Fold calibration ratio between early and late expansion routes	depth_independent	empirically_tested_and_independently_replicated	2,048	4	sha256:b58587667e05026f07be77365d7b81e121a46114aedb0cfde51b566d1ebb365b
497	Physics	SFT-PHYS-COSMO-SPATIAL-FLATNESS-001	Complete-partition spatial flatness and absent curvature remainder	depth_independent	empirically_tested_and_independently_replicated	2,048	4	sha256:f0302179aa36d55df5b54595b6ece20566ec9c5e7a3f00f9e302ef705f5f504d
498	Physics	SFT-PHYS-COSMO-COMPLETE-BUDGET-001	Depth-five Fold cosmic energy budget	depth_independent	empirically_tested_and_independently_replicated	2,048	4	sha256:616ac87b6340810e8dee1c13237b4de2569b7ee827ec9863e982b03ccbef8507
499	Foundation	SFT-FOUNDATION-EXACT-OPERATIONS-001	Exact positive foundational operations	depth_independent	independently_replicated	128	4	sha256:5c0c14bcd2aa4f1df7eba1618f59ee9aa56be028aa6a8aa78fcac4d28139970d

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500	Foundation	SFT-FOUNDATION-HALF-ONE-001	The forced half-One ground	depth_independent	independently_replicated	128	4	sha256:eef37a29bf86f0e880af72af355112f4f8fe9a97ba817ef71ef25d9e74883a1d
501	Foundation	SFT-FOUNDATION-FOLD-DYNAMICS-001	Exact Fold dynamics and fibre law	depth_independent	independently_replicated	128	4	sha256:ffaa7c3740dd747aecfc858391cec140dd6c1b3c8ab219ca798fccfc1785b675
502	Foundation	SFT-FOUNDATION-PRIMITIVE-MAP-UNIQUE-NESS-001	Mechanically enumerated primitive Fold-map uniqueness	depth_independent	independently_replicated	84	4	sha256:8ca5e163461025f367a87f8105208ec72e4a13679a881690a3420142a9518996
503	Foundation	SFT-FOUNDATION-DERIVATION-TRACE-001	Complete replayable derivation trace	depth_independent	independently_replicated	256	4	sha256:457414dd7e3a14e44b200b1ac329cdc9e2bb3527abe901c3255d7fb6f9233c5e
504	Foundation	SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001	Single fail-closed SFT admission law	depth_independent	independently_replicated	2,048	4	sha256:0e21ffcb217271ddfab4000901691761c2b7fd423b93d7f81ab757f0300c58c
505	Mathematics	SFT-MATH-EXACT-RELATIONS-002	Exact ratio, separation, reciprocal, threshold and cycle composition	depth_independent	independently_replicated	256	4	sha256:8141546a120b73aa7cac0cd843728bdf3122d1bf3c0048c132760c82fd103238
506	Mathematics	SFT-MATH-ORBIT-NUMBER-THEORY-002	Fold orbit number theory and exact prime-spectrum structure	depth_independent	independently_replicated	256	4	sha256:8725bce36fe77923288b99b1a978ef3bf84cac9ffe1b172fd67bd221527d19ff
507	Mathematics	SFT-MATH-LIMIT-CONTINUUM-002	Potential infinity, exact refinement and the continuum boundary	depth_independent	independently_replicated	256	4	sha256:a32c6d6ee891f97acadd53cbec922bbcad5a6ad8a928264bb654b4caaf997098
508	Mathematics	SFT-MATH-ALGEBRAIC-BALANCE-002	Positive polynomial-balance certificates for algebraic magnitudes	depth_independent	independently_replicated	256	4	sha256:ba975fdfe0c8946cb713eabcd9f48e77284941ec08fc257b9d3bb614b701540d
509	Mathematics	SFT-MATH-BOUNDED-N-BODY-002	Exact recurrence of every finite componentwise Fold configuration	depth_independent	independently_replicated	256	4	sha256:e355bc5aa649df8fe2daf21a3e0ae56d5582680edd7f867a751bb954fca224a4
510	Mathematics	SFT-MATH-FLOORED-FLUID-REGULARITY-002	Depth-independent discrete-gradient bound for floored Fold fluids	depth_independent	independently_replicated	256	4	sha256:6fb3f40f5631367f23bad6cda58c87b335d33fd212a7c9d37068d8b920ad65e0
511	Mathematics	SFT-MATH-PRIME-PAIR-CENSUS-002	Exact bounded Goldbach and twin-prime census	depth_independent	independently_replicated	256	4	sha256:52fcc090c336360dcdfd4b95919ffa5926d9d8fcd430ff6bb53d7d18cf5db1d6
512	Mathematics	SFT-MATH-RIEMANN-MIRROR-002	Unique half-One reflection axis and Riemann correspondence boundary	depth_independent	independently_replicated	256	4	sha256:27847629dbd2fb1d1c2f14d05017f8fb9395482d6eca001a24622e55364dfc4
513	Mathematics	SFT-MATH-COLLATZ-FINITE-CENSUS-002	Exact bounded Collatz census and contraction-control correction	depth_independent	independently_replicated	256	4	sha256:3bb3a9fe4790150369defc5b2dc8568fed100b688dbb6a9c1d5cb40c4f9a045
514	Mathematics	SFT-MATH-SELF-SIMILAR-CONVERGENCE-002	Exact self-similarity, chaotic rate and convergent-series laws	depth_independent	independently_replicated	256	4	sha256:0abf8d557108f37f4d9cfe735db3fd6e0d79a3d9405129f0a40ae86fbb606d5b1

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515	Classical Computation	SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002	Unrestricted native Fold Busy-Beaver law	depth_independent	independently_replicated	256	4	sha256:92715d1d41f9021e01c27ac9962f7fa8c172b1c5237dd623bbab6da5bc1d143d
516	Classical Computation	SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002	Fold-P equals Fold-NP law	depth_independent	independently_replicated	256	4	sha256:a6d19f95f567cd3d870b45330dbcb3b3387b5b3b47390c5ee1b92cbc5d2f6d74
517	Classical Computation	SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002	Arbitrary admitted Fold-circuit lower-bound law	depth_independent	independently_replicated	256	4	sha256:18e485a78cc263bd16ceae80afee117f5971e549802d60b696645652b596074d
518	Reversible and Quantum Computation	SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002	Unbounded finite Fold quantum fault-tolerance law	depth_independent	independently_replicated	256	4	sha256:97e061e60929c595adbc3706113e920f80a7b9ea224b3606893edd4c244d445b
519	Physics	SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002	Complete prime-sector force and mediator ladder	depth_independent	independently_replicated	256	4	sha256:1e22b2eda37dc8afe38d740cf10db286fb2e5f238704a22ccfee7066c8cbaa85
520	Physics	SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002	Electroweak Fold-fibre mixing law	depth_independent	independently_replicated	256	4	sha256:c15ae82a6cb030fc2b13b36ab6099bdbaf8faffb0092a7537e44aaa0d160db0c
521	Physics	SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002	Squared proton-to-Planck hierarchy	depth_independent	independently_replicated	256	4	sha256:d1c1f13e86e2d9aeaa5f6993456b5f594a303d75e446ea1d9c80d41906b984d9f
522	Physics	SFT-PHYS-NEUTRINO-PMNS-ANGLES-002	Exact Fold PMNS squared-support triple	depth_independent	independently_replicated	256	4	sha256:f15bedd63f3cddc56f4ca7397633b6081ad47086cf77496cfd834bb82f5a5287
523	Physics	SFT-PHYS-ELECTROWEAK-WZ-RATIO-002	Exact squared W-to-Z Fold relation	depth_independent	independently_replicated	256	4	sha256:a69443396f1fdf1f9beb11e65d2cc318b1c01dc5864e52d52a22eaebee1a4e89
524	Physics	SFT-PHYS-STRONG-RUNNING-DIRECTION-002	Self-sourced colour running direction	depth_independent	independently_replicated	256	4	sha256:c9628d8af62fc85936b91328ab2d539f4a5def412e3a61c4cf36c0b6d5a8d412
525	Physics	SFT-PHYS-NEUTRINO-CP-PHASE-002	Self-antipodal maximal Fold phase carrier	depth_independent	independently_replicated	256	4	sha256:66a61437fe9e5f7101002e716423413dbeb111125e7c9c4d540f179a8c1eb12f
526	Physics	SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002	Bare Dirac gyromagnetic Fold value	depth_independent	independently_replicated	256	4	sha256:2d4d2cf74bd6c8b589961f0ad65b4ebe3f614317493daa65c4024de4ef2ad66e
527	Physics	SFT-PHYS-WEAK-PARITY-FIBRE-002	Weak parity from the held Fold fibre	depth_independent	independently_replicated	256	4	sha256:5e33439d3af142c055b5455921783e65a846e8d3d9bbf588409022480f263454
528	Physics	SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003	Terminal exact on-shell electroweak Fold share	depth_independent	independently_replicated	1,024	4	sha256:263cf79e79ed1d7ba0246f84abe869084aa4db7b195dd0c70300d1dabdef72f4
529	Physics	SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003	Terminal charged-colour proton-to-Planck hierarchy	depth_independent	independently_replicated	1,024	4	sha256:affd1dc6754854b15d7f286291611e481f0572f50a11589dd88cb019289f88d7

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530	Physics	SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003	Blind PDG test of the terminal electroweak Fold share	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f4fd9f4991baf12de960ff990065313fa6292e4c23cda53995b2019f0e9acbc6
531	Physics	SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003	Blind CODATA test of the terminal proton-to-Planck hierarchy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6483314f23ac951735fa6b201281236b523f2ecbfc33e73813dd92d3501264b8
532	Physics	SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003	Complete finite prime-sector force and particle inventory	depth_independent	independently_replicated	4,096	4	sha256:ee6b271f050a867f8e8d5ddcd10e7dba7f7ccba227b4094efbd5c6e2b48741fb
533	Physics	SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003	Blind PDG anchor test of the complete force-sector formula	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:88a3d6273a4332a6eb1dbe565e812e59d6ec768c094c0e37f1f81b3bcc0c00ba
534	Physics	SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003	Half-One vacuum floor and oscillator spectrum	depth_independent	independently_replicated	256	4	sha256:288674203497f376e9e6dbd0cb8aba169f1f2343b3b7717dbac789c886759548
535	Physics	SFT-PHYS-VACUUM-ODD-RECURRENCE-003	Perpetually live odd-denominator vacuum recurrence	depth_independent	independently_replicated	256	4	sha256:dad4e023733464a86c1fcd9913bbe661064c10e67d270e47cc1430d3846a821
536	Physics	SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003	Vacuum polarization and running direction	depth_independent	independently_replicated	256	4	sha256:40f95e89489d596b961aeb5d234dca5938b41b0dd6de02ffae7da90e2d66d76
537	Physics	SFT-PHYS-VACUUM-INERTIA-UNITY-003	Exact vacuum-to-inertia unity relation	depth_independent	independently_replicated	256	4	sha256:3e1b000691688a8bec6ab6301c21ae05136256e08f10d0e0e36abb0dc602d71
538	Physics	SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003	Positive asymmetric vacuum-beat extraction	depth_independent	independently_replicated	256	4	sha256:de7c651c4eccf51ac10efdc04853bcd36fcb3ef0dcbe3ef58bba0cf52efc0e1d
539	Physics	SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003	Complete returned-cycle vacuum energy ledger	depth_independent	independently_replicated	256	4	sha256:a0ae88fcb49661c576a54741af33c09c88fb11d125f48e41edf2658b08ede835
540	Physics	SFT-PHYS-VALIDATION-VACUUM-FLOOR-003	Post-seal NIST test of the nonempty vacuum floor	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f14ff2733c67bd9100858be8410ccf0effba1632d77b481630baa6749930834d
541	Physics	SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003	Post-seal PDG test of the vacuum-polarization running direction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2faf46c5d43a2e2026a6ea4dc78cb52557592b262b6801d3ccda2c6a594b1d5e
542	Physics	SFT-PHYS-VALIDATION-VACUUM-INERTIA-003	Post-seal MICROSCOPE anchor test of vacuum-inertia unity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9dbc21cecc28efb90789178529c54474aa4077fbd1452065ef038cad34c25d2f
543	Physics	SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003	Post-seal NIST boundary test of vacuum-work extraction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ee9efc5016007be792ea5f6a8cce8f5efb6b7b45ef997c64709296d250b5922
544	Physics	SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003	Free-particle Fold phase and exact dispersion	depth_independent	independently_replicated	256	4	sha256:d49c884c63339615c07f1b03712b44675c84df8cf0ebd84299aabc23645cba82

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545	Physics	SFT-PHYS-DYNAMICS-POTENTIAL-EVOLUTION-003	Potential evolution and additive exact phase	depth_independent	independently_replicated	256	4	sha256:2c0f29888cbfe276f5c1b03d27dc809046841d57b2857fe60a2f2f43c3a341f7
546	Physics	SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003	Stationary Fold spectrum	depth_independent	independently_replicated	256	4	sha256:b144cf1ec79dc2f2086a89f90d3781f4165344e1170161469f2e800823e584e2
547	Physics	SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003	Two-hand exact Dirac square	depth_independent	independently_replicated	256	4	sha256:8781e3f2ddd1b3980b2ab928c942ccc4db1bccac6527d0723c0a797855991279
548	Physics	SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003	Full three-plus-one Dirac square	depth_independent	independently_replicated	256	4	sha256:8b58f344d672ea8b89164ee8c5e46e4faacbleda64aa4d0d682fa4931cd30a4f
549	Physics	SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003	Charge, Coulomb potential and Gauss closure	depth_independent	independently_replicated	256	4	sha256:f30356c4d6f5fc9478100b9626b8303088cead7845844e1dab28c744761b5197
550	Physics	SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003	Magnetic relativistic correction	depth_independent	independently_replicated	256	4	sha256:8039e9c6e59176d5f9a09a7796e95e95fdf2587b1abc870becfdd5d715fe45dc
551	Physics	SFT-PHYS-FIELD-LORENTZ-TRANSFER-003	Lorentz transfer partition	depth_independent	independently_replicated	256	4	sha256:353351d17f145e0f89d0db4a7250469284dd4bae4732c1961bf519bfcd5f211f
552	Physics	SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003	Planar Maxwell curvature closure	depth_independent	independently_replicated	256	4	sha256:7178eac96fce3c977c64c4e4a9d5e681f1e48a91d8561045c6020979a476460b
553	Physics	SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003	Three-space Maxwell curvature closure	depth_independent	independently_replicated	256	4	sha256:e9314b2725d52aa3136325aee8ba08bca5fbc41e0ee3ee19db391ff98b7c1dac
554	Physics	SFT-PHYS-WAVE-EXACT-OPERATIONS-003	Exact wave, polarization, interference and nonlinear operations	depth_independent	independently_replicated	256	4	sha256:bc9f7fff3079a79ca77a60d0680da1b8474ad13f0effedbf8b5cce9a75513e03
555	Physics	SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003	Floored finite-loop closure	depth_independent	independently_replicated	256	4	sha256:654937a341dfef2a5efc3ae1094a90b940e3535923bc1aa465d64e8930549853
556	Physics	SFT-PHYS-VALIDATION-DYNAMICS-SPECTRA-003	Post-seal dynamics and spectra comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b9d360bc08458cdb4507806883d1fd07fdf5bd1ce119192ee95b86ae6bf9c0b7
557	Physics	SFT-PHYS-VALIDATION-RELATIVISTIC-FIELDS-003	Post-seal relativistic and electromagnetic field comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be5db432d9abee0d45670c13606e3b5ab4b501ac626c457a458365c530c01c6e
558	Physics	SFT-PHYS-VALIDATION-OPTICAL-OPERATIONS-003	Post-seal optical-operation comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:73d24227b39bed29d9648706021b5d098732eb9e054568f70d5144bc29f3881b
559	Physics	SFT-PHYS-VALIDATION-FINITE-LOOPS-003	Post-seal finite-loop and renormalization-boundary comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78f1d28a0a4ca09b87ef08f5ebc95af026e2d90f372a48ddb5d735cdf21140b3

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560	Physics	SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003	Weak gravitational potential and Gauss closure	depth_independent	independently_replicated	256	4	sha256:35145c1a594d6707add50a3f2e72dd3a2162092390fd9b3db03eed38b79ee147
561	Physics	SFT-PHYS-SPACETIME-EXACT-INTERVAL-003	Exact causal interval	depth_independent	independently_replicated	256	4	sha256:c5560ae617b8b077c8d945e3f9b40475f2ca9c4b64028b4a7ea339753d936c4f
562	Physics	SFT-PHYS-GRAVITY-STATIC-CLOCK-003	Static Fold-covariant clock	depth_independent	independently_replicated	256	4	sha256:8c03d3fd2812c980509616560286fd28ee1b29400c611cde02ef129d15fa8cb4
563	Physics	SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003	Gravitational-redshift and acceleration equivalence	depth_independent	independently_replicated	256	4	sha256:6726f8bb1dd6a005b03f9058957774e3e97b2d14f7850901331494aefef7185c
564	Physics	SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003	Exact lattice curvature family	depth_independent	independently_replicated	256	4	sha256:f10d0261e22c0cdfbce4d91bd75653851064c132325530b1b57e2da5e06ee457
565	Physics	SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003	Gravity self-sources through field energy	depth_independent	independently_replicated	256	4	sha256:ce5bb88d7247a7c82c3aa1050b1e4a5261acb4e4cbc301489b504668a87c5fbe
566	Physics	SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003	Graviton polarization count	depth_independent	independently_replicated	256	4	sha256:708fffd5a309a9cee077c2db46aef2bf8374435ec2d7bdf4685ef1fb719d0d73
567	Physics	SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003	One-speed quadrupole gravitational propagation	depth_independent	independently_replicated	256	4	sha256:e0769c5998c353edbc1b4d0a6df76c0c409591e722297df3803cabe99e73ab1
568	Physics	SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003	Strong-field horizon and finite floor	depth_independent	independently_replicated	256	4	sha256:3c02d56ccea9efaed7a9fc9f065e27145f2b6597504a447e3b4ac623e61cc6af
569	Physics	SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003	Horizon information closure and reconstruction	depth_independent	independently_replicated	256	4	sha256:db27fe2ffddb49475498a04db13944c1443b625794229c3b22267e143520a2
570	Physics	SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003	Wormhole admissibility boundary	depth_independent	independently_replicated	256	4	sha256:ad75329a73a9f31385c1e0e991770bed8415bafa318be84afb8f278ca322d96d
571	Physics	SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003	Warp-support admissibility boundary	depth_independent	independently_replicated	256	4	sha256:b864f95be8c08840d7e5e99ab664e84404d6121fa17385ed3fe751d5a53e5ab9
572	Physics	SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003	Closed-timelike recurrence admissibility boundary	depth_independent	independently_replicated	256	4	sha256:101b8251369135bcf188c580c15f045d3f54cc502ad46813ded8b478b60b19cb
573	Physics	SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003	Post-seal gravity, clock and equivalence comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a78e700c5fd4bed0bafef60270e978e25166dcc20ddc31b7d8c62efb6daf15040
574	Physics	SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003	Post-seal gravitational-wave speed, polarization and quadrupole comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:627976ac082693a34548353be3a26eae93d6804108d549a4ab6411c0575f70ea

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575	Physics	SFT-PHYS-VALIDATION-GRAVITY-HORIZON-003	Post-seal horizon and information-boundary comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5fab7cea569fcb273a9b39aacdd824a37ed42f3939fd494517dcb62619a95752
576	Physics	SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003	Post-seal nonstandard-spacetime realization boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:58e0992e82b5379cb948eed83b8ea8babc83386599c5c8ab93eafb4705fb3
577	Physics	SFT-PHYS-MATTER-QUARK-INVARIANTS-003	Quark channel invariants and cover depths	depth_independent	independently_replicated	256	4	sha256:8dad6c72f49f916be9fd5d4aba1c3837e86795580ef246403098aed2a2b34b63
578	Physics	SFT-PHYS-MATTER-QUARK-CUBICS-003	Dual exact quark mass cubics	depth_independent	independently_replicated	256	4	sha256:e6fccd6c428a5ee8c9126ae25ef145d8c5780d69465d1bf878d3ff39b34969e3
579	Physics	SFT-PHYS-MATTER-QUARK-DRESSING-003	Terminal quark self-coupling dressing	depth_independent	independently_replicated	256	4	sha256:4156d17fa99d73374151d8abebba074fb43f2d9852ae3c3e15564f8a49dd7f6e
580	Physics	SFT-PHYS-MATTER-CKM-FIBRE-003	Exact CKM fibre alignment	depth_independent	independently_replicated	256	4	sha256:2cf6948eb88b328bc6efb24b394a37cb8c6d2c32044196fa39ae189488976ea3
581	Physics	SFT-PHYS-MATTER-CKM-PHYSICAL-003	Mass-root CKM physical mixing law	depth_independent	independently_replicated	256	4	sha256:beaa25910b1a8fb944c90bca43b8cf07fec997756bb3cf3b96724133eee18d4a
582	Physics	SFT-PHYS-MATTER-PROTON-ELECTRON-003	Proton-to-electron hierarchy reconstruction	depth_independent	independently_replicated	256	4	sha256:456305f867a1afc75a47afa355364f48132b901c4c1f0f7a7ace9a40be64b988
583	Physics	SFT-PHYS-NEUTRINO-SPLITTING-003	Exact neutrino splitting ladder	depth_independent	independently_replicated	256	4	sha256:037d94c3ae94a015aa5177ee829fcd77e1137f5cdb10cfb2a9476f7a4bf4f5b7
584	Physics	SFT-PHYS-NEUTRINO-POSITIVE-MASS-003	Positive absolute-neutrino mass structure	depth_independent	independently_replicated	256	4	sha256:049c0ce9ac0b05591b8636141be26c976a31f39aabb5941ad6aca5ffb60dd7fd
585	Physics	SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003	Physical PMNS and CP support	depth_independent	independently_replicated	256	4	sha256:6cc0b2828e8a304fd80c274f6e9a2e722cd1bebd68cff071f4c394bf25c872bd
586	Physics	SFT-PHYS-NEUTRINO-MAJORANA-003	Single-hand Majorana discriminator	depth_independent	independently_replicated	256	4	sha256:da6688fb7442a5d3f4e3aa39f745c04aebc87102af60186b6e1bcd4e7078345e
587	Physics	SFT-PHYS-NEUTRINO-ZERO-NU-BETA-BETA-003	Neutrinoless-double-beta noncancellation prediction	depth_independent	independently_replicated	256	4	sha256:3b7bc18e1160fe2a3633ce7cba15b9c3001f55ed73516ef9a6a6db474292f6fb
588	Physics	SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003	Electron/muon magnetic-anomaly structure	depth_independent	independently_replicated	256	4	sha256:e0a6844ac70a6e929c5e23ed576df2ea5c13eb4a05d33b148b2b3ad75a4c4f6f
589	Physics	SFT-PHYS-VALIDATION-QUARK-CKM-003	Quark dressing and CKM comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:42b3d23953c8914d3affd7385c6f52158dca86b01131a4748c45d00600480402

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590	Physics	SFT-PHYS-VALIDATION-PROTON-ELECTRON-003	Proton/electron precision comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c9c81674c4671965244d499e7846cfd6154e3b256db1aaf0e14ef32e58972292
591	Physics	SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003	Neutrino mass, ordering, PMNS and CP comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f7ce6f015aa32501a5e90ce65ee001c92b60d9c3543a9b062607c65aa5b84d8f
592	Physics	SFT-PHYS-VALIDATION-MAJORANA-ZERO-NU-003	Majorana and neutrinoless-double-beta status comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:67bc72ab10d4e08f78495417337900a4543ad146d4c03c543e86d5e2ecc6851a
593	Physics	SFT-PHYS-VALIDATION-MAGNETIC-ANOMALIES-003	Electron and muon magnetic-anomaly comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c203e50408fe3b0bcf653ce46fea4e72b8947ddc0c8103054b1db3f8ab44dfdd
594	Physics	SFT-PHYS-MATTER-BARYON-PHOTON-003	Baryon-to-photon Fold transport law	depth_independent	independently_replicated	256	4	sha256:2455218c2d7918d015bcacceffca6308da79a73beedf33f9ecd5a715413ceb
595	Physics	SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003	Common quark/lepton mixing construction	depth_independent	independently_replicated	256	4	sha256:8a8a498160b16bb65ebfc1071d404a5db6de08b6c9f3499765889ca17677b501
596	Physics	SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003	Depth-independent Fold mass-ratio family	depth_independent	independently_replicated	256	4	sha256:2bd73d71dcee6d09c2d427dale1ac79c1866ef75949c5571509fd37f8fd21168
597	Physics	SFT-PHYS-MATTER-MIRROR-MASS-CLOSURE-003	Mirror-closed lepton mass carrier	depth_independent	independently_replicated	256	4	sha256:bd01069f06d421e37672d6a58657824c529f0bf33ce2286dc723a40d1d7855e4
598	Physics	SFT-PHYS-MATTER-INTER-ENTRY-COUPLING-003	Mixing-row coupling return law	depth_independent	independently_replicated	256	4	sha256:c1ef281840dac57df5ac309768874c793c7880fcbbfe5299d1b4deff3f2531ea
599	Physics	SFT-PHYS-MATTER-GENERATION-DEPTH-003	Common generation return depth	depth_independent	independently_replicated	256	4	sha256:a120ad261a4308fc3008ccaab35d38939f6152f82c31f4f00c773582e3f2d726
600	Physics	SFT-PHYS-MATTER-CONFINEMENT-LIFT-003	Lightest-quark confinement lift	depth_independent	independently_replicated	256	4	sha256:53f2509cb96b422753fb8211e2d7fdb93b351d64fb85eb1a90a7dacc717e646a
601	Physics	SFT-PHYS-MATTER-CKM-TERMINAL-004	Terminal physical CKM transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b97a60bba913a679c25137427491abe5c13dc24c641d00125059ce6cb3db90fb
602	Physics	SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004	Terminal baryon-to-photon transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e5bd6eede834a53c25e7b33a41c4bf83a8eeb439e5a89de8fe87f8fd0997680a
603	Physics	SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004	Terminal composite proton/electron precision law	depth_independent	empirically_tested_and_independently_replicated	2,048	4	sha256:eb6a7b8b32dd79a9971ce3a64e1b7b29ebfca54bbb6264fe9be3984dd60fc5bf
604	Physics	SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004	Terminal exact finite turn projection	depth_independent	independently_replicated	512	4	sha256:95e58749591f916244b2fa09fc71e9f8fb13f9db848dc4c2c78ed1d74bd403f0

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605	Physics	SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004	Terminal exact electron magnetic anomaly	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:8f9f8c6663f56a16c11e0b5b5fc344926dc264e868ef6a40c47ef424f72655d8
606	Physics	SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004	Terminal exact muon magnetic anomaly	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:efa486a8ea1c432219f915d288265975fda3c9f0fec9b393c1570737a71b785f
607	Physics	SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004	Cubic atomic support and exact neighbour balance	depth_independent	independently_replicated	256	4	sha256:8fa8b8bb81ea31d715fae3c5f3ceb869b5e4d8afad6c959ff86f12ab2ea7471a
608	Physics	SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004	Depth-independent exact hydrogen spectral ladder	depth_independent	independently_replicated	256	4	sha256:ea9b3b9f34311e1f1f41962a5405ffad6250fcd74d75cf107a3b92d35dd2d464
609	Physics	SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004	Exact atomic gross, fine and first bound-return hierarchy	depth_independent	independently_replicated	256	4	sha256:a6239e61a184cf72e869fabe9c7d2dcc43366e30ef4bdfdf68ebd0ff2a0fd9d
610	Physics	SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004	Atomic transition, selection and field-splitting law	depth_independent	independently_replicated	256	4	sha256:34ea2b8ac5ee23d6cfe9e2ca37a190682c73830e53fb3b0c32ece650e7f8de2
611	Physics	SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004	Exact molecular rotational-vibrational spectral hierarchy	depth_independent	independently_replicated	256	4	sha256:234231a79ab030c84486fbee91cd79a5498ebb4f88c6467c366f38815ff681d0
612	Physics	SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004	Post-seal NIST comparison of the forced six-neighbour cubic support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0086a149cdbc14e727bfd35a6df637e53220728b2d137a8d442a6009dc4c586fc
613	Physics	SFT-PHYS-VALIDATION-ATOMIC-HYDROGEN-SPECTRUM-004	Post-seal NIST comparison of the exact hydrogen spectral ladder	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bb91ca7db989b8462ce2594703fed2705b738a25036f4c6498ef26b93dad47c
614	Physics	SFT-PHYS-ATOMIC-LAMB-SHIFT-TERMINAL-005	Terminal exact hydrogen Lamb-shift successor	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:40acfad3f08aa1358a859a5ab95388d74d055c7d708a9521702b27a2c172ab75
615	Physics	SFT-PHYS-ATOMIC-FINE-STRUCTURE-TERMINAL-005	Terminal exact hydrogen fine-structure successor	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:12eb3bdc8a9340c15fa621a3d37bee91c32fd1336491fa29ddd70f03c8bb1773
616	Physics	SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005	Terminal exact hydrogen hyperfine and twenty-one-centimetre successor	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:0e39c9521ebfe7ad1e66f06b203534155ee1762c660d907d118757cdd9dc346a
617	Physics	SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005	Terminal exact molecular rotational, vibrational and isotope spectroscopy	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:17e6e8ac57a97e1c23334300ebf9bcfddad8a5c9f0dca822119fef004c00762c
618	Physics	SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005	Terminal exact hydrogen reduced-mass and Rydberg-scale completion	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:5d7e0c49ec4455a01a886ceedfaa5522323d691247fb9455dd41afd9c158d90
619	Physics	SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005	Terminal exact atomic shell, filling and periodicity completion	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:d7f0cce03863186e4f97aa36ec68a00797edd4103268c44dc18faceb05c40cf7

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620	Physics	SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005	Terminal exact atomic transition-rate, lifetime and multipole completion	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:41ac193f7b4d9e904e3557ad5e01d9fc529acc0d54d8fd0e141b43a99b666b25
621	Physics	SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005	Terminal exact atomic Zeeman and Stark field-splitting completion	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:4c1b970f750e939d90fa57d37316a7fbaeb9883643fb582613aea733e38b9c4b
622	Physics	SFT-PHYS-NUCLEON-BINDING-TERMINAL-005	Terminal nucleon composition, binding dominance and mass ordering	depth_independent	empirically_tested_and_independently_replicated	4,096	4	sha256:f61d912f53959182fd1b5b3544dd750bd0beb095b4c72adc11323f375df90762
623	Physics	SFT-PHYS-HADRON-REGGE-TERMINAL-005	Terminal light-hadron multiplets and depth-independent Regge support	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:66bc011c6612641467f7deb8554279abbc55311f86f34efab51ac3fa2f718b29
624	Physics	SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005	Terminal residual nuclear interaction and inverse-mediator range	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:918406dee504d2f2b2ce5f178a98d88a3fb7cc3dd52fea8e948a675da034201d
625	Physics	SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005	Terminal zero-parameter nuclear binding curve and stability maximum	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:08d480f86053d84de80bb82879ba57fc11d0dbcba02557045244e7998cf a65a5
626	Physics	SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005	Terminal radioactive transition topology and exact half-life survival	depth_independent	empirically_tested_and_independently_replicated	1,024	4	sha256:6c9903242fbe62385d16d103f5d7b27a214f4e6d1e2adbee47f845daf62363f
627	Physics	SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005	Terminal fusion/fission direction, barrier and release law	depth_independent	empirically_tested_and_independently_replicated	1,152	4	sha256:fbbdba722e6fb27d738ad3cd1df1f62e4caa979038b212147120ca814f0e25fa
628	Physics	SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006	Fusion/fission yield ordering and distinct threshold carriers	depth_independent	empirically_tested_and_independently_replicated	5,832	4	sha256:1b13169930d9f4c22d75008ef7debe615a8c1dfdedee7a5f495b85c467444e50
629	Physics	SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006	Terminal deuteron binding, spin and dinucleon exclusion	depth_independent	empirically_tested_and_independently_replicated	2,592	4	sha256:c2a2ca11504bbe66760011c57dff2e8459f171c836b754d24b2ae3c200fa4cbc
630	Physics	SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006	Terminal Fold scattering, Rutherford angular law and Compton transfer	depth_independent	empirically_tested_and_independently_replicated	2,916	4	sha256:2c2e6212e5bb27a2b7647e8eca1583bc75775c886f2185885a7925d83e195b91
631	Physics	SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006	Terminal Fold decay widths, exact branching partitions and lifetimes	depth_independent	empirically_tested_and_independently_replicated	8,748	4	sha256:a4e14f20a7c1f6920115f0e5ad0b911b5462b690aad35ee79aac39b691d12d3f
632	Physics	SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006	Terminal Fold running functions and common-support convergence	depth_independent	empirically_tested_and_independently_replicated	8,748	4	sha256:b9263f585f01a2401261f465ad250f405a9914da5c53e57ba113f076aeb4f309
633	Physics	SFT-PHYS-ODD-LATTICE-ALL-REGION-OCCUPANCY-TERMINAL-007	Odd-lattice all-region occupancy and exact recurrence vector	depth_independent	empirically_tested_and_independently_replicated	2,916	4	sha256:185a5dfb16ba14910625b0e2e94a894e7f422cf8dfe65f824ab4f9f5759b07fe
634	Physics	SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007	Half-One pair synchronization and coupled-ensemble terminal law	depth_independent	empirically_tested_and_independently_replicated	2,916	4	sha256:c761476abce4bdf4d357022245a0d069f2ca5f30cb34ee917e8a2f7578a5c2d0

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635	Physics	SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE- -TERMINAL-008	Exact Fold expansion and information-rate carrier correspondence	depth_independent	empirically_tested_and_inde pendently_replicated	2,916	4	sha256:f0b74a8639f9e01be7b9adc31f83b4b24ed82f085e72c17736c38f030b5cae40
636	Physics	SFT-PHYS-COUPLED-MAP-CRITICALITY-TE RMINAL-008	Exact normalized coupled-map synchronization threshold	depth_independent	empirically_tested_and_inde pendently_replicated	2,916	4	sha256:845ceed3f993a0523e1f01a5ce9375d469e426be12539941c5e5fd5130b9534f
637	Physics	SFT-PHYS-ORBITAL-DIMENSION-STABILIT Y-TERMINAL-009	All-dimension orbital-restoration stability law	depth_independent	empirically_tested_and_inde pendently_replicated	2,916	4	sha256:9a067d10ece75b8300bec3c77193649313dbc2fcc116f721bd0fef54544a433c
638	Physics	SFT-PHYS-POST-NEWTONIAN-FIXED-POINT -TERMINAL-009	Exact gravitational self-source fixed point and correction contraction	depth_independent	empirically_tested_and_inde pendently_replicated	2,916	4	sha256:b0e30a2cb8fa9d9f2232df8f8790ab82d8fc2f8e6555326543e15fce2fb48bba
639	Physics	SFT-PHYS-SYMMETRIC-SOURCE-CONSERVAT ION-TERMINAL-010	Four-coordinate symmetric curvature-source conservation law	depth_independent	empirically_tested_and_inde pendently_replicated	2,916	4	sha256:79b12fc32760832dc52e87fcf8d743575ebe2f944c18e7cb24cbb41a6d8a5b22
640	Physics	SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERM INAL-011	Exact static spherical exterior coefficient, flux and clock law	depth_independent	empirically_tested_and_inde pendently_replicated	2,916	4	sha256:fca4c68a06026daea19121fd5b71d52bc98e2410be1f83eaeef13ff62600d880c
641	Physics	SFT-PHYS-QUADRUPOLE-RADIATED-POWER- TERMINAL-012	Exact quadrupole third-difference radiated-power law	depth_independent	empirically_tested_and_inde pendently_replicated	2,916	4	sha256:e43448d651ade64487e4cbd44620c4812ffafcb62d20bd71c9c3bde51cd b4db6
642	Physics	SFT-PHYS-STRONG-CARRIER-MASSLESS-CO NFINE-TERMINAL-013	Simultaneous massless, One-speed and confined strong-carrier law	depth_independent	empirically_tested_and_inde pendently_replicated	2,916	4	sha256:bae91c8dc0b88c6130260c7e10ea22e9433ca322bd5ad3a8f9077653315f57e1
643	Physics	SFT-PHYS-STRONG-FIELD-NONLINEAR-FIX ED-POINT-TERMINAL-014	Strong self-source iteration and finite fixed-point boundary	depth_independent	empirically_tested_and_inde pendently_replicated	2,916	4	sha256:641cdf7a00a56d3889b1a31731dff04c70dbe068465924f005d000d3fc100cdc
644	Physics	SFT-PHYS-COUPLING-ACCUMULATED-SEPAR ATION-TERMINAL-015	Finite accumulated Fold-coupling separation	depth_independent	independently_replicated	512	4	sha256:723ff785cf91a40c287d2f2b38394a63fd8bfaa7a0be7a34be2e9e973202f4bf
645	Physics	SFT-PHYS-DYNAMICS-SYMMETRY-ACTION-T ERMINAL-016	Finite Fold symmetry, conservation and least action	depth_independent	independently_replicated	512	4	sha256:3c158a5e6f5ad82f89435d0b00c6542d1230e20706f5cdadad0db202864d4075
646	Physics	SFT-PHYS-SCATTERING-PARTITION-PATH- TERMINAL-017	Exact two-fibre scattering partition and reciprocal mean free path	depth_independent	independently_replicated	512	4	sha256:b4b42d9e9e0a52a703a059c265cc94325c4f56907da3422c4cb83b0facd9efdf
647	Physics	SFT-PHYS-THERMO-LANDAUER-DEMON-TERM INAL-018	Exact one-distinction erasure cost and Maxwell-demon ledger	depth_independent	independently_replicated	1,024	4	sha256:b130990c39c87ac65a1f5728a4c471bc05f8f5720d0413feebfffb5b03760f88b
648	Physics	SFT-PHYS-THERMO-LANDAUER-EMPIRICAL- 019	Post-seal Landauer erasure and Maxwell-ledger comparison	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:eb264a10c1fe3666cf25a5e18beb9bab271534333bfdee25fadc0ffa9cc3a93e
649	Physics	SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TER MINAL-020	Preserved/broken Fold channels and inverse mediator range	depth_independent	independently_replicated	1,024	4	sha256:3ad2016a90d7adc2c1c5c5afdfdaf5510f73ade146e2ea2e0a097d9158621f54

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650	Physics	SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021	Fold baryogenesis dependency and retained-residue law	depth_independent	independently_replicated	1,024	4	sha256:738dd11360b3cb43280ce6b595023c6f6408166ab6ef96ad4f8e9a6d4592c7b4
651	Physics	SFT-PHYS-LATTICE-OPERATOR-TERMINAL-022	Exact Fold lattice operators, causal cones and phase modes	depth_independent	independently_replicated	1,024	4	sha256:2d54a071b1a994f26b3a05083d0d57ec33659b9d5430b17467f5dfd03d525fb8
652	Physics	SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023	Single finite Fold quantum-gravity composition	depth_independent	independently_replicated	1,024	4	sha256:a627c3db6aff944a007b53df0a06975454cec52c113e672da62a440ebfd191e7
653	Physics	SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024	Composite Fold-orbit topology and inter-universe transport boundary	depth_independent	independently_replicated	1,024	4	sha256:b03563b484c95269254dd2bb409ae42efbd2a71989eba3ac1b3dd8171c7e467d
654	Physics	SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025	Prime-sector standing-mode, interaction and period unification	depth_independent	independently_replicated	1,024	4	sha256:464f20ad41d716a52836a9d5952a62991e76b488a2ea70539ea6f4c0ee8c8f0e
655	Physics	SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026	Finite Fold colour-singlet positive excitation gap	depth_independent	independently_replicated	1,024	4	sha256:899e904600c1018041286e947de6acefe6c1afb50f1f8da202cfda587fa9568a
656	Physics	SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027	Post-seal lattice-spectrum boundary for the colour-singlet gap	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d85984ba94789e15d9fdb97588e63d318b014b3e84315ace56cb50c16cbeb2fe
657	Physics	SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028	Terminal colour-channel alpha-squared proton-energy fraction	depth_independent	empirically_tested_and_independently_replicated	3,072	4	sha256:41f17af1f94905b0e93d67d7dfc7a399fd9063a53e6db4d21d486e1a8c5f1673
658	Mathematics	SFT-MATH-SCIENTIFIC-CALCULATOR-003	Exact traced SFT-native scientific calculator	depth_independent	independently_replicated	1,024	4	sha256:f95ac235ac96429b36b10e8331478f110ed074e98a338c6f015d3f8874dc787c
659	Mathematics	SFT-MATH-SCIENTIFIC-CALCULATOR-004	Exact traced SFT-native scientific calculator with corrected enclosure parity	depth_independent	independently_replicated	1,024	4	sha256:2d243dbd08260c483e79009067051239f1304ad3998006c1ff82c301062ba28b
660	Physics	SFT-PHYS-PROTON-RADIUS-TERMINAL-029	Terminal probe-independent proton rms charge-radius relation	depth_independent	empirically_tested_and_independently_replicated	2,304	4	sha256:77a31b069cc1cdd4bdf84f3ca08fb4849ae33fddb02945727f1809d9e4c2373d
661	Mathematics	SFT-MATH-SCIENTIFIC-CALCULATOR-005	Complete exact SFT scientific calculator and cross-platform learning app	depth_independent	independently_replicated	8,192	4	sha256:75b0b5df0397c0aeaba035272561fd0e62949b0ef75580a153c8a109e84a06bb
662	Mathematics	SFT-MATH-SCIENTIFIC-CALCULATOR-006	Fully realised SFT scientific calculator and complete Mathematics law explorer	depth_independent	independently_replicated	1,024	4	sha256:addeb561438c9913313824ba43eb8fcec53c574f5a9853b3c9c1fac4e4e4b7
663	Mathematics	SFT-MATH-SCIENTIFIC-CALCULATOR-007	Accessible SFT-native scientific calculator application	depth_independent	independently_replicated	256	4	sha256:feeb1879d97191edc8827be6da5421a2c78aba56ba2d3c0fe5dbf0882aac9867
664	Physics	SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030	Common Fold scale axis and terminal electroweak transport	depth_independent	empirically_tested_and_independently_replicated	12,288	4	sha256:44d542f87d12844c02c321ac651008ae7ddd738a3b16e7d2c3d8700ec89eca55

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665	Physics	SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031	Depth-independent composite confining-sector orbit law	depth_independent	independently_replicated	2,048	4	sha256:eb8b53e1662db23eff80d371b5b6db094e3c7d4665b5ea0a6882ba3072e29ec8
666	Physics	SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032	Terminal cosmic component transport and expansion law	depth_independent	empirically_tested_and_independently_replicated	4,096	4	sha256:504257d441fad45a8f173fad45015c7d723caa7636c517e71d27fddc43a2c42d
667	Physics	SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033	Exact positive Fold velocity-composition law	depth_independent	independently_replicated	2,048	4	sha256:337fccb78afd2207d9976e8034d89d1db75ba9fed2f63c44a3b36ae0e90935a0
668	Physics	SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034	Post-seal Fizeau discriminator for Fold velocity composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:54dce92f903c6f102e4d39f3bdb826a89c05e69e80b6d6df66728711dbc03cf4
669	Physics	SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035	Terminal vacuum floor, cosmic density and scale-transport law	depth_independent	independently_replicated	4,096	4	sha256:c7b4777b12fc70628b0fd9a2f7d957274b61ede7f3624a665781364c8c9f7723
670	Physics	SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036	Post-seal Planck/CODATA test of the Fold vacuum-density scale law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6379f0d35a606d217e2351a35c42ae258aea71c52d8d20d2c4abeeff2 added8c202
671	Physics	SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037	Terminal thermal-history, freeze-out and recombination law	depth_independent	independently_replicated	256	4	sha256:25dda78644ebc04ad43cd3e416950ad76dae6b124ed023f41364f0661eae9c00
672	Physics	SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038	Post-seal thermal-history, nucleosynthesis and recombination test	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7440c78415e834656217aca301aaa8b6f987d6f795544eb5c03b0679970321f4
673	Physics	SFT-PHYS-INFLATION-GROWTH-TERMINAL-039	Terminal inflation, primordial-support and structure-growth law	depth_independent	independently_replicated	256	4	sha256:00aa52b47f8ebfe57fb5a90530a54614ba175d87f6708c69e454fc9fd888aa7
674	Physics	SFT-PHYS-VALIDATION-INFLATION-GROWTH-040	Post-seal scalar-support and tensor-support measurement test	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8e06d75a25206d5cce129a9afd2583a54cb1e070a0daf6eeeff92ece99a8d01f
675	Physics	SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041	Terminal finite radiation, acoustic, laser, plasma and Alfvén response law	depth_independent	independently_replicated	256	4	sha256:50974ab152438d8b84fe8220281e0d98737df0880f9a6384d50bac6060987b07
676	Physics	SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042	Post-seal blackbody, acoustic, laser, plasma and Alfvén response test	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d7dd8e6373027c10bd83aa9ff128ebabe3fdc60569b24ad597f9c22abadd2455
677	Physics	SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043	Terminal exact temperature, finite canonical equilibrium and fluctuation-response law	depth_independent	independently_replicated	256	4	sha256:cf536f7516eb1c85bb075500d9c1a0ebd64bddd544878f26a27424d08de25bf
678	Physics	SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002	Post-seal CODATA validation of the terminal charged-lepton invariant	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1df007d68f197e5a8a8a58dad28190bbaf10229ab2f8136365a25622eba5d5d8
679	Physics	SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002	Versioned post-seal exact Koide measurement validation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4e7a64b6751e5622c827612840751a26b40b61ee657bb011eded7ff7914aa776

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680	Physics	SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044	Post-seal acoustic and Johnson-noise thermometry test	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:60366227e00c50a3529d98538a3ca42d9620f4da30a174b9cfff96f1122097dbd
681	Physics	SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045	Terminal finite spin-statistics, occupation and Bose-condensation law	depth_independent	independently_replicated	256	4	sha256:f71da5b86f99d6569a1f33dc6fc37024cc5d458b12e625c5dbc4faf3c33ccda7
682	Physics	SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046	Post-seal finite BEC, Pauli-blocking and two-turn spinor test	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6c37132ae33b8f23b06e69b7034c5acd39144d51b2661d7e65f441b400604eb
683	Physics	SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047	Terminal exact phase-criticality, universality and turbulence-scaling law	depth_independent	independently_replicated	256	4	sha256:0601d19640943c4b99eb8cccf061e3115c520773eac5e762bf7c5b7440339b25
684	Physics	SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048	Post-seal critical-exponent, universality and turbulence-scaling test	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:075d304c85f48d6886ba7cb1fd44d5db25815cdf3a5d016502da1435654f1bdc
685	Physics	SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049	Terminal preparation-depth, support-uncertainty and joint-correlation law	depth_independent	independently_replicated	256	4	sha256:1560f2e0de3870abac2bdc6575aa9811c4dc013d5a6d705e729a832dc451b79f
686	Physics	SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050	Post-seal superconducting-circuit test of the Fold Bell factorization boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:92401779cd8580dbaf3962207ae25e0b7fc75787f91e27681d1ac290de403085
687	Physics	SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051	Terminal general Fold-fibre, particle-mode and generation-transport law	depth_independent	independently_replicated	1,024	4	sha256:18e4aac0792a01c613d31f4d980951b0e2da69107155c4e50967ed80c9eaa700
688	Physics	SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053	Terminal electroweak measured-value correction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e93f04515944a1951b78463db3083350fbc77e6d8506edbcdc13178cedfbc852
689	Physics	SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054	Common Fold scale measured-value correction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0514d2a2fe7ad505dfd99c494f0c6ac6214169791c09596d4657e32ee1c458d8
690	Physics	SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055	Terminal cosmic transport measured-value correction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e3e00a1c023f76cd0817c7eae86c0243ab10d338e552f6bae1f90a94a959380
691	Physics	SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056	Criticality and turbulence measured-value correction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7e437d81bee9d4c88b304a013c5c980fa9d0390f95acaf30c2dfb45b30a1602a
692	Physics	SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057	Terminal physical helium-isotope closure	depth_independent	independently_replicated	256	4	sha256:a3b3a44a0d3add7032680427c3b2b504147c0eb824d05fa5d044cef454c5ebc4
693	Physics	SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058	Thermal history and physical helium measured-value correction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7339aa07e1a9f52e8ef1fede03cbe17d38834362f1c9d63bb00b05deb6d49021
694	Physics	SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059	Terminal dimensional light-hadron Regge carrier	depth_independent	independently_replicated	512	4	sha256:f82510d61a2e1472c8d090f2fd1f63b53491bab9f875ce4ae27b0e0b6b5fc54e

#	Branch	Claim ID	Title	Formal closure	External status	Candidates	Controls	Receipt ID
695	Physics	SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060	Complete hadron Regge measured-value correction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:15c9daaa90d88c5193e5e3af4a5ca64a75c0418ee85797a2b90607c37a21cb06
696	Physics	SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052	Complete particle-generation, mass-pattern, mixing and longevity comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0a2ca2749eed1cdf6b92ab645ace007b3190ac5e5bb952e3a0cbdedff1c0804
697	Physics	SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061	Terminal dark relic, Smithion spectrum and flavour-transition law	depth_independent	independently_replicated	2,048	4	sha256:71a131119fffd152690ec772019177fb06599474cd5e89b60a62c1b9b69ec5762
698	Physics	SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062	Measured dark abundance and complete Smithion/LFV standing-test record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:51e623e6512223c23a0f0cae604ca556ed30fd15173236fffd1c889dd6424e27d
699	Physics	SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063	Terminal strong-CP alignment and proton/baryon stability law	depth_independent	independently_replicated	2,048	4	sha256:6bd69d2eb78150c11fefc82f512bdfd775eaa2f3a2667127407d7a34bd695657
700	Physics	SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064	Blind neutron-EDM and multi-channel proton-stability validation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:48e9dbebdb29b3902876a09cde2154eea3700904898930c1f12ab0a5e5ed9a09
701	Physics	SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065	Terminal displaced-ground and Higgs mass/self-coupling law	depth_independent	independently_replicated	2,048	4	sha256:3f84e4ede48a489bc78a8cc47eac935bbff3566d1e561c26ef0059d7bbcbfb1e
702	Physics	SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066	Blind terminal Higgs mass and self-coupling validation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f69729204538a9c0949fa5c4e3dfce51bece2ebe2d4e43163910d6192f9b523
703	Physics	SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067	Terminal stellar hydrostatics, galactic response and tidal-lock law	depth_independent	independently_replicated	4,096	4	sha256:24674dfa019c208012414ab3c587656ee404512f20d00ab44e26ee137be22ae2
704	Physics	SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068	Blind stellar, galactic and tidal terminal validation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2410fe8ac97e22ffea7303b50155129dfe6c914d84e549c3c9bb6d24b0672ad6
705	Physics	SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069	Terminal stellar nuclear chain, collapse and heavy-element law	depth_independent	independently_replicated	4,096	4	sha256:a41b28a935f54de5daabdbe77b6c1aba9286f174526b8feaa60829ec848e08af
706	Physics	SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070	Blind stellar nuclear chain, collapse and heavy-element validation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d439aaf47e94dcc7baba1031cf0ac05c446d03fbd089ed86654618d815c2032c
707	Physics	SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071	Terminal compact-object and horizon-thermodynamics law	depth_independent	independently_replicated	4,096	4	sha256:69152da01b20c6b1a7ab6ecfab4e44cf836d588cba4ea6f3b02d5827cb5a0b3f
708	Physics	SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072	Blind compact-object and horizon-thermodynamics validation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3df00caeeae7eb06d7b17539e9eb6282737e70c3a8c9a7208c83009b97acd124
709	Physics	SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073	Terminal gravitational-wave chirp, merger and ringdown law	depth_independent	independently_replicated	4,096	4	sha256:7add0b09e795aee79758286268637f442b40155f4af5c83a4a123f90e2150e19

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710	Physics	SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074	Complete empirical gravitational-wave chirp, merger and ringdown comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:80a9f13dc620d3cb2bf16597d54b63562aeac60ad1bab25d1327aaba8bf576
711	Physics	SFT-PHYS-GRAND-LOCK-TERMINAL-075	Terminal Physics Grand Lock and extension-open no-omission certificate	depth_independent	independently_replicated	4,096	4	sha256:ae18f67371c8e7054430935d6b5e5f3162f24cf9cba073769384bf7ba467d817
712	Physics	SFT-PHYS-VALIDATION-GRAND-LOCK-076	Complete Physics empirical-vector and adverse-result Grand Lock	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:93f4497d6f7ef3c477246079f62c21f96a7ae27fd9516fa876e9d413bbab569e
713	Chemistry	SFT-CHEM-ELECTRONIC-STATE-IDENTITY-001	Molecular electronic carrier and retained state identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:330b9bbf58741bbca7648d7bd1aadede56d70fbfdd5e427f0232d80f870d7a41
714	Chemistry	SFT-CHEM-ROVIBRONIC-COMPOSITION-001	Exact nuclear-electronic, vibronic and rovibronic state composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fea9fff1a16343aa9480d62cc7642e47dc55b5394a362845401e9ace942a1fbf
715	Chemistry	SFT-CHEM-ELECTRON-COUNT-SPIN-002	Exact molecular electron count and held-spin organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f9f7da16969c8222cb4a799c8e87a963f2b07d826d552567e35d15f0ceb17
716	Chemistry	SFT-CHEM-ORBITAL-SUPPORT-OCCUPANCY-003	Fold-native molecular support composition and fermionic occupancy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:726fe11c1cc65a69ba5db15cf70eb1113a760817d5c69e81b0b86e61615eb4cb
717	Chemistry	SFT-CHEM-STATE-ENERGY-ORDER-004	Exact finite ground and excited molecular-state ordering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7231405d08d5f63184bbc074f15578cdf5e505e2051d8ed82077f5b4e893c690
718	Chemistry	SFT-CHEM-STATE-SYMMETRY-DEGENERACY-005	Finite electronic-state equivalence, degeneracy and retained symmetry	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3fbf5236f491ba585d6d0baf70801dd92db65a7b258abb3c0d1b525da3e3fd8
719	Chemistry	SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006	Molecular exclusion and complementary electron-pair exchange organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6b56289c48e5b57443ae836a0b0dffad769c1a717a7d9d14e3a31e43112cb18c
720	Chemistry	SFT-CHEM-JOINT-CORRELATION-DISSOCIATION-007	Nonfactorizable molecular joint support and dissociation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f911d86a7563eee7f6589450425a7bcd5d19cb8e1defb782fc4ffbdaf4d0924
721	Chemistry	SFT-CHEM-MULTICENTRE-DELOCALIZED-SUPPORT-008	Connected multicentre and delocalized molecular support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0bbf92200254061e7a22f38a66abec14bf65c28f09f4d9fe38f38d2117994444
722	Chemistry	SFT-CHEM-MOLECULAR-STATE-TRANSITION-009	Exact molecular-state transformation and transition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d239abbc191bc662020f3315df0dd8b74f95ae196158f4907a47206b531f0e89
723	Chemistry	SFT-CHEM-SELECTION-RULE-STRUCTURE-010	Exact chemical observation-class and selection-rule structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eca9cf4ab69aeb92660506cf11b87e1f4bb7aa0e3a1fa27f05c61680b6a8725f
724	Chemistry	SFT-CHEM-CONFIGURATION-ORDER-PATH-011	Exact molecular configuration order, basin, barrier and path law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3860aa331987bef80cfe7be42265d5bb5eece24486e9e927f855e8b6fdf2bfaa

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725	Chemistry	SFT-CHEM-NUCLEAR-ELECTRONIC-COMPOSITION-012	Exact nuclear-electronic joint composition and isotopologue scale-separation law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f970c3ae1bbacf791154156bb1fba1dc1a743ad0c2742e8ea4739badab2fe16b
726	Chemistry	SFT-CHEM-RESOLVED-ROVIBRONIC-SPIN-COMPOSITION-013	Exact resolved vibronic, rovibronic and spin-state joint-composition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ed3419de39187595fd053d5f6e1e897f1994ebc8c4e562b322a5f5bd21977a1f
727	Chemistry	SFT-CHEM-MOLECULAR-QUANTUM-MEASUREMENT-REDUCTION-014	Exact chemical measurement-reduction law for molecular quantum states	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5fe377a187e37a0fe76eddfc954d2540d35223170b6e73ad7b99da813f481322
728	Chemistry	SFT-CHEM-OPERATIONAL-CLASSICAL-QUANTUM-CORRESPONDENCE-015	Exact operational classical--quantum chemical correspondence	depth_independent	independently_replicated	256	4	sha256:5a7dcf58b16c374f3d0ef8f30cdfb7a592a63dc15b370818ab8857f2a79eadca
729	Chemistry	SFT-CHEM-EQUILIBRIUM-BOND-LENGTH-001	Exact H2/D2 equilibrium bond-length relation	finite_complete	empirically_tested_and_independently_replicated	256	4	sha256:4a9b7cb81b8698229d13832eb27a5a17cdab9fb53f5fb1e244deefab5ec3ae5a
730	Chemistry	SFT-CHEM-BOND-DISSOCIATION-ENERGY-002	Exact H2/D2 ground-state bond-dissociation path relation	finite_complete	empirically_tested_and_independently_replicated	256	4	sha256:23224e2d802e317a0c49a3d2ae914fab62454843a2d660e0b027afb7f73793b92
731	Chemistry	SFT-CHEM-MOLECULAR-BOND-ANGLE-003	Exact symmetry-forced molecular bond-angle relation	finite_complete	empirically_tested_and_independently_replicated	256	4	sha256:744ccd57c01a23f29ae6f4785b6e30d4d1b685e2cc6868674a4bc9acdf11eef
732	Chemistry	SFT-CHEM-DIHEDRAL-TORSIONAL-STATE-004	Exact held-orientation dihedral and torsional-state relation	finite_complete	empirically_tested_and_independently_replicated	256	4	sha256:0a854aa8467aaa502b4160813108bff2c1638acb00a4ae4d48124f227b38272b
733	Chemistry	SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005	Exact molecular dipole organization and squared-magnitude relation	finite_complete	empirically_tested_and_independently_replicated	256	4	sha256:023e38bf6e35d0ae45934d92177d47236bd57f486ea73704ec632bbc95578147
734	Chemistry	SFT-CHEM-MOLECULAR-POLARIZABILITY-006	Exact molecular polarizability response and isotropic-composition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:926c85abc6d212f0264d91a7aef9609633f7a6c0a16bbcfb4cdb877020c6d20c
735	Chemistry	SFT-CHEM-MOLECULAR-IONIZATION-ENERGY-007	Exact molecular ionization-energy and terminal-state ordering law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e507ad134235d5ff6051762ee89a5321ac63ccf0cd1aad7c97cbf734a15ea2a
736	Chemistry	SFT-CHEM-MOLECULAR-ELECTRON-AFFINITY-008	Exact molecular electron-affinity state-order and magnitude law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6643867ae60a9d82388857fe0b1a21d1fded8ada313ee22c884dab97f9358a54
737	Chemistry	SFT-CHEM-VIBRATIONAL-FREQUENCY-009	Exact molecular vibrational recurrence-frequency law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b4073dd9088629d88d36ff88678f287dc7973aa06a39013bb2ad0a5445433dd7
738	Chemistry	SFT-CHEM-ROTATIONAL-CONSTANT-010	Exact molecular held-axis rotational-constant law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:63886feb3b4de292c6af6642a40d8e23d9f9d8cec031ad62ee28fe2b443c0292
739	Chemistry	SFT-CHEM-INTERMOLECULAR-BINDING-011	Exact intermolecular binding by retained state order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:84a558db081597ed77db5cc0dc4e19dd467ed57378a066985d7d20cd8ee47f7e

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740	Chemistry	SFT-CHEM-MOLECULAR-MAGNETIC-RESPONSE-012	Exact molecular magnetic-response relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cf1bf113ca5c59ba033a73dc62255bb85b7589e64a1767784326c112129bf1b3
741	Chemistry	SFT-CHEM-MOLECULAR-FORMATION-ENERGY-013	Exact molecular formation-energy relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f85de5707fd9bebc25ad8fc8dfdcbb8e927bef17fb3c2227ee37a11fbfb5cc
742	Chemistry	SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014	Complete zero-parameter cross-property molecular vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c3953d770e1837fc70536ba386e631a93a812b8e1008fe71d1635af1386999b8
743	Chemistry	SFT-CHEM-FINITE-MICROSTATE-SUPPORT-001	Finite chemical microstate-support law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ec8803582bbf3daeeaa2e7583a7d534153ea73225d80ebae365c30d3ddf95d572
744	Chemistry	SFT-CHEM-TEMPERATURE-CORRESPONDENCE-002	Chemical temperature correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1f16cb99330ed8077b47621df7cdf3122bb975c5e47d6320fc67cdac852ee6a
745	Chemistry	SFT-CHEM-INTERNAL-ENERGY-COMPOSITION-003	Internal-energy composition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6ebff46c4f5eb74fd58280b2336b0f99c69245a9835381caee0511c1ce0c4fcc
746	Chemistry	SFT-CHEM-HEAT-WORK-TRANSFER-PARTITION-004	Chemical heat and work transfer partition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3afeefa6b1dbde4d0a0000391a9865ad98d601460775ae79a5df5d9178494fb7
747	Chemistry	SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005	Chemical entropy and multiplicity correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be32db30904fd7974578c60b3e656371d428f718d2b9dec8097277d135f32ce1
748	Chemistry	SFT-CHEM-ENTHALPY-EQUIVALENT-STATE-006	Enthalpy-equivalent chemical state relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5b82f6813b158e70e196ccabedafa13a2c287064aa34d91a9eaa64282c8c698f
749	Chemistry	SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007	Free-energy-equivalent reaction direction law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f4cd9bc049a27671c9c722771d41956cc393a30419980ad2efa6b09ba7c62945
750	Chemistry	SFT-CHEM-CHEMICAL-POTENTIAL-EQUIVALENT-COMPONENT-008	Chemical-potential-equivalent component law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:766e2849ce3f2f88465ac16a1309c55925e2738ec60b53f5aa3ac9a10d3605bc
751	Chemistry	SFT-CHEM-ACTIVITY-NONIDEAL-COMPOSITION-009	Activity and non-ideal composition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dde514594020e0de01d67c6dac4357c5f46c427a3512e28aefb37f0e7595b2ca
752	Chemistry	SFT-CHEM-FUGACITY-EQUIVALENT-GAS-MIXTURE-010	Fugacity-equivalent real-gas mixture law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e767c09f6945e2ef91f63ff127007e663055b2d8215b06401e1d37f9db634ad2
753	Chemistry	SFT-CHEM-PHASE-RULE-STRUCTURAL-011	Chemical phase-rule structural relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e98e83e7f17426c14efde71543d97e559531afa80a123e73257b1be1bb500ad5
754	Chemistry	SFT-CHEM-ONE-COMPONENT-PHASE-BOUNDARY-012	One-component finite phase-boundary relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6ec44113a3770dcd2f4e8e479ded39ebf04ba4ee6303a5a8009b127f5e21dc1f

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755	Chemistry	SFT-CHEM-MULTICOMPONENT-PHASE-DIAGRAM-013	Finite exact multicomponent phase-diagram relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fbd07842719bd0ba06c1c54d03e5ad0376f3b1f790a937264a5fcf513b5f4393
756	Chemistry	SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014	Exact colligative composition-response law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f8c933f2777154b409d33968b254da03d8786cab4bb3dfc87e7686d5aef17177
757	Chemistry	SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015	Exact solvation and dissolution free-order law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b14e9d25bec95b67d11287094e2589aaad277f46a2d01302e5b1055014bfaf3
758	Chemistry	SFT-CHEM-MOLECULAR-DIFFUSION-RELATION-016	Exact counted molecular-diffusion law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f7dfd76d3091935073b5eac5cd205ceb53134e52ff85e583d02d3e83d22a9569
759	Chemistry	SFT-CHEM-VISCOUS-TRANSPORT-RELATION-017	Exact composition-bound viscous transport law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c8111e85027bf3cde2f2e532e313e118d1f88278dde6cc7b2094a36741a1099c
760	Chemistry	SFT-CHEM-THERMAL-CONDUCTIVITY-RELATION-018	Exact composition-bound thermal-conductivity law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:08666e0eaf14fe6fcf6a532601c27e29c5aa331301a04b6783d05d0685b2c66f
761	Chemistry	SFT-CHEM-COUPLED-MASS-HEAT-CHARGE-TRANSPORT-019	Exact coupled mass, heat and charge transport law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:25a5d2fe2a127f733017316c3bd650bc4a18d0fdf118e8de6fbcfaf007cf8e6b2
762	Chemistry	SFT-CHEM-ELEMENTARY-TRANSITION-RATE-001	Exact elementary-transition rate relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:64a3ca377719abfbc5af86b49607077dab82af1972e44a3c29ccf9831338190f
763	Chemistry	SFT-CHEM-CONCENTRATION-DEPENDENCE-RELATION-002	Exact condition-bound concentration-dependence relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4892f45dbce65c1aee2563912c23758797804f838aec48ee3fd5adfdb8bbf3fd
764	Chemistry	SFT-CHEM-TEMPERATURE-DEPENDENCE-RELATION-003	Exact condition-bound temperature-dependence relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6bc633600c0f5fdf8d6c3eed4c09ec23bff9eff88220c5aeea8aa9e5103fdbec
765	Chemistry	SFT-CHEM-ACTIVATION-BARRIER-VALUE-RELATION-004	Exact generated-path activation-barrier value relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:50c0543ffcc31753eb293d22583c527329ff998725e840721130fea1b6b1b707
766	Chemistry	SFT-CHEM-TRANSITION-STATE-EQUIVALENT-BOUNDARY-005	Finite transition-state-equivalent Fold path-boundary law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9eee2a251c9da6dafdab46390daca0bd590d4b567de17896f18e2b83a8116a13
767	Chemistry	SFT-CHEM-COMPETING-CHANNEL-BRANCHING-006	Exact complete-support competing-channel branching relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:21b108e96fa5c917f772307b1036cc4900f7d25ff1c144b9193a9899fc3d97eb
768	Chemistry	SFT-CHEM-SEQUENTIAL-MECHANISM-COMPOSITION-007	Exact complete-state sequential mechanism composition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be8cc14ef02150d594aa01d2ec92b07714665d9fbd8fd2be40c483caf5444bd
769	Chemistry	SFT-CHEM-PARALLEL-MECHANISM-COMPOSITION-008	Exact complete-path parallel-mechanism composition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bc2b4cf9b8725fda1618d867ddf5f00f72a1f0c1c175e212fe6ccb5b84a91f8

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770	Chemistry	SFT-CHEM-REVERSIBLE-KINETIC-EQUILIBRIUM-CORRESPONDENCE-009	Exact reversible kinetic-equilibrium correspondence law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6b09df5f93ff4357e250a41622fc4330df40fab5d5aab1dcda40cdaaa23f4f7e
771	Chemistry	SFT-CHEM-CATALYTIC-TURNOVER-CYCLE-FREQUENCY-010	Exact catalytic turnover and cycle-frequency law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0b28dd797f47297cfd8d3af3c3ab496f32cadda8a0a9600ce9110d9ef7e694d9
772	Chemistry	SFT-CHEM-DIFFUSION-LIMITED-REACTION-BOUNDARY-011	Exact diffusion-limited reaction boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9a0aab13e6029bd95143848a3cdc088e2b4a24251abd0cd8e3b06c5787ed929
773	Chemistry	SFT-CHEM-KINETIC-ISOTOPE-EFFECT-RELATION-012	Exact kinetic isotope-effect relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e78e85c742634420e497ccc0b312982f65edd547fa58e040c49c08fe8f80c1be
774	Chemistry	SFT-CHEM-REACTION-DYNAMICS-SCATTERING-PRODUCT-STATE-013	Exact reaction-dynamics scattering and product-state law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8faabcf7374b20bedc9288804c83aec6e4fea13c9c6a3a56504466db2a4b2a7
775	Chemistry	SFT-CHEM-COORDINATION-ENTITY-RETAINED-IDENTITY-001	Coordination entity and retained central-ligand identity law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bf8c0a082163ff1e531df0754a15edff05eb59c8c881f602af3bc4b581c030b6
776	Chemistry	SFT-CHEM-COORDINATION-NUMBER-INCIDENCE-COUNT-002	Exact coordination number from generated central-ligand incidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7b69ed0faf732d306aa10eac8ff74cb7e0cdad9debcb2c45b3b3ab75da4352f8
777	Chemistry	SFT-CHEM-LIGAND-DENTICITY-CHELATION-TOPOLOGY-003	Exact ligand denticity and chelation from connected donor-site topology	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0db5d8789b4be45a9c1579919598c1277b9f2b42e4c28b5bc4d8a218b32f8c4a
778	Chemistry	SFT-CHEM-COORDINATION-GEOMETRY-HELD-ORIENTATION-004	Exact coordination geometry from complete held orientation and adjacency	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bc7f8d5d5235f114984798515944893fc3be3c17a229192cb6e12728ce9d459a
779	Chemistry	SFT-CHEM-COORDINATION-ISOMERISM-EQUIVALENCE-005	Exact coordination-form equivalence and native isomer distinction classes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f10fd2dacc977ff082e2bd425c3a548ce350809e15baf1bd5f529f94a719898
780	Chemistry	SFT-CHEM-LIGAND-STATE-SPLITTING-006	Exact ligand-incidence state partition and balanced splitting	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:848c37693954460e03928ce0db1329df5fcb381ed8847ad7a56087d36bb383c5
781	Chemistry	SFT-CHEM-COMPLEX-SPIN-STATE-ORDER-007	Exact complex spin-state extrema, order and crossover	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:86e888fefab0bd9d8e195edeaf51cf5ec7e9822c849f49afbb0acbb6f4068c3a
782	Chemistry	SFT-CHEM-INORGANIC-COLOUR-ELECTRONIC-TRANSITION-008	Exact inorganic electronic-transition and selective-colour law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa45a1a796871c130839367088a65e6bfa18323f0da03bd63c834e5a79717997
783	Chemistry	SFT-CHEM-INORGANIC-MAGNETIC-STATE-009	Exact inorganic unpaired-support magnetic-state law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cabca6204189701f435c36647c70dd5ecc6dfe4e03a746ac984b2fe3ed6a7a6
784	Chemistry	SFT-CHEM-ORGANOMETALLIC-METAL-CARBON-BOND-010	Exact direct metal-carbon organometallic bond law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:52d096e861416551b25a99cb16f11af5879e0d00e360c200c5f3b3fd19ab9a41

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785	Chemistry	SFT-CHEM-ORGANOMETALLIC-ELECTRON-AC COUNTING-011	Exact organometallic electron-support capacity and accounting law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:39e83309e44945b8cf34698ec3 1d140041412cdd19451d214598fcf3c7c 5011b
786	Chemistry	SFT-CHEM-OXIDATIVE-ADDITION-REDUCTI VE-ELIMINATION-012	Exact oxidative-addition/reductive-elimination inverse law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:9dad1784e9aa6524686d39e0c7 57c65b9dba2355973f668470cf4639e9d 2316a
787	Chemistry	SFT-CHEM-INSERTION-ELIMINATION-PATH WAY-013	Exact insertion, migratory-insertion and elimination adjacency-trace law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:d78ae550b9001c0abdf4ae53a6 76d047c060d850bbfe81e021b74c2ce2d 800e2
788	Chemistry	SFT-CHEM-METAL-CLUSTER-BONDING-014	Exact finite connected metal-cluster and metal-bonding law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:700577fb0682ff8f314409c1d5 c5be7fbcacc966d351c9b4c3d7a3e0d4c 42ffa
789	Chemistry	SFT-CHEM-SOLID-STATE-LOCAL-COORDINA TION-015	Exact solid-state formula, local coordination and ownership law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:c4c74ec1c8d623ccb424542051 4693f5f61e324edc4642b83f108685aae c5cf3
790	Chemistry	SFT-CHEM-DEFECT-NONSTOICHIOMETRY-01 6	Exact defect chemistry and non-stoichiometry law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:a3ffe6b3e015e29ebb712dba07 c85b3d95b0ee8f54ed0a8ce86d808a718 da2e1
791	Chemistry	SFT-CHEM-INORGANIC-ACID-BASE-REDOX- NETWORK-017	Exact inorganic acid/base, redox and reaction-network law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:e70073623d4c8b9da5476274af 3b012bb74c0e85282d335c5b289bf4f45 c17eb
792	Chemistry	SFT-CHEM-CONJUGATED-SUPPORT-001	Exact connected alternating conjugated-support law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:a125efec090d44d39f62cb3e59 e7cc2a0cc9bab6174c74d4f4b654b3d63 fd602
793	Chemistry	SFT-CHEM-RESONANCE-EQUIVALENT-REPRE SENTATION-002	Exact one-carrier multiple-representation equivalence law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:f2c0e4ec6a2c1223b27e058314 66da02ceb92ff44b121759fa09ff673c9 6222d
794	Chemistry	SFT-CHEM-AROMATIC-RECURRENCE-STABIL ITY-003	Exact aromatic recurrence, support sequence and positive stability law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:5339c34ad1f819bbefa0ebed15 404358964ec90488544bdc060c5feff9e a9b28
795	Chemistry	SFT-CHEM-ANTIAROMATIC-NONAROMATIC-D ISTINCTION-004	Exact aromatic, antiaromatic and nonaromatic distinction law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:3ba25097797fbac99c2575eb55 54d30f6a475c4f74116b452716ba04c80 2953c
796	Chemistry	SFT-CHEM-CONFORMER-GENERATION-EQUIV ALENCE-005	Exact finite conformer generation and equivalence law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:9731b0ba4ed8c8186e8ba78f05 1da4a84543f8006fd7e34d9f8351c7f03 a03f4
797	Chemistry	SFT-CHEM-CONFORMER-POPULATION-ORDER ING-006	Exact conditioned conformer population and ordering law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:6668695f637557751274648697 02a2e9defabbd3bd60182cca72c614f12 985d1
798	Chemistry	SFT-CHEM-NUCLEOPHILIC-SUBSTITUTION- FAMILY-007	Exact entering/leaving carrier substitution family law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:7f79c743cc120a8dfb19266933 a69ef142f1cfc52806ae6677920b4f6f7 3b55b
799	Chemistry	SFT-CHEM-ELECTROPHILIC-SUBSTITUTION -FAMILY-008	Exact acceptor-driven aromatic substitution family law	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:e2c208084ac61cb0dc0ba6eeaf d12ad74b3cd556f27a40952c83213e408 7d548

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800	Chemistry	SFT-CHEM-ADDITION-REACTION-FAMILY-009	Exact complete addition-reaction family law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bcc6d8a50dce889570f775f89c2277b832be027ea697f31f1339535e1a42d118
801	Chemistry	SFT-CHEM-ELIMINATION-REACTION-FAMILY-010	Exact complete inverse-addition elimination family law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7dc26f50dfa404272ea47a9426cbda3c27f069ddb54120a1b4dcf8a589be1f27
802	Chemistry	SFT-CHEM-REARRANGEMENT-REACTION-FAMILY-011	Exact composition-retaining molecular rearrangement law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:905bd55a18ffe29d8484ad273fe9a6245121b3bf0b56f0166dd45fda91b3bdca
803	Materials Science	SFT-MAT-CRYST-QUASICRYSTAL-INFLATION-002	Exact quasicrystal substitution inflation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ac6f202a08f452225c0fe8a378d95f216de0c49b147d5a3b1c11a1a20fe242f0
804	Materials Science	SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002	Acoustic, optical and thermal mode limits	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:658c8a9808a71ff33f048f529b88e8189db27a9d4321aad19f9944df4249eab6
805	Materials Science	SFT-MAT-SEMI-RECTIFICATION-002	Exact p-n junction rectification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e71ca636880f795b1e68037e7ad6f62ce1388b4a8a8d6c6a05399847beaa836c
806	Materials Science	SFT-MAT-SC-ISOTOPE-RESPONSE-002	Superconducting isotope-response record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a55394460feb15801fb15a215f8e6c6ab5b9fb62523db33307c4f721c15dc1f6
807	Materials Science	SFT-MAT-MAG-FERRIMAGNETISM-002	Ferrimagnetic sublattice order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:42f94c9f6bfbf33931ff8f8dc7603dde8134e2b6afb7e2a0b22d96942f0a29a7
808	Materials Science	SFT-MAT-HALL-QUANTIZATION-002	Integer and primary fractional Hall classes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8bf14f74a78ae92fd3ff5878df726d92cbac72870b8af7f44601acf2b01397f5
809	Materials Science	SFT-MAT-TOPO-EDGE-COUNT-002	Exact topological boundary-mode count	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5b742a33b9d5b5822b2ff2c8db6375cbe0b9f3e605643f187fee05f43a921c57
810	Materials Science	SFT-MAT-BULK-WATER-RESPONSE-002	Hydrogen-bonded water bulk-response ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bbc61e7fbaa2d3cf5ce25e2015c3aa1713e03d741067f422760e797ab6a42774
811	Biology and Life Sciences	SFT-BIO-LIFE-ORGANIZATION-001	Living organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c81ed1467e76985489eb75341f8444d9b7dd70806158c065f76276adb038e5ff
812	Biology and Life Sciences	SFT-BIO-LIFE-CHEMICAL-BOUNDARY-001	Living and nonliving chemical-network boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ea8f6776c77a0aa597926fc3919a4fed35c47ec6f0cf0b63189b296c953efca0
813	Biology and Life Sciences	SFT-BIO-LIFE-AUTOCATALYTIC-CLOSURE-001	Autocatalytic closure and origin-of-life boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2ba082e116d52b085704fa2b521e74749aca028326a9a3714e7a7f3e343115ff
814	Biology and Life Sciences	SFT-BIO-LIFE-MAINTENANCE-001	Biological maintenance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b90688b59f2dcb1654c668f5068c08772bcea2c46ff7762f7db78d0b3588f60d

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815	Biology and Life Sciences	SFT-BIO-LIFE-IDENTITY-TURNOVER-001	Identity through constituent turnover	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a95e5dc8151d49e4dddab3a193e7b5a66f63d04a231d72e1a4064cc72c2755ab
816	Biology and Life Sciences	SFT-BIO-LIFE-DEATH-001	Loss of living closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:12c472570e01cb1d43d7d5f9446284d9c3c459dbbb4f7b756db26728c9e7d750
817	Biology and Life Sciences	SFT-BIO-COMPARTMENT-001	Biological compartment	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7803f76dac841eb4ca77f8f93b867ec6dab24c521c368692d2fe176de0bc2cef
818	Biology and Life Sciences	SFT-BIO-INSIDE-OUTSIDE-001	Inside/outside orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b40c7c3a9150268343bf0170331f0dd0ad6ad2e97e0a415d5a3c5ab03892b60d
819	Biology and Life Sciences	SFT-BIO-MEMBRANE-001	Membrane organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0f6b2cec6d4396ab957fc2f3ed3f4204b677170eccafaa50b9707631a77c2e12
820	Biology and Life Sciences	SFT-BIO-SELECTIVE-PERMEABILITY-001	Selective permeability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2ef57b37bde96ec4bee92de6174564ed8f70b4c72530d367c7316d628b2074c7
821	Biology and Life Sciences	SFT-BIO-NESTED-COMPARTMENT-001	Nested biological compartments	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4a92735e1f74f82ff0058013c136ef9ce2753437f6b822fcd61ac1ff6fe81111
822	Biology and Life Sciences	SFT-BIO-COMPARTMENT-PROVENANCE-001	Compartment lineage provenance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8db42cb9a92c6ddef570c53aed2672fd8f4714357daec1cd587e945d4955f26d
823	Biology and Life Sciences	SFT-BIO-METABOLISM-001	Metabolic network	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d6478476c899e8a4995d0332764480abbed0f3d6953144652993ba226b445415
824	Biology and Life Sciences	SFT-BIO-RESOURCE-UPTAKE-001	Biological resource uptake	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b967bbb1af45e237bdfb7a44afd92156b328f7821f5369aeb0a5bbe3e8366354
825	Biology and Life Sciences	SFT-BIO-METABOLIC-TRANSFORMATION-001	Metabolic transformation conservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0f1ee13bab8ff013eb1e629c8ea7f95d710817fb54744617342881f9e019ffa3
826	Biology and Life Sciences	SFT-BIO-ENERGY-COUPPING-001	Metabolic process coupling	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:01e85868d0ff5ebca68aba0cacb4521615fc96982027aab2a347154370309465
827	Biology and Life Sciences	SFT-BIO-WASTE-EXPORT-001	Metabolic by-product and waste export	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7fce9323e170ea5f1948f56c25331618289f899c9abd0a400aa51837ee30667a
828	Biology and Life Sciences	SFT-BIO-RESOURCE-CLOSURE-001	Finite biological resource closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:72758f467ea389cdb7b6a346645361f81f7cf04c4a8b07cf52f71331cfc21114
829	Biology and Life Sciences	SFT-BIO-INHERITANCE-001	Information-bearing inheritance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bda67ea3373d60bc39193d3e349cb4134dbc67a74c75ff195cdcec01ff9fa2ed

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830	Biology and Life Sciences	SFT-BIO-TEMPLATE-001	Template-directed copying	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:61253b914236e5d5311c56897776efd2be36082b311f1dbfda780aafc7d2dd04
831	Biology and Life Sciences	SFT-BIO-REPLICATION-001	Biological replication	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:786da8a317ddf2f962210ed732dc9cdd68011cf4045ea288a44fd5825bb159fd
832	Biology and Life Sciences	SFT-BIO-COPY-FIDELITY-001	Replication fidelity and error	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:02f163116212c08a6f3c238f7b007daa26b02948159a246ce7cbab7258c537aa
833	Biology and Life Sciences	SFT-BIO-HERITABLE-VARIATION-001	Heritable copy variation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bf30cba5ca7777237d9989a3a820981b8fb1bfddc05d0418e9d39187a215f50f
834	Biology and Life Sciences	SFT-BIO-REPLICATION-LINEAGE-001	Replication lineage	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ab8ac6e3ff4c35a7389870a76995ccd049e29980d43773686ee4efbcedc52391
835	Biology and Life Sciences	SFT-BIO-VARIATION-001	Biological variation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b954170326296f4004da08e53cc32ceafabd09e84aca0bd274cf5151d294a888
836	Biology and Life Sciences	SFT-BIO-HERITABILITY-001	Heritability relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4bf4352533206993a667ffb2502a0a73963659b877192d2f0bdf28c33fc7a99a
837	Biology and Life Sciences	SFT-BIO-DIFFERENTIAL-REPRODUCTION-001	Differential reproduction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4dbd6e4ba6ea52aa8b4ab09ea6a1804fbdc34ad7ed57014981982ce701960501
838	Biology and Life Sciences	SFT-BIO-SELECTION-001	Selection	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ad24aa695baf0f2105af2a4422093a3e61b41805ff42c8fbe133744d0967d4e3
839	Biology and Life Sciences	SFT-BIO-FIXATION-001	Fixation and loss boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5433bc60179895f6ea7c0aafb31fe07de2b418fad79ca23b33bcd503172cef
840	Biology and Life Sciences	SFT-BIO-DRIFT-BOUNDARY-001	Selection and drift distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:86df52d952985ced70d264231c72484aec2ee99bac822c5d9e3b70a7c8c3552d
841	Biology and Life Sciences	SFT-BIO-NUCLEOTIDE-ALPHABET-001	Nucleic-acid symbol alphabet	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:39e582a3d655540a7f6ff19c813e7ee9e246c97d49e062515b9cc50bfd5e0b80
842	Biology and Life Sciences	SFT-BIO-CODON-001	Triplet codon word space	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fd2c8f05f16667ba8eb4cee73b2a8fef5ed9fecaa0194b8991464dbe0dfb5940
843	Biology and Life Sciences	SFT-BIO-CODON-BOX-001	First-two-position codon boxes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:15585592b4edb63a9d8dacad3d81b9dc4f06782a239cdceff805007b48114794
844	Biology and Life Sciences	SFT-BIO-GENE-001	Gene as expression-bearing hereditary interval	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1f647ee1b8fdbf32fcb5fca9934a4d7b3dcd300fb73b4a069cd1fd3ddfdcc2321

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845	Biology and Life Sciences	SFT-BIO-GENOME-001	Genome organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6c0797aa33cb2bd27abc50954ed0ebdfc9daf9cae65ebba4cf1905f43bb77de0
846	Biology and Life Sciences	SFT-BIO-EXPRESSION-REGULATION-001	Expression and regulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c3d503b8e21516a9ef94990d71b37e651441b2b040e32be6e261a863eab6cf2d
847	Biology and Life Sciences	SFT-BIO-PROTEIN-SEQUENCE-001	Protein sequence identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3789bdf44006ea0f149389546b5ca94ed4f38fb4272ce37212c66ed9451965c71
848	Biology and Life Sciences	SFT-BIO-PROTEIN-FOLD-001	Protein folding relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e9c16744ae02fcd4e406a2a454c9eecd909027ec6aaa88693b2426b8ca29e09
849	Biology and Life Sciences	SFT-BIO-PROTEIN-ENSEMBLE-001	Protein conformational ensemble	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:47dd4ee2964bd7acfebe732314a5cb600499fa0523fb09ba6cf0a0c70dec2401
850	Biology and Life Sciences	SFT-BIO-PROTEIN-FUNCTION-001	Protein function	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:946afec8074bb304b554d02c2b91056966d13d930868d873a389d21ac5435252
851	Biology and Life Sciences	SFT-BIO-PROTEIN-INTERACTION-001	Protein interaction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fd81c6551ea90e53f7976f2426dac503dde3269368fac9798aa43f4776c5a94
852	Biology and Life Sciences	SFT-BIO-PROTEIN-QUATERNARY-001	Quaternary protein assembly	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3b62cc207a3f30a23e590c94e4b20247a59ee7006c187e74fc94e78d3cf180f0
853	Biology and Life Sciences	SFT-BIO-BIOLOGICAL-HOMOCHIRALITY-001	Biological homochirality	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1ab0ed8c242bddfe3a54ac4652b01a65cfc8197e5af16c9660b722c7982ead31
854	Biology and Life Sciences	SFT-BIO-CELL-001	Cell identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3a8ab83a75bac852b533de00acf6851ff4ecb7f6ca6177d5358a3b7b4692194c
855	Biology and Life Sciences	SFT-BIO-CELL-TRANSPORT-001	Cellular transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5f041a52c3628067d9741750665bb18a2bb4475055df0992be750af8748b824e
856	Biology and Life Sciences	SFT-BIO-CELL-SIGNALLING-001	Cell signalling	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:13c962f012563386ecb1b0368ac5c1a155f625035cec94afd26071e414dc5b57
857	Biology and Life Sciences	SFT-BIO-EXCITABLE-THRESHOLD-001	Excitable-cell threshold transition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4282cbeed88f9d120d7ce1489a7c96c4325cd966e1e56d2d86701c1fcb6065b
858	Biology and Life Sciences	SFT-BIO-CELL-DIVISION-001	Cell division	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:812e3a48b5dde43facb3f3b5491fe5b68ed0951ec8961c8ce8893945eae5965
859	Biology and Life Sciences	SFT-BIO-DYSREGULATED-DIVISION-001	Dysregulated cell continuation and cancer boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cfb86089a23b4c3240ff65c7275c09855138bb729d08c8367f93d5ef8929c2d7

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860	Biology and Life Sciences	SFT-BIO-MULTICELLULAR-COMPOSITION-001	Multicellular organismal composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:afb5f7805ecda6f297068f7532eb5683361d0dfd5b46f1d2175e8587a5bab3f4
861	Biology and Life Sciences	SFT-BIO-CELL-ADHESION-001	Cell adhesion and tissue boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0c854c871fd66f0b15e416e1da98cf127e8e562c49d095c2da11396e178f2e8c
862	Biology and Life Sciences	SFT-BIO-DIFFERENTIATION-001	Cell differentiation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:10fc2b713e757614a0b216a741cb8b11c7760e871f099e903e45d2eee24840ad
863	Biology and Life Sciences	SFT-BIO-DEVELOPMENT-001	Developmental lineage	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:66f84782d6c8a70c2ec7a9e98676aba00e65565588b01c40bd2374bcb01a8c1f
864	Biology and Life Sciences	SFT-BIO-MORPHOGENESIS-001	Morphogenesis	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:39da10edb31c18b4b00f4a69a4806aa5c46aa6543c1bb6d7e50d5c423e12f529
865	Biology and Life Sciences	SFT-BIO-REGENERATION-001	Biological regeneration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1181aa8c81e16a17dca6a612b879a6866d759cea7dc0eaae0a189475b20cf1bb
866	Biology and Life Sciences	SFT-BIO-ORGANISM-001	Organism identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5be8b1f41d5fabdd92a7319dbe75c1f71eefd1bb6a75781b354926d6132a234
867	Biology and Life Sciences	SFT-BIO-PHYSIOLOGY-001	Physiological process	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:efb61f71851c4c5a4af6e38bac1bb5a0524983ae7990d06a44899b1269dd0c6
868	Biology and Life Sciences	SFT-BIO-HOMEOSTASIS-001	Homeostatic regulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6d125e5d47a334da623d6b6859f89cfa85852efa2364a81b5875766c67bf151
869	Biology and Life Sciences	SFT-BIO-REPRODUCTION-001	Organismal reproduction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b391636734ce9f838d52feafefbfeeed4e809e4f70c14abb172373e77ace862
870	Biology and Life Sciences	SFT-BIO-LIFE-CYCLE-001	Biological life cycle	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:796de364657e92a20c6bc7717a9a9beb334122f2aa559261e9b5492651762c87
871	Biology and Life Sciences	SFT-BIO-SENESCENCE-001	Senescence and longevity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b24bee7ff30267308f7f6aee985f931e54b45b545774914a628bd79f4801acbc
872	Biology and Life Sciences	SFT-BIO-POPULATION-001	Biological population	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d92a8470023c0f5ae5f93f0fb8865af2986bd5bf5d09e164adb53124d11a52d
873	Biology and Life Sciences	SFT-BIO-SPECIES-BOUNDARY-001	Species and reproductive boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5594e8750cc5de11e2d96a74f5cc72a126e81b7e66efeffc35088866a24f35a9
874	Biology and Life Sciences	SFT-BIO-COMMUNITY-001	Ecological community	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ce593cef58bc0880814d28ed4efccb0bfedd325d83511767a36b752c8bfb68bbe

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875	Biology and Life Sciences	SFT-BIO-TROPHIC-NETWORK-001	Trophic and resource network	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0bca610fd19ea03b8711a6622e cefe413729d9e7cf890e336184baaea08 396a0
876	Biology and Life Sciences	SFT-BIO-ECOSYSTEM-001	Ecosystem organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7f75dd42da58bc7246f8bd7883 7d2b6562a655a5b6be15622c0c06e1826 ef006
877	Biology and Life Sciences	SFT-BIO-ECOLOGICAL-RECURRENCE-001	Ecological recurrence and stability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b670971a03f2f2d088debfb79 1529419052470c1f86eb22ba94c009f44 461b4
878	Biology and Life Sciences	SFT-BIO-ECOLOGICAL-NETWORK-001	Ecological network topology	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f0671939caa9ed6d2fe121453e 60df8452e681f31d5d37d871452b62052 2773b
879	Biology and Life Sciences	SFT-BIO-BIOLOGICAL-ALLOMETRY-001	Biological allometric relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d5ce1f4e35ed88884352ee720b bd29ca4dfdc2513e90bf70cdf210f2e72 3c596
880	Biology and Life Sciences	SFT-BIO-BIO-SPECIMEN-001	Biological specimen and sampling	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e8adfad0f9b420176de99c2d31 f622fb178b6ef766494cb9d9df2adb333 c14e1
881	Biology and Life Sciences	SFT-BIO-BIO-CONDITION-001	Biological condition and control	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fc6ea229bc38e9de68a4d0504d 386a2b90bc33b950bcd408c24c49c27e2 38700
882	Biology and Life Sciences	SFT-BIO-BIO-ASSAY-001	Biological assay	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0f51212a550b8a30526dc82853 d945b037c359b272dd8a66f96279ae307 1df5c
883	Biology and Life Sciences	SFT-BIO-BIO-UNCERTAINTY-001	Biological uncertainty and missingness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fe2987400977553ecfa40a7d3b 5816a5a4e960b84ea807c1b5d255f2453 bc37c
884	Biology and Life Sciences	SFT-BIO-BIO-CAUSALITY-001	Biological causal and associative boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:89b1826edcf95ef8571cf058e2 8a46253279e75125df8cc8a0f4e62ca16 2c883
885	Biology and Life Sciences	SFT-BIO-BIO-HANDOFF-001	Cross-branch biological handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d36e9dbd732ed6331adbad9f25 c6dca5ddb0b6d56511c1671e47920c351 47e5f
886	Medicine and Health Sciences	SFT-MED-PATIENT-IDENTITY-001	Patient identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:09736e67d12ee4fbcc725383cb 7ea3a5520b9bda3b357bbc9f490683c6f 98ad8
887	Medicine and Health Sciences	SFT-MED-CLINICAL-EPIISODE-001	Clinical episode	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:27e13f7926e948fa862ae54b50 7b32ff1ac9fa76e6ba1912419838817d7 712ea
888	Medicine and Health Sciences	SFT-MED-CLINICAL-POPULATION-001	Clinical population	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f7f8c7ed9561c9cc9237b64634 6748975c2a792a160478e5e8ea858a57b 3583f
889	Medicine and Health Sciences	SFT-MED-HEALTH-STATE-001	Health state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9ff35c90332ac776b3787783a5 b40d40842a4aa54b3fe305f1149797f75 a9199

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890	Medicine and Health Sciences	SFT-MED-CLINICAL-OBSERVATION-001	Clinical observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:db504374e985d2f673ac290218352c3544d02476eaf41bfda2fd61917181dbf8
891	Medicine and Health Sciences	SFT-MED-LONGITUDINAL-COURSE-001	Longitudinal clinical course	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:26e4dc2d17268c21b99d15a338cba7661cddb8e8b6c795a823de3bcfd4407ae5
892	Medicine and Health Sciences	SFT-MED-NORMAL-VARIATION-001	Normal clinical variation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0eda0a1eea6c39d68be5e906bf31525faeb83b204f2160c0814d2897da6212fa
893	Medicine and Health Sciences	SFT-MED-DYSFUNCTION-001	Dysfunction boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c428cc97163fb8b162d6db4be5ed92d78549d5ea8c8d4dcdf2b160ff059f6e7d
894	Medicine and Health Sciences	SFT-MED-DISEASE-001	Disease state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:02b64355610b3b30ed7dea14ae72efa4ab7f9d8652187947710a0e8da3abfc4a
895	Medicine and Health Sciences	SFT-MED-INJURY-001	Injury state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4a011252846c30ea2825619210ebed680f37a96dc46780f8f0a4a99e3e16b22a
896	Medicine and Health Sciences	SFT-MED-COMORBIDITY-001	Comorbidity composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ea094d0cb6741aacb8f2e06e50807c2d31ab85c447c6f2227f5708bf269f2bae
897	Medicine and Health Sciences	SFT-MED-SEVERITY-STAGE-001	Severity and stage	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:011c56e91bdcdfdeb96526cd2bf75f7e497b74ab880a324f8975eeb08c420004
898	Medicine and Health Sciences	SFT-MED-SYMPTOM-001	Symptom record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a2374682cdc923a9dd6cd88b4f541480746939507c83a8aa939213463823711b
899	Medicine and Health Sciences	SFT-MED-SIGN-001	Clinical sign	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:454e5785a37425db27f63a8eb134d4c472ec7a5e554068b6024e70d9fb139852
900	Medicine and Health Sciences	SFT-MED-CLINICAL-MEASUREMENT-001	Clinical measurement	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:faceaa239618ddd156ff77437bd4c6bfaf68461b65053e30dfcf4f733134bf71
901	Medicine and Health Sciences	SFT-MED-DIAGNOSTIC-CLASSIFICATION-001	Diagnostic classification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1e55f7167ea50eb3f99bbbc8e319501ddae4c1f9cd20b80c2db48bd486ee009
902	Medicine and Health Sciences	SFT-MED-DIFFERENTIAL-DIAGNOSIS-001	Differential diagnosis	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ddeb40383f8964a1693e542a975ecb3efa8949c2226d795de73731a0f374a7e2
903	Medicine and Health Sciences	SFT-MED-DIAGNOSTIC-ACCURACY-001	Diagnostic accuracy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:071608ddf71edeb394e5cecb7df8434d259579da3f992ac33abdef6c365399
904	Medicine and Health Sciences	SFT-MED-CLINICAL-ASSOCIATION-001	Clinical association	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ef723331144115bd77edf75ec8fed6e5bd30c2c3c8a88c7604bb1efd525eb9b9

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905	Medicine and Health Sciences	SFT-MED-CLINICAL-CAUSE-001	Clinical causal claim	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f327d41c2c5928c687ce971a46095fb36c57e502f8a1cc169158aefc30c65c97
906	Medicine and Health Sciences	SFT-MED-CLINICAL-MECHANISM-001	Clinical mechanism	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:16f410bd3c4969a561adca4caf550bf4303ee9247926ca1a52d2b45debd047fc
907	Medicine and Health Sciences	SFT-MED-CONFOUNDING-001	Clinical confounding	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c9c81f396a46438b2b273042713a0f6b054c5f2c801c4bef4df038796d1af3f7
908	Medicine and Health Sciences	SFT-MED-MEDIATION-001	Clinical mediation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:44248f3339ae7377b144a6f930ca9f13c6a6731d3d95a30eb547a3ebeb565e40
909	Medicine and Health Sciences	SFT-MED-EFFECT-MODIFICATION-001	Clinical effect modification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f2b3a00497b80f2bd576ed4562977e1f7c670db07871af69211a49c832bc7000
910	Medicine and Health Sciences	SFT-MED-RISK-001	Clinical risk	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f38143f22f98b85462cb732d929aa797d3108da56aa176a05dd3cce74eec36c7
911	Medicine and Health Sciences	SFT-MED-INCIDENCE-PREVALENCE-001	Incidence and prevalence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a1fc63617f42562e76e169efb9f4a3e4812b681880a3650329ccf1f2df295dd8
912	Medicine and Health Sciences	SFT-MED-PROGNOSIS-001	Clinical prognosis	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a0990310880d83648da40cc153e1ead7a75be9638e4ece60e71eca04505b5ace
913	Medicine and Health Sciences	SFT-MED-CLINICAL-OUTCOME-001	Clinical outcome	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:37df271433294315ed27c02bca4672fd5351ca7f32995bca32a6ce0e12a85722
914	Medicine and Health Sciences	SFT-MED-CENSORING-COMPETING-001	Censoring and competing events	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:10ccdc27c33fef37de264045ed5cc288dec3228e35a57309a4fabe0e7b6b2af6
915	Medicine and Health Sciences	SFT-MED-PROGNOSTIC-CALIBRATION-001	Prognostic calibration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e3702627ecea41f1f14fbeb6995e62d43c6aaf953ae95da03f5eeff1daee4e90
916	Medicine and Health Sciences	SFT-MED-INTERVENTION-001	Clinical intervention	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ef389261cfe2919237ee5a95ed42263554fa6271eb87e13685b1d43ed1ddcc
917	Medicine and Health Sciences	SFT-MED-COMPARATOR-001	Clinical comparator	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3b156a2ee733c40ac3eb86c1d0860fddfa3ed48e10a6dfd1718b2ac1d8af277
918	Medicine and Health Sciences	SFT-MED-COUNTERFACTUAL-001	Clinical counterfactual	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:931e3d7e244c7630ea57fb5f686c9bce2d14c9ff6ecea9f302e673b16ff073
919	Medicine and Health Sciences	SFT-MED-ALLOCATION-001	Intervention allocation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7dac7d6c996e2ed5a914b2d5ec6318b1d2be6730e56c3531dc99216f90bd6ea2

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920	Medicine and Health Sciences	SFT-MED-ADHERENCE-CROSSOVER-001	Adherence and crossover	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5076f9e466898ad2d677ffa86b6beb06ccba33afe2b1a274a3148795eebf42eb
921	Medicine and Health Sciences	SFT-MED-COINTERVENTION-001	Clinical cointervention	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:473077c4999a24d10246b71f4fa85aa1b9c316b0679752f66d09e649c840b0f3
922	Medicine and Health Sciences	SFT-MED-DOSE-001	Dose	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ac1d83eb8670e428a755b801006e386ef850299d00310595b34b68fdbbac76a5
923	Medicine and Health Sciences	SFT-MED-EXPOSURE-001	Clinical exposure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:14e2723b299a0ff56e2e35cbeba96dad77a4e278e00a4cb9a7f813acc3cd5a7c
924	Medicine and Health Sciences	SFT-MED-RESPONSE-001	Clinical response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3a051896a773bd06407d83654f7e55a5f64aca695299eace57d1e33edef69367
925	Medicine and Health Sciences	SFT-MED-DOSE-RESPONSE-001	Dose-response relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e36d17521cbda0f61edc1297b12eabaa07ccc6c7bd50fd95ac57f9d5fd4020cd
926	Medicine and Health Sciences	SFT-MED-ADVERSE-EVENT-001	Adverse event	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78905c3c202917110d452a8e1fbc36297d06a9ca41227d0bf6daf96f71bc184a
927	Medicine and Health Sciences	SFT-MED-TOXICITY-001	Clinical toxicity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:829bb46bb023e38579070b20897606b964d882dbf05322a2c70cd09c983644df
928	Medicine and Health Sciences	SFT-MED-EFFICACY-001	Efficacy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fad071eef2c92861ab7b6ef138adaa62012830e8c3708a35eb630d0946a4093c
929	Medicine and Health Sciences	SFT-MED-EFFECTIVENESS-001	Effectiveness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ec467dd4d013aa9d8322d5c1f82886b7de39ac767f26d6201a8278129ef983ce
930	Medicine and Health Sciences	SFT-MED-SAFTY-001	Safety evidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d11984f1d2f2cf293239f79421b260618766fdee59141d743c3ba784a8b3611a
931	Medicine and Health Sciences	SFT-MED-BENEFIT-HARM-001	Benefit-harm composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fdcc94f8f9d80f1cb1a1a62fdf23f9971938b470e0986f37c8d135223c65f1b
932	Medicine and Health Sciences	SFT-MED-ABSOLUTE-RELATIVE-EFFECT-001	Absolute and relative effects	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9935031e091e83e4d59f404b161de2fbc6f2eae61b99b5ef03bbe1b3f444151c
933	Medicine and Health Sciences	SFT-MED-TREATMENT-HETEROGENEITY-001	Treatment-effect heterogeneity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2c2ed8e3aa4b025c627d17df97ea84d966f9ce0378b34cc2e2e8843769438a3eb
934	Medicine and Health Sciences	SFT-MED-CLINICAL-TRIAL-001	Clinical trial	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b2207bbc4ad20a8369f19ed3fb999460dc87fc7dea9c248774b3063f7cc7fa

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935	Medicine and Health Sciences	SFT-MED-RANDOMIZATION-001	Randomized allocation in a deterministic model	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3699c5d458f2c3427c92193951e86e2976685527a59c01128d8d9bb12d3ad1ae
936	Medicine and Health Sciences	SFT-MED-BLINDING-001	Clinical blinding	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:02ee97e420fd490447b38cfa12c0234827e469d9ed5280749776035075167585
937	Medicine and Health Sciences	SFT-MED-OBSERVATIONAL-STUDY-001	Observational clinical study	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7a80fb51f8716b7b95fc3eabf2591ed77bb49c2b3ea96140162ceec838ba2015
938	Medicine and Health Sciences	SFT-MED-EVIDENCE-SYNTHESIS-001	Clinical evidence synthesis	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:91a1c5af78ab9f5415d436ec54c4fa6683fb9dbd364cdabb7d40d2999ec01a8a
939	Medicine and Health Sciences	SFT-MED-MISSING-REPORTING-001	Missing and selective reporting	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cdaf826443e597536993c81365b2978a96a2e4a68ca1d8cbd496694b47a8d484
940	Medicine and Health Sciences	SFT-MED-SCREENING-001	Screening	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:770e80dec3643f760475ced8d8d7b7ebb4b08f18f2fce76579dcf6bebd004168
941	Medicine and Health Sciences	SFT-MED-PREVENTION-001	Prevention	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b5f5c829fd5d0973df088228ef932b37b6d5b8b52e8ff75976985d4075882e759
942	Medicine and Health Sciences	SFT-MED-TREATMENT-001	Treatment	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:400c1338756c2f4c002d95b68607cb71f6e97a573e3a553d9106fde8e21c4c68
943	Medicine and Health Sciences	SFT-MED-REHABILITATION-001	Rehabilitation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:248785ef1ba749a5a71984ec0bcdfed368de479cee3f665aff962ee786cd87eb
944	Medicine and Health Sciences	SFT-MED-MONITORING-001	Clinical monitoring	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2940988359a18d6bcd422e2a4008e6b504f8d13208542f7b8c4701191f4cd163
945	Medicine and Health Sciences	SFT-MED-STOPPING-WITHDRAWAL-001	Stopping and withdrawal	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0f8125348a367df9c09b5efc52277455b1e2c4a560cb78784ef598f16444f157
946	Medicine and Health Sciences	SFT-MED-INDIVIDUAL-INFERENCE-001	Individual clinical inference	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4ff1fffee484941acd7ec9e3fb6e4392c7d42b1a8e5b236a003e0584d6817dd3
947	Medicine and Health Sciences	SFT-MED-POPULATION-INFERENCE-001	Population clinical inference	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b289ddf62ddadfe5bafb3b725fd180d9c22a97175463d0d68d03973382c747da
948	Medicine and Health Sciences	SFT-MED-TRANSPORTABILITY-001	Clinical transportability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:99f4642cbb83fb23bf102d691842d4df867e8df7b68a394f166fa040ebaf44f8
949	Medicine and Health Sciences	SFT-MED-SUBGROUP-MULTIPLICITY-001	Subgroups and multiplicity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eb9b3ff70f470dbdcf485e20f6c0924063e0af4f2ec14cc315d7caf442303679

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950	Medicine and Health Sciences	SFT-MED-CLINICAL-DECISION-001	Clinical decision boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:10acbaa77b0ffff08e501c884b6b91ac42f3a7883868308ffa2fa7c62979411d4
951	Medicine and Health Sciences	SFT-MED-CLINICAL-UNCERTAINTY-001	Clinical uncertainty	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c86b7c45809b6536ad297ecd36cd5ea0f33cbdf639da79841e6694e96c71eb66
952	Medicine and Health Sciences	SFT-MED-INFORMED-CONSENT-001	Informed consent boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:39a8cd25cdabbff7d3f96ed129ad05b4f2ecd91b89ad4bd6a8c0bc2352b7d407
953	Medicine and Health Sciences	SFT-MED-DECISION-CAPACITY-001	Decision capacity boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9de84a984e7b1bf1105c2ce95f7b363d79b21901a177fb9605e1c1a71004320f
954	Medicine and Health Sciences	SFT-MED-CLINICAL-PRIVACY-001	Clinical privacy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6780db2d12f9c570c9ee4cf2494ed173ba656f9731020f992d4971e517dcfb53
955	Medicine and Health Sciences	SFT-MED-DATA-MINIMIZATION-001	Clinical data minimization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dbc403cde9af573b23fca628854e07ffa7483be141ed4274e9ac4ac281b86d8
956	Medicine and Health Sciences	SFT-MED-CLINICAL-EQUIPOISE-001	Clinical equipoise and stopping boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:32be873f515a8ec20f5df36bc4cafa3d393c6defb5ace2fdeaf07b16d089dfdd
957	Medicine and Health Sciences	SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001	Clinical evidence handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:55e545140e2d2716d836eb426154f3ac606229eff3cd7fd063b00bddd1847b01
958	Consciousness and Cognitive Science	SFT-CONSC-OBSERVATION-ACT-001	Observation act	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ea4146344584c793e097676b01ff627ef0713056c9868d46728ace4aee5f5b40
959	Consciousness and Cognitive Science	SFT-CONSC-EXTERNAL-OBSERVATION-001	External observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2040b6d33047b2e19802b8da555b5e8fbe969f6d24657622f709d36a585898d5
960	Consciousness and Cognitive Science	SFT-CONSC-INTERIOR-OBSERVATION-001	Interior observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4aa084b58b750d57b37fc4d12db0acd8139333cb1a5adca3d9a8152dbaab175c
961	Consciousness and Cognitive Science	SFT-CONSC-OBSERVER-OBSERVED-001	Observer and observed distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d5028a0971e896774070564e9dc258ea26e75577a4c669fdb4dd0becddb513ee
962	Consciousness and Cognitive Science	SFT-CONSC-OBSERVATION-CLASS-001	Observation equivalence class	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0c57714ec9fdf16b3ed3bcf8e6f0ff99379916a3101d197c01d0b4cc3170fad0
963	Consciousness and Cognitive Science	SFT-CONSC-OBSERVATIONAL-MOMENT-001	Atomic observational moment	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:62c9ee5358e79067953f31b29a0b9dae74bc48e8cb27996a7fae1bab25869dda
964	Consciousness and Cognitive Science	SFT-CONSC-MEASUREMENT-CONSCIOUSNESS-BOUNDARY-001	Measurement and consciousness boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b55066cb71f9fc634e331bf893ed9f2b1602c21a334ecaf86bc3024eb0adec11

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965	Consciousness and Cognitive Science	SFT-CONSC-COGNITIVE-ACCESS-001	Cognitive access	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8ae8fa661981c0ee2f0c8b96061b087785a9fedd72b566607b72220a9e86609e
966	Consciousness and Cognitive Science	SFT-CONSC-REPORT-001	Report	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0c666d6d63b07968ea85db7c27e177e1cfff2596d1e68b3dadbf6c2d8c67ab9a
967	Consciousness and Cognitive Science	SFT-CONSC-PHENOMENAL-PRESENCE-001	Phenomenal presence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94e5ea251bf89ec65344079aee3a926c015023d3e4be515e85e8f9eac24aabd1
968	Consciousness and Cognitive Science	SFT-CONSC-REPORTABILITY-001	Reportability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:674f19e64fdb3a6485b49ec0c15890308f898025b86785056e9bd1b1901034f2
969	Consciousness and Cognitive Science	SFT-CONSC-UNCONSCIOUS-PROCESS-001	Active unavailable processing	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa334010554c126808560e78e8af4788981244496b9ff10a670b2f06cdf08be0
970	Consciousness and Cognitive Science	SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001	Altered-state evidence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bd8431fda7b9c23f2179b1f44fc3d624de0af2def23ea6443c70a2edcd9090de
971	Consciousness and Cognitive Science	SFT-CONSC-CONTENT-PLURALITY-001	Plural conscious contents	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e18cca317d9758e664c61de2384f6824e04ba53cd258f652d34219c7176ed90d
972	Consciousness and Cognitive Science	SFT-CONSC-BINDING-001	Binding relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ffc2d41dd6abb0cc20efa97823488619db0adba759a3118ed0f6713c3d883a16
973	Consciousness and Cognitive Science	SFT-CONSC-UNITY-001	Unity of experience	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:389adaf57aaa2bbe491ab44a470c6ab63913c9ff1529b530ebc7207e4791826c
974	Consciousness and Cognitive Science	SFT-CONSC-INTEGRATION-BOUNDARY-001	Integration boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:45a4d53110a1ab574c74fcd186be33dd10becf5e85807d819ec37802cccd5a98
975	Consciousness and Cognitive Science	SFT-CONSC-SYNCHRONY-BOUNDARY-001	Synchrony and binding distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ce19b1ec90c8508be91246272716bc450af0868ad37b983c3383ea3e961a988a
976	Consciousness and Cognitive Science	SFT-CONSC-FRAGMENTATION-001	Fragmentation and divided access	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3a6d372a2c715e2d34799dcdf73304b503a18216935fd424efcbb1112dcc75637
977	Consciousness and Cognitive Science	SFT-CONSC-PERSPECTIVE-001	Perspective	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eb898f58b07150cfc4a3fbf96f1e32d6cf56c45d5b2cbb2a3e3cf0cb7f4be4ee
978	Consciousness and Cognitive Science	SFT-CONSC-INTERIORITY-001	Interiority	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:59ae46e5a54732dea1b38a2466638e612f69076c72079035b27e14959270082b
979	Consciousness and Cognitive Science	SFT-CONSC-SUBJECT-CARRIER-001	Subject carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bfd669d8b5abef024f7cf001f7f1afeceac2b34a830eff777d1aab5d0cc61000

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980	Consciousness and Cognitive Science	SFT-CONSC-SELF-INVARIANT-001	Self invariant	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:02aca8e3d823e59800bacba232cd787d69a1430b79fe4742bb2a753d73b1df01
981	Consciousness and Cognitive Science	SFT-CONSC-FIRST-THIRD-PERSON-001	First- and third-person evidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5e56f6376aaab75f73c9ee3764f408ecf18f4a049bd9cb398dd900ca833a01d5
982	Consciousness and Cognitive Science	SFT-CONSC-PHENOMENAL-PRIVACY-001	Phenomenal privacy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ee914882836a80f76b9b8d676adc1ae6a56fedcae38ca9f9bfff30e0ab079c7b
983	Consciousness and Cognitive Science	SFT-CONSC-SELF-OBSERVATION-001	Self-observation closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:02090663f6c9634fd3b5c5982db8899596f6122d45d95d5716a64c92c86bf8c
984	Consciousness and Cognitive Science	SFT-CONSC-INTROSPECTION-BLIND-SPOT-001	Introspection blind spot	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8e4f5d01fd8763c983350d229332d74756945c465f7441ec7519e156275db91e
985	Consciousness and Cognitive Science	SFT-CONSC-SELF-HISTORY-RECONSTRUCTION-001	Self-history reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ba47a6dbbb4cde85fd6eae2714053d301cd59a3b78a7c530315ef3a5b8a24aad
986	Consciousness and Cognitive Science	SFT-CONSC-INTROSPECTION-LOSS-001	Accumulated introspection loss	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:88fe5ba6e920ccccc62d58ddf4f1a98bb57c0cd51f0be32a778a183778acfd7d
987	Consciousness and Cognitive Science	SFT-CONSC-CONFABULATION-BOUNDARY-001	Confabulation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:db98ee24f4b1d7e1113a7ab950e046b2d892719e453628622d63297a27edcc5a
988	Consciousness and Cognitive Science	SFT-CONSC-DETERMINISM-SELF-OPACITY-001	Determinism with self-opacity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cfcd06be185fa623913811f25f3f3d1dc0195004d8b558b7bb926561e0982066
989	Consciousness and Cognitive Science	SFT-CONSC-MEMORY-CARRIER-001	Memory carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:999b2b38351f2dd258de5e156760751843ea946e9371811d2da361b2bba9736b
990	Consciousness and Cognitive Science	SFT-CONSC-MEMORY-PERSISTENCE-001	Memory persistence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:de477d1ca5f7f41046fa03b8953099a1870dcaa2ef18d2098aaf06925ac3c2e7
991	Consciousness and Cognitive Science	SFT-CONSC-RECALL-RECONSTRUCTION-001	Recall and reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c0e26f00a9ba25ad7226fc20e7b322a093c1320bb6d9c826a5b1eca2250c7205
992	Consciousness and Cognitive Science	SFT-CONSC-TEMPORAL-CONTINUITY-001	Temporal continuity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4544cf07cbf0c158911a4421c3018716c6ff30ee80a02beb60ffaa0df7dd46de
993	Consciousness and Cognitive Science	SFT-CONSC-IDENTITY-CONTINUITY-001	Identity continuity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b163769e0b55ee4a410b66748c4adaeb7f0f393ff5bd3eca83d8ea6f04211af4
994	Consciousness and Cognitive Science	SFT-CONSC-CESSATION-001	Cessation of a conscious organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b2d4f5095c14973e38e56ead5b6995ab3c83be9531da93731d07fc8dda2168e9

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995	Consciousness and Cognitive Science	SFT-CONSC-SELF-REPRESENTATION-001	Self representation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:06fd96bbf8387d365f4438b36a8a7b0687a98abfaf03c3ed47705594ee635ce6
996	Consciousness and Cognitive Science	SFT-CONSC-SELF-APPLICATION-001	Self application	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d50621bc7c0281c71764076a04904467a1380c33f01c690ddb57ab89aea7e14
997	Consciousness and Cognitive Science	SFT-CONSC-SELF-MODEL-DEPTH-001	Finite self-model depth	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fedf1408e4bb4366acb97d0730e6694a54db77c963fb39783febae29be07b1eb
998	Consciousness and Cognitive Science	SFT-CONSC-SELF-SIMULATION-BOUNDARY-001	Self-simulation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e7b2bf0da20ba4bccbbde2e49d92b2f6969ca0493ff07828b2060a39a95b2eda
999	Consciousness and Cognitive Science	SFT-CONSC-SELF-VERIFICATION-LIMIT-001	Self-verification limit	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:69040b12d89729b15987e398e705106b8856a227486ed690177946a9eb68e62d
1000	Consciousness and Cognitive Science	SFT-CONSC-ATTENTION-001	Attention	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:18845f97cb140068ca487a993602a4544c3d694bca3cfeb0cdb13212f7d4dc91
1001	Consciousness and Cognitive Science	SFT-CONSC-ATTENTIONAL-FOCUS-001	Attentional focus	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eb4e37136deeacc996f7196b6bedbad37944cfa7bffd1147dfea428fb2af2454
1002	Consciousness and Cognitive Science	SFT-CONSC-COGNITIVE-AVAILABILITY-001	Cognitive availability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bec0c6573f14847771f42b24dbe30c2324b2df00c80b155f2c16c651cf90531b
1003	Consciousness and Cognitive Science	SFT-CONSC-ATTENTIONAL-CAPACITY-001	Attentional capacity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:abab6e31cd225514d568bb8e6d2b726d863c636c986c0dcd353253533d4b95e
1004	Consciousness and Cognitive Science	SFT-CONSC-ATTENTIONAL-SWITCH-001	Attentional switching	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:36465465bb499cf6a574fca72f51dac75ceff7aea9468a527b81392df3819d1
1005	Consciousness and Cognitive Science	SFT-CONSC-COGNITIVE-REPRESENTATION-001	Cognitive representation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:64a2bc87ed69af599d5e358a091168d68ee01586e10287a8ac1588acb1d8373a
1006	Consciousness and Cognitive Science	SFT-CONSC-INTENTIONAL-RELATION-001	Intentional relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4c922ac3925f7650429171266b751df818a11de9b79ba2ec9b67a0ff82058142
1007	Consciousness and Cognitive Science	SFT-CONSC-COGNITIVE-INFERENCE-001	Cognitive inference	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0b77d9f97335ea1233302841881bd5c20cb88a480a69ef9ce33548478bca94c1
1008	Consciousness and Cognitive Science	SFT-CONSC-PREDICTION-001	Prediction and anticipation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ffc5bbaacb7fc6518d80bb5fd61b0976fd2b627755101c215d3ab5e63ce3efab
1009	Consciousness and Cognitive Science	SFT-CONSC-EXPECTATION-001	Expectation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:17a7463a5071f2d4843c1f9cffb660715058985a9d172356660e5bcb76c0ea4f

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1010	Consciousness and Cognitive Science	SFT-CONSC-DETERMINED-AGENCY-001	Determined agency boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f20ff083e70c6c7b1213233f71f75477c21ac777e7d57d9f732a9e8bf811205b
1011	Consciousness and Cognitive Science	SFT-CONSC-REALIZATION-CARRIER-001	Consciousness realization carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:abe35f3545af5cc67893a77bc9a23cc1e97a0ac4765ab0424c1d3d9c4ffd1e78
1012	Consciousness and Cognitive Science	SFT-CONSC-STRUCTURAL-CRITERION-001	Consciousness structural criterion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d5171fe7997756895e49d665fd38db49b2fbd3f4589122138eb51f52472baaa3
1013	Consciousness and Cognitive Science	SFT-CONSC-SUBSTRATE-INDEPENDENCE-001	Substrate independence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:565998cccf378dd69d3224f9ccfa2b1b95f6886a9d15b622e434137530fd21ef
1014	Consciousness and Cognitive Science	SFT-CONSC-FEEDFORWARD-BOUNDARY-001	Feed-forward boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e76094074db809a15f6e61da570a8639eab552a8d99e7310d803ca225574b977
1015	Consciousness and Cognitive Science	SFT-CONSC-REALIZATION-EQUIVALENCE-001	Realization equivalence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dd471a02b716b60a9942ddf871124fe08234de904eb409b286010ce264f289a1
1016	Consciousness and Cognitive Science	SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001	Artificial consciousness evidence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e6c34fdb37b02b797d59fbefc8488984a6fbd8f46e720e1fcd44ddc3462d7dd
1017	Consciousness and Cognitive Science	SFT-CONSC-QUALITATIVE-DISTINCTION-001	Qualitative distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eac356504041299f12e5dd6e305f2b454f5eaedd6f2d1592631c354054c319e4
1018	Consciousness and Cognitive Science	SFT-CONSC-QUALITATIVE-IDENTITY-001	Qualitative identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1000dcb26fca83d03ad6a9568ca43837b001c9dff70ef46e17f2862a01cdeff3
1019	Consciousness and Cognitive Science	SFT-CONSC-QUALITATIVE-SIMILARITY-001	Qualitative similarity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fe83c25f3a26a538795cf89316b0406ca5adc66540c5fa8f4002ff5a72eee28
1020	Consciousness and Cognitive Science	SFT-CONSC-QUALIA-RESONANCE-001	Qualia resonance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:08b4634f1e95b62bfff9429582ee1bc7bfe0527320a7dd733de597fd4017d4b59
1021	Consciousness and Cognitive Science	SFT-CONSC-QUALIA-RECURRENCE-001	Qualia recurrence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a1678c978869491e0f316bfe9d46eb4b36248afb4e58a8ad9ad09e8d427c9130
1022	Consciousness and Cognitive Science	SFT-CONSC-QUALIA-COMPOSITION-001	Qualia composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7865854cb5cd18c836c602e91bdeaf1dca40f0ffc54ba76039819c9954fafcee
1023	Consciousness and Cognitive Science	SFT-CONSC-CROSS-MODAL-QUALIA-001	Cross-modal qualitative binding	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6db5ce7ce9393018e365451896a07dc3fa2bc15ee574e9e2809d2c89c593090a
1024	Consciousness and Cognitive Science	SFT-CONSC-RED-STIMULUS-BOUNDARY-001	Red stimulus and experience boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6b3c57163aaf0b0ce1009040c55b4c20375d03416b677eb0855e695f3822410d

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1025	Consciousness and Cognitive Science	SFT-CONSC-RED-QUALITATIVE-IDENTITY-001	Red qualitative identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bb195ff50462932f27db6de2dd26798d8157138a4630227db0b7c74f20c3124d
1026	Consciousness and Cognitive Science	SFT-CONSC-RED-OF-RED-001	The red-of-red self-reidentification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:69b23e1cd2dcd9545059c9a8528a3c8085ea04f816ad1cf6c8366fd437f4542d
1027	Consciousness and Cognitive Science	SFT-CONSC-RED-RECURRENCE-001	Red recurrence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2ea40f0832299ee54c30e0aa3f200adc3ee53965d3a4979416674260d160ad29
1028	Consciousness and Cognitive Science	SFT-CONSC-RED-CONTROLS-001	Red-of-red contrast controls	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c76466e545afa551a0afb93163b6bf337a5d1a523dbb28df74d7a3c1b3803c55
1029	Consciousness and Cognitive Science	SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001	Red-of-red empirical boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cdb9dec823a5d95b78a89e3e8605d369e1f26ea20908aee781247bc85a5d21b4
1030	Earth and Environmental Sciences	SFT-EARTH-SYSTEM-001	Earth-system identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b2c266ed350d33a40b236d95727e55fcd8b38e7537898576bab94041c0391053
1031	Earth and Environmental Sciences	SFT-EARTH-RESERVOIR-001	Earth-system reservoir	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a54878c96ce4f428686e4ad22ba9dc7e81f23af5205ec5cc4c9b4741b5b24d85
1032	Earth and Environmental Sciences	SFT-EARTH-BOUNDARY-001	Earth-system boundary and interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6e6e0c9566e336744b56a604ccfdab5908911453fd262150562d76b8066e19a6
1033	Earth and Environmental Sciences	SFT-EARTH-STATE-001	Earth-system state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a790fb6e2df910548a230c32b674891cc2fed9a66e07d92608e7f2b04f8f2c3a
1034	Earth and Environmental Sciences	SFT-EARTH-OBSERVATION-001	Earth observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8b6a8ee6f0aab80b7ea1e9d737248cbc01f20142e1c3de318fbf33b866231b69
1035	Earth and Environmental Sciences	SFT-EARTH-ABSENCE-MISSINGNESS-001	Absence, non-detection and missingness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2e5cf46bef731f8ef72f0b0bc6cb746ff7024b67d45c808e41bda4bc36f1efa4
1036	Earth and Environmental Sciences	SFT-EARTH-ENERGY-BUDGET-001	Earth-system energy budget	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:de5ea4b0a621a9a47f9f0f272702584353ffa09185cdb951a7b3e55b90963b36
1037	Earth and Environmental Sciences	SFT-EARTH-MATTER-BUDGET-001	Earth-system matter budget	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:34942b19640ee03f34cdae7d7ea01bb258668feb7291a4f03487786ee80fe34e
1038	Earth and Environmental Sciences	SFT-EARTH-INFORMATION-BUDGET-001	Earth-system information budget	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:14aa00f28ff6c71073c77ad1aca80c55d42b311ba49c189c0d0d2cc8e5c0a564
1039	Earth and Environmental Sciences	SFT-EARTH-STOCK-FLOW-001	Stock and flow distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2a21e28b8288bdd73f95a06bc6950e7376cc707c5c6df04b99641f0d5b57484f

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1040	Earth and Environmental Sciences	SFT-EARTH-SOURCE-SINK-STORAGE-001	Source, sink and storage roles	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:00938fce626f4c74780542f114907934621c2b42350f40a5a4372c43e9e4dc9b
1041	Earth and Environmental Sciences	SFT-EARTH-BUDGET-CLOSURE-001	Budget closure and residual	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:61658ad6107b07e20188fdd35f26c03f8f06bd248b11619f2bde057c081ad167
1042	Earth and Environmental Sciences	SFT-EARTH-GEOLOGICAL-MATERIAL-001	Geological material identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:701d119a2a29bfa4979b7fe0cdf7dec6b83e1d3063f69bb73e344d777b266ae9
1043	Earth and Environmental Sciences	SFT-EARTH-GEOLOGICAL-LAYER-001	Geological layer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:116ab73eab4c1f79ef6119f2ca671f234e4a678551a5b77e39cfa8aee9aae177
1044	Earth and Environmental Sciences	SFT-EARTH-STRATIGRAPHIC-ORDER-001	Stratigraphic order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:41f826d3bb75313530365272560df0e48149f66522a01f8300bcbf154afa3752e
1045	Earth and Environmental Sciences	SFT-EARTH-GEOLOGICAL-CYCLE-001	Geological transformation cycle	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:82a3a688e27dc2d90df87dba81ef18f300524956d4c02c3de5bf267fab8ff3a1
1046	Earth and Environmental Sciences	SFT-EARTH-GEOLOGICAL-PROVENANCE-001	Geological provenance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6326865fe7807b2645a6a402c7b87f32ccf7cccaa8cb9446f72865a2cccae46f
1047	Earth and Environmental Sciences	SFT-EARTH-EARTH-HISTORY-001	Earth historical record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5bc50a62a3e5eb2354c01a146acd248a2b55437433c916ea0dd80e1da029a887
1048	Earth and Environmental Sciences	SFT-EARTH-INTERIOR-COMPARTMENT-001	Earth interior compartments	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:054996179e5ec7a8745973f9c68fa170b60e1565eed2e2f65fbe2dab1da4509
1049	Earth and Environmental Sciences	SFT-EARTH-INTERNAL-HEAT-001	Earth internal heat sources	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3357bcff272a96cb45d8f3d98e7694e6258ade8b1571f6d215bafc9c5a3082a6
1050	Earth and Environmental Sciences	SFT-EARTH-INTERIOR-HEAT-TRANSPORT-001	Interior heat transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5f0c7741ccc3031a6da52d220ad80cd144380214ba556a5de084c06e47ffa4fe
1051	Earth and Environmental Sciences	SFT-EARTH-CRUST-MANTLE-BOUNDARY-001	Crust-mantle boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9d2618cb35c99fe7687503ae36f2493e9caaf3d5f89db4335d9ce42e72ab0236
1052	Earth and Environmental Sciences	SFT-EARTH-PLATE-BOUNDARY-001	Tectonic plate boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d9dcd8021473f468bc7cf0569f708fa1f18b10c402308a0868b76509d0b6dfcb
1053	Earth and Environmental Sciences	SFT-EARTH-TECTONIC-MOTION-001	Tectonic motion and deformation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dc57e68ba400a59ff492830f106e034b6bb73aae7d79a4045179583c7ad596c0
1054	Earth and Environmental Sciences	SFT-EARTH-STRESS-STRAIN-001	Geophysical stress and strain	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6286cab2f1d868af4443f1b473770b1312c9f2580d77c0472be2666f316113ef

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1055	Earth and Environmental Sciences	SFT-EARTH-RUPTURE-001	Earth rupture process	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cbbfa3cca4c6fd0471e81a76c31b8cd236534de3b39132a35d755233fd42c8ea
1056	Earth and Environmental Sciences	SFT-EARTH-SEISMIC-WAVE-001	Seismic propagation record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8e147c5eec7c1286f7b4d84e4a4ea05850e943b83707fa28dc461dee999b9952
1057	Earth and Environmental Sciences	SFT-EARTH-EARTHQUAKE-CATALOG-001	Earthquake catalog	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0b92793e1b4f37ec12872ebc82d8bba312eae97e40f4fc73d51ccfe23f23f152
1058	Earth and Environmental Sciences	SFT-EARTH-QUAKE-MAGNITUDE-FREQUENCY-001	Earthquake magnitude-frequency law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e17ee1c4fb75debd7ac64a4cba308926415ef124df4f45908bf51f83084b60d6
1059	Earth and Environmental Sciences	SFT-EARTH-VOLCANIC-PROCESS-001	Volcanic process and eruption record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5ae1862b28dafa2c35858c64775482d17fe6b572a82c855a31e85dee9e104c3e
1060	Earth and Environmental Sciences	SFT-EARTH-WATER-RESERVOIR-001	Water reservoir identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f2eee3e63ecba04faed36bcf828c203aead5d1b3a4d49ed749ff3000afc86af7
1061	Earth and Environmental Sciences	SFT-EARTH-WATER-PHASE-HANDOFF-001	Environmental water phase handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7f037ea8c6d5a29616899ca27bf8d35c7eaa003b016ce1b146ccb3bd42855f1
1062	Earth and Environmental Sciences	SFT-EARTH-HYDROLOGIC-CYCLE-001	Hydrologic cycle	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c984bb56ec2018dae267b635ac5e35e860451a99d5f3526b5ac1bca70842fe49
1063	Earth and Environmental Sciences	SFT-EARTH-GROUNDWATER-001	Groundwater system	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:061ec0ea77701328ac3b8326a9d75aa69b3eb66fd308bd487cf5870c94687139
1064	Earth and Environmental Sciences	SFT-EARTH-SURFACE-WATER-001	Rivers, lakes and surface-water networks	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:532157a251356bed60f616f20dc1d9548c0fe3c71416a5f00ce41bc02c999a79
1065	Earth and Environmental Sciences	SFT-EARTH-CRYOSPHERE-001	Cryosphere state and transition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:185ff3a0c30cbf254e05c303479804c40909fffbda79d0b8f2128ca1f2c6a426b
1066	Earth and Environmental Sciences	SFT-EARTH-ATMOSPHERE-001	Atmospheric system identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:328297e6b5903b149bc20f75b35537c5eeac3a851795aa86d273eec580d3dd6f
1067	Earth and Environmental Sciences	SFT-EARTH-ATMOSPHERIC-COMPOSITION-001	Atmospheric composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0bc028829580795fd5d7152cf74d8d55cf930eb481a1ceef65c064a407a4062a
1068	Earth and Environmental Sciences	SFT-EARTH-ATMOSPHERIC-RADIATIVE-BUDGET-001	Atmospheric radiative budget	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6d88b381038fc08752beff62eafb1942936de895f479cd772025b710ed2c884b
1069	Earth and Environmental Sciences	SFT-EARTH-MOISTURE-CLOUD-001	Atmospheric moisture and cloud organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7b957974667ff8c2dadcb9cb82a88a1c993ed86ad3f73da5f4c9ff9bf7fd1cc5

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1070	Earth and Environmental Sciences	SFT-EARTH-ATMOSPHERIC-CIRCULATION-001	Atmospheric circulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0cdc723b6c39a6e6ee1414ca050b4482461094dc3fb69e90468584da713f27b5
1071	Earth and Environmental Sciences	SFT-EARTH-WEATHER-001	Weather state and forecast boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fddb62c115f6dd565cf4c008720a7f8619170ce4104c2a5017f15ceafe7502bb
1072	Earth and Environmental Sciences	SFT-EARTH-EARTH-IONOSPHERE-RESONANCE-001	Earth-ionosphere resonant cavity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:089d63910614824c5d2a9424a2308b2ceb4cb6a0381ec7fd89a5d1d77fcfd8ae
1073	Earth and Environmental Sciences	SFT-EARTH-OCEAN-001	Ocean system identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e84ef511820f3cd37b7f8df20793111ca7f971e47f3ed9b0835934d9ff09deed
1074	Earth and Environmental Sciences	SFT-EARTH-OCEAN-COMPOSITION-001	Ocean composition and salinity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:84914bfb33d1be20ed6a57dbbbaeb1c7802636d6dcf8fd6dabb6c766aaff8f2b
1075	Earth and Environmental Sciences	SFT-EARTH-OCEAN-STRATIFICATION-001	Ocean stratification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fa688fb11daf7c0e6dce1047f26d6e26db9148c390cf939e78479b77261d41
1076	Earth and Environmental Sciences	SFT-EARTH-OCEAN-CIRCULATION-001	Ocean circulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:63f2c8b81e8b1f448db38a1cce67d952dc03a5f6f0855d269cec74486b54a6fa
1077	Earth and Environmental Sciences	SFT-EARTH-AIR-SEA-EXCHANGE-001	Air-sea exchange	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:47570f3a637629341d5ecc1de42037f109c5c6e3aef57870e9e0964d1a998a8
1078	Earth and Environmental Sciences	SFT-EARTH-COAST-TIDE-001	Coastal and tidal Earth-system response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0ecc3db9c0519d9908e662e6dbf15c3b0fff8221b69b55fbc05f12a5b0a88f3b
1079	Earth and Environmental Sciences	SFT-EARTH-CLIMATE-001	Climate state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d99699708929f2ac543e9559642735ff08de2dd5b5e4adb589a1880831cd945e
1080	Earth and Environmental Sciences	SFT-EARTH-WEATHER-CLIMATE-TIMESCALE-001	Weather-climate timescale distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fde55140e6f9ad1022bfad630103a55469f2bbd884740799fa9d09093dc8a1c5
1081	Earth and Environmental Sciences	SFT-EARTH-CLIMATE-FORCING-001	Climate forcing	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8336679d3056e872210936ddf1d326bec025eed573cc4bc7c2cbd60bda5dc7de
1082	Earth and Environmental Sciences	SFT-EARTH-CLIMATE-FEEDBACK-001	Climate feedback	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d0293c869cef7f636ae317f35e7e03e169fcdbece06bd3b524df427d1939a43
1083	Earth and Environmental Sciences	SFT-EARTH-CLIMATE-VARIABILITY-001	Internal climate variability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5348ca53751d27ca4262738ebecfe7a166730424e91eb42bd46c9647dc950dc6
1084	Earth and Environmental Sciences	SFT-EARTH-CLIMATE-ATTRIBUTION-001	Climate detection and attribution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5fe64d4e0896c6844feb6762a7a74cee285c7a2104488379a5cdf9c543ad4f4d

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1085	Earth and Environmental Sciences	SFT-EARTH-EARTH-SYSTEM-TIPPING-001	Earth-system tipping boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:af4ee0a52a6067727c14b16376c790f8338df98389581f09f063583e069fd1eb
1086	Earth and Environmental Sciences	SFT-EARTH-CARBON-CYCLE-001	Earth carbon cycle	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a0ccc2c74781615163b5773ed202120c3aac7d1941dad35734a11598251392f4
1087	Earth and Environmental Sciences	SFT-EARTH-NITROGEN-CYCLE-001	Earth nitrogen cycle	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0db8e35b2dda3e781c0aa20c5fc1727f0861e19640e5f214d1ba4e6163707e45
1088	Earth and Environmental Sciences	SFT-EARTH-PHOSPHORUS-CYCLE-001	Earth phosphorus cycle	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:49344c713351324ec81cbd41d0f6363b370d81d4ebc257c45b30664f743c599e
1089	Earth and Environmental Sciences	SFT-EARTH-BIOSPHERE-EXCHANGE-001	Biosphere-Earth exchange	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e074043cb8f77ff8031ebb3d85be04d9d1d4a2772c7449a39cb63f9729d5e19
1090	Earth and Environmental Sciences	SFT-EARTH-COUPLED-ECOSYSTEM-001	Coupled ecosystem-Earth organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c0ca6cd5d9c6bf0d091bd1331e7c2da327e807b75fcc6386c8c11f636db6ab18
1091	Earth and Environmental Sciences	SFT-EARTH-BIODIVERSITY-ENVIRONMENT-001	Biodiversity-environment boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9eca1b410e560534940ec34d28887c1c46ff8e9ebcde85549c8ac1f1ac0b0945
1092	Earth and Environmental Sciences	SFT-EARTH-POLLUTANT-IDENTITY-001	Pollutant identity and boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f70ce5119a12f8ad926fd279bbe6fcb49b650d666e604dff2f6e586b7d1f993
1093	Earth and Environmental Sciences	SFT-EARTH-SOURCE-PATH-RECEPTOR-001	Environmental source-path-receptor chain	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:29ff886be12a107ecbcbe035eb5573a52bc661557faa4c0bfafd9ff7f28a8727
1094	Earth and Environmental Sciences	SFT-EARTH-ENVIRONMENTAL-TRANSPORT-001	Environmental contaminant transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4406c9886170dbcd2c3b7a8348b855d249b42b605272afc2ba4a02ae4cde94c0
1095	Earth and Environmental Sciences	SFT-EARTH-ENVIRONMENTAL-TRANSFORMATION-001	Environmental transformation and persistence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:90f90906ca26785c8256f3e093512560e27bfb09ca386250231eb0f66bc24fb8
1096	Earth and Environmental Sciences	SFT-EARTH-ENVIRONMENTAL-ACCUMULATION-001	Environmental accumulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78aa279eb4f757e454c1d5148fc6722520c26e5ddcccb1a094b4272883fb3d48
1097	Earth and Environmental Sciences	SFT-EARTH-ENVIRONMENTAL-QUALITY-001	Environmental quality state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1c71e9649e5baf46adbe3c1690b297a15cd85754ef8f0fd5cd59b37f3b3319e9
1098	Earth and Environmental Sciences	SFT-EARTH-FIELD-OBSERVATION-001	Field observation and sampling	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0a0767dc855b71b885537b3fb2f2fa2e4811e9f14219e6bc5ea4bc68cc251f28
1099	Earth and Environmental Sciences	SFT-EARTH-REMOTE-SENSING-001	Remote-sensing observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:826728a3fa2756f2cf9c9e97da5b1a7245d5e6c1cedaed03b98404a90013be2

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1100	Earth and Environmental Sciences	SFT-EARTH-PROXY-RECONSTRUCTION-001	Proxy and Earth-history reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d02ad77618b5f20aca6067b7a28ef8002c34b40d80166a08f1ed588df9e68c90
1101	Earth and Environmental Sciences	SFT-EARTH-MODEL-DATA-ASSIMILATION-001	Earth model and data-assimilation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0589bd2c187b02828686eaf6e38bfaaedaf1b29aa9bd0f194a739594d06b824f
1102	Earth and Environmental Sciences	SFT-EARTH-FORECAST-VALIDATION-001	Earth forecast validation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6a4d0b638c170dbb8dd13c7f253637e3b1c4127806a2e1f68fab74a05e49563d
1103	Earth and Environmental Sciences	SFT-EARTH-HAZARD-RISK-HANDOFF-001	Hazard, risk, resilience and disciplinary handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:27d3e0cf47bb4199bbd260c9643df2afd0ea24975e8799ab5ada6ca9a9985eed
1104	Astronomy and Cosmology	SFT-ASTRO-SOURCE-001	Astronomical source	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5473ed30f43a3ee1099bf327c08e196a9abcd888adb958b68d7c85d51681e958
1105	Astronomy and Cosmology	SFT-ASTRO-OBSERVER-001	Astronomical observer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f714bc90e2a60620a46297a6839c7a5b3dbf02d7cd9e51f7b1c81071fb7d0d7a
1106	Astronomy and Cosmology	SFT-ASTRO-SIGNAL-001	Astronomical signal	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c0417bce86220840e95fcec8e05c9888b85857274b388151c95c1a744e721b1e
1107	Astronomy and Cosmology	SFT-ASTRO-ANGULAR-POSITION-001	Angular position and reference frame	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5baee62478fbcd17834102db63297cb4c72efcd2029f90652f9fc24d14588ae8
1108	Astronomy and Cosmology	SFT-ASTRO-DISTANCE-001	Astronomical distance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:163fa182412a88c4e90f0006888a2d78ff4d2fb5d579fe938a5c6e521035aa54
1109	Astronomy and Cosmology	SFT-ASTRO-SELECTION-001	Survey selection and non-detection	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2c1a77d0fbc0a3ccd3180f637b9067c4f89fdd1fb2eaf2094bbfd74cf5bcb430
1110	Astronomy and Cosmology	SFT-ASTRO-OBJECT-001	Astronomical object identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3aa8c2aa9a6e07c3a509e0f9ddfda46f777bbf74e489bf35ab967a0c7e165d08
1111	Astronomy and Cosmology	SFT-ASTRO-CLASSIFICATION-001	Astronomical classification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:38676430baf965fd0716b3e5386b3271651d850aff8c29d925e7fe0b9c831e56
1112	Astronomy and Cosmology	SFT-ASTRO-POPULATION-001	Astronomical population	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0a62514b2e54766ac4b30eedccd4c6ce170a42980849b27b1ddda55d3fa31441
1113	Astronomy and Cosmology	SFT-ASTRO-MORPHOLOGY-001	Astronomical morphology	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7e5ce0ee426488dc2e6d78aa5c6d1fca06d6cdc2cccf3f0bd56b3eafa0fe44dc
1114	Astronomy and Cosmology	SFT-ASTRO-COMPOSITION-001	Astronomical composition inference	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:26cc87a10853fecab01b32b1fa0e693cfbf86fd0cbec19091ee16fc2fd5d17d

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1115	Astronomy and Cosmology	SFT-ASTRO-CATALOG-001	Astronomical catalogue	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:acdd861f1b9ddfb158859329b8062e970f19f77f497b673fe8caf0c4272f68bc
1116	Astronomy and Cosmology	SFT-ASTRO-FLUX-LUMINOSITY-001	Flux and luminosity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:244626352d46e53db6f803dce4d8de4b5a9fc01a0c4a48c9ee7c74863b9f3f0a
1117	Astronomy and Cosmology	SFT-ASTRO-SPECTRUM-001	Astronomical spectrum	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3c2ae757898fd4ca1063639281403d361117113843e1a4bd69144ffa55655594
1118	Astronomy and Cosmology	SFT-ASTRO-REDSHIFT-001	Astronomical redshift	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d3148d125f87843df676f13fcd10f8c9c05baa3afe48398d6743fa616d5606c2
1119	Astronomy and Cosmology	SFT-ASTRO-MOTION-001	Astrometric and radial motion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bbea1c97377efd636dc90e8df5d9963b2b6f55a59422efc302d0a0172897de8f
1120	Astronomy and Cosmology	SFT-ASTRO-VARIABILITY-001	Astronomical variability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:91d8d1c51df7727d0da50e2ebc76099744b8faca1c4dafd29fc99b962f72aeb
1121	Astronomy and Cosmology	SFT-ASTRO-PERIOD-TRANSIENT-001	Periodicity and transient distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d4a7ebf3b590fe9908e73b46d032cbcb0e34ef99ad286ab3ff5644df05ac1b6a
1122	Astronomy and Cosmology	SFT-ASTRO-STAR-FORMATION-001	Stellar formation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1687511f165df988b1a70cb12f64fda63bc0a8236daab845dd92d2f7a1723756
1123	Astronomy and Cosmology	SFT-ASTRO-STELLAR-STRUCTURE-001	Stellar structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:745723e85c03326d670183c817d64823618a9576ecd1efb2ace244d954efa0b5
1124	Astronomy and Cosmology	SFT-ASTRO-STELLAR-ENERGY-001	Stellar energy generation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:677612669b5fd145ef82ad8bd915e8ef433e4a8c491ffc1b3467467d9f4db940
1125	Astronomy and Cosmology	SFT-ASTRO-STELLAR-EVOLUTION-001	Stellar evolution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:452f1a61686abff87da13d1345308ad9722fd3c30e2db5ed71e63dd359a48f42
1126	Astronomy and Cosmology	SFT-ASTRO-STELLAR-POPULATION-001	Stellar population	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:238a4d5f7c1962eea3c0039c054810fe25aa55a0f57218b3ac96885f99bce77a
1127	Astronomy and Cosmology	SFT-ASTRO-STELLAR-ENDPOINT-001	Stellar terminal state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cc3186110029ec8264d7cfff7583e93cd8cb599853d2645374bc9c7377ad2f8
1128	Astronomy and Cosmology	SFT-ASTRO-PLANETARY-SYSTEM-001	Planetary-system identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c246e3a9b32c67d57ac4ec24cf388171bc7f56b403018ce48d1834def18b7d70
1129	Astronomy and Cosmology	SFT-ASTRO-ORBIT-001	Astronomical orbit	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:058bed515488bd9ed3e3f8e3a7f6cbe218fef342910a86800cf6f40202f4be1

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1130	Astronomy and Cosmology	SFT-ASTRO-RESONANCE-TIDE-001	Orbital resonance and tidal state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7fb8371722eb7ced1d07e8e9617f2a368b04dd05adda017db98648d339afa87b
1131	Astronomy and Cosmology	SFT-ASTRO-PLANET-001	Planet and satellite classification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cc44ff010d51d2f18e183a9642d1c1f1bc3525ea658bd18e92973b259c69c90e
1132	Astronomy and Cosmology	SFT-ASTRO-EXOPLANET-001	Exoplanet observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b45e633c526545940df83343788a26f671c0061ddd899a9fae70251d37564314
1133	Astronomy and Cosmology	SFT-ASTRO-SMALL-BODY-001	Small-body population	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dc8a555a924c996226270750247dc628b5b4d7741b40acd27df07fd8015d85b9
1134	Astronomy and Cosmology	SFT-ASTRO-ISM-001	Interstellar-medium identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0cec1bfffce82ee255eab56babbd859767b0618e7791c555b77d4a3cc8a0467e4
1135	Astronomy and Cosmology	SFT-ASTRO-DUST-001	Astronomical dust	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:213ab20c5ab76aa3761d091100564d21726e2112b6030f48c4d30a319b19f67d
1136	Astronomy and Cosmology	SFT-ASTRO-MOLECULAR-CLOUD-001	Molecular cloud	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:05313d27771f58964ebe43b714f0f74af495152b0ad8fc9362c3a0d5e9376fae
1137	Astronomy and Cosmology	SFT-ASTRO-CHEMICAL-ENRICHMENT-001	Cosmic chemical enrichment	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9cd403fe963694ee2817abb48d9e1c87cb7b38b49e61eb90c21efff8726bc85be
1138	Astronomy and Cosmology	SFT-ASTRO-STAR-FORMATION-ENVIRONMENT-001	Environment-to-star-formation relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:28da28a0062a3873e96d08d96610ca8b8af37164f59251da7046dc4a27df6293
1139	Astronomy and Cosmology	SFT-ASTRO-FEEDBACK-001	Stellar feedback	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1a44a2f07fadd9785f8d7e1b50f7eba9c43106c84cb5663b014133d5dbe5bedc
1140	Astronomy and Cosmology	SFT-ASTRO-GALAXY-001	Galaxy identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:39520fdbde70a6f44e6e576a76668c0770b8b6e5a092add65a5b688f55d7135c
1141	Astronomy and Cosmology	SFT-ASTRO-GALACTIC-STRUCTURE-001	Galactic structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8c55b5bfe11c3d144090c351c4852344af828c2ec7fffe05e207fc6d015617f5
1142	Astronomy and Cosmology	SFT-ASTRO-GALACTIC-DYNAMICS-001	Galactic dynamics	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:09221a0166047713578ad34a5ba48d2748ab3b857e0ac7b181766c37110d408d
1143	Astronomy and Cosmology	SFT-ASTRO-TULLY-FISHER-001	Luminosity-rotation scaling	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c3828b9cd507fa3d9b9abf663f4aefd6b4348b414d424262b843d05a45aa295e
1144	Astronomy and Cosmology	SFT-ASTRO-GALAXY-EVOLUTION-001	Galaxy evolution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b8cdebfea3ab8986671702da489624ec8c20478bae24ce8bcfe4841aa4e2dd98

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1145	Astronomy and Cosmology	SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001	Groups, clusters and environment	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:83121167316eb91e192b2ecaabdf73cee80c7db739a5cc1d3d0267f1c288483
1146	Astronomy and Cosmology	SFT-ASTRO-COMPACT-OBJECT-001	Compact-object identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d72a9707b68e99d69093885946b890ebdb3d08dffdb8e1a9c0d166335c90ec6
1147	Astronomy and Cosmology	SFT-ASTRO-WHITE-DWARF-NEUTRON-STAR-001	Degenerate remnant classes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8270a0c8773e17094236ae656d4872538b39c7c3153f3c9149a7cba014896dd0
1148	Astronomy and Cosmology	SFT-ASTRO-BLACK-HOLE-001	Black-hole candidate and horizon evidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7764fe13a9fb7b3d43062fec060acf9379bf00321f0bbcab8439813fbf3ff9ec
1149	Astronomy and Cosmology	SFT-ASTRO-TRANSIENT-001	High-energy transient	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c5eac33d788970c7c587923786815268fcb4c635f2f2469e16d08dc378136291
1150	Astronomy and Cosmology	SFT-ASTRO-MULTIMESSENGER-001	Multi-messenger association	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:98f412670daf803b95eadc7246a7a5749e56caaf065a2266f008cd4be971e921
1151	Astronomy and Cosmology	SFT-ASTRO-COMPACT-POPULATION-001	Compact-object population	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6b6de3d5331b1c9e197bada2a99f2fb8c4685f13c83714b7d0d178a7ecf6c17d
1152	Astronomy and Cosmology	SFT-ASTRO-COSMIC-HOMOGENEITY-001	Large-scale statistical organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7e928a24126ea2671b1fddb0fd3d38e85e0fece770f5d759573517f9205b18ea
1153	Astronomy and Cosmology	SFT-ASTRO-COSMIC-WEB-001	Cosmic-web structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:473d2d8bf8837e40ef8530f0aedd56aff5cfe10a3f0f59b146be1a0ea85711b
1154	Astronomy and Cosmology	SFT-ASTRO-EXPANSION-RECORD-001	Cosmic expansion record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d08db815fc6f99661fd7e78f5f67a40e4734053c672d5cb5d053bde2f349bd75
1155	Astronomy and Cosmology	SFT-ASTRO-DISTANCE-LADDER-001	Distance-ladder composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:09bb53ce938c971978590248d20c8241f162a071f7a820fa660cfefbb82d2139c
1156	Astronomy and Cosmology	SFT-ASTRO-LENSING-001	Gravitational-lensing observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa79bb09787cd670d87fd4a7ddfadf08bba448490936a24b6e7600d46bac534
1157	Astronomy and Cosmology	SFT-ASTRO-EXPANSION-HISTORY-001	Expansion-history inference	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f49381be08c9ea93843dd3b65e22c9cd26a9fbaa329cd6c3ab085d55503b5d36
1158	Astronomy and Cosmology	SFT-ASTRO-THERMAL-HISTORY-001	Cosmic thermal history	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c2f4e22ae5b613c94534375ee96900a2751ff1cfc8f8ea35c3ae74c004907018
1159	Astronomy and Cosmology	SFT-ASTRO-CMB-001	Cosmic microwave background	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7118549efc4e4f62f0a3f11bd35f87cc36682d2f557515aeaa40d8b0b3d5a557

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1160	Astronomy and Cosmology	SFT-ASTRO-ACOUSTIC-STRUCTURE-001	Background acoustic structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:897cd20cee5b797d7833de676b37dbd87a27ad30282b26fc64357198366bbea4
1161	Astronomy and Cosmology	SFT-ASTRO-PRIMORDIAL-ABUNDANCE-001	Primordial abundance record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7cf55eed144d2fad2357efe5c54fdc92510151d15e73c07321723395df4883fb
1162	Astronomy and Cosmology	SFT-ASTRO-RECOMBINATION-001	Recombination and photon release	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:55693de3cc33c46c54235ea2ed160fb034a7bf81a5af2ce2ad5a6c5f7d709843
1163	Astronomy and Cosmology	SFT-ASTRO-INITIAL-PERTURBATION-001	Early perturbation record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2cfecdf1565cc5f5bcf85154beb1e17120cbe8188b01418c605e77934803ac2f
1164	Astronomy and Cosmology	SFT-ASTRO-DARK-MATTER-EVIDENCE-001	Dark-matter evidence class	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:da36b528cee229d263a4537a6886b66fb0c5e7448bf0f8d08e32158f31945981
1165	Astronomy and Cosmology	SFT-ASTRO-DARK-ENERGY-EVIDENCE-001	Dark-energy evidence class	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c65dbbb54d1c96f9c2c02534f42eac5c4523f3c8ef09256f33013b3764437f8f
1166	Astronomy and Cosmology	SFT-ASTRO-COSMIC-HORIZON-001	Cosmic horizon	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:973035e63119a5e728bb88f8606b633621a1f28d5c3a63a50e925504c79e42cd
1167	Astronomy and Cosmology	SFT-ASTRO-TOPOLOGY-001	Cosmic topology claim	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0040739a2ddeb6c99077b83d3fe789f427c38f159274e8f9495274ee1b7fa0e7
1168	Astronomy and Cosmology	SFT-ASTRO-MULTIVERSE-001	Multiverse admissibility boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a117824bcf01755b83dc742009d2aad4339370fc5bf9c9626a1d3b8c7e494b6c
1169	Astronomy and Cosmology	SFT-ASTRO-ULTIMATE-FATE-001	Cosmic ultimate-fate inference	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:49730cfa0cd10c71cf80dd286733833e51dee356deca785b1fa0ee10e82867f6
1170	Astronomy and Cosmology	SFT-ASTRO-LAW-HISTORY-001	Universal law and cosmic history distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c3a3d30c99d99d46c7a26df839bec5d9cd41351cd48e5ab40c5ed9cb0bb58796
1171	Astronomy and Cosmology	SFT-ASTRO-MODEL-OBSERVATION-001	Astronomical model-observation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:887f1ed59a17938a9603af5790b6e13dbcb1c63cdd768242a0a89dcfb19b8991
1172	Astronomy and Cosmology	SFT-ASTRO-STANDING-PREDICTION-001	Standing astronomical prediction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b83b752df3a1e802277ac2e5eaeef8205baeebfd00d827236838e8a57cca1b93
1173	Astronomy and Cosmology	SFT-ASTRO-UNEXPLAINED-001	Unexplained astronomical observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7af6264c33e27e20954db0642d0051f8e9b14af8664c8446daf278ed238bbcf2
1174	Astronomy and Cosmology	SFT-ASTRO-ASTROBIOLOGY-001	Astrobiology and technosignature boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1b06beceb6d9a84c63a86aef6d02baf535a9244f99301d68a82778fae3d6e585

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1175	Astronomy and Cosmology	SFT-ASTRO-ASTRO-HANDOFF-001	Astronomy scientific handoffs	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d8b4042aa2faf71d32266ffcd173e2d2462772695741ae0f290d55d0ba6a883
1176	Social and Collective Systems	SFT-SOCIAL-AGENT-001	Social agent	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f2f609e054ce3520a7f646913f90e57d7dc53aaaf77ba0556570d0ff27c17ebd
1177	Social and Collective Systems	SFT-SOCIAL-RELATION-001	Social relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:146c15af99c64010363b325bf1d0df656f6680492ca32f2fce517b83adf8b7cf
1178	Social and Collective Systems	SFT-SOCIAL-GROUP-001	Group identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ea38d831ac38a601d2aaedcfa5153d810f8ca7d31b6227d420da9b8c6956f387
1179	Social and Collective Systems	SFT-SOCIAL-COLLECTIVE-001	Collective state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f0866c38e5fc8d9c9dc0190ddb5a8994de2b679bddf4406a43141629c5ee9cda
1180	Social and Collective Systems	SFT-SOCIAL-ORGANIZATION-001	Organization identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:422c2814a0efc03176f88be51639b782eb4bf45afb9cd4dbf9bdd83660547e9e
1181	Social and Collective Systems	SFT-SOCIAL-POPULATION-001	Social population	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:176f9f0a8f0c176a6beca49d36d60c4310ba772a229681fc2eb7a7c3f6187ece
1182	Social and Collective Systems	SFT-SOCIAL-OBSERVATION-001	Social observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9e5f48e91a6767b794854f5c461addb7d9016687a4721cc0970c23fa475f47a6
1183	Social and Collective Systems	SFT-SOCIAL-ACTION-001	Social action	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:99446d01ce954f8802de92a9839ef59588352022b9b9b8665cc6a5ecd70d0c82
1184	Social and Collective Systems	SFT-SOCIAL-MESSAGE-001	Communication event	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f3674a401671a9f62232868be604578e8d23a1b629982ac17a0a58250f24450e
1185	Social and Collective Systems	SFT-SOCIAL-SHARED-RECORD-001	Shared social record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f56e1e73f396479164cdf9c2e4688d84f875d812d4f97c17ce23efbe47ae6ee1
1186	Social and Collective Systems	SFT-SOCIAL-COMMON-KNOWLEDGE-001	Common-knowledge boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3b5736ee7fbc8d85b03e2edad2e4b2a5447fb2baa06d55f9e704e3b01eb099fbd
1187	Social and Collective Systems	SFT-SOCIAL-COMMUNICATION-NOISE-001	Social communication noise	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c609ce1483a76a7a83d8f28c5fd7170640788f6afef041858b81e20aa23150e
1188	Social and Collective Systems	SFT-SOCIAL-COORDINATION-001	Coordination	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:332c7b9b943206b9afa4b5b83ce4fc3c08cf81653588e2c217d2966cfc0fcd6d
1189	Social and Collective Systems	SFT-SOCIAL-COOPERATION-001	Cooperation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5c58d6b54a1282c18b43b1785af6f6f7689a713dee415eb607df14171fc37ea5

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1190	Social and Collective Systems	SFT-SOCIAL-COMPETITION-001	Competition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6b6d00922b1c86a7c0c38187646d9ec0c62b74075afe7fd0a9b05cc064d6488b
1191	Social and Collective Systems	SFT-SOCIAL-CONFLICT-001	Conflict	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0c2dfd4f8e85ce16ff07f28527f9e08bb0d4677836a294403856f3dc50d04780
1192	Social and Collective Systems	SFT-SOCIAL-BARGAINING-001	Bargaining	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b452e58367879549904f8cc507672a5ef1bbae7e43052b92ec7365f31d52ead9
1193	Social and Collective Systems	SFT-SOCIAL-COLLECTIVE-ACTION-001	Collective action	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a7185420f280dd7879b17b1b675bcbd8920bf7d9bcd752b60733ccf468ae2760
1194	Social and Collective Systems	SFT-SOCIAL-NORM-001	Social norm	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:81ed8141611660faf49244ba772291a6a232970c571f398f99fd19b9bf11bf7b
1195	Social and Collective Systems	SFT-SOCIAL-RULE-001	Social rule	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e7ed142368911acc1f16f02f17d142061e03ea2114d5c3afa799e2345403f4b7
1196	Social and Collective Systems	SFT-SOCIAL-INSTITUTION-001	Institution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b9e33b4de5564358ded810f93b3298bea090786567f936270ba09ba3569c9918
1197	Social and Collective Systems	SFT-SOCIAL-ENFORCEMENT-001	Enforcement	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b998039d03a7d316d7dbfea6e71a7b83d5a0774f84727d8b47839b3c174a84f2
1198	Social and Collective Systems	SFT-SOCIAL-COMPLIANCE-001	Compliance and resistance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c9e98286efce43ca1d05239813f71d18955c9d9420d1071237c72d6422895580
1199	Social and Collective Systems	SFT-SOCIAL-INSTITUTIONAL-CHANGE-001	Institutional change	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a3453592fdd413eeda7249449dec8f49cacb169203d83c7888ed55df399c3116
1200	Social and Collective Systems	SFT-SOCIAL-TRUST-001	Trust relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7d449223848c03df9221d66e6483af9123afd8bf01787c1c24e51500b5d9c9b3
1201	Social and Collective Systems	SFT-SOCIAL-REPUTATION-001	Reputation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6fac43db587b30f5f66791a478d0c848b583c8dfb18e4ed5b10e16daace7ef9
1202	Social and Collective Systems	SFT-SOCIAL-TESTIMONY-001	Testimony and credibility	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4b52d15532ed596150734988a9cfd8ce768f6190152d912d5420a4fb4468c0b5
1203	Social and Collective Systems	SFT-SOCIAL-COLLECTIVE-KNOWLEDGE-001	Collective knowledge	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fcfe2f23cfc7590e92f561b84763b94bc6b65e5b19c6efa5a317cd329c2c4af1
1204	Social and Collective Systems	SFT-SOCIAL-EXPERTISE-001	Expertise evidence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bbde304ac7029a5c12d6256e624245f3fbfc8aeb6e6629f72843188ac0541ee0

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1205	Social and Collective Systems	SFT-SOCIAL-MISINFORMATION-001	False and unsupported social information	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8b2d26d118ad33a5180de1fa62f95ba04009100be7cf9f438d2e7f03f3b83f51
1206	Social and Collective Systems	SFT-SOCIAL-EXCHANGE-001	Exchange	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:df46ce2287ea5a75408e543fe7a36fd714b95f8fd11c611b97a3d1875c600a58
1207	Social and Collective Systems	SFT-SOCIAL-VALUE-001	Social valuation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:122e9236d3b52979be9fe329ea942a6fd431eb52b6a405859ce2bd5cf36b5a84
1208	Social and Collective Systems	SFT-SOCIAL-ALLOCATION-001	Resource allocation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:92cc1799ee389a64befd0ba46bc3ff916ead60a0b03f32aecafd9bef4a4c9136
1209	Social and Collective Systems	SFT-SOCIAL-PRODUCTION-001	Collective production	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ae53c6cd5ece19bc49289d53a331d7283bedc3c36f6b065fe93cdf63a1df41a7
1210	Social and Collective Systems	SFT-SOCIAL-DISTRIBUTION-001	Distribution and inequality	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94cd29960f46745e6f2ac0a7dc569aeef6d1918e36572172dfc28005de2bd6a
1211	Social and Collective Systems	SFT-SOCIAL-MOBILITY-001	Social mobility	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c5ed76c9b77ad263fb8bd1c66d620aa54b73c373e4fdbb0aa7111d912b219823
1212	Social and Collective Systems	SFT-SOCIAL-POWER-001	Power relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4c0a80d595e1a96fcdf44f40e8970537a863559f2c0af12e7d1f52726586349f
1213	Social and Collective Systems	SFT-SOCIAL-AUTHORITY-001	Authority	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0ef12b8ddff302feeee5b78871251d4fc32f15a1a4339cbb8e6276356ffb202c
1214	Social and Collective Systems	SFT-SOCIAL-DEPENDENCY-001	Social dependency	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c1c1151da996efecdc23edb781a9fbaf4077c38bc76b903d61c99218f53f2163
1215	Social and Collective Systems	SFT-SOCIAL-EXCLUSION-001	Exclusion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a706086b718bc0c37342a9586801a6b6e53528d197bbd7e592a47249c98d23b4
1216	Social and Collective Systems	SFT-SOCIAL-STRATIFICATION-001	Stratification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bdace614acbf29d3c8007db7999b6fdfe0cf4079dd7c93d4785cc2aa5127a5da
1217	Social and Collective Systems	SFT-SOCIAL-GATEKEEPING-001	Scientific and institutional gatekeeping	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9d1f3fb1cc3eb7127d43c77fd7f279c9f22c1c8f02e50075e4aae47f0589b4c2
1218	Social and Collective Systems	SFT-SOCIAL-SOCIAL-NETWORK-001	Social network	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d731ec9187e36749fd901e73d419bf40d6f0c5757d5b5db3d97b44eab87c5bc
1219	Social and Collective Systems	SFT-SOCIAL-DIFFUSION-001	Social diffusion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:67f740214a3cb3f17d5980370451e1a0229775652bfb7552b53a50c04fd9708b

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1220	Social and Collective Systems	SFT-SOCIAL-SYNCHRONIZATION-001	Collective synchronization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:80e44dc1ceefce9f9a613530d8d23f3772b0bb3748587e4035c9302b56b8d294
1221	Social and Collective Systems	SFT-SOCIAL-CASCADE-001	Collective cascade	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:217adf4d391053fa69d18f11faa90e665f18c827863e7eb1bb41ad447b770c03
1222	Social and Collective Systems	SFT-SOCIAL-COLLECTIVE-TRANSITION-001	Collective transition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2d9c7a6452a51c3691d4ff3f6db6294a483b0723fd357c8e21430ae1a474f510
1223	Social and Collective Systems	SFT-SOCIAL-SOCIAL-CYCLE-001	Socio-economic cycle	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9959bc1bdebd09230f2219c1b2122637585365e4eca719e1e11ff44ecfbba248
1224	Social and Collective Systems	SFT-SOCIAL-CULTURE-001	Culture	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e80d173192171fcbe88fd26c3c57e7a3191d5ed3d4ca98db62ff54531705b593
1225	Social and Collective Systems	SFT-SOCIAL-SYMBOL-001	Social symbol and meaning	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2954e69231b375ec0440a69da946ab51192107b05360efbbafe3eba1debca9f84
1226	Social and Collective Systems	SFT-SOCIAL-LANGUAGE-COMMUNITY-001	Language community	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:27f3aec512e0ec3ae29a7826f51de3f4676d678d60865565dcf1e3ea595a536c
1227	Social and Collective Systems	SFT-SOCIAL-EDUCATION-001	Education and knowledge transmission	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c3e3422783196463271d69df0f596ae0d30a3b0485b0ba8f0723b8787e235e0d
1228	Social and Collective Systems	SFT-SOCIAL-MEDIA-001	Media system	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d746622ff5a470f4d18e70033d6bbf1a9ca137736df7efca8193c286527fbe1
1229	Social and Collective Systems	SFT-SOCIAL-COLLECTIVE-MEMORY-001	Collective memory	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:787f1074b105c114d5e3506c01c9a626a39f67ac8a73f37ee6f75893632723e2
1230	Social and Collective Systems	SFT-SOCIAL-MICRO-MACRO-001	Individual-to-collective composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cff9adcd8d1aa01d9fe52c2fe7b60bd3275ac72a813de37b7ffcc9cf2ae99eb0
1231	Social and Collective Systems	SFT-SOCIAL-MACRO-MICRO-001	Collective-to-individual constraint	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f67cabe097f5789ea4637b873b80b854246c72619c73ef360a3cf62538e7f06b
1232	Social and Collective Systems	SFT-SOCIAL-EMERGENCE-001	Collective emergence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:539df59af1716049ceaeec48ce23f8df321d8041e8ce99f262e5517c81ea252c
1233	Social and Collective Systems	SFT-SOCIAL-MULTI-AGENT-001	Human and artificial multi-agent collective	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:60e78a62c2a13e7aca9740c5c8c924fddaf5177f7b747adcfca487785ccf4067b
1234	Social and Collective Systems	SFT-SOCIAL-COLLECTIVE-DECISION-001	Collective decision	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ebd6ba8b976fd3197bf996059944d8bcbead597c2e03840e9f07c0a3e356861b

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1235	Social and Collective Systems	SFT-SOCIAL-AGGREGATION-LOSS-001	Aggregation and ecological inference boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7ff9a069d479701ef5f32b29468908f4476ca0bcd77572f62d413cdd13c32765
1236	Social and Collective Systems	SFT-SOCIAL-HISTORICAL-EVENT-001	Historical social event	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5d2c8091d96cfda76c054ea5ab140b4534508d6d7f8d2c64d102934786854533
1237	Social and Collective Systems	SFT-SOCIAL-CONTINGENCY-001	Historical contingency	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:79be04ee3a298f8ceca4cf7827cb30d905f125227f92f742158487be5c98a0c1
1238	Social and Collective Systems	SFT-SOCIAL-CAUSAL-INFERENCE-001	Social causal inference	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2ed40b2f01083326f219d9aca6f455c1d6585f185b410b751a7e4bcf18bb737e
1239	Social and Collective Systems	SFT-SOCIAL-INTERVENTION-001	Social intervention	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa62a0c0d10360df013c9b20d813fac4383cb3aff2e18fc51cd9e785bdbfe315
1240	Social and Collective Systems	SFT-SOCIAL-REPLICATION-001	Cross-context replication	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dd32b12f5464a8ce1964f7e13f600dcc4394a95554cee8702add49c94872de4a
1241	Social and Collective Systems	SFT-SOCIAL-ETHICS-BOUNDARY-001	Ethical and value boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3cfb0486415cdab2c5ce460390aab1782820d675af4b08e782e175442c5e13b3
1242	Social and Collective Systems	SFT-SOCIAL-MATH-HANDOFF-001	Mathematics and graph handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fdc6cffbd20780ccaf6c91ae1fc4f054fc9f6357e3612bc6d6d7a869cd5b599b
1243	Social and Collective Systems	SFT-SOCIAL-COMPUTATION-HANDOFF-001	Computation and distributed-system handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:de0cc7b5d201335a2cfa310fc1bc03d706361bd52e26a6eee479e7e16b30a210
1244	Social and Collective Systems	SFT-SOCIAL-BIO-CONSC-HANDOFF-001	Biology, consciousness and individual cognition handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7a26afef3725b313ba83e780dc752a863110a77f0e03c86e9d2e6ae177eaa0f
1245	Social and Collective Systems	SFT-SOCIAL-MEDICINE-PUBLIC-HEALTH-HANDOFF-001	Medicine and public-health handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f5748560880bdbbc3ad54e5dd3457d3df4707505ed574df7d6b319f7cc29595
1246	Social and Collective Systems	SFT-SOCIAL-EARTH-HANDOFF-001	Earth and environmental handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:949369f3908bb753e62f2b5017113fc3d1b5ab7d7d62ab6c26366ce55a227aa0
1247	Social and Collective Systems	SFT-SOCIAL-ENGINEERING-HANDOFF-001	Engineering and policy implementation handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bfe9b4bbad15adb6a0d7bac2472d99f035c9ac7810b25c365d92b6f5b74b1f87
1248	Engineering Translation	SFT-ENG-REQUIREMENT-001	Engineering requirement	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d576e9c15d258d3c64253f4e8f05ed95a275c544a6db286dc23b6c3007b70316
1249	Engineering Translation	SFT-ENG-FUNCTION-001	Required function	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c3cde1442d2ae6fbed97c59b287da2c790327f965736295951dd68cb1ac50206

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1250	Engineering Translation	SFT-ENG-CONSTRAINT-001	Engineering constraint	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0b201dc8237be12a2290a8d7113ce6f834d1a17d7d0289889b81130e30b89f0b
1251	Engineering Translation	SFT-ENG-OPERATING-BOUNDARY-001	Operating boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3447dce9ca5104791f130d5cf2b4612259d9b5ad5e73cd45bce488c29270fca5
1252	Engineering Translation	SFT-ENG-USE-CASE-001	Use and misuse case	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e82fd4f8d04b79886ae1b2da4d9ae232d34bc07029efc5a5a9c3dab431ea50f5
1253	Engineering Translation	SFT-ENG-ACCEPTANCE-BOUNDARY-001	Requirement acceptance boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cd154a063434c15b09f39717c0853d04b08f9df9a24ebe975c1ad557545d6db4
1254	Engineering Translation	SFT-ENG-COMPONENT-001	Engineering component	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1a028469b13f45730ec5a6a1c75966b87229f40e7da12df43f434750d1421eac
1255	Engineering Translation	SFT-ENG-INTERFACE-001	Engineering interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:37e9f45a48de371bc058941dbc7f3a9d883632e2f78aeea8ff47bb915878cba3
1256	Engineering Translation	SFT-ENG-ARCHITECTURE-001	System architecture	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7deffa065dcfc825ec565444d246ff656c6b3f10e75b7bb56efe3682c14aeb9c
1257	Engineering Translation	SFT-ENG-SYSTEM-001	Engineered system	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5cd0b2eb968f28707d85b6d60ce3977816de74319cc342f3a5c18776a1ec497a
1258	Engineering Translation	SFT-ENG-MODULARITY-001	Modular decomposition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4927d9b6e8f6fdda5ec599f2badaf1090ad1baa9c03a8110d338d96873c3fa54
1259	Engineering Translation	SFT-ENG-INTEGRATION-001	System integration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d2b8103d27c920f149a191509764f1e24ad891b58516fc00bed2c59c8872cd41
1260	Engineering Translation	SFT-ENG-RESOURCE-001	Engineering resource	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:264d3cd9efaaceea723aae48a58392418af0a8a11c6fcdc6bdf65fec0e0ec1e3
1261	Engineering Translation	SFT-ENG-CAPACITY-001	Capacity and load	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3963d278a9dde0a0c4f1c137720a3335d73c5557527eb06a8c99e84cccc16796
1262	Engineering Translation	SFT-ENG-EFFICIENCY-001	Engineering efficiency	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1160782f2e7324856d0fe651e21519afd0368a759e827edd0e790b9b146c72b3
1263	Engineering Translation	SFT-ENG-RELIABILITY-001	Reliability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dff52b462bd4a1e798eccae2125213e8dd3341bd2aabd73935967952ad3ab060
1264	Engineering Translation	SFT-ENG-AVAILABILITY-001	Availability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4c1848211e1cce6edb226b77981fb85e148de4025bdf865727f40cb1c51f2647

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1265	Engineering Translation	SFT-ENG-FAILURE-MODE-001	Failure mode	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:099c3b3650fe1fe4e8b8b05eaa25d413a880079796df52aea10404c61c92b12d
1266	Engineering Translation	SFT-ENG-MEASUREMENT-001	Engineering measurement	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1608c9591220707596f110129e8e2067a7af9be542c55bc33c5c220b95208558
1267	Engineering Translation	SFT-ENG-CALIBRATION-001	Calibration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:deca4c00620d9e161f9255ca3d1b0baa74856c5e48964c32b1ac6d9be5fa0b9c
1268	Engineering Translation	SFT-ENG-TOLERANCE-001	Tolerance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4ba2c2f415d392ae660698be08d0c89c2567d84b3f76e47be776a8cfe5fbfd2b
1269	Engineering Translation	SFT-ENG-TEST-001	Engineering test	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5957316c89ac4ea53b2ac9314e2dfbd7f1353a162dd94743f8a08da293e69276
1270	Engineering Translation	SFT-ENG-ACCEPTANCE-TEST-001	Acceptance test	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b12c54d4d4f0836beb387a1d91fb21dec030c3d945d33f5058ea8ace80eff987
1271	Engineering Translation	SFT-ENG-MEASUREMENT-UNCERTAINTY-001	Measurement uncertainty	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:45a61872d6bea9c23660d4b649ae931ce6c00a23c36e1150c4941aa749098863
1272	Engineering Translation	SFT-ENG-ALTERNATIVE-001	Design alternative	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cc6584f3217d9d2a73f7914dd439848c0307390734cbb3420abd421e209802d4
1273	Engineering Translation	SFT-ENG-TRADEOFF-001	Engineering trade-off	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dbb7ef3b75368e8e8eddb02c80dd7397f13edf88ffacc3f8dfadbb9746eca0ac
1274	Engineering Translation	SFT-ENG-OPTIMIZATION-001	Bounded engineering optimization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:acfdc9809f00fd4f3c22a9dda5e1d3a58d4a9cacb2f8e64aa92ec87bc02db2ae
1275	Engineering Translation	SFT-ENG-MULTI-CRITERIA-001	Multi-criteria decision record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a050fbf706e359069160b952cb3a7e60442b6cb499251554190d3337ad71a55f
1276	Engineering Translation	SFT-ENG-ROBUST-DESIGN-001	Robust design	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0d0782c2fd5ddd62c75ceb0b924c32f028f03e8cbe8f120f60a02dff8a9a2016
1277	Engineering Translation	SFT-ENG-DESIGN-FREEZE-001	Design freeze and change control	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8a48d8e7705c051d36098baa026a8d7aa3fee6522b9ac2f1a0399f577127006d
1278	Engineering Translation	SFT-ENG-CONTROL-001	Control action	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d4509325413d99106b345719a0e9e1274199be38329a4de49297b3b483c914e7
1279	Engineering Translation	SFT-ENG-FEEDBACK-001	Feedback loop	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7ea37de19f49afc9b337d3e7bb3fb9ae98b109adc65faa3bf222abdf96deae3

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1280	Engineering Translation	SFT-ENG-STABILITY-001	Operational stability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e8358a5b9b7cfeb32c863a943ecd1eb31836146432fc2d1fa18a4388b749aa7
1281	Engineering Translation	SFT-ENG-OBSERVABILITY-001	Engineering observability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aac08906b9ea57a0e48ca7f1da0ea10f952914cfa6737f39d35a11e5cf552eba
1282	Engineering Translation	SFT-ENG-CONTROLLABILITY-001	Engineering controllability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:71a66eb5016f1a8d2e40ad11705b561255a2c1b9816025c64093e7eee9a5fea8
1283	Engineering Translation	SFT-ENG-FAULT-RESPONSE-001	Fault detection and response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e56021c6ea58b1e74425c159fbaf64a2350d1f04f65de97fbf31adc41a39322b
1284	Engineering Translation	SFT-ENG-HAZARD-001	Engineering hazard	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:440b90261973a86e637ed8c25a2991f332aaad7eabbbf3f16dc730162b4ab58af
1285	Engineering Translation	SFT-ENG-RISK-001	Engineering risk record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dc952d2c85cf201ff0cdef942030347e25f5842a6fdeec510e6a3a5bcd57ae21
1286	Engineering Translation	SFT-ENG-SAfETY-001	Safety case	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:68e77efa7e20c3522e74a3f7ca910bc7d0c6baeaf95b5f9f0e9a3d66f7c3fec2
1287	Engineering Translation	SFT-ENG-FAIL-SAFE-001	Fail-safe operation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9134a2270e5cf3534e086e680450f7008be5f32965c812a138e227ced6323261
1288	Engineering Translation	SFT-ENG-RESILIENCE-001	Engineering resilience	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1765e8ffe9b823048f0ec4b8bebc88653b82e9d60e1a37886db1dcb1bd16ddba
1289	Engineering Translation	SFT-ENG-DEFENCE-IN-DEPTH-001	Defence in depth	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:940904d1b6fa7f7bd79be5a92c7ee6f90ebad6d2a28812c1c102fe4a5f4cf259
1290	Engineering Translation	SFT-ENG-VERIFICATION-001	Engineering verification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:393c68fee236e0f4ec90c0e2c998e690154236e8073ff2653e1d1e3bb72799ec
1291	Engineering Translation	SFT-ENG-VALIDATION-001	Engineering validation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a8c982119f3bb9f4887036daa9f5d02ccc6b15a438b7cbf6ee9cad7d12a68f22
1292	Engineering Translation	SFT-ENG-TRACEABILITY-001	Bidirectional traceability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3c212e3f3108cf8129b362ef95fbd6a044c441058748f8be3d50fbee1da7d52
1293	Engineering Translation	SFT-ENG-REPRODUCIBILITY-001	Engineering reproducibility	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ef072bf0eab7aa89aee2642c033dfc38adb51ca0ff69321dfbba4fa62dfcd276
1294	Engineering Translation	SFT-ENG-INDEPENDENT-CHECK-001	Implementation-distinct check	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:927d1b0f871cb92e06ae180a9c1fbcf6b2fe0a024135fb487fbf41a9b082b364

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1295	Engineering Translation	SFT-ENG-E2E-001	End-to-end verification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7a4acab46f4caa1d114918096648f62bb3a71347b35d2345e1a0353f1fa67b6e
1296	Engineering Translation	SFT-ENG-CROSS-PLATFORM-001	Cross-platform behavior	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a009ad70a8663a428d971e9cfd37765344ed0cff220fb991a83801e955b66572
1297	Engineering Translation	SFT-ENG-PORTABLE-DATA-001	Portable data and record format	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a34ca2385523846818968bfb4f0faf86b00b8ca6205291cd0ea54fa82bfe2d5d
1298	Engineering Translation	SFT-ENG-ACCESSIBLE-INTERFACE-001	Accessible human interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:239e2b3219d5815d69bdd260db4406dee4d0494204d6156600f9c15a348e0ddc
1299	Engineering Translation	SFT-ENG-LIGHTWEIGHT-DEPLOYMENT-001	Lightweight deployment	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:abeb57586a62a61b447c72cac396427666a79fcf0cafaba3f0d9c3b29e72569f
1300	Engineering Translation	SFT-ENG-ONE-COMMAND-001	One-command public verification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94b593700a97b650125e459e4e75a3600542aed875d7d75d986183080e19d4c8
1301	Engineering Translation	SFT-ENG-LEARNABLE-USE-001	Learnable public operation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1e8205c6d619a0b70399e6a4c68a28a5df3997933a6ace803aa5e1f648f2a2d8
1302	Engineering Translation	SFT-ENG-CONFIGURATION-001	Configuration identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b447c208278ac2fef2764deafc27e128f167bedbb7e17211b5e6b528791847de
1303	Engineering Translation	SFT-ENG-MAINTENANCE-001	Maintenance action	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c2373257572192e77d76ba6c52eff6123c52e8a90abaad8a4e17489c9b86db65
1304	Engineering Translation	SFT-ENG-CHANGE-IMPACT-001	Change impact	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:30f84346b9018ccfa62f832883de26beb5bb7c39e9ebae32e5dc02db3daf9814
1305	Engineering Translation	SFT-ENG-DECOMMISSION-001	Decommissioning	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:45be484f60e18ab440a2f956881937e921b3d52181cac0a7d16d49f8d82d4ccf
1306	Engineering Translation	SFT-ENG-LIFECYCLE-001	Full lifecycle record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6eb58d7716a9da0c2a5b865c272db63eb090b31b56dd73890f9f5792604ea985
1307	Engineering Translation	SFT-ENG-SUSTAINABILITY-001	Bounded sustainability claim	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7a3510262c0aa4807573001ccfa38c1cf5e12f56a62977aefc85abce1cf09c42
1308	Engineering Translation	SFT-ENG-LAW-TRANSLATION-001	Scientific-law translation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4d67b3f68c70fbf95c1a9f852e93782710b7821d02bc1e309cb7503c394faa1c
1309	Engineering Translation	SFT-ENG-PERFORMANCE-001	Design performance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ed77a06d3f29985acf91f2b2f67a5d396fa0f25a56b1d2f1409b6a16de48e05a

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1310	Engineering Translation	SFT-ENG-IMPLEMENTATION-FAILURE-001	Implementation failure boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d190bd5feedd40c3b9abae516143bec91fb27ccbfa4b6f6e6ef8e902845c3c417
1311	Engineering Translation	SFT-ENG-PROTOTYPE-001	Prototype evidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8f19503903dc28e39c772205296e92fb67a25ae0f4d3fd2de378e24d8e69bf5f
1312	Engineering Translation	SFT-ENG-SIMULATION-001	Engineering simulation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c88333739def6fda69b8d90fc4337eaab45bc453ac7b5fe67f24d9dc8fe4e0a0
1313	Engineering Translation	SFT-ENG-DEMONSTRATION-001	Engineering demonstration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dec0c9f70709d6b4c1a799f9d27e8624d4a67af314594331ef174990aadbf1f47
1314	Engineering Translation	SFT-ENG-ANOMALY-001	Engineering anomaly	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1f0ad93282eef64f4d4fdd8e69f88c5fc50a8c614880548c2fc5e38bec522a54
1315	Engineering Translation	SFT-ENG-SCIENCE-RETURN-001	Return of anomaly to science	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:34e0866bd57b15b2fb248cfc397b9ab24eaf242b57014034536f2d222be3a00a
1316	Engineering Translation	SFT-ENG-SOFTWARE-HANDOFF-001	Software and computation handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:45f670866891a5c3228a55768fae7ef15cf44776aae1e92a7f4a1fee5b4e7a
1317	Engineering Translation	SFT-ENG-PHYSICAL-HANDOFF-001	Physical sciences handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3c5e2bcfc0353b573de31a1f50e5b60e1983a348e9712b6ada307d889c5bbcf8
1318	Engineering Translation	SFT-ENG-BIO-MED-SOCIAL-HANDOFF-001	Biological, medical and social handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3feb705c20067692e2e8d4166858d32ca429a9144902b0fb43bedbce3418cacf
1319	Engineering Translation	SFT-ENG-V4-HANDOFF-001	SFT-derived self-hosted V4 handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aedcf633fb62d3dc4fd8a18c5fbfed692746a0d2f57fd8b41ed2d87244544ded
1320	Chemistry	SFT-CHEM-PERICYCLIC-REACTION-FAMILY-012	Exact cyclic-transition and relative-orientation law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:608a48d3e1be90c6dbafdc6fd68fa731841c9c7b8a69cca351b05e54689a871f
1321	Chemistry	SFT-CHEM-RADICAL-REACTION-NETWORK-013	Exact initiation, propagation and termination network law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:97ec902a0830f6578f2ca981be588a9906b4fffd0d7eaa7a277e29c778f790860
1322	Chemistry	SFT-CHEM-SELECTIVITY-COMPLETE-DISTRIBUTION-014	Fold chemo-, regio- and stereoselectivity with complete product distributions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:765110af963aac06676adda2ffc9873d77d02848f7ae3c0c7c4095aa8d4ed000
1323	Chemistry	SFT-CHEM-PROTECTING-GROUP-REVERSIBLE-STATE-015	Fold protecting-group reversible state and exact restoration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f78de0d43da42ba65f04328fb2b46e04bda108b69952c52500c1736da1b0681
1324	Chemistry	SFT-CHEM-RETROSYNTHETIC-DECOMPOSITION-RECONSTRUCTION-016	Fold exhaustive retrosynthetic decomposition and exact forward reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2a460669175e8ea2c8b4c29de14025a2f470ad886c2534d9c71004c8ca9980ed

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1325	Chemistry	SFT-CHEM-HALF-REACTION-IDENTITY-ORIENTATION-001	Fold half-reaction identity and held transfer orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b62e7c5b5b4ea98b691734a7099344965bc555c1b0f8e2d4f9ec5e6433a2cd1a
1326	Chemistry	SFT-CHEM-ELECTRODE-POTENTIAL-CHEMICAL-RELATION-002	Fold electrode-potential chemical relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8df16ac7856b666f091b5e72f6f00e7291e042d4043619f34d8a2b29b6e8b675
1327	Chemistry	SFT-CHEM-CELL-POTENTIAL-COMPOSITION-003	Fold cell-potential composition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:35ddc2ae557f0174e49a4ef763c668e56d90dd93c6574514ee8f398a0a72fbae
1328	Chemistry	SFT-CHEM-CONCENTRATION-DEPENDENT-POTENTIAL-004	Fold concentration-dependent potential law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1eea2b72f83d08bbffdc8291adb1c830764410ba065d8d056ca12a8bdf5bfce
1329	Chemistry	SFT-CHEM-ELECTROCHEMICAL-WORK-REACTION-DIRECTION-005	Fold electrochemical work and reaction-direction correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f50794cbe0530156b52fb4cbaccf18b7d9d570dd8d4ad254bd4a325cba757057
1330	Chemistry	SFT-CHEM-ELECTROLYSIS-PRODUCT-AMOUNT-006	Fold electrolysis and product-amount law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:95fcb8396d3792c917e768e341f501e27ab8e3392a38f6fedf6e16e020eea2df
1331	Chemistry	SFT-CHEM-IONIC-CONDUCTIVITY-RELATION-007	Fold ionic conductivity relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e49c9f8ca3137d6b1618bc56d96e16e55354569f0a6dee3f404af2dec1b670bd
1332	Chemistry	SFT-CHEM-IONIC-MOBILITY-TRANSFERENCE-008	Fold ionic mobility and transference law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:df5f677264d69df93f29920c45789f8264a359214049c983d7a137b763ea7682
1333	Chemistry	SFT-CHEM-ELECTRODE-REACTION-RATE-009	Fold electrode reaction-rate law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:85b5ca8c3be8347f4175c93f9804dae6a843c619ddaaa3900d170b3b7c570c0
1334	Chemistry	SFT-CHEM-OVERPOTENTIAL-POLARIZATION-010	Fold overpotential and polarization law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5ffa4d639d74352f13b1a56b0464c7fe5ed2aa3f27c4f4091db16346e33fc1b4
1335	Chemistry	SFT-CHEM-DOUBLE-LAYER-INTERFACIAL-CHARGE-011	Fold double-layer interfacial-charge organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:113855e15da76e59f3f7b907613c113c661f80c6d1fa5051a234e893998cda18
1336	Chemistry	SFT-CHEM-CORROSION-REACTION-NETWORK-012	Fold coupled corrosion reaction-network law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa576b702605524bb8518fc3148afd07d094598abddfe116f24c045459a0a7ea
1337	Chemistry	SFT-CHEM-ELECTROCHEMICAL-STORAGE-HANDOFF-013	Fold electrochemical storage ownership and handoff law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ed5079835f482468f5b5dc7eab3d87ecc8412b18b99374216e300a1dcfb02692
1338	Chemistry	SFT-CHEM-NUCLEAR-CHEMICAL-CARRIER-001	Fold nuclide chemical-carrier law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:86fdd019ae04bf7dc01fb957f5c136b658218aaecab526898530a2461965824f
1339	Chemistry	SFT-CHEM-RADIOACTIVE-CHEMICAL-TRANSFORMATION-002	Fold radioactive chemical-transformation network	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d9f532123edcae0e686f4a4ccfbdacc43605f56dfddc3719b6f603c298d5687f

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1340	Chemistry	SFT-CHEM-ACTIVITY-AMOUNT-TIME-003	Fold activity-amount-time law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e332f2d356c8111e51ad00d890645a4fa7b8be0e149d4f5f9d707e71d86d5e0b
1341	Chemistry	SFT-CHEM-RADIOACTIVE-BRANCHING-CHEMICAL-YIELD-004	Fold radioactive branching chemical-yield law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:85705f51398e1561e5be35cf9f9806fa5878679dac6c5abaaf16435f2fe3ee6c
1342	Chemistry	SFT-CHEM-RADIOCHEMICAL-EQUILIBRIUM-005	Fold transient and secular radiochemical-equilibrium law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9639309d02ccbeaa77794ad8e25ec0a28d0714ec5ce3e0a1a0a27c2c7a695cdb7
1343	Chemistry	SFT-CHEM-ISOTOPE-EXCHANGE-006	Fold isotope-exchange reaction law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a94b6eb90aef2a3a5d04dd4c3135c1f43fd086d6c826078c91a09f2dcee42a5e
1344	Chemistry	SFT-CHEM-EQUILIBRIUM-ISOTOPE-FRACTIONATION-007	Fold equilibrium isotope-fractionation law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0ba2c2105037fa8f4bf1adba65b7bb8322c063726a0d379ebac9121a2ee2e44c
1345	Chemistry	SFT-CHEM-KINETIC-ISOTOPE-FRACTIONATION-008	Fold kinetic isotope-fractionation law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f9b52ef14fc8be4e08d25067f9632d8048bc99c10416aa2e38d980b637b851f6
1346	Chemistry	SFT-CHEM-RADIOTRACER-CUSTODY-INFERENCE-009	Fold radiotracer custody and inference law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d5a8c7ac0e12d9c4213e247e3cf32ecc25b7c62fa5205d7b634d2c8b3adc5fe9
1347	Chemistry	SFT-CHEM-RADIOCHEMICAL-SEPARATION-DECONTAMINATION-010	Fold radiochemical separation and decontamination law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eb846cb0c2554f3bd873aa45f902039b1b71aeb7d79c3de52825322b67d01873
1348	Chemistry	SFT-CHEM-FISSION-PRODUCT-CHEMICAL-DISTRIBUTION-011	Fold fission-product chemical-distribution law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d3f311c2d0ccb2b91ce59d58f202b1c4dd1252a58315f66fad7fc7b0b51b1376
1349	Chemistry	SFT-CHEM-RADIATION-CHEMISTRY-REACTION-NETWORK-012	Fold radiation-chemistry reaction-network law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ad00dad43f0a51d743cca24641be1ecab903298cc0e95290f78091432c2a8088
1350	Chemistry	SFT-CHEM-ANALYTICAL-ACCURACY-TRUENESS-001	Fold analytical accuracy and trueness law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e96550c2bcb1a718b8bcffbac6e1f45c9a920e926701e2132cf1637123820ccf3
1351	Chemistry	SFT-CHEM-ANALYTICAL-PRECISION-REPEATABILITY-002	Fold analytical precision and repeatability law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:189762298384f7f436f22c3a9f6ef858a6333ae7bcd3d4c498ce01b5f713812b
1352	Chemistry	SFT-CHEM-ANALYTICAL-SENSITIVITY-003	Fold analytical sensitivity law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:157e7368c283e846a5dbfee9a656fdea0db746654f0037104ba9985a2647fca2
1353	Chemistry	SFT-CHEM-ANALYTICAL-DETECTION-QUANTIFICATION-004	Fold detection and quantification boundary law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d2d81bc678902abb4709f33583789ec28190133e7c3c5e1c6a91c87ff911ed68
1354	Chemistry	SFT-CHEM-ANALYTICAL-SELECTIVITY-INTERFERENCE-005	Fold analytical selectivity and interference-matrix law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:68ab2e8497202157e6cbd1ab8b8303cd46671384123312c65c4595230f1c28bf

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1355	Chemistry	SFT-CHEM-NMR-CHEMICAL-SHIFT-006	Fold NMR chemical-shift relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:07525c04e8cfea4d1d1044356d602447ededeeef6335f2949216987d8c1482677
1356	Chemistry	SFT-CHEM-NMR-SPIN-COUPLING-007	Fold NMR scalar spin-coupling relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e7f9cf5d8813ce2c7e445d785d8914f60e566db7e2c787649bb2cd94bcf7c366
1357	Chemistry	SFT-CHEM-NMR-RELAXATION-EXCHANGE-008	Fold NMR relaxation and exchange law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cf3149e0d7e99648e36cdf8db7ee0d35e3acd938fbbcc3cc9578bb015da018fe5
1358	Chemistry	SFT-CHEM-RAMAN-TRANSITION-INTENSITY-009	Fold Raman transition and intensity law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3a31c40ef538e5177cd4f20be15f0ed5fccee352a787ae53854be9d84301568c
1359	Chemistry	SFT-CHEM-FLUORESCENCE-YIELD-LIFETIME-010	Fold fluorescence yield and lifetime law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:db992b28c2c1144acf643fcf9c1cbc7fea7c72b6ad92c99cb01c0f4430772c63
1360	Chemistry	SFT-CHEM-PHOSPHORESCENCE-INTERSYSTEM-011	Fold phosphorescence intersystem law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7402f1ff97095dd5342c8fec51864a26e957882bcb2ed377f3991353955af400
1361	Chemistry	SFT-CHEM-INFRARED-LINE-INTENSITY-012	Fold infrared line-position and intensity law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f02f6937e976f9e1f538adeba977dd325215403f87020176180fbf5920f1b426
1362	Chemistry	SFT-CHEM-UVVISIBLE-LINE-INTENSITY-013	Fold UV-visible line-position and intensity law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bdba363b0775c4185cef2caa2f190c6b8ecd358a1bd7fbcc186fea28841dd724
1363	Chemistry	SFT-CHEM-MASS-ISOTOPE-FRAGMENT-VECTOR-014	Fold mass, isotope and fragmentation spectrum law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ba5cd4595e4c3ceecfe6ea9a8c23330df8a8259df70711bc22e290122edc64e9
1364	Chemistry	SFT-CHEM-ROTATIONAL-SPECTRUM-LINE-015	Fold rotational spectrum line law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e3079d4d40dad58b944ef2b5b17ef06bff8285d7d2a102358fb2019b92ef4942
1365	Chemistry	SFT-CHEM-XRAY-DIFFRACTION-STRUCTURE-016	Fold X-ray diffraction chemical-structure law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5ee17101561f8ab2362d74ff43b67bbfd64664300a7e8c2723462537ce64b621
1366	Chemistry	SFT-CHEM-ELECTRON-NEUTRON-DIFFRACTION-017	Fold electron/neutron diffraction correspondence law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cf768f015cdb8251479ab5617c55d02a8637a5be053c313f15f9024fbd3f4553
1367	Chemistry	SFT-CHEM-CHROMATOGRAPHIC-RETENTION-RESOLUTION-018	Fold chromatographic retention and resolution law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:288a23013dd25c34ddfffb33fa425241df15bc5f904557f41d8d9de548a56a550
1368	Chemistry	SFT-CHEM-ELECTROPHORETIC-MOBILITY-SEPARATION-019	Fold electrophoretic mobility and separation law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:84321bf139917e3acddc89ace7c14511e4ab04364b65bc03c22a92b7b07c61e1
1369	Chemistry	SFT-CHEM-ELECTROANALYTICAL-RESPONSE-020	Fold electroanalytical response law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0a47e880278b7797759fd54efe124803eebb0d4b2cddb69e752b4134f00f21f

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1370	Chemistry	SFT-CHEM-MULTIMODAL-MOLECULAR-IDENTITY-021	Fold multimodal molecular identity reconstruction law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ce1108185e2b7ad7e02149909424887e4173c6c4c8068043a198a7ccdd87f0b2
1371	Chemistry	SFT-CHEM-ANALYTICAL-PERFORMANCE-BUDGET-022	Fold complete analytical performance and uncertainty budget	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bdb6476cc6078def755b0a43e42de3ff7b6d18cf4c50783057ef2f9c9ad878de
1372	Chemistry	SFT-CHEM-CANONICAL-MOLECULAR-GRAPH-ENCODING-001	Fold canonical molecular-graph encoding law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9358e99b84fdf4738c170a8167e30bf44becc2ec010ed5ba04755efff83256b51
1373	Chemistry	SFT-CHEM-MOLECULAR-GRAPH-ISOMORPHISM-002	Fold molecular-graph isomorphism and identity law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e1895597e606243629700724394c81c68ad21c8f4c28595236293cf3267c2fc3
1374	Chemistry	SFT-CHEM-CHEMICAL-SUBSTRUCTURE-RELATION-003	Fold chemical substructure law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:81874d352c9c124722c3902bfff20deb33032b47f643d7bb26065a2b23e6638f1
1375	Chemistry	SFT-CHEM-CONSTITUTIONAL-ISOMER-ENUMERATION-004	Fold constitutional-isomer enumeration law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9c7ef55ae880cd0da55070786eb517262dc6d84da61d57180eb92088e7395705
1376	Chemistry	SFT-CHEM-STEREoisomer-ENUMERATION-005	Fold stereoisomer enumeration law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:749cbd4b1f5b0ef10a5af2ad36f7f4751ad217458b80c01ad6e2eda40f6340ab
1377	Chemistry	SFT-CHEM-CONFORMER-ENUMERATION-006	Fold conformer enumeration law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3a8960886113dbba8b92ba068a31a01faa620f7e1fa3f66b125dd30a0aa9644b
1378	Chemistry	SFT-CHEM-REACTION-GRAPH-GENERATION-007	Fold reaction-graph generation law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0a51b0dedcd06539bd1e935aa395bc2a443cc15a4c8af99d033c79c7be20b0b
1379	Chemistry	SFT-CHEM-ATOM-MAPPING-REACTION-BALANCE-008	Fold atom-mapping and reaction-balance law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:377ac33f481e02a143540d265734e6754944095bb986f0577191b755fd46fc7
1380	Chemistry	SFT-CHEM-MECHANISM-SEARCH-PROOF-TRACE-009	Fold mechanism-search and proof-trace law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f9d9df0b6c21854dbda0889cd90926a3b88bf180d49dbf3677e3b3f202d75d9d
1381	Chemistry	SFT-CHEM-EXACT-MOLECULAR-SIMILARITY-DISTINCTION-010	Fold exact molecular similarity and distinction law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:97898c7179d8b2b877d76ac80c08c81960fc0b23d454a616632cc1b67c1d3648
1382	Chemistry	SFT-CHEM-DATABASE-IDENTITY-PROVENANCE-011	Fold chemical database identity and provenance law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:287ab043a520733650c816cff1436f8890349b10ff721e8990a102a6c7cd40e3
1383	Chemistry	SFT-CHEM-SYMBOLIC-PROPERTY-EVALUATION-012	Fold symbolic chemical-property evaluation law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:763febb44d5029ce32b130fce8f17669da1805dbbbe533ff892b33e61234dd83
1384	Chemistry	SFT-CHEM-PREDICTION-UNCERTAINTY-APPLICABILITY-013	Fold chemical prediction uncertainty and applicability law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:188e0bf716fd6ca34b210699b19054bb71fddd051f7efd37e473d1b198f9c69e

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1385	Chemistry	SFT-CHEM-CLASSICAL-QUANTUM-ALGORITHM-CORRESPONDENCE-014	Fold classical and quantum chemical algorithm correspondence law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:05f6e8c3de78a3172c1a3cd27412b0d93a9849171572c1a5a633dc7ec3f4dd19
1386	Chemistry	SFT-CHEM-DEGREE-OF-POLYMERIZATION-001	Fold degree-of-polymerization law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:065acfb40cb3eb8e7fb3598c77311df64f5c191fab1c24194e1bf1e960225041
1387	Chemistry	SFT-CHEM-NUMBER-AVERAGE-MOLECULAR-SIZE-002	Fold number-average molecular-size law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2074e8e1207661559765c1d88e51a4665a4b76b8536020332a437b08988ac534
1388	Chemistry	SFT-CHEM-MASS-WEIGHTED-MOLECULAR-SIZE-003	Fold mass-weighted molecular-size law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f176593b59c0cf7b53b5ef492515908a9ac47ac2b26f712874bd3c7d7b0c6151
1389	Chemistry	SFT-CHEM-POLYMER-DISPERSITY-004	Fold polymer dispersity law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d32e9721f4366db655e96e401760ab71b4c49e6ce0bba8717c7073b6a7424203
1390	Chemistry	SFT-CHEM-CHAIN-GROWTH-POLYMERIZATION-NETWORK-005	Fold chain-growth polymerization-network law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b480d01f119976bedf205450bca2e715f2dfffb93d68068dd43ac228f451a8c9d
1391	Chemistry	SFT-CHEM-STEP-GROWTH-POLYMERIZATION-NETWORK-006	Fold step-growth polymerization-network law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0278e5e0e9f6a83640a7863751d0fd0f205a8ed657ca3e526fafdb2b5a2d2d66
1392	Chemistry	SFT-CHEM-COPOLYMER-SEQUENCE-COMPOSITION-007	Fold copolymer sequence-composition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d6bb8a00838a12fccec6bbe408e5ff8d71ff540a6548aa2ad478c7cc405e9d08c
1393	Chemistry	SFT-CHEM-POLYMER-ARCHITECTURE-008	Fold branched, star and network architecture law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4047581cfcc8c92646c81323ff2ae72697c673a3a745b681fc38a18b4e821318
1394	Chemistry	SFT-CHEM-CROSSLINK-GELATION-BOUNDARY-009	Fold crosslink and finite-gelation boundary law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f18e59a809621d8da060d00487821d16b807aacd236d6282cfdcab7da5c933b7
1395	Chemistry	SFT-CHEM-POLYMER-CONFORMATION-SIZE-010	Fold polymer conformation and size law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ebdb8895ec6882a4812462c711c1a4c33df8badafa4a201ae0046188f3552bbe
1396	Chemistry	SFT-CHEM-POLYMER-PHASE-TRANSITION-011	Fold polymer phase and transition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ee2dba715ca2a3b640a1e2c95156ff25817ad663a1a5f5e243760e2882f5f701
1397	Chemistry	SFT-CHEM-POLYMER-DEGRADATION-NETWORK-012	Fold polymer degradation and depolymerization-network law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bcc3b5f0f762be039c57e36ca19b4b76640c911843e2217d0b54ff303ca31f09
1398	Chemistry	SFT-CHEM-POLYMER-MATERIALS-HANDOFF-013	Fold polymer Chemistry-to-Materials handoff law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a69103e6715074cfdb9843ef831a9ad775fc5f19039bb7b3a761e052361a1b07
1399	Physics	SFT-PHYS-UNIFIED-CONSTANTS-OBJECT-077	The Unified Constants Object	depth_independent	independently_replicated	4,096	4	sha256:e6112bcfa663b3d24727bd03688491f55fdb5fc32359542b92acee7e551fbac8

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1400	Physics	SFT-PHYS-TESLA-BOUNDED-CAVITY-078	Exact bounded-cavity recurrence and discrete mode law	depth_independent	independently_replicated	256	4	sha256:52164b45a1cd9eac61d3fd2850d10dd144a0fc6d2699520bd10d166325f3126e
1401	Physics	SFT-PHYS-TESLA-ODD-QUARTER-WAVE-079	Exact odd-quarter-wave and odd-harmonic law	depth_independent	independently_replicated	256	4	sha256:3f1341a5b993e042ffcea6cc7220100e8ee8ab98c59aaafc3694426381e30c83c
1402	Physics	SFT-PHYS-TESLA-LONGITUDINAL-TRANSVERSE-080	Common recurrence with distinct longitudinal and transverse modes	depth_independent	independently_replicated	256	4	sha256:3c7be179aa4e991f5bf66fcb926a51b3660f9d11e9c70216ea0cf4e75fe91a09
1403	Physics	SFT-PHYS-TESLA-RESONANT-TRANSFER-081	Exact resonant connected-path transfer and complete energy ledger	depth_independent	independently_replicated	256	4	sha256:33dbb3b402878cfd1553f2eaf0d25d1df3194bd29cfe390d43e7a13facde069c
1404	Physics	SFT-PHYS-VALIDATION-TESLA-RESONANCE-FAMILY-082	Complete measured Tesla-resonance family and power-boundary comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:270786783cb68b4737d35faf5385079767c22036b03dc8f83965bb1eb2467b92
1405	Physics	SFT-PHYS-VACUUM-LOCAL-RESONANT-DRIVE-083	Exact local resonant vacuum-state drive channel	depth_independent	independently_replicated	256	4	sha256:3ffdf3606dd8f27146c3e0bb203c3758e85780049e8c60e074841d567bfc1e0a
1406	Physics	SFT-PHYS-VACUUM-INERTIA-COVARIATION-084	Exact driven vacuum-to-inertia co-variation law	depth_independent	independently_replicated	256	4	sha256:a956088e2454d1594a7a04ca4f6050816bca30f59a32aae52dcafd1cb9617474
1407	Physics	SFT-PHYS-VACUUM-INERTIA-POSITIVE-FLUOR-085	Finite-depth positive lower bound on driven vacuum and inertia	depth_independent	independently_replicated	256	4	sha256:0a7552d2128afa000bc701c2199f3e7efdc1c5091bd09d612b3826c900c6e23
1408	Physics	SFT-PHYS-VACUUM-INERTIA-COMPLETE-LEDGER-086	Complete vacuum-drive, inertial-response and restoration ledger	depth_independent	independently_replicated	256	4	sha256:3fc114314e2065848eca08536e13323688be410b9693a064921e2c8c62cb8b57
1409	Physics	SFT-PHYS-VALIDATION-VACUUM-INERTIA-DRIVE-FAMILY-087	Complete measured vacuum/inertia-drive correspondence and apparatus boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ae878e3fe8666a5471b9d19a2f67ab235b05f2187f3e9934523cc0e1661f4ca1
1410	Physics	SFT-PHYS-PENTA-COMPLETE-PHENOTYPE-088	Complete penta sector phenotype	depth_independent	independently_replicated	256	4	sha256:11362854f921cda3962cfed7578380b32cca2663c85de8528d4aa17895fe20c4
1411	Physics	SFT-PHYS-HEPTA-COMPLETE-PHENOTYPE-089	Complete hepta sector phenotype	depth_independent	independently_replicated	256	4	sha256:57d1a7b380211d15d035e2853e74fa81dc7cd150f4acade9fa55aald8efb7a1a
1412	Physics	SFT-PHYS-PENTA-BETA-SLOPE-090	Independent sector-5 beta-slope law	depth_independent	independently_replicated	256	4	sha256:df6eeb2137767de6ae27e4dde9a6f4655629411248f30136c5cbdf82c3df9198
1413	Physics	SFT-PHYS-HEPTA-BETA-SLOPE-091	Independent sector-7 beta-slope law	depth_independent	independently_replicated	256	4	sha256:bcdca84f9778bcf437fc0fe253f13f2ab4672db1414d5e1ccc78c7cc38607e
1414	Physics	SFT-PHYS-CATEGORY-CLEAN-PARTICLE-CENSUS-092	Category-clean complete particle census	depth_independent	independently_replicated	256	4	sha256:3d96ec5a6eb16daa8bc8971bfdb34d876fa76fb082da1737583c7eb209ec1e06

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1415	Physics	SFT-PHYS-NO-EXTRA-SECTOR-PARTICLE-BOUNDARY-093	No-extra-sector and outside-particle falsification boundary	depth_independent	independently_replicated	256	4	sha256:aee65ca87af0ec74235a3e62c475d6254301ef623dbbcbef5c1582aab0bb9cd9
1416	Physics	SFT-PHYS-SMITHION-INTERACTION-SEARCH-SIGNATURES-094	Exact Smithion interaction and search-signature law	depth_independent	independently_replicated	256	4	sha256:5e1ffcf9a06c962f4d8301714cf57d5a6e306884cf9c4c4fcfed4b43365c101b7
1417	Physics	SFT-PHYS-VALIDATION-NEW-SECTOR-COMplete-FAMILY-095	Complete measured new-sector, particle-census and Smithion search record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f13335a3bfdd7a395689ff113676a6c3287d90c7b4c8e49eac53730d5c5f2bd
1418	Classical Computation	SFT-COMP-CPLX-CONVENTIONAL-TRANSLATION-003	Conditional conventional decision-family transport law	depth_independent	independently_replicated	256	4	sha256:2b389d7cdae40491b9aeb9808b617aa660969daf44b047b15d29c481f0ac5419
1419	Chemistry	SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001	Complete Smithium synthesis-conservation grammar	depth_independent	independently_replicated	256	4	sha256:c26635123b4b59798b649c11d3980194cf216242ebbaceb7bec83bdbebcca12e
1420	Chemistry	SFT-CHEM-SMITHIUM-DECAY-CHANNEL-LEDGER-001	Complete Smithium decay-channel coordinate ledger	depth_independent	independently_replicated	256	4	sha256:b274737acd5b5f638d7cecdf4f7f79c42b46a50a11d130ff40381d56586ac90a
1421	Chemistry	SFT-CHEM-SMITHIUM-LIFETIME-BOUNDARY-001	Smithium lifetime identifiability and mandatory-halt boundary	depth_independent	independently_replicated	256	4	sha256:fe1d8b934a1edd046d578e632368981fabbb92944aa4fa7bc48668190af091377
1422	Chemistry	SFT-CHEM-SMITHIUM-ION-OXIDATION-LADDER-001	Exact Smithium positive-ion and oxidation ladder	depth_independent	independently_replicated	256	4	sha256:477fbfde3f3abfbc9e30d51fe49fef3e6cccd92e77b99a4bfbc05bd27b023a61
1423	Chemistry	SFT-CHEM-SMITHIUM-SPECTROSCOPIC-CLASSSES-001	Smithium spectroscopic transition-class law	depth_independent	independently_replicated	256	4	sha256:11ce87ed6b65a993ec7b1923871cca267490ef000025521848d382a2977ef9ab
1424	Chemistry	SFT-CHEM-SMITHIUM-CHEMICAL-SEPARATION-001	Complete Smithium chemical-state distinction and separation boundary	depth_independent	independently_replicated	256	4	sha256:5f1e5ccee4cc5dbf7fc3a11c37abc9979770ffc441ee45b4e766ed60f3b16a4c
1425	Chemistry	SFT-CHEM-SMITHIUM-JOINT-DETECTION-001	Complete Smithium joint-identification protocol	depth_independent	independently_replicated	256	4	sha256:4c17d9f41d1cb558e31419f8d036a8fcf05ee7f87e06372958d5b341ed78487b
1426	Chemistry	SFT-CHEM-VALIDATION-SMITHIUM-COMplete-FAMILY-001	Complete post-seal Smithium external-status record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b5e611832ae9f3fe36316d0168851e5783e504dd3bf4d42c36b399bb9ba1dcb5
1427	Biology and Life Sciences	SFT-BIO-ORIGIN-AUTOCATALYTIC-IGNITION-002	Exact autocatalytic ignition boundary	depth_independent	independently_replicated	256	4	sha256:e07997c9f3ec0c8f2be6b29934c61d42d734ac7aaddc78f6f79881cd54a0d4e1
1428	Biology and Life Sciences	SFT-BIO-HOMOCHIRAL-AMPLIFICATION-002	Finite homochiral selection and amplification	depth_independent	independently_replicated	256	4	sha256:5b241bd519c0c0788d904e7b653e9ae7ef1bb2662f85024a6968396adc3f98c4
1429	Biology and Life Sciences	SFT-BIO-SOMATIC-GERMLINE-ORBIT-SPLIT-002	Somatic transient and germ-line recurrence split	depth_independent	independently_replicated	256	4	sha256:d77cbf85bd158f6a70faa860256c865845143a9cedff2b3ab5843f878d6e50e2

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1430	Biology and Life Sciences	SFT-BIO-NEURAL-HALF-ONE-THRESHOLD-002	Half-One excitable event threshold	depth_independent	independently_replicated	256	4	sha256:72d8c1967e3d995b261efbcabfb48daa3d00aac52ad26fe834303629b0d5becb
1431	Biology and Life Sciences	SFT-BIO-DIFFERENTIATION-LOSS-CANCER-002	Differentiation-loss cancer mechanism	depth_independent	independently_replicated	256	4	sha256:1dd834f90e219fa72857887fda5d9d67b5face8ef25c857c5791333e7db6b70a
1432	Biology and Life Sciences	SFT-BIO-BOUNDED-ORBIT-ECOSYSTEM-002	Exact bounded-orbit ecosystem stability	depth_independent	independently_replicated	256	4	sha256:bf616e9c330e60b71bd316b9ced8a5111ef41606c52d2ec71a31d4d7ee38bb17
1433	Biology and Life Sciences	SFT-BIO-VALIDATION-PRIOR-MECHANISMS-COMPLETE-FAMILY-002	Post-seal Biology prior-mechanism family comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0feadaeed5e67f849fb8552c20397cddd8de3abc4eccff11766545a9bac3a360
1434	Medicine and Health Sciences	SFT-MED-PLACEBO-EXPECTATION-FIBRE-002	Exact expectation/body Fold fibre	depth_independent	independently_replicated	256	4	sha256:84338832d2057f854707262e2c59af07a41c65451c3d823567596d98185ed555
1435	Medicine and Health Sciences	SFT-MED-PLACEBO-AVAILABLE-STATE-BOUNDARY-002	Available-state placebo/nocebo boundary	depth_independent	independently_replicated	256	4	sha256:183d1cc98d5d834ec6b5b44b90c76b5dfed458ed591da348166d868824e8a594
1436	Medicine and Health Sciences	SFT-MED-PLACEBO-OBJECTIVE-REPORT-SEPARATION-002	Objective physiology and report separation	depth_independent	independently_replicated	256	4	sha256:b80dc8fea8258a86a5017270ce9fd9205354705a9ba4faa9295fd2f67929910c
1437	Medicine and Health Sciences	SFT-MED-VALIDATION-PLACEBO-NOCEBO-COMPLETE-FAMILY-002	Complete post-seal placebo/nocebo physiological comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d871692e08a433860c43f9a808efaf3af427e39ca0c49ce0064c08fa1d391f3
1438	Consciousness and Cognitive Science	SFT-CONSC-SYNAESTHESIA-DIRECTIONAL-LOCK-002	Directional synaesthetic Fold lock	depth_independent	independently_replicated	256	4	sha256:827a629893b47f98576e3dac63d66528a3838db90389f274c042fe9ed186a01d
1439	Consciousness and Cognitive Science	SFT-CONSC-NONORDINARY-THREE-QUALITY-ORBIT-002	Three-quality held orbit and nonordinary boundary	depth_independent	independently_replicated	256	4	sha256:8ccf5643bbf358627986218d2b9291cd2dde37f6ec2cc938bde54fe85c72aa7
1440	Consciousness and Cognitive Science	SFT-CONSC-SLEEP-DREAM-PERIOD-TWO-002	Exact sleep-dream period-two Fold cycle	depth_independent	independently_replicated	256	4	sha256:96e900518c135df090379516e5793e174c96a8fab7e59d6030254c965fbee1f51
1441	Consciousness and Cognitive Science	SFT-CONSC-CESSATION-LOCK-ANCHOR-002	Cessation lock and invariant anchor	depth_independent	independently_replicated	256	4	sha256:8662c2274a82541f6c313b81263e388801e591ebc72f93bc0bdf511fe15c2d7
1442	Consciousness and Cognitive Science	SFT-CONSC-VALIDATION-NONORDINARY-COMPLETE-FAMILY-002	Complete post-seal Consciousness return comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c5a0e4799ef79ce5ab7a4fdb9c975f5f3f3988c979804f79a090eb8a7b1ac2eeb
1443	Earth and Environmental Sciences	SFT-EARTH-TIPPING-FOLD-LOCK-002	Normalized Earth tipping Fold lock	depth_independent	independently_replicated	256	4	sha256:3a8ebb9a7cc49c18c5615b8744e02128bcd6414c2300362b1f8458e2b78eec9a
1444	Astronomy and Cosmology	SFT-ASTRO-SOLAR-RADIO-UNIT-RELEASE-002	Solar/radio unit magnitude-frequency release	depth_independent	independently_replicated	256	4	sha256:edf91022ff2d28e6865c002825eb0e508f2215b912e2bd66b5b161cbe0272958

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1445	Astronomy and Cosmology	SFT-ASTRO-ATOMIC-BURST-COMPLETION-002	Atomic Fold burst completion	depth_independent	independently_replicated	256	4	sha256:caedea69c14b9220fea53c58cefabbac44224e2dbceb634737c9cb072b521b66
1446	Astronomy and Cosmology	SFT-ASTRO-PLANETARY-BINARY-LADDER-002	Depth-seven planetary binary ladder	depth_independent	independently_replicated	256	4	sha256:316542613f55d9f07e05141976cc889c47c787a61886f0aee26f6612db844a4e
1447	Astronomy and Cosmology	SFT-ASTRO-LITHIUM-SEVEN-ONE-FOLD-DEPLETION-002	One-Fold lithium-seven depletion mechanism	depth_independent	independently_replicated	256	4	sha256:f295654ffdde16ad962d59c6af7ed9808f82b902a78f1c55785e01acea42ccb4
1448	Astronomy and Cosmology	SFT-ASTRO-VALIDATION-PRIOR-COMPLETE-FAMILY-002	Complete post-seal Earth/Astronomy prior comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:22c0111a58263f844a79e781e80a54bc4f52412aa86f3519b2fa9ba83b344b27
1449	Social prior-return continuation	SFT-SOCIAL-PREFERENTIAL-FLOW-INEQUALITY-002	Exact preferential-flow inequality	depth_independent	independently_replicated	256	4	sha256:04a60e9ee095d1a90f1a9802b18ab38a4a98ee3ef7c5072d87db95aeed474896
1450	Social prior-return continuation	SFT-SOCIAL-CONSENSUS-POLARIZATION-LOCK-002	Consensus/polarization Fold lock	depth_independent	independently_replicated	256	4	sha256:02d9e32e8d89c68b88668db175e2d8de4cdacba0f3adaef2a40b3fe2b0d3ee56
1451	Social prior-return continuation	SFT-SOCIAL-DISSIPATIVE-PERIOD-TWO-CYCLE-002	Dissipative social period-two cycle	depth_independent	independently_replicated	256	4	sha256:fcfeb4b70b7ccc725e116368b85180508377d5cad166860b6a1c72d88a666d07
1452	Social prior-return continuation	SFT-SOCIAL-VALIDATION-EXACT-COMPLETE-FAMILY-002	Complete post-seal exact Social comparison	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6ee94ca2079585d1d8cb8eb995bed82c5a4206bd2ef3d2eca1e2c61fc838744c
1453	Engineering Translation	SFT-ENG-TESLA-RESONANT-TRANSFER-PROTOCOL-002	Tesla resonant-transfer engineering protocol	depth_independent	independently_replicated	256	4	sha256:d5cc4adc3231b51af5e107ea7de0cb32d343245955a6ce9d2390d8af532163be
1454	Engineering Translation	SFT-ENG-VACUUM-INERTIA-RESPONSE-PROTOCOL-002	Driven vacuum/inertial-response engineering protocol	depth_independent	independently_replicated	256	4	sha256:afe48fe26787fd9d8ac066b5568ce0cb8799828c690818b8bf4445f591abale4
1455	Engineering Translation	SFT-ENG-VACUUM-BEAT-RESTORATION-PROTOCOL-002	Vacuum-beat transfer and restoration protocol	depth_independent	independently_replicated	256	4	sha256:b9889a36c4ed3f9192d7b832dc7d831dd6d0fced428632192721d1d433223769
1456	Engineering Translation	SFT-ENG-SECTOR-FIVE-SEVEN-DETECTION-PROTOCOL-002	Sector-five/seven blind detection protocol	depth_independent	independently_replicated	256	4	sha256:c0c460e08c6164471534ee3ceb0ed2d549c70d6887d293f978c5ed140683a095
1457	Engineering Translation	SFT-ENG-SMITHIUM-SYNTHESIS-IDENTIFICATION-PROTOCOL-002	Smithium synthesis and joint-identification protocol	depth_independent	independently_replicated	256	4	sha256:b56e36103ed992219a22c4d9bedfaa034aedb02e98f8217d082f40ebcfafc7e9
1458	Engineering Translation	SFT-ENG-NOVEL-TRANSLATIONS-COMPLETE-FAMILY-002	Complete prior-return engineering protocol family	depth_independent	independently_replicated	256	4	sha256:b29fcbcb46dbf68ed79bedc13a7aa2fb391ba487875c8fb21b973d5093fc1ea5
1459	Engineering Translation	SFT-ENG-CONSCIOUSNESS-PLACEBO-CROSS-BINDING-PROTOCOL-002	Open consciousness, placebo and cross-binding protocol	depth_independent	independently_replicated	256	4	sha256:52856443290a7ff3f0a058bf50aa8533462b7acf743be21afa85edd484edade

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1460	Engineering Translation	SFT-ENG-NOVEL-TRANSLATIONS-NO-OMISSION-ADDENDUM-002	Engineering novel-translations no-omission addendum	depth_independent	independently_replicated	256	4	sha256:ad3233db90a4586713e027eefb3d4a23a54aa652c88e89111091f487b88cd96a
1461	Cross-Branch Synthesis	SFT-SYNTH-PRIME-VACUUM-ORBIT-IDENTITY-001	Prime-orbit and live-vacuum period identity	depth_independent	independently_replicated	256	4	sha256:ad7554655177df6725acdd078e39ab657c698bdd17ff077738a5f0f7ab0f3011
1462	Cross-Branch Synthesis	SFT-SYNTH-COMMON-LOCK-IDENTITY-001	Typed common Fold-lock identity	depth_independent	independently_replicated	256	4	sha256:4dc97a4b0a47521c1213c7ab3d30d599d2ce846a1498d81c1ff98b63eb902346
1463	Cross-Branch Synthesis	SFT-SYNTH-COMMON-DESCENT-IDENTITY-001	Typed folding-fixation-optimization descent identity	depth_independent	independently_replicated	256	4	sha256:d2113f3da386018cf680511930dc801e1b7244427d0e7d6f7582df2fbd99cef27
1464	Cross-Branch Synthesis	SFT-SYNTH-WAVE-MODE-RECURRENCE-IDENTITY-001	Longitudinal/transverse common wave recurrence	depth_independent	independently_replicated	256	4	sha256:3b94171e62328740112fd58ae63c8319f4ad5e8b72cbb34026acaf4ea9244d56
1465	Cross-Branch Synthesis	SFT-SYNTH-FOLD-SECOND-HARMONIC-IDENTITY-001	Fold doubling as the second-harmonic operation	depth_independent	independently_replicated	256	4	sha256:4281f6bff0926a61934cccabefddecc6db72f5b82aa30a72f4359fc5bfb88f63
1466	Cross-Branch Synthesis	SFT-SYNTH-VACUUM-PERIOD-DIVISOR-PREDICTION-001	Vacuum-mode multiplicative-order divisor prediction	depth_independent	independently_replicated	256	4	sha256:520ce58d41c26b324b43c0f9b9583aec100e0ac52021043a0e8984eb95a2f628
1467	Cross-Branch Synthesis	SFT-SYNTH-POSITIVE-OBSERVABLE-ABSENCE-BOUNDARY-001	Positive-observable and structural-absence boundary	depth_independent	independently_replicated	256	4	sha256:ce4cca05ba35d8fd00bbf184a5d87d8f6e6bf1e860cc37162e0d440b37892d47
1468	Cross-Branch Synthesis	SFT-SYNTH-TESLA-CORRESPONDENCE-ASSEMBLY-001	Typed Tesla Physics-Earth-Engineering correspondence	depth_independent	independently_replicated	256	4	sha256:3c7b77c5765f09318d8b0b37d49a94a364f3626a66f5607113c1436c5fba64ff
1469	Cross-Branch Synthesis	SFT-SYNTH-UNIFIED-CONSTANTS-ASSEMBLY-001	Cross-branch Unified Constants Object assembly	depth_independent	independently_replicated	256	4	sha256:3b4d24afe61e66693d2056a91ad48c3f1bb5fb436e45cc22348361af64f04fe6
1470	Cross-Branch Synthesis	SFT-SYNTH-PREDICTION-FALSIFICATION-LEDGER-001	Complete premeasurement prediction and falsification ledger	depth_independent	independently_replicated	256	4	sha256:e20dad87a471121fb67574eb1932b03f7a8ad2ef93012d4b5aab5eaf9a87ba68
1471	Cross-Branch Synthesis	SFT-SYNTH-ONE-OWNER-NO-OMISSION-LEDGER-001	One-owner V1/V2/V3 no-omission ledger	depth_independent	independently_replicated	256	4	sha256:869786a12a6829f20eccfcd9e5ccd00f82c6a9f31155bd587214fb5d6a383c2c
1472	Cross-Branch Synthesis	SFT-SYNTH-ROOT-TRACED-TERMINAL-ASSEMBLY-001	Terminal root-traced Cross-Branch Synthesis assembly	depth_independent	independently_replicated	256	4	sha256:4f3c4b8625c56128750070e8fa15fef1d4af831bb19f12c09b0b912c2a791e
1473	Chemistry	SFT-CHEM-VALIDATION-MOLECULAR-GEOMETRY-VECTOR-001	Substantial molecular geometry validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3fe068e9578065309505f4bd2b6b10c86331611573781be3c8fa75650ffa6f32
1474	Chemistry	SFT-CHEM-VALIDATION-THERMOCHEMICAL-VECTOR-002	Substantial thermochemical validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1b3d75c0246ea717bcd61969f05ed282f94f8eeb0d24e8b3de03092c828693ad

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1475	Chemistry	SFT-CHEM-VALIDATION-EQUILIBRIUM-VECTOR-003	Substantial equilibrium validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b9759323b6410152ed69b482f2a2687dd0b4c3835f2075ddcc1ab2a4cd379af7
1476	Chemistry	SFT-CHEM-VALIDATION-KINETIC-VECTOR-004	Substantial kinetic validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:662842be7a53f0a99166310a507a1b9da3a2928248ac355ef5f20e89947a6d39
1477	Chemistry	SFT-CHEM-VALIDATION-SPECTROSCOPY-VECTOR-005	Substantial spectroscopy validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9cc81bb257a521fb272f5e4fe832f9cb8beb471edfdabf8303d523b84ab3a270
1478	Chemistry	SFT-CHEM-VALIDATION-ELECTROCHEMICAL-VECTOR-006	Substantial electrochemical validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4b0221d819d6fc23faa0cbd3c9c2948f441546b4344f8bdce8b39789dba8b897
1479	Chemistry	SFT-CHEM-VALIDATION-INORGANIC-COORDINATION-VECTOR-007	Substantial inorganic and coordination validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:98548c90b133401da439ea2860dd337c23e05f24dfcaba2896742aa4f179d870
1480	Chemistry	SFT-CHEM-VALIDATION-ORGANIC-REACTION-VECTOR-008	Substantial organic-reaction validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0d3727e75b924eb416297d3822bcba72c734c24ac08ae6514bcbeb9c39a61995
1481	Chemistry	SFT-CHEM-VALIDATION-POLYMER-VECTOR-009	Substantial polymer validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:03f3a14da345577d0ee5823ac7cc232f682ca86eaa271d9003229c895112cfc9
1482	Chemistry	SFT-CHEM-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010	Cross-source Chemistry reproducibility vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:043d3795d9b725736b67badcaf95545999e67162868c74f7a0fa4b3a066dff5e
1483	Chemistry	SFT-CHEM-VALIDATION-ADVERSE-OUT-OF-BOUND-VECTOR-011	Complete adverse and out-of-bound Chemistry vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:68919fc8d349fd2cd6b1b9bfbcb3792289a792ecc12800e660c7c1db3d67b4853
1484	Chemistry	SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012	Chemistry empirical Grand Lock	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be4b72344aeafcd342ef43b7660e19fe73fcd6cda30541ca198debcc72f994a9
1485	Chemistry	SFT-CHEM-MATERIALS-OWNERSHIP-HANDOFF-001	Chemistry-to-Materials one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:da67f6404c234aeb9db221306817032248d20f581db03732828bafed718fdbe
1486	Chemistry	SFT-CHEM-BIOLOGY-OWNERSHIP-HANDOFF-002	Chemistry-to-Biology one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:37d9b00972e3dd38e15e09ed6de31b355990e32b6d1044d1b4f7e83487c56ade
1487	Chemistry	SFT-CHEM-MEDICINE-OWNERSHIP-HANDOFF-003	Chemistry-to-Medicine one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4b264c200dae449b3bebcf5d8af248097a539a72e2c411204f133b0215bfc35
1488	Chemistry	SFT-CHEM-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004	Chemistry-to-Earth and Environmental Science one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1b37f27441880372331632b8bbf9e7856610b2a8155b08f6ae9a18aed70cd2a
1489	Chemistry	SFT-CHEM-ASTRONOMY-OWNERSHIP-HANDOFF-005	Chemistry-to-Astronomy one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:355716812aceedef1e07165e27f12f265628a8be4993e37976cb743068c1d1a9

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1490	Chemistry	SFT-CHEM-CROSS-BRANCH-ONE-OWNER-COMPLETENESS-006	Chemistry cross-branch one-owner completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:19a909f22474460812b3821628c27dbe0e7c24f8cf56919a0012cd618cde839e
1491	Materials Science	SFT-MAT-CRYS-DIFFRACTION-AMPLITUDE-001	Exact diffraction amplitude and intensity ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:66b25d6e80686a090e0f5bb0e9c95cde9568726a998edcf6389ce22abfcae5739
1492	Materials Science	SFT-MAT-CRYS-STRUCTURE-FACTOR-002	Finite structure-factor composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a5389501fdf3f002a63f0852699d9aa5eaf2def492ebb793a555af20d788f7a3
1493	Materials Science	SFT-MAT-CRYS-TEXTURE-ORIENTATION-003	Polycrystal texture and orientation distribution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ccc88e8824914311822eb8eae03ea58811d1fd7bf93fea9f91a810f096a2a8ef
1494	Materials Science	SFT-MAT-CRYS-SHORT-RANGE-DIFFUSE-004	Short-range order and diffuse-scattering relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c567f9ebd825af83bca30b576eb97adc7b6cee2e9c032e48af8ca6e818b57be
1495	Materials Science	SFT-MAT-CRYS-STACKING-FAULT-DIFFRACTION-005	Stacking-fault sequence and diffraction consequence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e240c408ba701401889e6d105471e556c07b63778a70c74bc8ec6d5660177167
1496	Materials Science	SFT-MAT-CRYS-TWIN-DOMAIN-006	Crystal twinning and domain relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a400404257782ed601e29462864cff39aa505b407a7327bf1f16f37e65865b09
1497	Materials Science	SFT-MAT-CRYS-MODULATED-INCOMMENSURATE-007	Incommensurate and modulated structure organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:313ced02f25e6e053754c7fcd26e889ec8babb7640db65351c95fd2f1c1cdab2
1498	Materials Science	SFT-MAT-CRYS-PAIR-DISTRIBUTION-008	Total-scattering pair-distribution reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5af3a5acad0cf0ecf3c601c2db4549f59423fdd26cd799ede24ad34664753557
1499	Materials Science	SFT-MAT-MICRO-DEFECT-POPULATION-001	Defect population and site-fraction balance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:902c083fc7c51628739a79389dae656223864ca0966bb5dd7f28a6ff7e66cff7
1500	Materials Science	SFT-MAT-MICRO-DEFECT-MIGRATION-002	Defect migration and retained path	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:313402a897492a0bc284213931b53f8418db5afd04b5e943dbc27a94e6c11e58
1501	Materials Science	SFT-MAT-MICRO-DISLOCATION-REACTION-003	Dislocation reaction, climb and cross-slip	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:89293aa1943ddd99f7c67de12a3d8e3391cd0f6303d7333338148ef8551c3d0
1502	Materials Science	SFT-MAT-MICRO-GRAIN-GROWTH-004	Curvature-driven grain growth	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dfc9ebf5aaa6044a4121ec7b94cd92ea9c79f1f47d6453258b637d3a5efdfec6
1503	Materials Science	SFT-MAT-MICRO-BOUNDARY-SEGREGATION-005	Interface and grain-boundary segregation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ea1fdda4308216474a7b3bafb4c82b8b26e08ade0458f2fbb9f5c29842ed00134
1504	Materials Science	SFT-MAT-MICRO-PRECIPITATE-INCLUSION-006	Precipitation and coherent-incoherent inclusion boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c1165612f8dec6355601012603eff1c402cfc1d793ae01fd5c3d850bd0c72b1b

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1505	Materials Science	SFT-MAT-MICRO-COARSENING-TRANSFER-007	Ostwald-type coarsening as exact carrier transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1c4b55d223e02a856d7ec3484f9db556792b27b8b6c9e36e1db8c1efdd4049ec
1506	Materials Science	SFT-MAT-MICRO-INTERFACE-MOBILITY-008	Interface migration and mobility record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ab5d106f5833f889f7251c0d603ff9f2616b6250b2c7f181a1a8553305576d5d
1507	Materials Science	SFT-MAT-MICRO-MULTISCALE-CORRESPONDENCE-009	Microstructure-to-bulk multiscale correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ceeac26aabb0b5e76fef5e2e0f234d22ac74a228da008d6a685d43163623c7da
1508	Materials Science	SFT-MAT-PHASE-FRACTION-LEDGER-001	Complete multiphase fraction ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fdb02cf19624df051b3716fc053c3bc646376a2afbdddcl8b09d6dc3415421c1
1509	Materials Science	SFT-MAT-PHASE-TIE-LINE-LEVER-002	Tie-line and lever-partition relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:76bf325991a7c5071cc2e173d9e6b9a523f8e95cedc5fd3e13d9bdbbed103850
1510	Materials Science	SFT-MAT-PHASE-COMPONENT-HANDOFF-003	Component potential and phase-coexistence handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:02c5a4d91cad16088babbf0a32333ab8c4950ff6b2f519d16a3cdd31d34eb2ab
1511	Materials Science	SFT-MAT-PHASE-METASTABLE-RETENTION-004	Metastable-state retention boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7c50a76f079d3017e36ac5f0b281cb70a3a3359fcb64e6259f0f0b52ba6c5226
1512	Materials Science	SFT-MAT-PHASE-SPINODAL-INSTABILITY-005	Spinodal-equivalent instability organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8c984394ff7fd2c1c89630eb0e8a4be3ebe1699700699e20aee68de755386788
1513	Materials Science	SFT-MAT-PHASE-MARTENSITIC-006	Displacive and martensitic transformation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2b3d48180790e9a789f5bc4b71f17578b0931d32264772538c413bc40ef5a2ee
1514	Materials Science	SFT-MAT-PHASE-RECONSTRUCTIVE-007	Reconstructive transformation path	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d50bf0ad012aabb49da0e88e5c0b82cd35e9e4d71e2897974a34a653d9776611
1515	Materials Science	SFT-MAT-PHASE-ORDER-DISORDER-008	Order-disorder transition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6370db6cbde2d909482bdedef0ebda821fe9caf134555297275a058acdf653c7
1516	Materials Science	SFT-MAT-PHASE-GLASS-ARREST-009	Glass-transition and kinetic-arrest boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:144c6de994caf84feffc382ac0402dfa0bf200d0d0fb1e7172ec02354e2712d8
1517	Materials Science	SFT-MAT-PHASE-TIME-TEMPERATURE-010	Transformation kinetics and time-temperature path	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:71edbf8a3e8928061e68a084735b8b1584ca55d77bfcd1924b466206d6ed2e23
1518	Materials Science	SFT-MAT-MECH-TENSOR-STRESS-STRAIN-001	Tensor-resolved stress-strain response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:75ee953ca94f9c3df746a63cd62d94ec034df39948be0e474fe8666f61cfc4ae
1519	Materials Science	SFT-MAT-MECH-TRANSVERSE-STRAIN-002	Transverse-to-longitudinal strain relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6da865c282e26407adf565ed67d845b00a50f9cdf6b654764e90974f230c1e96

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1520	Materials Science	SFT-MAT-MECH-VISCOELASTIC-MEMORY-003	Viscoelastic memory and recovery	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:553f84e6d310b623a0c7a4ba93e55634866255fba42e2e3c7e9986dd2803e3c9
1521	Materials Science	SFT-MAT-MECH-VISCOPLASTIC-FLOW-004	Viscoplastic flow relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:799d82588b4d636407becfcbcd95c697a1a68ea15178be65b30bb5b5517984437
1522	Materials Science	SFT-MAT-MECH-YIELD-PATH-005	Yield-surface and loading-path boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5f252fe1aa5ac3debb6d6f9e63d8738e4736715e5dd458ed4ad3a0cc2ff6f06a
1523	Materials Science	SFT-MAT-MECH-WORK-HARDENING-006	Work hardening and retained deformation history	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9ac5c14209b96d41254394f844de68749fb9ac484d9d5657ad7182a9bd77f827
1524	Materials Science	SFT-MAT-MECH-FRACTURE-ENERGY-007	Fracture energy and toughness ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cb8423938cc668b23e699e5659bbfa492ffd835fd7cda1da0ec58c43d40258bd
1525	Materials Science	SFT-MAT-MECH-CRACK-GROWTH-008	Stable and unstable crack-growth relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f12803c9e26564d123a26760f7250e727637707dcccdaaf0b86cae3261a9efa6
1526	Materials Science	SFT-MAT-MECH-FATIGUE-009	Cyclic fatigue initiation and propagation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d81cfe5aebce710cd17d91ced8742f0ce3e7596d21dec2880daa1916d4f037b
1527	Materials Science	SFT-MAT-MECH-CREEP-RUPTURE-010	Creep mechanism and rupture-time custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fde173039ba0f1d8f74bb5cba12c8b774943cffd9ee14be7bec852e41e1f3e94
1528	Materials Science	SFT-MAT-MECH-IMPACT-011	Impact and high-rate mechanical response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8a8153930d7dd7d03553d9964c49e9f2ad78645d461525bd73f75bd9399b7fb4
1529	Materials Science	SFT-MAT-MECH-FRICTION-CONTACT-012	Friction and contact-state relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:23017fea0efff13d900b3690c75560f00929caf5cf6910cec388d88b177045b91
1530	Materials Science	SFT-MAT-MECH-LUBRICATION-TRIBOFILM-013	Lubrication and tribofilm organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be276516d621b0d48b943e10c062d1f583a7106c302c1ae63ac99c1d5c04315a
1531	Materials Science	SFT-MAT-MECH-RHEOLOGY-014	Rheological flow and relaxation classes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b38317d23bcc37b7ffe7dbf46629754d041fa15a27dc33f70f774cbeeeef9b6
1532	Materials Science	SFT-MAT-THERM-DIFFUSIVITY-001	Thermal diffusivity relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:203cd80e00d1c50650d093112e8231834b7fe241a34552f88d05c3402f2f625a
1533	Materials Science	SFT-MAT-THERM-BOUNDARY-RESISTANCE-002	Interfacial thermal-boundary resistance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8892aa549888cf5c8e45886e5b893074c167bf4480ebe3313adf30ae90bd7e71
1534	Materials Science	SFT-MAT-THERM-PHONON-MEAN-PATH-003	Phonon scattering and mean-path ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:76a926b4634a25262369ade1a043e5dfbb4abc62139b3f5fc7d05fe1d5326895

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1535	Materials Science	SFT-MAT-THERM-RADIATIVE-TRANSPORT-004	Radiative thermal transport in materials	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0c4494cc76fe6ed5ed829a4f289158669fac95e271e47bd3d97db42ce4b770c4
1536	Materials Science	SFT-MAT-THERM-THERMOELECTRIC-BOUNDARY-005	Thermoelectric coupled-response performance boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f448e0cc638e3d7fd5ad7f72245002d2215a386f08e74b71c34ad5ba3d2f5cb6
1537	Materials Science	SFT-MAT-THERM-PHASE-STORAGE-006	Phase-change thermal-storage ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:db2cc00f176e3b82dd10915ca448451fdb759fa1897e514b16678b193df63179
1538	Materials Science	SFT-MAT-THERM-SHOCK-FATIGUE-007	Thermal-shock and thermal-fatigue boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:384a57b2c1e8032b7ee57c398fbbae7cbdc24b349d7781ab2eabe9724fb7568f
1539	Materials Science	SFT-MAT-ELEC-CONDUCTIVITY-RESISTIVITY-001	Electrical conductivity and resistivity relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f36daae94e69e01b10297efefd4da77b2d0431de6b372029d751d0197229759
1540	Materials Science	SFT-MAT-ELEC-MOBILITY-CONCENTRATION-002	Carrier mobility and concentration separation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:af184916c5fa1e1d858363839ad505f93d0034903d7bbeca988845562fa85e10
1541	Materials Science	SFT-MAT-ELEC-HALL-RESPONSE-003	Hall-response carrier ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c6147980136418e1e89200bae0e4574a2ac2d5f10db011ec4b7e69e701377ac3
1542	Materials Science	SFT-MAT-ELEC-DIELECTRIC-LOSS-004	Dielectric permittivity and loss relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4d86ad5d9cc531086b23f5b8db6c82191dcfdbab230d639b9aa242dda8aeb9fca
1543	Materials Science	SFT-MAT-ELEC-IONIC-TRANSFERENCE-005	Ionic conductivity and transference	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cfc2c55ab5621ddc6c4fd2d6b9dce3f44984316a49a2cfc156e5cd520740257
1544	Materials Science	SFT-MAT-ELEC-MIXED-TRANSPORT-006	Mixed ionic-electronic transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:75d2701859b7a6b04ba078ad5bba2cb0d9c519d675f0f4bd74362995d2988215
1545	Materials Science	SFT-MAT-ELEC-FINITE-BARRIER-TUNNELLING-007	Tunnelling through a finite material barrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:223f4c84743d38002d7031222be1d66f5892b50f1119bc655902a8468b155c74
1546	Materials Science	SFT-MAT-ELEC-BAND-ALIGNMENT-008	Band-alignment and interface offset	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9b3afc5ae9b18eaf8dd83976578802ae51f2d67b80f3cf0e170228bc5511d4bc
1547	Materials Science	SFT-MAT-ELEC-CARRIER-CONFINEMENT-009	Heterostructure carrier confinement	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0afccf622730db4176a14b347ee8d4e453e1eb52fb78bc04835549031d00805
1548	Materials Science	SFT-MAT-ELEC-DEFECT-TRAP-STATES-010	Defect and trap electronic states	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0a9b20a06741a4d825d5b6b8eae58e1df3cd01650b72bfbefb6a913d214d5efdc
1549	Materials Science	SFT-MAT-ELEC-SCREENING-DEPLETION-011	Charge screening and depletion length	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7f783979da059859e675242ec28ed6a5da90c706599e0163fe1b7f3d229cf3ef

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1550	Materials Science	SFT-MAT-ELEC-ELECTROCHEMICAL-INSERT-ION-012	Electrochemical insertion and material-state response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:39e1ed13fdf4eda1887c6c54107dbeb519158eae317691102248899dc4553333
1551	Materials Science	SFT-MAT-MAGSC-PARAMAGNETIC-RESPONSE-001	Paramagnetic response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7228fa5be9be3cd341e17c494182a86aa8e4e0344042bd1b8d8839c53e50cccb
1552	Materials Science	SFT-MAT-MAGSC-DIAMAGNETIC-RESPONSE-002	Diamagnetic response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cc74750ce36474675622217b4bacd5aee0bc00a2646cdf6db44f58677b98f8b5
1553	Materials Science	SFT-MAT-MAGSC-SPIN-GLASS-FREEZING-003	Spin-glass freezing and history	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9e7d4eccbc5c5cecc5bcf575cae702c4f66ff7f824bd0c341e4721bd6067621
1554	Materials Science	SFT-MAT-MAGSC-DOMAINS-WALLS-004	Magnetic domains and walls	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:28ced43e9c0880b8e94dd7a0c29677e8fe89922a635b0f65a596ab7d63d39013
1555	Materials Science	SFT-MAT-MAGSC-HYSTERESIS-LOOP-005	Magnetic hysteresis loop ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fd139ec657173ab8253dac9b667f604b6e7d1a74ba65b5b8b98969e7ff447cba
1556	Materials Science	SFT-MAT-MAGSC-MAGNETOCRYSTALLINE-ANISOTROPY-006	Magnetocrystalline anisotropy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:490b9b5f874289adb99425c0825d9972735bc9c9c23a7a3c58dbd3d9d3e856f6
1557	Materials Science	SFT-MAT-MAGSC-MAGNETORESISTANCE-007	Magnetoresistance and field-response relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ac5452c0a039eba0a778441d5fec468e9a7a6b65a25aa41b90e27dc443b7e4d3
1558	Materials Science	SFT-MAT-MAGSC-SPIN-TRANSPORT-RELAXATION-008	Spin transport and relaxation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ede3249d7281586061ae6f48c4a98ee7f162e7ec2b2fa9fffff24f88c1a813464
1559	Materials Science	SFT-MAT-MAGSC-SC-CRITICAL-FIELDS-009	Superconducting critical-field organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b36a35fac9671a641f6dadca15f42f257904ab2a56290630255acb5bf0dcb1d5
1560	Materials Science	SFT-MAT-MAGSC-SC-VORTEX-PINNING-010	Superconducting vortex matter and pinning	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:298911f8bf33212f62692879d476eed5f5da9b31a636e259ac8b2146f435f1c3
1561	Materials Science	SFT-MAT-MAGSC-SC-COHERENCE-LENGTH-011	Superconducting coherence-length boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9213d8e6499d8185965e0ecc3112214c416cc13414f39f895969cd989a849756
1562	Materials Science	SFT-MAT-MAGSC-SUPERFLUID-CRITICAL-FLOW-012	Superfluid excitation and critical-flow ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2e9849264a6199c9db15770d10fa9b5670e746881a1527f5b9c2d0e179a8a3ef
1563	Materials Science	SFT-MAT-OPT-ABSORPTION-EXTINCTION-001	Optical absorption and extinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78e7589d839869adf80751b806c75f11edeeea9633b1cceao0c35a467a384862
1564	Materials Science	SFT-MAT-OPT-REFLECTION-TRANSMISSION-002	Reflection and transmission ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e9433b2af8c835782e3ac6d25a87fc616a81b6f5d63ae2b327eb7090334ec361

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1565	Materials Science	SFT-MAT-OPT-LUMINESCENCE-YIELD-003	Luminescence and quantum-yield custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ab79045539d925089303124b8d0c6d3d7c7702e426c7f8b9395cc32e0f36b716
1566	Materials Science	SFT-MAT-OPT-LIGHT-SCATTERING-004	Elastic and inelastic light scattering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fda29881226f097c034019cc90dd00bcd1475e12806abad570a54fca70ec18b2
1567	Materials Science	SFT-MAT-OPT-BIREFRINGENCE-ANISOTROPY-005	Birefringence and optical anisotropy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:77178a25bcf4f585bba258b8416f8b5e57b722e3b0b50b5ff545a44fc1d9f0d8
1568	Materials Science	SFT-MAT-OPT-NONLINEAR-MIXING-006	Nonlinear optical mixing response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8beecce8dcc03c64d1efe44d3a8224f09353eea80704812a0f76a2abc60f99684
1569	Materials Science	SFT-MAT-OPT-WAVEGUIDE-CONFINEMENT-LOSS-007	Waveguide confinement and loss	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0612df0cfeb508bcb99f1d101daf475c8e411d2927018f3d3c68f4d3e9caf842
1570	Materials Science	SFT-MAT-OPT-PHOTONIC-GAP-DEFECT-008	Photonic-band-gap and defect mode	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a6461f40cf7c98b313a2375e2afe77729a9ed808be09e026934bade054a4ba5b
1571	Materials Science	SFT-MAT-OPT-PLASMONIC-RESPONSE-009	Plasmonic collective response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1bdcbb18e974e7dae69c290e6242c01465f6c2dd8674443397237b45dcd22145d
1572	Materials Science	SFT-MAT-OPT-EXCITON-DYNAMICS-010	Exciton generation, transport and recombination	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:03749083f3b44d3ca6eddddcfc97b0a75c07b96e0c3f003cde9b8e661bda217b
1573	Materials Science	SFT-MAT-CLASS-SOLID-SOLUTION-ALLOY-001	Solid-solution and alloy phase organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f4a93b8b79815bde21f22600d14a65da1a6105808d3cf945dcd5f59a5b13ddba
1574	Materials Science	SFT-MAT-CLASS-INTERMETALLIC-ORDER-002	Ordered intermetallic compound organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1f77f6063469a0c456146a5807612afb55c10ce801620968ec04cdcf6d32e28f
1575	Materials Science	SFT-MAT-CLASS-HIGH-ENTROPY-BOUNDARY-003	Compositionally complex alloy boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f6044a1580b64db186e3620881616f7177d0531ce8dfaa5254423642c16ce3a3
1576	Materials Science	SFT-MAT-CLASS-REFRACTORY-UHT-004	Refractory and ultra-high-temperature material class	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ba549f60fab7b88361b73e97ca179007f76e86adb8339ecbe112925545a3d2ca
1577	Materials Science	SFT-MAT-CLASS-CEMENTITIOUS-CONCRETE-005	Cementitious and concrete composite organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6e559584d473ceb17f1c5e067355a5644dac42b49005f243e91d5d978f7d342
1578	Materials Science	SFT-MAT-CLASS-FIBRE-REINFORCED-006	Fibre-reinforced composite load transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:de902521804f45f5fe36e980d229e69a83923ed65294f07bf163a679b3160e62
1579	Materials Science	SFT-MAT-CLASS-PARTICLE-REINFORCED-007	Particle-reinforced composite load transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9e2c00016a582eef5e25f25c067c38be0d7bbd076631c2944a9b1e35b4697103

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1580	Materials Science	SFT-MAT-CLASS-METALLIC-GLASS-008	Metallic-glass organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9718b54af7ca52008c6b43bb05c60a5d7d61c78870d87ccc83bda63ee479e8d1
1581	Materials Science	SFT-MAT-CLASS-CERAMIC-SUBCLASSES-009	Structural and functional ceramic subclasses	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dc16234399a1b13aaf6968f48602c508258b0e132e7904b58a8968af45c23b11
1582	Materials Science	SFT-MAT-CLASS-POLYMER-SUBCLASSES-010	Thermoplastic, thermoset and elastomer distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3377c28b097ac207e79b8530c9b73c5b43bf49f40ce8e9c73c7f33d84cc516fa
1583	Materials Science	SFT-MAT-CLASS-FUNCTIONALLY-GRADED-011	Gradient and functionally graded materials	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3beed22d7a6345510be495ca931c777d4c7a075bc6c844d641ef4f3e20340818
1584	Materials Science	SFT-MAT-CLASS-ARCHITECTED-CELLULAR-012	Architected and cellular materials	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f7eec41372016f198c06805f29fc2f3fa759d6fe3e6ed83191ec861df994ffcb
1585	Materials Science	SFT-MAT-SOFT-COLLOID-AGGREGATION-001	Colloidal stability and aggregation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:85d4ac6dba37c7352a3597f8080d2495312ca0aaa60d8a142326bd50905a0256
1586	Materials Science	SFT-MAT-SOFT-GEL-PERCOLATION-002	Gelation and percolated soft network	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa1dc8180c723c0ce409d49ef2938fd857c1ea1bc9c73dfd8859bed9f917ba09
1587	Materials Science	SFT-MAT-SOFT-FOAM-DRAINAGE-003	Foam cell and drainage organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:872101dd10667aabe677ec1e25d3ce53f0b9bf998691193f289a7317f2d00926
1588	Materials Science	SFT-MAT-SOFT-LIQUID-CRYSTAL-ORDER-004	Liquid-crystal orientational order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:572d0c1d48995f676100ff7bb7e8620ba36f5effaaa5f72bcf7b9f40f2618ca0
1589	Materials Science	SFT-MAT-SOFT-EMULSION-DROPLET-005	Emulsion and multiphase droplet organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:69bf1637f23b87e943680c331d3db3c661d86710380382338e79f30ba10540b8
1590	Materials Science	SFT-MAT-SOFT-MEMBRANE-THIN-FILM-006	Membrane and thin-film soft matter	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cc20019289aed6302bcd561f3f0a3333e641a7351847cb081d802da7dd6fb27
1591	Materials Science	SFT-MAT-SOFT-GRANULAR-FORCE-CHAIN-007	Granular packing and force-chain support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c9420e94b3f100db2e85a69f274253f95f8fa58e8b7e697e892f39c6986951b8
1592	Materials Science	SFT-MAT-SOFT-JAMMING-BOUNDARY-008	Jamming and unjamming boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2951fbb4b789a7dc97aa9425ea21ac58ed55904975180183865d15185a7229e5
1593	Materials Science	SFT-MAT-SOFT-STIMULI-RESPONSIVE-009	Responsive and stimuli-sensitive soft materials	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dbabe6f570d78223954004a21806f541b639b186c7674d8482cf7bb995b5a53e
1594	Materials Science	SFT-MAT-SOFT-ACTIVE-NONEQUILIBRIUM-010	Active-material nonequilibrium organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa19b4d8a60ac4e7fab62abab36c855e77df8c11914ee66649b1d0b0b13d5df8

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1595	Materials Science	SFT-MAT-BIO-BIOCOMPATIBILITY-INTERF ACE-001	Biocompatibility as a material-interface boundary	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:a10aa7bcb625a83b2d3ebf9dd8 e4ff28df0ae2d5d74ab13268cb4ef711b 8bd2f
1596	Materials Science	SFT-MAT-BIO-BIORESORPTION-DEGRADATI ON-002	Bioresorption and degradation ledger	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:c09dea6fb0753c440817413d6c 725e4766f0e60a5a0edd97b1ae4b07fdf 61d99
1597	Materials Science	SFT-MAT-BIO-SCAFFOLD-POROSITY-CONNE CTIVITY-003	Tissue-scaffold porosity and connectivity	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:1fe3a1e81c362902e3c4071d0e 9dde9d04e9361616fa13f22c91314c874 4f559
1598	Materials Science	SFT-MAT-BIO-CELL-MATERIAL-ADHESION- 004	Cell-material adhesion handoff	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:9b7533748ea7aab0c1fb99c70e ffd14297d8563dbd71e9ddd7db7bcf1b8 e4c3d
1599	Materials Science	SFT-MAT-BIO-MECHANICAL-MATCHING-005	Mechanical matching at a biological interface	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:240cdfc025d172d1b3bf2421f5 bf8a9f503e4b0ce16c22b4e959f56f6bf 0429e
1600	Materials Science	SFT-MAT-BIO-CONTROLLED-RELEASE-006	Controlled-release material transport boundary	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:10c4443b381f4482ad2889e538 708109711794e1c8c8aac77d6c7d66dac 4d3a1
1601	Materials Science	SFT-MAT-BIO-MINERALIZED-ORGANIZATIO N-007	Mineralized biological material organization	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:bd00986cccd114e52724965d3b 2775686ee13a2353b6a88760f34404a1b 4bd07
1602	Materials Science	SFT-MAT-BIO-BIOFABRICATED-IDENTITY- 008	Biologically derived and biofabricated material identity	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:f39a1f35f33f68a03fb28adf80 abbc665c69e85eae15894b03819db2fa2 69af4
1603	Materials Science	SFT-MAT-NANO-SIZE-SHAPE-DISTRIBUTIO N-001	Nanoparticle size and shape distribution	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:c74d6d61481d890ee9570f1a2a e244149babdc2e4c76293cd4b1ae18377 5800a
1604	Materials Science	SFT-MAT-NANO-NANOWIRE-CONFINEMENT-0 02	Nanowire and one-dimensional confinement	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:96ceb3232e229582f916a4da6f ad2e9aba46d973d6b4f17e110e98743bc 49fab
1605	Materials Science	SFT-MAT-NANO-LAYER-STACKING-003	Two-dimensional layer and stacking organization	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:2e6d121c23ede76ba7490885b4 19daf8f70d23a15921228978606bb5b0f ba16e
1606	Materials Science	SFT-MAT-NANO-QUANTUM-DOT-CONFINEMEN T-004	Quantum-dot finite confinement	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:237de22de91f905cf55e4788c9 fab3bf34a8a4b026cdc46f817a87b7 83d9d
1607	Materials Science	SFT-MAT-NANO-SURFACE-VOLUME-DOMINAN CE-005	Surface-to-volume dominance relation	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:57c22baf89837b4006bf20685e c4f7400457949c477ec442f2e4e8b5231 07293
1608	Materials Science	SFT-MAT-NANO-PHASE-MELTING-BOUNDARY -006	Nanoscale phase and melting boundary	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:d70a707f104b7d602449be424e f12ff523c1295ca6b76932e189ca72ec5 51393
1609	Materials Science	SFT-MAT-NANO-QUANTUM-COLLECTIVE-STA TE-007	Quantum-material collective-state classification	depth_independent	empirically_tested_and_inde pendently_replicated	256	4	sha256:1a99613267e755337aa3ddb196 6d3851081db5cbd2d294e7f0f28e6c10d d5d26

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1610	Materials Science	SFT-MAT-NANO-MOIRE-SUPERSTRUCTURE-008	Moiré and twisted-layer superstructure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d26bccbedef31d38d5be6b79bf fe940e6f1ca6b351580a7e2f8a9f9cf75 47294
1611	Materials Science	SFT-MAT-NANO-NANOCOMPOSITE-INTERFAC E-DENSITY-009	Nanocomposite interface density	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:675222df1d62c1af5aa1b5f570 7fd2248cc9f568357492d0d0956e67a8a fdf03
1612	Materials Science	SFT-MAT-NANO-AGGREGATION-DISPERSION -CUSTODY-010	Nanomaterial aggregation and dispersion custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a2089b82b67287f189bc0e9d76 a8145ab1916c4c24225ac9cb0d8c74e9f 5d5f6
1613	Materials Science	SFT-MAT-SURF-FREE-STATE-ENERGY-001	Surface free-state and energy relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dcde8fab4311e8d383a5fc687d 9c3c15ef81a1662865ea82e0910a06330 8ca37
1614	Materials Science	SFT-MAT-SURF-WETTING-CONTACT-ANGLE- 002	Wetting and contact-angle custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:52ee8e2b2f42259b02e6094cf3 801ebd59ac9f08ccc8264f0657f5f66a9 aad06
1615	Materials Science	SFT-MAT-SURF-ADHESION-SEPARATION-00 3	Adhesion and work-of-separation ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fa0b6946cc725c0f96c2e7d20f 4ab6187a483b55320e41deab1fb77a7a9 7078f
1616	Materials Science	SFT-MAT-SURF-COATING-SUBSTRATE-004	Coating layer and substrate organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2227007ad00495a284644907a6 91daabadbbdff9f1ef27e4635654eede3 a5f1b
1617	Materials Science	SFT-MAT-SURF-ROUGHNESS-SCALE-005	Surface roughness and scale boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1ad9133ffa04881642cba04195 5c66a4edb1d1ccff397831018cdbab53d 5c2e3
1618	Materials Science	SFT-MAT-SURF-REACTION-CATALYSIS-HAN DOFF-006	Surface reaction and catalysis handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bd13a0ba76ceb3706a1f1d6c4b bd9ab2664a093d9d4c276d921158bce33 f6105
1619	Materials Science	SFT-MAT-SURF-TRIBOFILM-RETENTION-00 7	Tribofilm formation and retention	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aea10406336c2c7ba505903bb8 910588f5c56eabf7144124b4b409d3bb1 c0d4d
1620	Materials Science	SFT-MAT-SURF-DELAMINATION-008	Interface fracture and delamination	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:14e6d55a48f7c856bbb681901e 5d3f582ff4bf88da0c03a77b6263273e2 30610
1621	Materials Science	SFT-MAT-DEGR-OXIDATION-SCALE-001	Oxidation scale growth and transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:947c31475711cd7e1f7e394a21 15b1e7b9a672215ad17189bc8e8350d90 9d116
1622	Materials Science	SFT-MAT-DEGR-CORROSION-PATH-002	Corrosion-rate and electrochemical-path ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:063ae7ade3f8e9e018cfb6234f 834fa07b29f0e95a37e16a828c32a1545 c603b
1623	Materials Science	SFT-MAT-DEGR-PASSIVATION-BREAKDOWN- 003	Passivation and film-breakdown boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:26287756ed1abe3b51b51a1843 d3b7145221786dac8dac086c1fb3f66f1 19f24
1624	Materials Science	SFT-MAT-DEGR-STRESS-CORROSION-004	Stress-corrosion cracking	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a92186f2b4df0f75f0ba19cff3 6f22bf11df5011f42dfcfa996c20b22b b74dc

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1625	Materials Science	SFT-MAT-DEGR-HYDROGEN-EMBRITTLEMENT-005	Hydrogen uptake and embrittlement	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:79fe6b792a07b688c72d746dd58adf3e184f676f7c1beddf2aedd8327c6301b2
1626	Materials Science	SFT-MAT-DEGR-WEAR-MODE-DISTINCTION-006	Abrasive, adhesive and erosive wear distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cffc36d8c727d6e0d1a3d9fdcc174ef4e52a0e559797d92f8259ae3022b44dda
1627	Materials Science	SFT-MAT-DEGR-RADIATION-DEFECT-RECOVERY-007	Radiation-defect accumulation and recovery	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:58d0099c0a28fd3d4305be6797698fafb636dd31092304be72e4e006355868e3
1628	Materials Science	SFT-MAT-DEGR-PHYSICAL-AGEING-008	Physical ageing and property drift	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b3c2d79260adfe051f83e284fe36363388d044927b59ebbf4f66b2b8680ae5531
1629	Materials Science	SFT-MAT-DEGR-WEATHERING-009	Environmental attack and weathering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c42fcb92ad214f709df497f726ebbdaf784842512437bbc215b8dbecb88c4545
1630	Materials Science	SFT-MAT-DEGR-SERVICE-LIFE-EVIDENCE-010	Service-life and failure-time evidence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cce897586e6bcd409ad936e2c6284691fcc6770e1156b471e3fff70e45a5fcfe
1631	Materials Science	SFT-MAT-PROC-CASTING-HISTORY-001	Casting and mould-filling material history	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:91c3072765ec33f4d5a0e2d1f2ee177e2a2df1c875ce5952b45e2f0eb8d06d03
1632	Materials Science	SFT-MAT-PROC-FORMING-TEXTURE-002	Thermomechanical forming and texture	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:82aa2ea4e778763837d0ffaaac35e0383964a6c79a26692c98e6e3a57640c2bd
1633	Materials Science	SFT-MAT-PROC-MACHINING-DAMAGE-003	Machining-induced surface and damage state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:997502316d6b4dc58060c367cbdca5f1e02a50ca3393f10c9bc05f73e1a2711b
1634	Materials Science	SFT-MAT-PROC-ADDITIVE-BUILD-004	Additive layer-build and melt-pool history	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e7d4f7875ee957bedb5077f32565db67a385cff2c55ec22edff2d739c7bd4b3d
1635	Materials Science	SFT-MAT-PROC-THIN-FILM-GROWTH-005	Thin-film deposition and growth	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7815bd4368b4df46fe44298998ceb8ec9db851a8f141edd185f437cfcd780bfd
1636	Materials Science	SFT-MAT-PROC-EPITAXY-MATCHING-006	Epitaxial growth and lattice matching	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:218ceb30fdd566cc5fd54f49afe4ec745fb1c1a2f0b9b4b75745a35e8df65c0b
1637	Materials Science	SFT-MAT-PROC-JOINING-INTERFACE-007	Welding, brazing and joining interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:abdac12922fc77dc89ad444cbb393a2db849d7fedbce5524dee039018d0c7a35
1638	Materials Science	SFT-MAT-PROC-POLYMER-ORIENTATION-008	Polymer processing and orientation history	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:06ff2fe86a03a7ea13581eb586e697af6c7c75cd71aca2229af4fdc4176a8695
1639	Materials Science	SFT-MAT-PROC-POWDER-COMPACTION-009	Powder processing and compaction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:791d6c4a1d40f5e1f204718863833dc59f1ebe757f5511f318aaea9b7fbd9e1

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1640	Materials Science	SFT-MAT-PROC-WINDOW-PROVENANCE-010	Process-window provenance and reproducibility ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0ad7b92e5c8d24cdfc2ee2617314ee031fbf333bf5e0b18f35e9809fa34d261f
1641	Materials Science	SFT-MAT-COMP-DATA-REPRESENTATION-001	Exact material-structure data representation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5cb145faa284f47b049c493613a77ec34d81ce7663ab2936f6646fd7c894fc92
1642	Materials Science	SFT-MAT-COMP-STRUCTURE-PROPERTY-002	Structure-property computation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:119129d06d51dbe1a5ca49a20a356dacb43a10a74faea9feda46b0205c446805
1643	Materials Science	SFT-MAT-COMP-FINITE-SIMULATION-003	Finite numerical material simulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f87320573329f585ad42f15acd55ac03c449594922b20a79986553d9086acba
1644	Materials Science	SFT-MAT-COMP-MULTISCALE-COMPOSITION-004	Multiscale model composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9eaf30305b42d6083076be451c45d63bc720441ebf1ac310a03aedfa68ae35b9
1645	Materials Science	SFT-MAT-COMP-ERROR-PROPAGATION-005	Numerical stability and error propagation in materials	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0e666132b5a7ea08e840c4c151af4249d039012740824d80a4a221df4f638b02
1646	Materials Science	SFT-MAT-COMP-INVERSE-PROBLEM-006	Inverse materials problem	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c13fcf6f8de3af8b86bc8de025abd71956b434ca3ae754301e90d8829321bec6
1647	Materials Science	SFT-MAT-COMP-LEARNING-BOUNDARY-007	Machine-learning materials inference boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7b230e12ee2be28170850c668a9ae42e26541908878b91a938a1f1c649f07da9
1648	Materials Science	SFT-MAT-COMP-DATABASE-PROVENANCE-008	Materials database identity and provenance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ba44476128acc010777e8bbbc793004d9c167b221bb93a6f808a8c2d3bfe61839
1649	Materials Science	SFT-MAT-COMP-PHASE-FIELD-009	Phase-field computational correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4f46b3aba70bc6f456cbb7151de65fd380d8efa5a2eb386c0be4eb5096c2990f
1650	Materials Science	SFT-MAT-COMP-MOLECULAR-DYNAMICS-010	Molecular-dynamics computational correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:75edb43db5151fd1f75084e2d7af29940612577167b1aba0bfb552d0715d5333
1651	Materials Science	SFT-MAT-COMP-ELECTRONIC-STRUCTURE-011	Electronic-structure computational correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:636d9371895a27cf6fae2e5f7a2b050441799b66b3c9d65b42116ffc0af89c5b
1652	Materials Science	SFT-MAT-COMP-SIMULATION-EXPERIMENT-012	Simulation-to-experiment validation ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eaf13f8ed5c6bc5816bfc32ebadb35bf75f26a473a6faa77dc11f344afffbca
1653	Materials Science	SFT-MAT-EXT-HIGH-PRESSURE-STATE-001	High-pressure material state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:656183d7e8ecf4a9c8d09efd8e40eea8821392f36595186f9d934599149cb4bb
1654	Materials Science	SFT-MAT-EXT-HIGH-TEMPERATURE-STATE-002	High-temperature material state	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b4e46b616c93e8ba397d849a5ab3a5b82a6f219b6b65c206f8d9e0b6029c004b

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1655	Materials Science	SFT-MAT-EXT-CRYOGENIC-RESPONSE-003	Cryogenic material response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:861c495f2b5e062f82a7ba415d4e1b1a8fe02e9864298b7d09d86b2fa416ce2f
1656	Materials Science	SFT-MAT-EXT-ELECTRIC-FIELD-RESPONSE-004	High-electric-field material response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:32f74faa03625c03d3d7d740b1a4a3fa629b35eb9ccca3bc9a407b7ae11f02c
1657	Materials Science	SFT-MAT-EXT-MAGNETIC-FIELD-RESPONSE-005	High-magnetic-field material response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6c065f9b43a2507ea5416c4a6cb705da77d0cbf2e636ca1a3f679297e3915a0c
1658	Materials Science	SFT-MAT-EXT-SHOCK-RESPONSE-006	High-strain-rate and shock response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c3d8d24832ea1d7bab078386b61d78dcb11bcde6e53a05775e4e700b5b97705
1659	Materials Science	SFT-MAT-EXT-RADIATION-RESPONSE-007	Extreme-radiation material response	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9378b3b6d4e9a2a2413e9d856f063be97bfc02e2cf7c289c9b9660efed86305d
1660	Materials Science	SFT-MAT-EXT-COMBINED-PATH-CUSTODY-008	Combined-extreme condition and path custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f297de783de640f02297dbff2e2ce71955f5f557e873e78b5d89ce99f68f10e5
1661	Materials Science	SFT-MAT-SUST-EMBODIED-LEDGER-001	Embodied material and energy ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:37e497bdc37c1ca2dc454761261dff4f68f4d667889276415617f44d9295c187
1662	Materials Science	SFT-MAT-SUST-AVAILABILITY-BOUNDARY-002	Critical-material availability boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1620043d743b49ff48e8468332379828b793c8ce100fbd796947cdd52e71790e
1663	Materials Science	SFT-MAT-SUST-REUSE-REMANUFACTURE-003	Material reuse and remanufacture	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d820f6b69450e79f26b2c5b0d75cde77f118713828dd2eb8b3f2eb61c380c52
1664	Materials Science	SFT-MAT-SUST-RECOVERY-YIELD-004	Recycling separation and recovery yield	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4047e724adcc58e2a359e7a7a144564ded1f9bd34b050ad8409191613492def9
1665	Materials Science	SFT-MAT-SUST-CIRCULAR-FLOW-005	Circular material-flow organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3be057d6b5c1faf6d8c88f7fe6f6fc11eab06366f270d65e12d7ea40bd3c8f03
1666	Materials Science	SFT-MAT-SUST-DURABILITY-EXTENSION-006	Durability and life-extension relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b31769a1805cc3dbebbd6bbeceb83225dff21a6a7249cce49dc8bccdd6181e5934
1667	Materials Science	SFT-MAT-SUST-TOXICITY-HANDOFF-007	Material toxicity and health handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4e27fad89061063b3f0d96fb80c982511ad0d38396c647ea04773e77156c91b0
1668	Materials Science	SFT-MAT-SUST-SUBSTITUTION-FUNCTION-008	Material substitution and function preservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a76f6c73192aa6a94b613d47e653cf11d5be96453ae427e6a58fba1dae652ffb
1669	Materials Science	SFT-MAT-SUST-END-OF-LIFE-CUSTODY-009	End-of-life fate and residual custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c722e8ed79cdea2dbbdf7dd45e64e9e33d6c0fe63c0d7b283a24e72f1b0f580

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1670	Materials Science	SFT-MAT-VALIDATION-CRYSTALLOGRAPHY-DIFFRACTION-VECTOR-001	Crystallography and diffraction validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4ad9b7e3ae07359ffedfa94f3fb5489dee448ef95b63fb028df48f1097aa3428
1671	Materials Science	SFT-MAT-VALIDATION-DEFECT-MICROSTRUCTURE-VECTOR-002	Defect and microstructure validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:07c5e25cfd0077d4efbd0652068b1890a861cbffcfabc9826553b97d443dd3
1672	Materials Science	SFT-MAT-VALIDATION-PHASE-TRANSFORMATION-VECTOR-003	Phase and transformation validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:064539aa81cfee8f3d9a96eed1c554a7da11049ff07c78a34ac0bc42dba14c13
1673	Materials Science	SFT-MAT-VALIDATION-MECHANICAL-TRIBOLOGICAL-VECTOR-004	Mechanical and tribological validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2ceed395c1dafeda88a1280beebb9046503570b4dc994ccd98e7af625f7eca1
1674	Materials Science	SFT-MAT-VALIDATION-THERMAL-TRANSPORT-VECTOR-005	Thermal and transport validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f4f3ec66e67611e9c08643a7d846ac79c47b8964688ae2ab5ba05317a73c15f9
1675	Materials Science	SFT-MAT-VALIDATION-ELECTRONIC-IONIC-DIELECTRIC-VECTOR-006	Electronic, ionic and dielectric validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:857f130a81d18da493390bd38c96dcc82521f45f01dd594259b36c9f2157fb1f
1676	Materials Science	SFT-MAT-VALIDATION-MAGNETIC-SUPERCONDUCTING-TOPOLOGICAL-VECTOR-007	Magnetic, superconducting and topological validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:136971745a3094bd8b155e164990a41b172b99d33bb7217ea211be34cdc06765
1677	Materials Science	SFT-MAT-VALIDATION-OPTICAL-PHOTONIC-VECTOR-008	Optical and photonic validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:20d2650407f790688fc8c8deae94315a17deb8235834956653bf5e970a755bbb
1678	Materials Science	SFT-MAT-VALIDATION-MATERIAL-CLASS-PROCESSING-VECTOR-009	Material-class and processing validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f8fd9d1cd1bcf8cbefc282a747f11f0fdf688f47aa9a0e743b14d5ffda451b35
1679	Materials Science	SFT-MAT-VALIDATION-CROSS-SOURCE-REPRODUCIBILITY-VECTOR-010	Cross-source Materials reproducibility vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5fdf955b9220bfff8d9fe0fce5754df31ff5b7db027b18c6c520233b4c5cc1c7
1680	Materials Science	SFT-MAT-VALIDATION-ADVERSE-ABSENT-OUT-OF-BOUND-VECTOR-011	Complete adverse, absent and out-of-bound Materials vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6c3a8fb1a8b0444d37a25fbb55244b4c5fb9130e991f8eb98d0cedf46e17c070
1681	Materials Science	SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012	Materials empirical Grand Lock	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dbb75828d061c43824bfad454e100d0502d76b5ebc3ee74e9dd03cbcd8967268
1682	Materials Science	SFT-MAT-ENGINEERING-OWNERSHIP-HANDOFF-001	Materials-to-Engineering one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be4f56a688538024ce7e0f47e3484316d97b382760f01ab93d795169959f5312
1683	Materials Science	SFT-MAT-BIOLOGY-OWNERSHIP-HANDOFF-002	Materials-to-Biology one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f24161d22e77e070e6bd4b052173fe4072996c01e3a29c7755c1dfa8d3252293
1684	Materials Science	SFT-MAT-MEDICINE-OWNERSHIP-HANDOFF-003	Materials-to-Medicine one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cf60886a12c3826e6f3fbcfbd12f4e46b86e24bd573d8696e203e967bd2409f3

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1685	Materials Science	SFT-MAT-EARTH-ENVIRONMENT-OWNERSHIP-HANDOFF-004	Materials-to-Earth and Environmental Science one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cec7220113f95cacdefc5b3154df9194d625bc0f52c73000b227279ed5981cee
1686	Materials Science	SFT-MAT-ASTRONOMY-OWNERSHIP-HANDOFF-005	Materials-to-Astronomy one-owner handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d3ab4763a02984075b5d379e1ef1425b5de071c0bf584800a66465d52cc566b9
1687	Materials Science	SFT-MAT-CROSS-BRANCH-ONE-OWNER-COMPLETENESS-006	Materials cross-branch one-owner completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7ca27841a01815e2ff3402430fcb3c11115962de21e7c1d07d27005a60a5d7e
1688	Mathematics	SFT-MATH-ARITH-GENERATED-SUCCESSION-001	Generated whole succession and exact induction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d3229eb70862bcc3a5d03f42c9abfc650ab90781d45fc2d5090abf8322d2ca08
1689	Mathematics	SFT-MATH-ARITH-JUNCTION-ADDITION-002	Addition and disjoint-junction arithmetic	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ecbb32a37fb10022315429bb39d88f49437fc856396d0b8db42fca49b2032493
1690	Mathematics	SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003	Multiplication and complete pair-cell arithmetic	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e25cde93469d6d813be366bc228e55cb13e704a965e60711ab396c6f7a36f476
1691	Mathematics	SFT-MATH-ARITH-DIVISIBILITY-GCD-LCM-004	Divisibility, common divisors and common multiples	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e02b5bc7444746ceaf5a115853d012583abae63804f6531b39727fc504424e7e
1692	Mathematics	SFT-MATH-ARITH-QUOTIENT-REMAINDER-005	Exact quotient and oriented remainder	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4b082b2a5fd4bed326e2ebaa180ff23aa3d7bcb0475947732bedf9d65f821af6
1693	Mathematics	SFT-MATH-ARITH-PRIME-IRREDUCIBLE-006	Prime and irreducible whole structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d74f03c498a284dcb6a576786ce45a09c87dd921ac78db3ce71bcd54247bc018
1694	Mathematics	SFT-MATH-ARITH-UNIQUE-FACTORIZATION-007	Unique finite factorization certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:12882efe837fb967097e5f5540d48077cd12b48bb0c37662fbd6e0052c797ac
1695	Mathematics	SFT-MATH-ARITH-CANONICAL-FRACTION-008	Canonical exact fractions and common refinement	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:45b93a4d1bb7a137599e21c74e7e9522054f6b741433365d957dd06fc703a3a1
1696	Mathematics	SFT-MATH-ARITH-CONTINUED-FRACTION-009	Finite continued-fraction correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b18a07076f76362a892ce1ed995e8b67e1af5c41ecdde90b16ea34a35dfa886d
1697	Mathematics	SFT-MATH-ARITH-CONGRUENCE-010	Residue classes and congruence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dfc025633b4bc068fab1d51dd8596eaf531d428a8b8d83d5c49b015b1311b62a
1698	Mathematics	SFT-MATH-ARITH-COMPATIBLE-CONGRUENCE-011	Compatible congruence composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f15dd91147438a821fcedb1cd0abcab032151e1393fea1766f02d97fe5d4d64f
1699	Mathematics	SFT-MATH-ARITH-VALUATION-012	Prime-power valuation and divisibility depth	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3f73120fde18553beedeec0de54e9ee4834b9eb9ebe19976ddb38b251d1bfffac

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1700	Mathematics	SFT-MATH-ARITH-DIOPHANTINE-ENUMERATION-013	Diophantine relation enumeration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2a078b638cd00e98a59fb4ccd6a47e9b691c4a713710bdbb4112062441c41085
1701	Mathematics	SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014	Recurrence laws and exact sequences	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a8a9c91bf09055eafe7c8946783af33d8cbf408241c7a0cc65b022fc8a85393d
1702	Mathematics	SFT-MATH-ARITH-GENERATING-FUNCTION-015	Finite generating-function correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:18143d49ddbc35dfaa29e6333b0529d9c93c269483c06206f79a2bd465d2bbfb
1703	Mathematics	SFT-MATH-ARITH-PARTITION-COMPOSITION-016	Whole partitions and ordered compositions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:322408091620c6dfed8b960578f5744a202f7f05536bc0b348f9e62cccb1b9fb
1704	Mathematics	SFT-MATH-ARITH-ARITHMETIC-FUNCTIONS-017	Arithmetic functions and divisor ledgers	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a0126b838d3f636788373c0a95adc8ff48e8a9d9b1d7c0797aae6403fda8473e
1705	Mathematics	SFT-MATH-ARITH-PRIME-DISTRIBUTION-ENCLOSURE-018	Finite prime-distribution and growth enclosures	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2d9cb180af9554925c3ff156e2b05d1545b9dba2206a24b60c0e17023e0e9fe2
1706	Mathematics	SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001	Polynomial identity and exact root isolation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:37fd63da2da1d8bff586bea60bebad91ef43d0115437e9efd7e360dbf05b94f0
1707	Mathematics	SFT-MATH-ALEXT-ALGEBRAIC-BALANCE-002	Algebraic-magnitude balance certificates	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:41100b9b02e60bc1321f4ffa7590f3f2c5ef6223f3983e323a4177d53bde4a4e
1708	Mathematics	SFT-MATH-ALEXT-EXTENSION-TOWER-003	Exact finite extension towers	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b4db0d57fd93d9489c8d94fe9bae1167b9ffa29c775db04f909dc9c707cdbc922
1709	Mathematics	SFT-MATH-ALEXT-FINITE-FIELD-004	Finite-field correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3e417f42c092e58b78ceda9477a4d4c3303b8a1cb0896c6d3623f387edc24824
1710	Mathematics	SFT-MATH-ALEXT-GALOIS-ORBIT-005	Finite Galois-orbit correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2dff8bb2232a8572a0c20e34c05fc873f83f2e9cb74d008cc32b7b8ff2d3d26b
1711	Mathematics	SFT-MATH-ALEXT-CYCLOTOMIC-CORRESPONDENCE-006	Cyclotomic and root-of-unity correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:81f843b80e1f07e44b1f212eeef448f604aef89255cfe08dd26ff5fd8c5c253
1712	Mathematics	SFT-MATH-ALEXT-HELD-PAIR-COMPLEX-007	Held-pair complex-number correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ccaaa6636f523d2e6bc6cd13996d49a0c748dbc4378568188f6e307de4bfc48d
1713	Mathematics	SFT-MATH-ALEXT-REAL-ALGEBRAIC-ORDER-008	Exact real-algebraic ordering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1492db9e71b8fbd66da325cd1c27997d144df3e1674818eda01f3b9e11547d2b
1714	Mathematics	SFT-MATH-ALEXT-PRIME-ADIC-VALUATION-009	Prime-adic valuation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c38b6cbec7e61e8fbfa7be7ca0a1ab4eb3af572576521a37c9cf03acde3bb413

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1715	Mathematics	SFT-MATH-ALEXT-TRANSCENDENTAL-BOUNDARY-010	Transcendental and nonrepresentability boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:674c5e710359cd7da9382006ad39c9539591fc9f0ba8d7bd9728ca3eb804a010
1716	Mathematics	SFT-MATH-COMB-COUNTING-LAWS-001	Product, sum and bijection counting laws	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:afca8ab23f33cbad39e5aa6e51d5df24bfddd946cfa5568f46e6c781b5503d4f
1717	Mathematics	SFT-MATH-COMB-PERMUTATION-COMBINATION-002	Permutation and combination enumeration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9b95112577a4816525ad072bf79acf3c8a573c98847872d16c79f5161d11625a
1718	Mathematics	SFT-MATH-COMB-INCLUSION-EXCLUSION-003	Inclusion-exclusion with complete overlap custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5d43c44b342b930315faa5159e3ab7ec026fc538d9b3f77bdd56ff7826ce098e
1719	Mathematics	SFT-MATH-COMB-PIGEONHOLE-OCCUPANCY-004	Pigeonhole and occupancy forcing	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c40a635564aae5ef85402c7b69e06ec78ed0024112d2084b58fcc0a2ccf7e004
1720	Mathematics	SFT-MATH-COMB-RECURRENCE-GENERATING-005	Recurrence and generating-function counting	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ba8380cefcf6fd4b157b57674588bacdf9c5565bc3e848f3638882df1bc3a1f
1721	Mathematics	SFT-MATH-COMB-PARTITION-INCIDENCE-006	Integer partitions and Young-type incidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2d097fe60b69c55a86accb8d417c083c3bdbe07041e0eb058081887689943664
1722	Mathematics	SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007	Extremal finite-set systems	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e01c8059812a3e920d2f722ef9f47e2e3375ae255da534d20c5efcab589e17c2
1723	Mathematics	SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008	Probabilistic-method correspondence without ontic randomness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d650fb44b26a7778b1431e877854dbcbaed78f8612c57d47d6e80118fc5f37d6
1724	Mathematics	SFT-MATH-COMB-DESIGN-INCIDENCE-009	Design, block and incidence structures	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d1e03a25a63c3099f5d2a8f414ad7815c8471db3b30581e48a08d58a39dcf982
1725	Mathematics	SFT-MATH-COMB-CODING-PACKING-010	Coding and packing combinatorics	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b5ede9884b69fb563c6774eacd7bd3df38cd803bb164cb775e000e7d1cb8c2ac
1726	Mathematics	SFT-MATH-COMB-RAMSEY-FORCING-011	Ramsey-type finite forcing boundaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2a8c17b72cdefc363e819e2019edbf9aecf13d12ea569b64b0bee4dfa8f01cf2
1727	Mathematics	SFT-MATH-COMB-SPECIES-COMPOSITION-012	Species and compositional enumeration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c7ab86cf76ff13e5c47488f640d8d34afa30fb2008fc16e4a273562b3f4e2c98
1728	Mathematics	SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001	Graph identity, adjacency and isomorphism	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:489635c31e29a277498395071437690e60c00fd3287af337754ed86cd2d1c9a8
1729	Mathematics	SFT-MATH-GRAPH-PATH-REACHABILITY-CYCLES-002	Paths, walks, reachability and cycles	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:164d8d5708f6238d6c7864f2b2261dea4a4097d12c335f68d56765cea778cc01

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1730	Mathematics	SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003	Connectivity, cuts and flow support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:383274c9576c7f228b386a5eb969fb995d051ca71d963c8068014712304be26c
1731	Mathematics	SFT-MATH-GRAPH-TREE-FOREST-SPANNING-004	Trees, forests and spanning structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f81889343d13708d3cc085df21f33b4636619dea455cbf58cd4df9abada48e57
1732	Mathematics	SFT-MATH-GRAPH-PLANARITY-EMBEDDING-005	Planarity and embedding correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9b310cbbfe57e1eca969de9a2e73c7136ce86428d741a10cff9e7e1170fa24c4
1733	Mathematics	SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006	Colouring and constraint partitions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:361fb44c8f989543dacde61ce4b3d03f6f6096ab9fba773664b339627f275671
1734	Mathematics	SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007	Matching, covering and packing	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:283c65166202c674101581fbcf1b718bab3f5ad45ddcf38b4375f34480fa11f9
1735	Mathematics	SFT-MATH-GRAPH-DIRECTED-CAUSAL-REACHABILITY-008	Directed graphs and causal reachability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a3ea7e43de3c0a16186153b9855e4b9f90acc8b096a2767bc84d0002b4f5e66c
1736	Mathematics	SFT-MATH-GRAPH-WEIGHTED-NETWORK-EXACT-PARTS-009	Weighted network correspondence with exact parts	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:95711a8c91c539d061e323813786738ecf6f65fb4d800ce7a7619e1a64bc993
1737	Mathematics	SFT-MATH-GRAPH-HYPERGRAPH-HIGHER-INCIDENCE-010	Hypergraphs and higher incidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9ce82e3125329f6738cca2ab5b89ef6bbd92f237af6cf1619160351d5d220b69
1738	Mathematics	SFT-MATH-GRAPH-MATROID-INDEPENDENCE-011	Matroids and independence systems	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6bf043d86d7d1d7a8a860124bf51cdb5f9bb780586d937880377b4fe0175c65e
1739	Mathematics	SFT-MATH-GRAPH-RELIABILITY-FAILURE-CUSTODY-012	Network reliability and failure custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a2827845f35854c3fc3774078d4e794b651ee10ec65995f2cbb14a31c50f72ab
1740	Mathematics	SFT-MATH-GRAPH-SPECTRAL-CORRESPONDENCE-013	Spectral graph correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dfd637109db6dfdc9a1ebd98df9ce928782ddec159f8310428d7154178d4ce5
1741	Mathematics	SFT-MATH-GRAPH-DYNAMIC-TEMPORAL-NETWORK-014	Dynamic and temporal networks	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fb1a3980a9e3273950f1ad3d9426d228287085c3cba62162735538726bde0626
1742	Mathematics	SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001	Vectors as exact generated coordinate carriers	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:61163023459ed332dfc906e44bcb647778587e2db0c85e32b9390334de8675e9
1743	Mathematics	SFT-MATH-LINEAR-MAP-COMPOSITION-002	Linear maps and composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b0ff81fb4f50d49ba8e11c1e0dc7f414ff7711bf5ba1ec4d70cd9af6cdf7f5fe
1744	Mathematics	SFT-MATH-LINEAR-MATRIX-ROW-OPERATIONS-003	Matrix representation and exact row operations	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a069bdc5521cb9dabf4978a906631c9f795c332771bfd6ee7f7bad56b0d37e07

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1745	Mathematics	SFT-MATH-LINEAR-RANK-NULLITY-004	Rank, nullity and retained distinctions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a111ld9fdee741f94a9f57268078c4f8935bdc8fcddec9419a2f858d67418bc9
1746	Mathematics	SFT-MATH-LINEAR-DETERMINANT-ORIENTATION-005	Determinants and orientation custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1eb50aad30609be2adbf13fd7589e8fdb3013d5a4e0e25292b4c27d2cda0587d
1747	Mathematics	SFT-MATH-LINEAR-EXACT-SYSTEMS-006	Exact systems of linear relations	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:df388656cff7ee3d4afa27c2c5efb2dad840fb0abfbdbec457eb2eb6f3e22a10
1748	Mathematics	SFT-MATH-LINEAR-BASIS-DIMENSION-007	Basis, dimension and change of basis	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f9cc18b562f233e53210a5788786195759321f2d9484faa0188a05c46c192de9
1749	Mathematics	SFT-MATH-LINEAR-INNER-PRODUCT-METRIC-008	Inner-product and metric correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bf412fb7caf2d4fe721ab9d01381a552a490d1b789b8469eb534a287de9da355
1750	Mathematics	SFT-MATH-LINEAR-EIGEN-INVARIANT-SUPPORT-009	Eigenvalue and invariant-support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f4dd00fe4be2cb27fec953298d56913b6299c2c17cf34cec00a7fcd1d71ccb33
1751	Mathematics	SFT-MATH-LINEAR-RATIONAL-SPECTRAL-ENCLOSURE-010	Exact rational spectral enclosures	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78bfa64c5f59b2314c1137839cdb8e8cef9778d056a094837ec553a1c5ddffc2
1752	Mathematics	SFT-MATH-LINEAR-MULTILINEAR-TENSOR-PRODUCT-011	Multilinear maps and tensor products	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9a0648891c36fa2004a4130cf182de9f9aed20f746ed391c30dd8b657c6a2b53
1753	Mathematics	SFT-MATH-LINEAR-TENSOR-CONTRACTION-012	Tensor contraction and index custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:144d72725402644b443e0674b9a7d36c0ac3759527fb55ae4a664fbf31be1016
1754	Mathematics	SFT-MATH-LINEAR-EXTERIOR-SYMMETRIC-013	Exterior and symmetric composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7514349b966d1a39444e5f841909f6e00a777c31d426db7a12a91da6e9013351
1755	Mathematics	SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014	Finite-dimensional operator decomposition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:52a0d942eb72cefec9a693af78189036fe049d1da9a5ab8b2f1c85c61ce74a
1756	Mathematics	SFT-MATH-ALG-MAGMA-CLOSED-OPERATION-001	Magma and closed operation structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c5e93e05277f7feb172cc3a7003518d9210ad7be5ad8771ba88a6dd397f067b80
1757	Mathematics	SFT-MATH-ALG-SEMIGROUP-ASSOCIATIVITY-002	Semigroup associativity witnesses	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cf47d05b924a577416959ddc022d305c6aa02509046c37239b125f1e0fe13ab4
1758	Mathematics	SFT-MATH-ALG-MONOID-IDENTITY-003	Monoid identity witnesses	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:beb8f7c7965680020a19391958ee25ec44998ca6b33feb9978f7adfd7a1fe5
1759	Mathematics	SFT-MATH-ALG-GROUP-HELD-INVERSE-004	Group inverse as held reversal	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94ea42e4036ab10f3e1eb8cdab2e194e18a1a053c664f1070fbac6a7a7c73152

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1760	Mathematics	SFT-MATH-ALG-PERMUTATION-GROUP-ACTION-005	Permutation-group action	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4fe452d169159708a9cb65c9f42edfba611f0d86fc6143026a5122b839fba278
1761	Mathematics	SFT-MATH-ALG-QUOTIENT-NORMAL-SUBSTRUCTURE-006	Quotient and normal-substructure correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d468d9fb67765ecaa8c7b3f2e1e0404a27e07bbdb5843f943e01c62ac2d4f7b6
1762	Mathematics	SFT-MATH-ALG-RING-DISTRIBUTIVE-007	Ring distributive structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2207ec02691fbc1989e0ae93e324ebb034047996b14277b521774ae634cac4fd
1763	Mathematics	SFT-MATH-ALG-INTEGRAL-DOMAIN-008	Integral-domain boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:69176d3a233e9eaa708fbbf5517c30b9cb3b7148f4f6cf12562d35d788b4f6b1
1764	Mathematics	SFT-MATH-ALG-FIELD-EXACT-DIVISION-009	Field correspondence under exact division	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d1e1bc506263760193b0b99b66e3f7669ec9a2581ab2e4e10bb3f5ac656dd7ee
1765	Mathematics	SFT-MATH-ALG-MODULE-SCALAR-ACTION-010	Modules and scalar-action structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bb21a886d6f54b9c232c9e4c6abb37c78d20372f1b3aa8c3a30b7e9795cc5fdf
1766	Mathematics	SFT-MATH-ALG-COMPATIBLE-ALGEBRA-PRODUCT-011	Algebras and compatible products	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2170e5d46c51d5cac7879b84b7466bad4f891683985fa16d920f0a0f1f4b3a6f
1767	Mathematics	SFT-MATH-ALG-IDEAL-QUOTIENT-012	Ideals and quotient structures	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:51014e51baa2db97fcb3c486c521aa813b08e33541a38c791abf465958ecfc3e
1768	Mathematics	SFT-MATH-ALG-REPRESENTATION-ACTION-DECOMPOSITION-013	Representation and action decomposition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:57cbe383561e1219e2ebae0731dba20a8a73b7f2961313cd20ff89c3af3ca2e8
1769	Mathematics	SFT-MATH-ALG-EXACT-SEQUENCE-HOMOLOGICAL-014	Exact sequence and homological correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:22db8ecef3c0225e195f7d08ed2011e187177df110bb9220cd2aa76d4337380
1770	Mathematics	SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015	Universal algebra and identities	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:88a76935879afa29f6f647396afa0711a5626baae4107b68e4538a57f03c1a73
1771	Mathematics	SFT-MATH-ALG-OPERADIC-COMPOSITION-016	Operadic algebraic composition interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c45a860ee26cca756e9d1120e40d217d8255f5d68495e8cd9b3b275311081f7a
1772	Mathematics	SFT-MATH-ORDER-PREORDER-QUOTIENT-001	Preorder and distinguishability quotient	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:62b8a349524faaad77513549e980c4f3d9f3960dfeaa5df04bd0f7338e4cb
1773	Mathematics	SFT-MATH-ORDER-PARTIAL-ANTISYMMETRY-002	Partial order and antisymmetry	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:90737e2dd2e996533513d1d6b2dedb0453101ced5dbf1972fd12672d7eb09ee2
1774	Mathematics	SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003	Conditional total order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fa8f185e14ec8d4e621b22db29f13bcf50d9c9a3b1101e434b148b6bc25ac5f3

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1775	Mathematics	SFT-MATH-ORDER-MEET-JOIN-LATTICE-004	Meet, join and lattice structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ef972980b97f3eae7b7060fcca7ce8f6b2d49f86cabcf3ddd23852195c7b44d
1776	Mathematics	SFT-MATH-ORDER-DISTRIBUTIVE-MODULAR-005	Distributive and modular lattice correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c67a3e785599e9dd49003227fef98d588bc99d2a18d7d267e1ea1e7646eacf89
1777	Mathematics	SFT-MATH-ORDER-BOOLEAN-COMPLEMENT-006	Boolean-like complement correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bab7103a46409d68b5275b3e60e12a6b7457763dbbc8cbf5b18d859d44d652f2
1778	Mathematics	SFT-MATH-ORDER-CLOSURE-SYSTEM-007	Closure operators and closure systems	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8c63e67ef5d6b6d785dbfb1bcebe0df202af2908650236db7902c13e7ef0171a
1779	Mathematics	SFT-MATH-ORDER-GALOIS-CONNECTION-008	Galois connections between orders	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e68701f93b8bc3b1b2994f1701d31f2a56e3ed49d54306f13f605e7b9cfb2fd2
1780	Mathematics	SFT-MATH-ORDER-DOMAIN-APPROXIMATION-009	Domain approximation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94ae3c06d59a7cc4fd1de2f6db35ceb1fad52c343e8622360f94e14f65ec5087
1781	Mathematics	SFT-MATH-ORDER-MONOTONE-MAP-010	Monotone maps and order preservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:053130750729ea82310b9c55b17d1d9113d05666aaac6d2639b86863c497b8d5
1782	Mathematics	SFT-MATH-ORDER-FINITE-FIXED-POINT-011	Fixed-point existence on finite orders	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d81be77c0466b0b27fc0c703c9e40c284f2052c4663d8ae491bf23f7e1f2e7bc
1783	Mathematics	SFT-MATH-ORDER-COMPLETE-LATTICE-BOUNDARY-012	Complete-lattice correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c058a971d9c63aea9690da203a2bafd33ffa85c3b456ed893c0fbc483d1df96
1784	Mathematics	SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001	Point, incidence and exact coordinate identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6e21c474249b5f7416086cffdfa67b0dc82ae3b3b5ec3e085f966a51923489c2
1785	Mathematics	SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002	Finite Euclidean-distance correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:476f77266580be2d60d3958a6ac59279ff4c0f30757ec7a0034dc792cf4a0b3b
1786	Mathematics	SFT-MATH-GEOM-AFFINE-INVARIANCE-003	Affine combinations and affine invariance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:10e679a7ee169bd7d4db58480e7b08a4db29a2bcce9a0071775a143ea077fbf6
1787	Mathematics	SFT-MATH-GEOM-PROJECTIVE-PERSPECTIVE-004	Projective incidence and perspective	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa1b4a590e43ae08efb84276484f8e9bf0bd6fa76611eea1d1dff07ccb996a88
1788	Mathematics	SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005	Convex hulls and separation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6c66327e91dff8df94b11eddc42778bb8ab3d86e174401333decdbd3d033aa5
1789	Mathematics	SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006	Discrete geometry and lattice polytopes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5f1da049f9226f536b49893ca4fa13e2f7f454f53bbcb2f6ec3ec46ec650e7d1

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1790	Mathematics	SFT-MATH-GEOM-POLYHEDRAL-EULER-INCIDENCE-007	Polyhedral faces and Euler-type incidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f7150f1b221d7610f49c7e175d12578d3c5e646cf4834b803f2fd87254335532
1791	Mathematics	SFT-MATH-GEOM-COMPUTATIONAL-PREDICATES-008	Computational geometry predicates	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:88bd71da009f553ffc0ea404616ad3a4269dfcd0c2e3d233ed1f24b8abb30774
1792	Mathematics	SFT-MATH-GEOM-ORIENTATION-INTERSECTION-009	Exact orientation and intersection tests	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:24596a0663ce622201d42208bb517558807fa73ee356f40d32ea7466a6045b93
1793	Mathematics	SFT-MATH-GEOM-ALGEBRAIC-SOLUTION-SET-010	Algebraic-geometry solution-set correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cfa6ae0f410c735a5d4a2a782ae80bd70e149682beec68d45bb58fa3a740b1d2
1794	Mathematics	SFT-MATH-GEOM-DIFFERENTIAL-FINITE-CHART-011	Differential-geometry finite-chart correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f3e05eae5cba1877fe920845cf8b489897d358f61b57737840d3266a4d89fb6
1795	Mathematics	SFT-MATH-GEOM-CURVATURE-FINITE-TRANSPORT-012	Curvature by exact finite transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c402bd060dc95d06eeef218f4ecbbeaa4cb2d907e73d3d0208703f4cbeba24f34
1796	Mathematics	SFT-MATH-GEOM-METRIC-GEODESIC-013	Metric geometry and geodesic correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:09605a920fde7f770e6951a2b9d2d1ec641a89dd55fb2a7c00ad3ae9c7766007
1797	Mathematics	SFT-MATH-GEOM-FRACTAL-SELF-SIMILAR-014	Fractal and self-similar geometry	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:695c16a32dd6cc22ac2f0fca7817a89088b58cd1a88b7f6d1c7b3d1c7d104a60
1798	Mathematics	SFT-MATH-GEOM-PACKING-COVERING-TESELLATION-015	Packing, covering and tessellation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f0e9e9bfb4de8cc2f7419c8a96325a040497b09b4cfa3457fc6fe4b2799ec25
1799	Mathematics	SFT-MATH-GEOM-TRANSFORMATION-GROUPS-016	Geometric transformation groups	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f601584092d818d2180fa5b6b32aff5089152b8a2b29d31a826ad2f106658c2e
1800	Mathematics	SFT-MATH-TOPO-OPEN-SET-SUPPORT-001	Open-set correspondence on generated supports	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:087501b54410e5b0cc2582c58f56e0ae422ff60ae8e112669ad6257904b828f6
1801	Mathematics	SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002	Continuity as distinction-preserving transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:573c965f1d7c4ca35f06606ef420cbd243824da0a33ef5239ed3eeda9b39bc11
1802	Mathematics	SFT-MATH-TOPO-COMPACT-FINITE-SUBCOVER-003	Compactness finite-subcover correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:093d2d6a7de58e145ed538bc8ca98e21197f57236c9895ecc688896aef4f0da1
1803	Mathematics	SFT-MATH-TOPO-CONNECTED-COMPONENT-004	Connectedness and component structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:15563c18cf9cb4909fed4801546848b883a4a532e5b92a19d7f865abb66942a
1804	Mathematics	SFT-MATH-TOPO-SEPARATION-FINITE-005	Separation properties on finite observations	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c5e7396091b5d6f67fbda3a6b709c3c2e0a65a549376de10eb271a3bc52e5f40

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1805	Mathematics	SFT-MATH-TOPO-PRODUCT-QUOTIENT-SUBSPACE-006	Product, quotient and subspace topology correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a82e2923c6b54b8bc343741e75582056e2c37a5058376ca947c0c07d15cff42b
1806	Mathematics	SFT-MATH-TOPO-SIMPLICIAL-INCIDENCE-007	Simplicial complexes and incidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8ab47bcdcb0b0a32fffe001651b90b87a33462ee55f203ffa30300664c917ee346
1807	Mathematics	SFT-MATH-TOPO-HOMOTOPY-PATH-DEFORMATION-008	Homotopy path-deformation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e9a150b12a6b2c46ae0e816f04f943af98073118526fbdc446d96bf870f1d580
1808	Mathematics	SFT-MATH-TOPO-FUNDAMENTAL-CYCLE-GROUP-009	Fundamental-cycle and group correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:70093215c70da4bb5253edf8b7a016a6788c5158c670ce96b24413a952a81ff4
1809	Mathematics	SFT-MATH-TOPO-HOMOLOGY-BOUNDARY-010	Homology and boundary composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:01fa67cf3367f6ba1401e220a59a69e8e81be36b53ae713f95149e6638f3286b
1810	Mathematics	SFT-MATH-TOPO-COHOMOLOGY-DUAL-OBSERVATION-011	Cohomology and dual-observation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:daa2f48449e9a9b802fa8193d8a7298683680fc55118d1963caa2ecf2037475a
1811	Mathematics	SFT-MATH-TOPO-MANIFOLD-FINITE-ATLAS-012	Manifold finite-atlas correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:06b0bd43fcc4ff33c4522a44cd3425d14dab1d0c833e8f5993f553567abf6865
1812	Mathematics	SFT-MATH-TOPO-KNOT-LINK-INVARIANTS-013	Knot and link finite-diagram invariants	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5896d19c43946aaf7ea803a46d656d83877e3a33bfe835fb25652e7d6b0a0cc0
1813	Mathematics	SFT-MATH-TOPO-PERSISTENT-FEATURE-CUSTODY-014	Computational topology and persistent-feature custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cb7338023bc31ad5a784693d1b348ba04c7808a83dd59d02d8bb91602176fe4a
1814	Mathematics	SFT-MATH-CALC-FINITE-DIFFERENCE-001	Exact finite difference and local change	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:507dcb047bbe74bb826a5e6ed551568a04827638d7e02341b616ed21b9441a74
1815	Mathematics	SFT-MATH-CALC-HIGHER-DIFFERENCE-DEGREE-002	Higher finite differences and polynomial degree	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:06e93214674f5a4c1ee6bea0c9670845143bb157b4562a64c80028e916a297fd
1816	Mathematics	SFT-MATH-CALC-ACCUMULATION-SUMS-003	Accumulation and exact finite sums	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:64ca281a0dcc6cc417fdc9a401001f9ca15e65053726e2f599a1b8dbfd183d1e
1817	Mathematics	SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004	Fundamental difference-accumulation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e6035a82d486ed59b79d120ed341ccaeef17ab11c8e492b828ce58dec5f17579
1818	Mathematics	SFT-MATH-CALC-PRODUCT-COMPOSITION-LAWS-005	Product and composition difference laws	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0b230f54b347912d403b5661a468156a48a9d79b28c09053b666552ff4980873
1819	Mathematics	SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006	Rational enclosure convergence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:09ae9608a9587eadc2d1fe6741cd1840ccb16bf9d0cb4eb8cedb0ca11b35bf24

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1820	Mathematics	SFT-MATH-CALC-DERIVATIVE-SHRINKING-PARTS-007	Derivative correspondence through shrinking exact parts	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0df975977b5212414b32cf5991f27778c811c0d19b9c4640a377ef2748077376
1821	Mathematics	SFT-MATH-CALC-INTEGRAL-REFINEMENT-SUMS-008	Integral correspondence through refinement sums	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9dae94ea1c1bb7f5cc021fc5a5e54daa799b2b96cefbbab47d0ce3c089b95e3d3
1822	Mathematics	SFT-MATH-CALC-MULTIVARIABLE-DIRECTIONAL-009	Multivariable difference and directional change	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cd9c565f4ff93e6039bb09ff0e461c6f3a0ebb1201efc18962274d45d27eb633
1823	Mathematics	SFT-MATH-CALC-DIVERGENCE-FLUX-010	Discrete divergence and flux correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d5e9fald6b35ecd29795a2de9d3edda147d6cb081b7eb4bda68b5734e410bfb
1824	Mathematics	SFT-MATH-CALC-VARIATIONAL-STATIONARY-011	Variational difference and stationary structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:def0ef022d70876425bc65b54b4bdd196317d3acb4961a76190a959ca54a02dd
1825	Mathematics	SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012	Continuum-limit admissibility boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f5917674c7af39fbe08ef5fad2ea7bdd850218b38b72af99fa3d3c75b81b348b
1826	Mathematics	SFT-MATH-ANAL-SEQUENCE-CONVERGENCE-001	Exact sequence convergence certificates	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ab97f2a13f8e91555f3f654b3d1fce419aaca2e71b6964550a971abe5aa5073b
1827	Mathematics	SFT-MATH-ANAL-CAUCHY-SUPPORT-002	Cauchy-type generated support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:75b3fde9dbd4c94ec57c3214b75bc6956a83af140ea077502b3613a8f2161cb2
1828	Mathematics	SFT-MATH-ANAL-COMPLETENESS-CORRESPONDENCE-003	Completeness correspondence without completed continuum	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:096b603a26d7963ec79cbd3a02212e6051b0183db3a19933de1a1bfbfe1dbfe7
1829	Mathematics	SFT-MATH-ANAL-SERIES-REMAINDER-004	Series convergence and remainder enclosures	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9ed9ee5292bcc297c39b1079e4de4067f14d8b23cb72f38fb0be39ae8e78bf14
1830	Mathematics	SFT-MATH-ANAL-POWER-SERIES-TRUNCATION-005	Power-series finite truncation custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b1c09495ffe4a58ae36b3f567a88ed80ddb8ef6a08deba96278aefc38688a0c8
1831	Mathematics	SFT-MATH-ANAL-FUNCTIONAL-SPACE-REPRESENTATION-006	Functional-space finite-representation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:084a51e26bac975e955191ce1370b9f77e05fa555c1685d6b7b93879e027177c
1832	Mathematics	SFT-MATH-ANAL-NORM-SEMINORM-METRIC-007	Norm, seminorm and metric correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:07d66c5b288d72dcfae3829b10af6fa129b261c4171007ad7ea387dbe1d56eb2
1833	Mathematics	SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008	Bounded and compact operator correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0770111ad74222f7d56d86d7798010faa49c9ae065c1b83282848d5792005267
1834	Mathematics	SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009	Harmonic and Fourier finite-support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:963056642837c7794301e5fb6787e9a61f3d61eba474eeaae6be0b67ad63449f

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1835	Mathematics	SFT-MATH-ANAL-TRANSFORM-INVERSION-010	Transform inversion on generated supports	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d5504ccb27de0398792565ba73813476b426919a9c01fe1490c799478bb b2946
1836	Mathematics	SFT-MATH-ANAL-CONVOLUTION-CORRELATION-011	Convolution and correlation identities	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e9d6cfff64a2340ae44d6c448e4c3f2f21b13b5401fb80abff6a9c15a3f6f0a7
1837	Mathematics	SFT-MATH-ANAL-ORTHOGONAL-BASIS-EXPANSION-012	Orthogonality and basis expansion correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be232594554c9502d6329fe8b147c2ad5c13efb6f0aa44a1fb6cce82be5be7ec
1838	Mathematics	SFT-MATH-ANAL-DISTRIBUTIONAL-WEAK-OBSERVATION-013	Distributional and weak-observation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:45c9743262487c3681ba1a4d8b290e0a4ccee29cdae06ef19dfe4e2d66f203a
1839	Mathematics	SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014	Nonlinear analysis and contraction boundaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:beaae3861d939920aa65210f3279a53acd8d0416166a75c5f5caa7758ad0d638
1840	Mathematics	SFT-MATH-ANAL-COMPLEX-HELD-PAIR-015	Complex-analysis held-pair correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5d105963b77ef66e5184e30d31c9be4a513602bd2544b90ec59258defa6fad5d
1841	Mathematics	SFT-MATH-ANAL-SPECTRAL-MEASURE-016	Operator spectral-measure correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2b900f3b430de0fcb8c0be9b27c90e8fa9d99b870d3cb197a23db3ad0449a7a5
1842	Mathematics	SFT-MATH-EQN-ORDINARY-DIFFERENCE-001	Ordinary difference-equation structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4fc9003cf83d90dcf955003cca7001e1547e5d7e1602af161870186ae1fcf66b
1843	Mathematics	SFT-MATH-EQN-ORDINARY-DIFFERENTIAL-002	Ordinary differential-equation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3c2ca60887577a62c86587512cec9143f1cb08293ac1e0267f4a4c5466be5511
1844	Mathematics	SFT-MATH-EQN-PARTIAL-DIFFERENCE-003	Partial difference-equation structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78591458da34ac31e6419e6670802ae5100128e808c08b987cf5b34cc3324e59
1845	Mathematics	SFT-MATH-EQN-PARTIAL-DIFFERENTIAL-004	Partial differential-equation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6e4130baf805346be5ab289307002290c2f9b6833686c72a85f987be1e25e51a
1846	Mathematics	SFT-MATH-EQN-BOUNDARY-INITIAL-WELL-POSED-005	Boundary and initial record well-posedness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:201f70839a8f800c9b62f6fc4cb957e2d3330533c87533ea6d83b19d30027900
1847	Mathematics	SFT-MATH-EQN-INTEGRAL-CORRESPONDENCE-006	Integral-equation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:82ade58665afc230884f81bd940a5ef39596e8c5015fbb861185f237547aladc
1848	Mathematics	SFT-MATH-EQN-FUNCTIONAL-STRUCTURE-007	Functional-equation structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:23a0f72d36f5d6653020722360e364cd157fdce162fb6eaf881713a12936432d
1849	Mathematics	SFT-MATH-EQN-RECURRENCE-SOLUTION-SPACE-008	Recurrence-equation solution spaces	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:23579f5dbdd465547c8cce50c194a49101b72e8c30cd8dbd336ad962e8d2e802

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1850	Mathematics	SFT-MATH-EQN-GREEN-RESPONSE-009	Green-response finite correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9dcd5a3ee4736a56455470116d2c72297f47e59f886ce1950e123bdcf2565225
1851	Mathematics	SFT-MATH-EQN-CONSERVATION-WEAK-010	Conservation-law weak correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4fbb4a7f64ef3694210546883ed3742b90e0a11c995f921e5d3e9269b94522d3
1852	Mathematics	SFT-MATH-EQN-STABILITY-PERTURBATION-011	Stability and perturbation enclosures	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b5862177e99aea7b7d3f1f03e5d062b99b4072dcfde36163dc8ed7f07544c36f
1853	Mathematics	SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012	Existence, uniqueness and blow-up boundaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a5b9056f828669049f5d02eb1c9493657a229b7da2454f7cea9353b1b7665579
1854	Mathematics	SFT-MATH-MEAS-FINITE-SUPPORT-WEIGHT-001	Finite measure and exact support weight	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:14d25ad66d068b2319cec859ce7d94be7d66cefda1b4721b2ec5719b124b3e1f
1855	Mathematics	SFT-MATH-MEAS-DISJOINT-ADDITIVITY-002	Additivity on disjoint generated support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c2c5612d75dce3a55857b1e506e47f3b86496117aabcc4e83cb8b0b6d5a8aaf0
1856	Mathematics	SFT-MATH-MEAS-OUTER-COVERING-003	Outer-measure and covering correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0ee9c394f47d86ec4a0cb0e8812e82aaa1df94bf7739b6360a4e0fb257a1a95e
1857	Mathematics	SFT-MATH-MEAS-MEASURABLE-BOUNDARY-004	Measurable-boundary correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fc75d7acb7ac063472a62ef61784d3b2464536f8938410cea68282dd69f5936d
1858	Mathematics	SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005	Exact integration over finite support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d0b85a891e495ff46031d96111dd3c424206599b9def8efaa67a60bb8929fc04
1859	Mathematics	SFT-MATH-MEAS-REFINEMENT-SUM-INTEGRATION-006	Refinement-sum integration correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:22c4fe37ebac4cc637920ff3872ba28f8ed7011a639014be2aaa6760ac71581b
1860	Mathematics	SFT-MATH-MEAS-PRODUCT-CONDITIONAL-SUPPORT-007	Product measure and conditional support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eaffbc9318e599b402acbdba1cc51cd4dd49fb27e7f009d1d17fb2c09befc06f
1861	Mathematics	SFT-MATH-MEAS-HELD-ORIENTATION-SIGNED-008	Signed-measure replacement by held orientation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5d84b3dbf7725037c1ec28db61b83139d79504335292b4357a74539f98745017
1862	Mathematics	SFT-MATH-MEAS-DISTRIBUTION-OBSERVATION-009	Distribution and generalized-observation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f72e9d297fdc79cbd7e4ffa47e81d163a42965e347157648cefef1ea11934a22
1863	Mathematics	SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010	Convergence-of-measures finite witness boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a4f38ba621f479ce435299b06099cc716d53db25ee31cb5307d354573e3c1992
1864	Mathematics	SFT-MATH-PROB-SUPPORT-CORRESPONDENCE-001	Exact support-based probability correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f7e63c215ef725afbaed9e153dc0e9afbac2ae30fa90237c27aaa18c463d71e

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1865	Mathematics	SFT-MATH-PROB-CONDITIONAL-BAYES-002	Conditional support and Bayes correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3de32cd0f017cd9829917e109c48bf5510084b96451c28f5f1af69c3059ec3d8
1866	Mathematics	SFT-MATH-PROB-INDEPENDENCE-FACTORIZATION-003	Independence and factorization witnesses	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c9f382194e21c74dc4838e29a2e6edf7e17fc470b5a171283dc88a7f94004469
1867	Mathematics	SFT-MATH-PROB-EXPECTATION-004	Expectation as exact support accumulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b87c57beb9072c476180b813248201c2b22f3f7f98112c1e3069194cba7836d3
1868	Mathematics	SFT-MATH-PROB-VARIANCE-DISPERSION-005	Variance and dispersion without negative magnitudes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c20397f6b4dc7c33323fb18b25e87efcca673139404ef7103796a07479418a9a
1869	Mathematics	SFT-MATH-PROB-FINITE-DISTRIBUTION-006	Finite distribution families	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4e219f4e7826997fb21e65bba1e98f924b6ab6e847ef699dea331542e76eba86
1870	Mathematics	SFT-MATH-PROB-LARGE-COUNT-007	Law-of-large-count correspondence under deterministic enumeration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e9c8cd3b6e51c3a20e1640ce3b5815b87771a4f2b94f1d99e6b5aa02878d078c
1871	Mathematics	SFT-MATH-PROB-CENTRAL-LIMIT-ENCLOSURE-008	Central-limit enclosure correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1f4ece1dc9dcffc4ab540bcf7cd3dcf29d5bda76255a47b96ec80f3286808977
1872	Mathematics	SFT-MATH-PROB-ESTIMATION-SUFFICIENT-RECORD-009	Estimation and sufficient-record structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b10c05e8a85e8e31388d8b3fd77717a82dbb17564bb8841164daee3def78682f
1873	Mathematics	SFT-MATH-PROB-CONFIDENCE-CREDIBLE-REGION-010	Confidence and credible-region correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f264d01b3cb82e643590415a53b9ea4b9b0bac5e818de89426b4ceba2e0242b2
1874	Mathematics	SFT-MATH-PROB-HYPOTHESIS-ERROR-CUSTODY-011	Hypothesis testing and error custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:21aadee13d2df1e2737ad5c4cd7ba4ee10b6636b3810cf6458450ae67cd060b6
1875	Mathematics	SFT-MATH-PROB-LIKELIHOOD-EVIDENCE-RATIO-012	Likelihood and evidence ratios	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2a7a74ff1432e3aa098187148f14b63b47977082f22f28848036374dcb971858
1876	Mathematics	SFT-MATH-PROB-BAYESIAN-UPDATE-013	Bayesian update as exact conditional support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c239b75f04ae9b064f3a6178ed377874ca563585c6e751956e1ea995fcfc0b5d
1877	Mathematics	SFT-MATH-PROB-FINITE-STOCHASTIC-PROCESS-014	Finite stochastic-process correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:686defbd4442ac15c71a3bc349175da32ba506f8cfcec565be22b6813e24d50a
1878	Mathematics	SFT-MATH-PROB-CONDITIONAL-CONSERVATION-015	Martingale-like conditional conservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:467057cb14cfa1b348de39f0916d99f257be84b5c7ce10de85fd3932e5fb6e32
1879	Mathematics	SFT-MATH-PROB-IDENTIFIABILITY-016	Statistical identifiability and nonidentifiability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dfel62067f03fbfafd04d35e28dfc9de7e45e7e4ff14c9de674c9e7d3721d40

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1880	Mathematics	SFT-MATH-OPT-FEASIBLE-OBJECTIVE-ORDER-001	Feasible set and objective-order structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a13f3cdb8b9906ebb0792c32ddaf516caf949ad77bf00c032b8782cf7c15cc7
1881	Mathematics	SFT-MATH-OPT-FINITE-EXTREMA-002	Exact minimum and maximum on finite support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:74b8e6f8afd76a490c34438b20b360709ad22fbe39887630bd2b72e79d8fa4a2
1882	Mathematics	SFT-MATH-OPT-PARETO-DOMINANCE-003	Pareto dominance and multiobjective order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0d581ec125db13d24f596f7cdbd8f656a244678a63441c9d71c17a24e36a2e52
1883	Mathematics	SFT-MATH-OPT-LINEAR-PROGRAM-004	Linear-program correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6323878558e3eb3fb4d0d8b2726bdba5fdcc0ab9a40aa57d9cd54748cc4e30da
1884	Mathematics	SFT-MATH-OPT-INTEGER-COMBINATORIAL-005	Integer and combinatorial optimization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bc3ef7947c9f9f489114c474b3c03d08c674e9ea91aef48a1e509b0088cf8a22
1885	Mathematics	SFT-MATH-OPT-CONVEX-CORRESPONDENCE-006	Convex optimization correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e39890906deedccecea769f9e59b8e73d8a82683178f00471f69ce7cb347cc64
1886	Mathematics	SFT-MATH-OPT-DUAL-CERTIFICATE-007	Duality and certificate structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:62c27ebb5829a3dad3f7fab9647f4a7bdccc2bfdb1241574b8c9eb4cec f8625
1887	Mathematics	SFT-MATH-OPT-VARIATIONAL-CORRESPONDENCE-008	Variational optimization correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fee2c183cf7b4f34029a2606dddf3491a9b3c85ecb6d2f2d44b9afa49446ed91
1888	Mathematics	SFT-MATH-OPT-DYNAMIC-PROGRAMMING-009	Dynamic programming as compositional optimization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94a6507475f5b899b0280eda449cdac26c23fafe729038b5c35eb257b3a1f4c1
1889	Mathematics	SFT-MATH-OPT-GENERATED-STATE-CONTROL-010	Optimal control on generated state paths	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:359e651b661d88d22d7602a0249f90c1b82a16d3eb9485f9ed59c3183bed13bf
1890	Mathematics	SFT-MATH-OPT-GAME-EQUILIBRIUM-011	Game equilibrium finite correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:df50833d973d90601e8d47ddca56172fc4e914d0e8fa925b1e9121759ee7a738
1891	Mathematics	SFT-MATH-OPT-DECISION-LOSS-012	Decision theory and loss ordering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:92b812df5237a10f034bf74974cb664dd25c0074fdd4c6186c7f802896823f6a
1892	Mathematics	SFT-MATH-OPT-FLOW-SCHEDULING-013	Operations-research flow and scheduling	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0f7c3d3f7d1c43f2abf11d7c21970c7f941446520ab41893ef35f49dac67aa29
1893	Mathematics	SFT-MATH-OPT-ROBUST-BOUNDED-UNCERTAINTY-014	Robust optimization under bounded uncertainty	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:03925760f19f50746faf61b93c7b164b58d424f52d585f6bcd32f87c21e42fb
1894	Mathematics	SFT-MATH-OPT-APPROXIMATION-GAP-015	Approximation guarantees and gaps	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:28224fd21eff9d40657483e3aceb4019904f841d9786da73c2ec59d114a16a08

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1895	Mathematics	SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016	Infeasibility and unboundedness boundaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e8e4a46874d1eb0f42b931300512437b8f1015e1de41ebc4c1a1b4a63a9b6080
1896	Mathematics	SFT-MATH-DYN-STATE-ORBIT-001	State maps and exact orbit structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aeb9921e8e40d6d59f313577666482b543ef5624f9061cbf22b2f79c176b6633
1897	Mathematics	SFT-MATH-DYN-FIXED-PERIODIC-002	Fixed points and periodic cycles	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9ae62db5ffdd31c07fea2a614bbbe9e5ba587bde5a38daa921fe0ba58c7e9a70
1898	Mathematics	SFT-MATH-DYN-RECURRENCE-RETURN-003	Recurrence and return-time structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9c18019e9f20d1d1b810243cd7d08875312668c2bc99c216adeceb8e22f9c664
1899	Mathematics	SFT-MATH-DYN-INVARIANT-CONSERVED-004	Invariant sets and conserved records	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2df3a6cbc61017a6eef83e76253575b6bfff962865cdc1002c7f77b00d2fba89c
1900	Mathematics	SFT-MATH-DYN-STABILITY-ATTRACTION-005	Stability and attraction correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:520a680b3349364ef71665582583a1f3bced6cd9411a4fd11f9de14a8adba730
1901	Mathematics	SFT-MATH-DYN-BIFURCATION-DISTINCTION-006	Bifurcation as finite distinction change	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8c84d60f21d282875750592142e7f749065029afeac13da0dc9ad38b11ecec93
1902	Mathematics	SFT-MATH-DYN-SYMBOLIC-SHIFT-007	Symbolic dynamics and shift correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d5d7b98dac7aa6e91704f5a82154ce0e5498d0e00fc67b1db7afa2a87e79cbb6
1903	Mathematics	SFT-MATH-DYN-EXACT-SENSITIVITY-008	Chaos through exact sensitivity witnesses	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:73d7f19de7ba19d7987c99282739818524e9b29c547ad8b7a21838768e9f6f2a
1904	Mathematics	SFT-MATH-DYN-ERGODIC-AVERAGE-009	Ergodic-average finite correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7f8777dd8e7f792ca65148f4a8015f93681cdd890936577821113af0b6d8b052
1905	Mathematics	SFT-MATH-DYN-HAMILTONIAN-REVERSIBLE-010	Hamiltonian and reversible-map correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ffa0ecb0ca362a3a6bd0a2c0033e0033b4addf071ca9c2a1316125debcd19474
1906	Mathematics	SFT-MATH-DYN-DISSIPATIVE-RETAINED-LOSS-011	Dissipative dynamics and retained loss	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:013899ed0283322e5ec156198a2845d00447a602ebe224bbaefc5e984d6cf950
1907	Mathematics	SFT-MATH-DYN-COUPLED-NETWORKED-012	Coupled and networked dynamical systems	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c525a01d63f6eeb62552b3058ed34808c261032dae7b3553d61adfabf872780e
1908	Mathematics	SFT-MATH-LOGIC-PROPOSITION-DISTINCTION-001	Propositions as generated distinctions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f85f9e844e3ec3f9581c1d9302dbea00e5791f1c4d11df0a41e3ef0bce2bbbf7
1909	Mathematics	SFT-MATH-LOGIC-INFERENCE-CONSEQUENCE-002	Inference and consequence preservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:153966835981d27d4115b986d3dfa5519147893171304aa9224ace83a36e2f94

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1910	Mathematics	SFT-MATH-LOGIC-SOUND-COMPLETE-CORRESPONDENCE-003	Soundness and completeness correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:340d97010a3dea95fd4339a6ad816685c90466027c866d62e45d4760ac2692c7
1911	Mathematics	SFT-MATH-LOGIC-PROOF-OBJECT-CHECK-004	Formal proof objects and checking	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e1d02e00afb9f4bf37ce78131e933afe4a71ef346cafe5ac815e8788d6ec4aa
1912	Mathematics	SFT-MATH-LOGIC-QUANTIFIER-FINITE-SUPPORT-005	First-order quantifier finite-support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:53083a9fef5c6ab7df19ad3708e04c0c47900087047ac48893d4163c5db51cf1
1913	Mathematics	SFT-MATH-LOGIC-MODEL-INTERPRETATION-006	Model and interpretation structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c0ebbe707ae0dca10b94d1b69b5122c3e881de668df7bee5bbafef8d9678c03e
1914	Mathematics	SFT-MATH-LOGIC-COMPACTNESS-BOUNDARY-007	Compactness correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f79e69c848156a49ac98efd70d86967895b4d2af23b30ea9d35b65f2099d888
1915	Mathematics	SFT-MATH-LOGIC-DECIDABILITY-INTERFACE-008	Decidability and computability interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1fc8005aaf5441ebd0362d84d9b1e0a6521f9306dcdcfbfc0299abe3f549f4ced
1916	Mathematics	SFT-MATH-LOGIC-INCOMPLETENESS-SELF-REFERENCE-009	Incompleteness and self-reference boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a93e6a3917e8b56002318a4cb9bdc381988b70aa3431a940655ed4d4371f794
1917	Mathematics	SFT-MATH-LOGIC-FINITE-COLLECTION-010	Set-like finite collection theory	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:66e9c9d47b3364b32c328570ccb83fa0a7f094cc7adac043ccb30c42e25c9799
1918	Mathematics	SFT-MATH-LOGIC-SIZE-BOUNDARY-011	Class, universe and size boundaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ae93deb5f11e14001d83c1afd20e2cd8e90246dd8dc4d9a0c605f4b9877d5823
1919	Mathematics	SFT-MATH-LOGIC-CONSTRUCTIVE-CORRESPONDENCE-012	Constructive and intuitionistic correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be1fa0f7ed9298f961e099db5213007d9c7a556e85756207024813bcea2cee41
1920	Mathematics	SFT-MATH-LOGIC-MODAL-TEMPORAL-013	Modal and temporal logic correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b2428380ee59ac674bd574be852e1168aaa1a392cb49f8e8879f82ae34cb95e5
1921	Mathematics	SFT-MATH-LOGIC-MANY-VALUED-014	Nonclassical many-valued correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:571b131341ecd1ad9d2fc2f3b131487f8e060e1a05e51af274793f8d3327a562
1922	Mathematics	SFT-MATH-LOGIC-NORMALIZATION-015	Proof-theoretic normalization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5af25a62e0f550c27d7dde1ff7314806f74aa97c6908f98c8e9753bad2654b76
1923	Mathematics	SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016	Foundational consistency and self-verification limits	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5717d448290a1af7fbbd8ad6a85c473c68b28abce0799711809f5538cb14d055
1924	Mathematics	SFT-MATH-CAT-OBJECT-ARROW-COMPOSITION-001	Objects, arrows and typed composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:71c69fc458b52088e7c1dc218452b99d29ef3a4e9634d60c261a7a69cfbb0394

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1925	Mathematics	SFT-MATH-CAT-IDENTITY-ASSOCIATIVITY-002	Identity and associativity witnesses	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:807ed10ff532e9c607cd05e5b289cae65eae44fef96cfec772ce1eaf6bd64c89
1926	Mathematics	SFT-MATH-CAT-FUNCTOR-PRESERVATION-003	Functorial structure preservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:45effaf6d8da98e393865ea391af69e936c277b99614dac177e4f0d14d850900
1927	Mathematics	SFT-MATH-CAT-NATURAL-TRANSFORMATION-004	Natural transformation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5dead80b1c237c637d35f6c43379a65812c2117954f77d5e9390cd3806c3e0e1
1928	Mathematics	SFT-MATH-CAT-PRODUCT-COPRODUCT-005	Products, coproducts and universal constructions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7d7a98b46c34532e47b6a37fcd19c87883fe3b009a0fc6059173309712bac821
1929	Mathematics	SFT-MATH-CAT-LIMIT-COLIMIT-006	Limits and colimits on generated diagrams	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:66e3d06fc4a7b84928e089f02b4a5e7151461520b6c96327b5fbf9627f2eb11e
1930	Mathematics	SFT-MATH-CAT-ADJUNCTION-007	Adjunction and paired universal maps	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bc77d0449dd3b2b490aff2c45dc27e9552a8798c49bad32cfff6dc021dd25b247
1931	Mathematics	SFT-MATH-CAT-MONOIDAL-TENSOR-008	Monoidal composition and tensor interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d46e43677dbab4705723767d4134bba4f59a0c6b60e6d45ae97057bfcfa7c5b
1932	Mathematics	SFT-MATH-CAT-CLOSED-INTERNAL-MAP-009	Closed structure and internal maps	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f68143ffe290ff07df3f325ad9af23893914b6dc1e8b02018f5ee5bf8fa9f8fd
1933	Mathematics	SFT-MATH-CAT-TYPE-DEPENDENT-RECORD-010	Type correspondence and dependent records	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ff7742d564cdc04c75e692c5642100b378945b08fcdcd9d11c0150af30e3a71e3
1934	Mathematics	SFT-MATH-CAT-SHEAF-LOCAL-GLOBAL-011	Sheaf-like local-to-global custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:73d7dd3a53e88a69cd63d2147afab60c9fce360c065b5f813ffc82b508bdffbfa
1935	Mathematics	SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012	Operadic and higher-composition boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:212418e46bfbec20e1f9b404cd43fb40fc13d180de2ab99bc0a8971017fd2197
1936	Mathematics	SFT-MATH-NUM-EXACT-REPRESENTATION-ROUNDING-001	Exact numerical representation and rounding custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8e0f6ddf5bc90fef46eefc32a535fa9fae3d8fc45d3a0f0dd93f657d8c1836c4
1937	Mathematics	SFT-MATH-NUM-INTERVAL-RATIONAL-ENCLOSURE-002	Interval and rational enclosure arithmetic	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8c13117d2033b4657174a69f0dae9093d27277e74ea89abe10609bb463cc5072
1938	Mathematics	SFT-MATH-NUM-TRUNCATION-DISCRETIZATION-ERROR-003	Truncation and discretization error	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:259b72ccea62db62192f6788cc51d7c5e54c21e36eb8a5099148c0f1b6b98b38
1939	Mathematics	SFT-MATH-NUM-FORWARD-BACKWARD-STABILITY-004	Forward and backward stability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6929ba74b44f626e8ea7e3b5af82fd3c97e5a4375aa27f1ba050eab41fa2b9be

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1940	Mathematics	SFT-MATH-NUM-CONDITIONING-SENSITIVITY-005	Conditioning and sensitivity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d426bb53ef24834bf45d6ff67e99cbcd3d4b1eb55e300e460138fe2907ab1b34
1941	Mathematics	SFT-MATH-NUM-CONVERGENCE-ORDER-BOUND-006	Convergence order with exact bounds	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eb7d30b58833016ecf2928e7ddc354f8d5e428bfe0ef3ca80a56a815d3706757
1942	Mathematics	SFT-MATH-NUM-ROOT-ISOLATION-EQUATION-SOLVING-007	Root isolation and equation solving	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cdde0b0bd96c982f20d5f2caa6227c8dcc5a56338f8efd7e8a08fff00df49ecd
1943	Mathematics	SFT-MATH-NUM-LINEAR-SYSTEM-SOLVERS-008	Linear-system exact and approximate solvers	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8b2b0f1ba1f2834f192150342fb2c319be37ff34289dbc4150edf24e092d4a0a
1944	Mathematics	SFT-MATH-NUM-INTERPOLATION-APPROXIMATION-009	Interpolation and approximation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e751920f65b81f5cc2649abdfcd11aa0c1dfae51c10f4d888d4415716d4d2b09
1945	Mathematics	SFT-MATH-NUM-QUADRATURE-ACCUMULATION-010	Quadrature and accumulation enclosures	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:648b6d3a1e7459deb29c93bf36fbb8f428b743ff0045f05f36cac4f6712a1910
1946	Mathematics	SFT-MATH-NUM-DIFFERENTIAL-EQUATION-CORRESPONDENCE-011	Differential-equation numerical correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:00a35c5c34967fc4c562146e871ba053dd4af162aef69f61a6122d5222fd789e
1947	Mathematics	SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012	Verified computation and certificate extraction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0696fffb60057ca05592a019bc74437c233e398d8e3ed8633d13122917999105d
1948	Mathematics	SFT-MATH-SYMB-CANONICAL-EXPRESSION-001	Symbolic expression identity and canonical form	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1fc6dbd6b91cb7dac6d70badd5ab5bee41c63de626a6abd2a076fb40f3f2a407
1949	Mathematics	SFT-MATH-SYMB-SIMPLIFICATION-PROVENANCE-002	Exact simplification with provenance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9cf05501f0c8844ef8a38799f148078b8458cc4e42582114034220672d1183cf
1950	Mathematics	SFT-MATH-SYMB-POLYNOMIAL-FACTOR-EXPAND-003	Polynomial factorization and expansion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:532d8d3fe185773057827ec65c5ede613f674b27e0ffdbefdc49a9a419be9ff
1951	Mathematics	SFT-MATH-SYMB-EQUATION-SOLVING-004	Symbolic equation solving	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:30c7bffc4bfa23e1bb4bbb549fd04f89c9cb1f72be967384b53be228a266d015
1952	Mathematics	SFT-MATH-SYMB-REWRITE-TERMINATION-CONFLUENCE-005	Rewrite termination and confluence interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ab851259d84f0c5b760b72589e8b6695d2b3e44efc11dd4b503e839d4647b37f
1953	Mathematics	SFT-MATH-SYMB-GENERATING-FUNCTION-TRANSFORM-006	Generating-function transforms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8c88175914327cf3786f3f8b238fa3ec9d4b5a872ac254beafc9f759d7f0732a
1954	Mathematics	SFT-MATH-SYMB-FOURIER-LAPLACE-CORRESPONDENCE-007	Fourier and Laplace correspondence transforms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78a40cfd8bef40d4c2979e7fcf85c1ff31f0b49925a73452d2bd3619bbf8e731

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1955	Mathematics	SFT-MATH-SYMB-SPECIAL-FUNCTION-RECURRENCE-008	Special-function recurrence representation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:86d0c48ae5b12195f37496629db40327bed840d8bbfe1fea05830f68540f6152
1956	Mathematics	SFT-MATH-SYMB-THEOREM-SEARCH-BOUNDARY-009	Automated theorem search boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f30bfc00ab43dc67ac366865b093161f74729273113b2e2a6361b45e80901069
1957	Mathematics	SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010	Constructive witness and certificate generation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fda3df9ae9119867ecd3cef93e2affb43dc0dd8596f37043767c6a45b3cf617a
1958	Mathematics	SFT-MATH-XINT-INFORMATION-HANDOFF-001	Mathematics-to-Information exact-structure handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6ffddc69383435d029cf280ab5d376838c88e0747f4d47ab9bb5db8157b66f68
1959	Mathematics	SFT-MATH-XINT-COMPUTATION-HANDOFF-002	Mathematics-to-Computation model handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3c14d83ea82bd2caaadccd4e8f83f2d9cfa2ceca3bd433ec8a9036382e43b450
1960	Mathematics	SFT-MATH-XINT-PHYSICS-HANDOFF-003	Mathematics-to-Physics quantity and geometry handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a8886f024818e308656ead69f4cd4aaab32762944ace0d4e6bc2892a6f8062de
1961	Mathematics	SFT-MATH-XINT-CHEMISTRY-HANDOFF-004	Mathematics-to-Chemistry structure handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7d8ff0f49eec26d921585ff950f2f0a0361059515d99ef868dac5147078db211
1962	Mathematics	SFT-MATH-XINT-BIOLOGY-HANDOFF-005	Mathematics-to-Biology organization handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ccdcf95d6ac30220d49853eed9c196f15fdc6a4cf578a555811714099d5dc038
1963	Mathematics	SFT-MATH-XINT-SOCIAL-HANDOFF-006	Mathematics-to-Social inference handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b404e098c202044f424847255966dc5938d54a3ee47c96cb75031d6074c6216f
1964	Mathematics	SFT-MATH-XINT-ENGINEERING-HANDOFF-007	Mathematics-to-Engineering calculation handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:26cb9935ef8d7b4deeecc3475b0347694990abe929c4b3dd8d123afa8a1f6a98
1965	Mathematics	SFT-MATH-XINT-ONE-OWNER-IDENTITY-008	Shared mathematical identity without duplicate ownership	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ae8b83e55ba8150ddafe5fc01cae15e7e7d880da1556b1946704ef737060cd80
1966	Mathematics	SFT-MATH-VALID-ARITHMETIC-ALGEBRA-001	Arithmetic and algebra complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f7793e7fbdbdaf18783503febe565e6a2f87706cf66df942752963520d0d7378
1967	Mathematics	SFT-MATH-VALID-COMBINATORICS-GRAPH-002	Combinatorics and graph complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a8db169be7fb482b1aa32a8a38915111fc9663128124af60a00528023449460
1968	Mathematics	SFT-MATH-VALID-LINEAR-ALGEBRAIC-003	Linear and algebraic-structure validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:923631359a6f87d1ffd3e994ea3f27d4789e01732aa0efaeaeaa462bb3407667
1969	Mathematics	SFT-MATH-VALID-ORDER-GEOMETRY-004	Order and geometry complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c2640e5e3f66fe60919039e8c8a2f775c2f913c951f5cec77bc9b8924b5d317f

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1970	Mathematics	SFT-MATH-VALID-TOPOLOGY-ANALYSIS-005	Topology and analysis complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ad81f85b9b3f94edde0402935da86a255c85544a99f4180c07651921d2e471a0
1971	Mathematics	SFT-MATH-VALID-EQUATION-MEASURE-006	Equation and measure complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:65d0fbfca9c893839c453cca668a393d086cc8407c5b6b9b6bceda06ed2be95d
1972	Mathematics	SFT-MATH-VALID-PROBABILITY-STATISTICS-007	Probability and statistics complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c9179fff3bc77f7f13003ce25ed099bf0e6f557d2cbe7606dab7ab66c7dc68fd
1973	Mathematics	SFT-MATH-VALID-OPTIMIZATION-DYNAMICS-008	Optimization and dynamics complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ca131191272d09d2ab262915de7f0b11ee564d86a207e61494b6761cc341426d
1974	Mathematics	SFT-MATH-VALID-LOGIC-COMPOSITIONAL-009	Logic and compositional complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6afb8b8f15ef954c25d276f3b9e654737b824419855ef250a8e6fa7100873657
1975	Mathematics	SFT-MATH-VALID-NUMERICAL-SYMBOLIC-010	Numerical and symbolic complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2e74a2af4ae929e1b550db41ab49a11f01b5da2a65e5a8ae2c43a6187a70c817
1976	Mathematics	SFT-MATH-VALID-ADVERSE-BOUNDARY-011	Adverse absent unresolved and boundary vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e24be1ab2cbf3a5f4a04ffaa42e4149e1976508eb8cbe259f63a4556082f0432
1977	Mathematics	SFT-MATH-VALID-EMPIRICAL-FORMAL-GRAND-LOCK-012	Mathematics empirical and formal Grand Lock	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:822ec7056f4e3dcef00c42fa941fd1fc2f65746649f4f760a119defcda6fd237
1978	Mathematics	SFT-MATH-HAND-DOWNSTREAM-ONE-OWNER-001	Mathematics one-owner downstream handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d02a4080f51e56bf936273065de8571ce061fb2475ae408e7ea01690d4fa8b1c
1979	Mathematics	SFT-MATH-HAND-MEASUREMENT-BOUNDARY-002	Mathematics measurement-boundary handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a962af816adb6cc2256a1f75e91ae932fcdb2ac1bcaade669bb9ea46a008734c
1980	Mathematics	SFT-MATH-HAND-FORMAL-EMPIRICAL-003	Mathematics formal-to-empirical handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:72d97f8cb655a93d62def7be0f336521322e6b4e7ebf3d6e67fe6dad8853e069
1981	Mathematics	SFT-MATH-HAND-CONVENTIONAL-CORRESPONDENCE-004	Mathematics conventional-correspondence handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:970d5376a5beb8c4d8fdd387c12dda90c72e00671e71ea4ef4ee2cbd4289ad38
1982	Mathematics	SFT-MATH-HAND-OPEN-EXTENSION-005	Mathematics open-extension handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:55240c646e44e399889211389bc995a80dd78abc23904c0c6a866ec42d02cd80
1983	Mathematics	SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006	Mathematics cross-branch one-owner completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d66b4c2094c3e340398bce5fb89aa88a873dfba5f19c67892cc598184f68709d
1984	Information Science	SFT-INFO-SYMPREP-ALPHABET-GENERATION-001	Canonical alphabet generation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e882bc27084f4e2a64e33620edc12d084a952447e6dc9f097b672e5e730644c4

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1985	Information Science	SFT-INFO-SYMREP-SYMBOL-IDENTITY-DISTINCTION-002	Symbol identity and distinguishability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cdcbf2734f3510b2b9cef328d695c9cefff9467419b6bc2822a3d9f308112e18
1986	Information Science	SFT-INFO-SYMREP-CODEWORD-PARSING-003	Codeword formation and parsing	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a382c2a8a75540d864aef2b338e235014c97adb731402dd745a24f27a92daa45
1987	Information Science	SFT-INFO-SYMREP-PREFIX-UNIQUE-DECODING-004	Prefix, uniquely decodable and instantaneous structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6a565d3313fe90878e104066b71a405fce310317b0346215bbc972eab7a6d4c3
1988	Information Science	SFT-INFO-SYMREP-GRAMMAR-REPRESENTATION-005	Grammar-constrained representation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1873a6a732469f5a8999951879ddb6c7a4c27304f311c3d3cea4558fa62b64bf
1989	Information Science	SFT-INFO-SYMREP-EQUIVALENCE-ISOMORPHISM-006	Representation equivalence and isomorphism	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a3f427723ba87a3f3489fe87c4d8e17aa8e628a0425b69a2f5c27da04ac12ddd
1990	Information Science	SFT-INFO-SYMREP-CANONICAL-NORMALIZATION-007	Canonicalization and normalization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f2885b4d6a1e043e5985511eb510c58db514ad30c8c16042adb7c6883b75859
1991	Information Science	SFT-INFO-SYMREP-VARIABLE-LENGTH-BOUNDARY-008	Variable-length representation boundaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c35239fea7e66aeaf96bf2fbee3df1c4c35033b0e25534b298dbdc6369fd555a
1992	Information Science	SFT-INFO-SYMREP-PRODUCT-ALPHABET-009	Composite and product alphabets	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f79e99c7b376f9a44004bb229c7e159817e7a35c769d771cca253fb83202e170
1993	Information Science	SFT-INFO-SYMREP-HIERARCHICAL-TYPED-SYMBOL-010	Hierarchical and typed symbols	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5ed07dc03dc62e4cbebe78449e0b483fef0a1e530bbb6c912906f6b2964c565b
1994	Information Science	SFT-INFO-SYMREP-CONVERSION-TRANSDUCTION-011	Representation conversion and transduction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:89ac0f4f8a9bb94981cd6b6da8c73af5ccf54948541e82412455c0ab7c76e65d
1995	Information Science	SFT-INFO-SYMREP-AMBIGUITY-ALTERNATIVES-012	Ambiguity detection and retained alternatives	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:de6d10b1160ea411911b7d761c195d707003a588de3d0d675163d6e9e51c1844
1996	Information Science	SFT-INFO-SYMREP-FINITE-SUCCESSOR-013	Finite-alphabet successor law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2410fd01a379abcb7091074c413976ebb71e8d3321d5447a8de823f6dc9af7ae
1997	Information Science	SFT-INFO-SYMREP-COMPLETENESS-014	Representation completeness and no-omission certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8808d673f54abe8890b5d37c2cf9501726d2953e082b6a3ce7ff14fedcee7899
1998	Information Science	SFT-INFO-RECORD-DATA-IDENTITY-001	Data item and record identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:188f5c8571a552f7a2e11ed331c20c752c20230612f7cc29b0c3576415caddfc
1999	Information Science	SFT-INFO-RECORD-TUPLE-RELATION-002	Field, tuple and relation organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9c96a61da557fa61aafc050738815f5e903f444f2db6ac7537217a709bdf5a1c

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2000	Information Science	SFT-INFO-RECORD-METADATA-003	Metadata as retained interpretive record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:25e38daf05a46551ff43aab2b9e61f9528960e9e3afd3b13bafd1e31ec212f81
2001	Information Science	SFT-INFO-RECORD-SCHEMA-TYPE-004	Schema and type custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f362adb1701f98f3aa6146adc30d98464694e832fbc910ee863d4fc61e000951
2002	Information Science	SFT-INFO-RECORD-PROVENANCE-005	Provenance chain composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7cbaa15ee285db6986d62029723d73cf3e72fe9878a354a7de4f65737c0d42e9
2003	Information Science	SFT-INFO-RECORD-INTEGRITY-006	Integrity and tamper evidence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:69fbb5e96b439ffdb77c551ace0e0c6fe920db4fb5bd6f413bbcca8e800b92a
2004	Information Science	SFT-INFO-RECORD-VERSION-REVISION-007	Version and revision identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:775b2a7f578c366836cb191538788f8b585f798f44586f48aed3f14faca63f57
2005	Information Science	SFT-INFO-RECORD-ABSENT-MISSING-UNKNOWN-008	Missing, absent and unknown distinctions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:89e4d009dfb6d60b36f34052373f73d7c772582f8d82dd5c56f4017fb2b2af11
2006	Information Science	SFT-INFO-RECORD-DUPLICATE-ALIAS-009	Duplicate and alias resolution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8bd9429bb1f87358f9b7e1d76bf5f024432dd4f6b986b9615295445d17745cfb
2007	Information Science	SFT-INFO-RECORD-LINKAGE-IDENTITY-010	Record linkage and identity boundaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3af4d67f4d641bdec40b47a8f234af19a9844f29f1b21b05d42fe4b80cf1f957
2008	Information Science	SFT-INFO-RECORD-DATASET-COMPLETENESS-011	Dataset completeness and omission ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7298a45fcb8497b3bdc6739512bc51f83ce6bcc53dc1f7b99f4227d3550b74
2009	Information Science	SFT-INFO-RECORD-CUSTODY-REPRODUCIBILITY-012	Record custody and reproducibility certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:939f47572f8806dc4409f625369f53c39b0ee387ebc34c046d82c3428491eeb9
2010	Information Science	SFT-INFO-SOURCE-SUPPORT-001	Finite source-support generation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d4cf50615da67505bdb9be8da733e118d30cb4087d6882633f5173b86f73f44
2011	Information Science	SFT-INFO-SOURCE-SEQUENCE-ORDER-002	Sequence information and retained order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:96609683f424ffcc3875ff030c519a90547bf86d3830775f9d912a7de84d6a6b
2012	Information Science	SFT-INFO-SOURCE-PROCESS-TRANSITION-003	Process information and transition custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:29aaaf0bd2c2447744f6d34374759ca64c232527960213b45d8b15b39b76f28d
2013	Information Science	SFT-INFO-SOURCE-SPATIAL-ADJACENCY-004	Spatial information and adjacency custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e77634cca7e04f0ec50c336b75dbde83045540816a1a34a79afcdea51590a3c3
2014	Information Science	SFT-INFO-SOURCE-NETWORK-PATH-005	Network information and path custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:035f5f93b5209dc1a07b7b77fe3f93b94b493beca2ddf443a74934c569ab85ea

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2015	Information Science	SFT-INFO-SOURCE-REFINEMENT-COARSENING-006	Source refinement and coarsening	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:80d59f7d08df01bde1b7283271c33458c7c17c4d3c1d91a4727dd1a9ae5b5d08
2016	Information Science	SFT-INFO-SOURCE-STATIONARY-SUPPORT-007	Stationary-support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:34709bf730aae360faaeaa589331b12f01444495cb1dd574fbf08684f2a4155
2017	Information Science	SFT-INFO-SOURCE-NONSTATIONARY-SUPPORT-008	Nonstationary-support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d9e4deda247cb25d96458cf6b0e02604b352c0a6acb7be6c75729923ba2b87e2
2018	Information Science	SFT-INFO-SOURCE-MEMORYLESS-009	Memoryless-source correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:77ba00dd7077d887cc2d386bb7bc287190642f89baf53b526676ba344410b904
2019	Information Science	SFT-INFO-SOURCE-FINITE-MEMORY-010	Finite-memory source structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:47256b3ea98d459ace50ed78c82e267fa2ccd3536809e7d10f40724db5cc8b30
2020	Information Science	SFT-INFO-SOURCE-JOINT-COMPOSITION-011	Joint-source composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0342e263902de2bc7898782d759b278f18f9fcd9ae6dbb4a68eeee6d363bd514
2021	Information Science	SFT-INFO-SOURCE-DEPENDENCE-COMMON-SUPPORT-012	Source dependence and common support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d548d93e0d37da8ac037cac7406f0828cba867aa0b6379cd563ae3e1a6312064
2022	Information Science	SFT-INFO-SOURCE-SUCCESSOR-013	Source extension by exact successor	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:20ce59552d421058110dc981a1ba4783d236c0c264b8dd5fd34dfcbb8aa80449
2023	Information Science	SFT-INFO-SOURCE-COMPLETENESS-014	Source completeness and boundary certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ecfa32db51e739c58d4dd241a4406a1fe1d09a1fb689c3a4e98230d3f03b4c05
2024	Information Science	SFT-INFO-MEASURE-DISTINCTION-COUNT-001	Distinction-count information measure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9439d2704d1942362e1ab678342db62738c3d78f6de66baf8d10f20c9f6e0784
2025	Information Science	SFT-INFO-MEASURE-COMBINATORIAL-QUANTITY-002	Combinatorial information quantity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9dc794f87f65d1f7e6d6b4aa989b0d36aa45ba851af5f466a5b41e99d6efd451
2026	Information Science	SFT-INFO-MEASURE-OPERATIONAL-COST-003	Operational discrimination cost	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8f164e185a31b3f03f1910a4955707e00fa261b5f5d39adaa5fde94581e4b483
2027	Information Science	SFT-INFO-MEASURE-DESCRIPTION-LENGTH-004	Description length on a fixed grammar	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4a5376dd6e2283698cbfbd0d6c84ba93afee70339f93f274ced62ae37a6a47d9
2028	Information Science	SFT-INFO-MEASURE-ALGORITHMIC-BOUNDARY-005	Algorithmic-description correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fa512d0c822f7f1a3fe46b6868ce46f84765f015a78e8beeffabe15a999d1388
2029	Information Science	SFT-INFO-MEASURE-PARTITION-REFINEMENT-006	Partition refinement information order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e09105740ebd9d57f1e50a4d25f63ccd056a322e0563420790978b5bab16b577

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2030	Information Science	SFT-INFO-MEASURE-PRODUCT-ADDITIVITY-007	Additivity on independent product support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fe66cd4675b04dd5a7d1a55443e011b097509ad9dd03fcf97a9cd1e31178131
2031	Information Science	SFT-INFO-MEASURE-SHARED-SUBADDITIVITY-008	Subadditivity under shared support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e094503f68b90836697b64cd3c27e061d4be8248322a4e18429a44e161d5dadd
2032	Information Science	SFT-INFO-MEASURE-COARSENING-MONOTONICITY-009	Monotonicity under observation coarsening	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3c02f0a645253b849b2b9bb524388b03628b57fa58d335c709fb091dc5a26b54
2033	Information Science	SFT-INFO-MEASURE-BALANCE-LEDGER-010	Exact information balance ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f9e9e4248e852e3851cccdcaf11f02f49b5da05df2e4d85c2b7a1a3dc5fc30d
2034	Information Science	SFT-INFO-MEASURE-RELATIVE-011	Relative information between observations	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:796a221d4c72c016b019feedbb84f9da2af8de73664bde26f22538613f04d686
2035	Information Science	SFT-INFO-MEASURE-DIVERGENCE-012	Information divergence correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6a0996db7794d1ee473d4b42c50c096ff1dd60e88a43c924ea7d33d739db39db
2036	Information Science	SFT-INFO-MEASURE-GEOMETRY-013	Information geometry on exact parts	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9b99c3d97aab0b792f05c5b22ad69b3dc00068fde9d921553d0171e481fc0741
2037	Information Science	SFT-INFO-MEASURE-MULTISCALE-014	Multi-scale information decomposition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:67be345678ea8083fcfd99c189de049c6e12e9f14e42a7ace547a5c397ff2294
2038	Information Science	SFT-INFO-MEASURE-UNIT-CUSTODY-015	Measure conversion and unit custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:61431200deb07c07eb2dd1ea2c53fcc9e7ed4eba34cf20a763ab8f74214723eb
2039	Information Science	SFT-INFO-MEASURE-COMPLETENESS-016	Information-measure completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f267f72a1b16ab7d14d45c1e7004001a33e721f13ee3f289fad4acea7a5e41b3
2040	Information Science	SFT-INFO-SIGNAL-ORDERED-RECORD-001	Signal as an ordered observation record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e1480fb2ec65d8ff917eb9fb8d628251f72c320eb84708049a8e4e34651f4bec
2041	Information Science	SFT-INFO-SIGNAL-AMPLITUDE-SUPPORT-002	Amplitude-label and support separation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa715eb732002cf90cef0de1f03151147633926b6434c5dcba15efbe43bd4c2
2042	Information Science	SFT-INFO-SIGNAL-SAMPLING-SELECTION-003	Sampling as retained position selection	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:94cdf76f9079431270f155fbadf45b3ef98b44993cee48fac81977947a728c2d
2043	Information Science	SFT-INFO-SIGNAL-FINITE-SUFFICIENCY-004	Exact finite sampling sufficiency	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3b89ab82ee1792ce20128af5fba7ad5affa5cbe2bd071f54521e544bdb73e3e6
2044	Information Science	SFT-INFO-SIGNAL-ALIASING-005	Aliasing as closed distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:57d7937c2c41f4cc234024e13df5472eb9621dfe199cc8bb861f1abe3a4a8d70

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2045	Information Science	SFT-INFO-SIGNAL-QUANTIZATION-PARTITION-006	Quantization as observation partition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5fd34246d405251f93db300ed1bf0ca5e3736f577a5611affcb85d7b99970739
2046	Information Science	SFT-INFO-SIGNAL-QUANTIZATION-ERROR-007	Quantization error custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c1241d38a332b2e4173d249e367f3e847686d1cbc7eb2cedce6e7c7285a07541
2047	Information Science	SFT-INFO-SIGNAL-RECONSTRUCTION-008	Reconstruction from complete retained samples	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:772cb633ffc72b3fcfd3a2f72e66422e37f7fcc76152f849db9dfdb27c888ed9
2048	Information Science	SFT-INFO-SIGNAL-INTERPOLATION-BOUNDARY-009	Interpolation correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c94acd30ab94af39f242d95d247db33eab69c577ba3715432874ae6ce70b885a
2049	Information Science	SFT-INFO-SIGNAL-TURN-TRANSFORM-REPRESENTATION-010	Transform representation of finite signals	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b01be7af191f4e0d72fddb44b51ed469b7f604f9d955961e5fb9e122dbfe8a6a
2050	Information Science	SFT-INFO-SIGNAL-TIME-FREQUENCY-011	Time-frequency support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2b959f97bc880c0763c1843f4b90b8d30b5c35ceb366b57151df9f0f9eb9666e
2051	Information Science	SFT-INFO-SIGNAL-SPATIAL-MULTIDIMENSIONAL-012	Spatial and multidimensional sampling	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bb24e7bb1f2f3480a436c9387b1b143aabad5f8e4d9e2d0227c6bd24f8d96eab
2052	Information Science	SFT-INFO-SIGNAL-PROVENANCE-013	Signal-to-record provenance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:08248d2357f25c3ddb67623827f0f99f6d779402477bb72288090459bc56c0be
2053	Information Science	SFT-INFO-SIGNAL-COMPLETENESS-014	Sampling and reconstruction completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:078dcfb462d56aa57863f88a97a9a5f588170274c668c6302f17f82cff0ffff2
2054	Information Science	SFT-INFO-COMP-LOSSLESS-RECONSTRUCTION-001	Lossless code and exact reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:75c800388923e6076a00a11487aed2e59bb3d2f30ad730e0eab0d1d9b9c8f542
2055	Information Science	SFT-INFO-COMP-PREFIX-TREE-002	Prefix-tree compression structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b5cacf1e07d7c25c5d9ff67b23f16edebc86a9ae88f28a9ebbf824e35338fa9
2056	Information Science	SFT-INFO-COMP-DICTIONARY-003	Dictionary compression structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9c85ce9dbd865fa64f25eb57bac02d32b65155e26a6b88a7e34be8e2a083225a
2057	Information Science	SFT-INFO-COMP-RUN-RECURRENCE-004	Run and recurrence compression	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b4a528eb3ab8ced04357e955ddb95a589902d586e753ed655926b40b02ca5a96
2058	Information Science	SFT-INFO-COMP-TRANSFORM-005	Transform compression correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ecb0fee425b7b45949d54d5cd7dd2b2573b22a0cf4c4d5fd2061cc3532eb26d
2059	Information Science	SFT-INFO-COMP-SOURCE-MODEL-BOUNDARY-006	Source-model compression boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:742ad4e7eb89095f56faf0252f1c8c852124a83c341b75c06b61c13197aac2c6

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2060	Information Science	SFT-INFO-COMP-REDUNDANCY-007	Redundancy as retained excess distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f850128e41642d307bf48ce21ea4391f943a081398cd427276ab4d6cc04606ce
2061	Information Science	SFT-INFO-COMP-MINIMUM-DESCRIPTION-008	Minimum description within a fixed grammar	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:42735a957fd33924e42a8b258840f35eca3e5247b9a40a15d437211cea8dd463
2062	Information Science	SFT-INFO-COMP-LOSSY-COARSENING-009	Lossy compression as declared coarsening	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a1ccda333ad3d3550b18ada3704c9ecc26d2104c1010dd9fd1a574d1819edbe4
2063	Information Science	SFT-INFO-COMP-DISTORTION-010	Exact distortion relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:11f671de205911db7f08466a3ebefed3d2a35f2d77c374030e0d495a29ac30e7
2064	Information Science	SFT-INFO-COMP-RATE-DISTORTION-BOUNDARY-011	Rate-distortion correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0f827a2d676d98d73c209c65c79885d8e50a7a6c099a75516866f15a1b4bd54d
2065	Information Science	SFT-INFO-COMP-SUCCESSIVE-REFINEMENT-012	Successive and layered refinement	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d2446fb05bf0591c284f9caf0cdb471bfdcef08f4e5f68d4c77f8841a92cc7a9
2066	Information Science	SFT-INFO-COMP-SIDE-INFORMATION-013	Compression under side information	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:82fece7fd52b43897da962940500e75ced12d34679f4e47e070cba852f766c94
2067	Information Science	SFT-INFO-COMP-COMPLETENESS-014	Compression completeness and adverse-control certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c8eb7350bc06e0e4c3b2ff9c1c3bbfcbcb2dd84fd48c09ed457ccf8c6f9a3d644
2068	Information Science	SFT-INFO-CHAN-INPUT-OUTPUT-RELATION-001	Channel as an exact input-output relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9b6d6289033145ac5a9fda6fd1bcd4fa3562793b11a757faa6e32b2e4104580c
2069	Information Science	SFT-INFO-CHAN-DETERMINISTIC-TRANSPORT-002	Deterministic channel transport	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:22e6d3a79fed38097a99abb5528d2ea2871c46408c454e1ee66cd500f5ec4fa3
2070	Information Science	SFT-INFO-CHAN-OBSERVATION-EQUIVALENCE-003	Observation-relative channel equivalence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d2c46be7dd3eae8df49ca261ec873c568ce21aed7e20072d166643fed55a75d5
2071	Information Science	SFT-INFO-CHAN-SINGLE-USER-CAPACITY-004	Single-user channel capacity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:53834d9a99266ce3af489d5ded0e1685a02188d4dbbfff07ada9a30e4fd2c7d
2072	Information Science	SFT-INFO-CHAN-RESOURCE-CAPACITY-005	Resource-bounded channel capacity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f8912021f96bde3a55df4b3ee714d73e80e3fe2a88441fa5d0cf123727ad293
2073	Information Science	SFT-INFO-CHAN-NOISELESS-COMPOSITION-006	Noiseless channel composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:88b8cbb3c5dc90be0d264f9bc30f027729aeac0ec680897eade8ee56f68d9bb1
2074	Information Science	SFT-INFO-CHAN-CASCADE-BOUNDARY-007	Cascaded channel capacity boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bldee0c023acc5b27d8c1cee1f22d8e2b4f31bd9c5d5297d94e97516c7e96049

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2075	Information Science	SFT-INFO-CHAN-PARALLEL-COMPOSITION-008	Parallel channel composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a668da33982d6faeea901f928f58ed885205c0e8b5cf761eeeca1a1610a0ad22
2076	Information Science	SFT-INFO-CHAN-FEEDBACK-009	Feedback channel correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a2680a6acd2ea50085b0136025dfbc6cddc68fce71c2daea4a5a18be49d39c40
2077	Information Science	SFT-INFO-CHAN-MULTIPLE-ACCESS-010	Multiple-access channel support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8e3b9e23e378669305eb5758f7c5c12037ca9e4db86bac395618464dc96ff007
2078	Information Science	SFT-INFO-CHAN-BROADCAST-011	Broadcast channel support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:215e043f018808b473b08f9113b2900b1948a728324f1990fdaffb48b4aa6969
2079	Information Science	SFT-INFO-CHAN-RELAY-012	Relay channel support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:df62ef27fa5b444b13adccfc7c2f9537b756a724591476dc885662cf562b0d6f
2080	Information Science	SFT-INFO-CHAN-INTERFERENCE-013	Interference-channel support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:431428161759b67f87b339a89148bfe1602ec4be735ea07b7308cf3096896224
2081	Information Science	SFT-INFO-CHAN-BIDIRECTIONAL-014	Bidirectional channel support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4e22c4332825787a8cbfffb3676c73f37edc72076625863e07cfbbfc0137704bd
2082	Information Science	SFT-INFO-CHAN-NETWORK-CUT-015	Network channel and cut boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:41d79da23d5b96253aabb7370098d045e18418d1e0f2242676731cdacf2fe74e
2083	Information Science	SFT-INFO-CHAN-FINITE-USE-SUCCESSION-016	Finite-use capacity succession	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c7531830adf2893dd2b011aa3d8aaf18925a189bce01a40a665498597988a2df
2084	Information Science	SFT-INFO-CHAN-SIMULATION-017	Channel simulation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d7266c17d27cc6c694d8357154b0b4ed73a21be9e2c4ca9aaa351577e86abaffb
2085	Information Science	SFT-INFO-CHAN-COMPLETENESS-018	Channel-family completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e3c2296a39a176c18bd68576bd1f623c3eb8e9757321974d2bc2f16a57ffaeal
2086	Information Science	SFT-INFO-NOISE-DISTINCTION-CLOSURE-001	Noise as unrecorded distinction closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2c59fcb12bb0cd1f79cba8728fe3740b4e7e53d8890175ff1f2dc3dc966dd8fc
2087	Information Science	SFT-INFO-NOISE-SOURCE-OUTPUT-MISMATCH-002	Error as source-output mismatch	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:20fe9bbcc22854b2caf0877cb0da418006b97f0f4d3a940c7eddd588a206e3d7
2088	Information Science	SFT-INFO-NOISE-PATTERN-SUPPORT-003	Deterministic noise-pattern support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ca36620686e6ba42acc03080cda9f0e4e3468b429ed9b20ae7792558202d47f0
2089	Information Science	SFT-INFO-NOISE-COMPOSITION-004	Noise transport through composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f411d0dd446f5e8aa8fec496d859398646ec02613011b5d75d7d57d8fe9cc9d1

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2090	Information Science	SFT-INFO-NOISE-DETECTION-005	Error detection condition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ff443b32efff524ab75589e0b024af5fd9561d6ea07801e84c12a8a08b03563d
2091	Information Science	SFT-INFO-NOISE-LOCALIZATION-006	Error localization condition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:25f5f7d192ad767fd4f5ee6d74b4392c2b73b0273c4bce8fdbba616b3e33ae5a4
2092	Information Science	SFT-INFO-NOISE-ESTIMATION-007	Error estimation condition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:80035d32fec7e6432c5dab60ca700b90ce1c7cc157fa2e044e8a2fac1bd83a64
2093	Information Science	SFT-INFO-NOISE-ERASURE-SUBSTITUTION-008	Erasure and substitution distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b7bd41ec0ec0c0bef55b91fb73331a1f0d3bcced6873707b656343ca52da8130
2094	Information Science	SFT-INFO-NOISE-BURST-CORRELATION-009	Burst and correlated error structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bcd898d47724b7c7de94e045b234bf837df6bc470adc054329c5995b76d24b6
2095	Information Science	SFT-INFO-NOISE-ADVERSARIAL-SUPPORT-010	Adversarial error support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5a50c16fcc7ecd835dcf922434f8e4ccdfb25c89cfc384af932d0eea79792ad
2096	Information Science	SFT-INFO-NOISE-BUDGET-LEDGER-011	Noise budget and exact error ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c291041e45974beb105ec02a65ae1f50562ef81e75edd429ac13cd02e70a5417
2097	Information Science	SFT-INFO-NOISE-COMPLETENESS-012	Noise-family completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be51f31b7e7846f52494eb93132aa6ca588b62e40e282f9750304563011314ee
2098	Information Science	SFT-INFO-CODE-INJECTIVE-REPRESENTATION-001	Code as injective representation relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8eedb307c63266882a05b5847914ffd99607833496076f4e64cffe8938c3291
2099	Information Science	SFT-INFO-CODE-SEPARATION-CORRECTION-002	Minimum separation and correctability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bad76227c373f70cbf44a67af4d068dc22bfa62392550b69f95d2ac1025e8904
2100	Information Science	SFT-INFO-CODE-BLOCK-STRUCTURE-003	Block-code structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:905e921ab972c4348f0aea9764c9945d1159b7b8c1d4942624809b15144dd2c7
2101	Information Science	SFT-INFO-CODE-LINEAR-CORRESPONDENCE-004	Linear-code correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b283fd299b33287ea5c2a445ae2f8f1c17966bca215641f97c9b2629fcb8e62
2102	Information Science	SFT-INFO-CODE-PARITY-SYNDROME-005	Parity and syndrome records	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bb6fce2cb9f68313d7ace83002d0da5f299aac40b0e8b8f2b0a21d806375af59
2103	Information Science	SFT-INFO-CODE-REPETITION-MAJORITY-006	Repetition and majority structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ac46498d78be4d3779b05e688659877eb01456d859eecd51858cf493a9d47678
2104	Information Science	SFT-INFO-CODE-ERASURE-CORRECTION-007	Erasure-correcting structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:85e62758213d1c9a18f068bdfd412c13a237be46f00a426f65d0f0a75452d398

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2105	Information Science	SFT-INFO-CODE-SUBSTITUTION-CORRECTION-008	Substitution-error correction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:506db5408d690b248e30fa4d958785d0aa07699594aa62dfc0e815bd136858e6
2106	Information Science	SFT-INFO-CODE-BURST-CORRECTION-009	Burst-error correction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:101335c069192c7d670895d0862b72a8d4f2e77ed4f5158644fc57b51c56ffe2
2107	Information Science	SFT-INFO-CODE-CONVOLUTIONAL-CORRESPONDENCE-010	Convolutional-code correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:72487799dc7d67f12cdbe3f6789fed882aaa259d11071a12a656c74e7ec277c2
2108	Information Science	SFT-INFO-CODE-TREE-SEQUENTIAL-DECODING-011	Tree and sequential decoding	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d815342fed0191dc4195e8cd0134a851bb3acb86cc5820b5dee5b32c4af33551
2109	Information Science	SFT-INFO-CODE-PRODUCT-CONCATENATED-012	Product and concatenated codes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:290655a4d8a33e94bb941d8829a425a03ebbd12b83673edce73edd086b9b6edf
2110	Information Science	SFT-INFO-CODE-SPARSE-CHECK-013	Sparse-check code correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:486e1fce3d13ba85cdfb02a817d17ff810de95ed0ec2bf7652109105dba55a3
2111	Information Science	SFT-INFO-CODE-NETWORK-014	Network coding structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2e044cbe04a0e7547d1ca01b3ac08ba58a9fe6b7aa39aafa94cb578b9fb7e646
2112	Information Science	SFT-INFO-CODE-LIST-DECODING-015	List-decoding boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1ab737f33f40f9d470ce5f700f24a0c6a39cfdf531a153b42c6cfe6aa569323d
2113	Information Science	SFT-INFO-CODE-ADVERSARIAL-BOUNDARY-016	Adversarial coding boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:48ca65c2688eaf13fc9142b6a5a0b8539006ef92a83e6f9a3154168dfa09f1fb
2114	Information Science	SFT-INFO-CODE-COMPOSITION-RATE-017	Code composition and rate custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:de4ee63380b53b014cb1a58b74555f20c6ff56fd3fd12400dae8c6a569acbccl
2115	Information Science	SFT-INFO-CODE-COMPLETENESS-018	Coding-family completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3229c52eba142879eb841c1149af7581c4ffa8b1bed3702a9f4bfe4eef53cd5c
2116	Information Science	SFT-INFO-REL-JOINT-SUPPORT-001	Joint information support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:00ba20826fa7886f7011c7e6e0984c59f5813a4a84e2ebfe55ad990bcc6c4519
2117	Information Science	SFT-INFO-REL-CONDITIONAL-RESTRICTION-002	Conditional information by exact restriction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9823edaf1f8c1c55b687027fdfa5fccaa80b3feeb6d156dea59641b5d2015518
2118	Information Science	SFT-INFO-REL-MUTUAL-SHARED-003	Mutual information by shared distinctions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f88c636c56eb29db885aef3de87e6e335ddca74a4ce8f5b4ef597c3137ce10ee
2119	Information Science	SFT-INFO-REL-CHAIN-RULE-004	Chain rule for retained information	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e81a57da54a45508607392bd274da76eb730b4f2d0dc4e9ae4df33ada9aec2c4

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2120	Information Science	SFT-INFO-REL-DATA-PROCESSING-005	Data-processing monotonicity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9307af947c9efb9363a3b5c28c8d40f4e699ccd949f4c08f5b3e883e80418e75
2121	Information Science	SFT-INFO-REL-CONDITIONAL-INDEPENDENCE-006	Conditional independence support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dda9fdc3e893dd8e503cc7630fde3eadb73fda5d7e125d56ff9f9f85cc22083efb
2122	Information Science	SFT-INFO-REL-COMMON-INFORMATION-007	Common information correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dae36a9b6029365ebe6ec148303a56e91d1a971082cfb9f46aed353ced0a4a2c
2123	Information Science	SFT-INFO-REL-DIRECTED-INFORMATION-008	Directed information on ordered records	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:933936e04c713c7e0b37eb9b883ed4d9bb5ab8fd523288db234566e9f74da58f
2124	Information Science	SFT-INFO-REL-INTERACTION-009	Interaction information correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fd5b38b110dce54b1ccd9496204f79aa354525dd69d49c55cbc15c8d0966172
2125	Information Science	SFT-INFO-REL-MULTI-TOTAL-DEPENDENCE-010	Multi-information and total dependence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3613adf822c9822cb6d05e3411ed89b1de4cbffd0595524c5b1d1cc2de28a167
2126	Information Science	SFT-INFO-REL-PARTIAL-DECOMPOSITION-011	Shared, unique and synergistic decomposition boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f66b4ac88b7fa7a98574bc566451ed5034078fe587f4ac0663848461f0d198e
2127	Information Science	SFT-INFO-REL-CAUSAL-FLOW-012	Information flow on causal paths	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bdc251e3f98224ab9c7a08c616e51fda4f449bb626984661ed201517c6df8f66
2128	Information Science	SFT-INFO-REL-REPRESENTATION-CHANGE-013	Relative information under representation change	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e0e7ae12c6ccaadfb7281fdde62bd2cf442dd4928905cb1a09ccdce3bf107f25
2129	Information Science	SFT-INFO-REL-COMPLETENESS-014	Relational-information completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:224b10cd37c046fa40ce6981d82d55bbabedfe79b44dfca0213159c94898ef6
2130	Information Science	SFT-INFO-COARSE-SUFFICIENT-RECORD-001	Sufficient record for a declared observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6c95f59d6ede0708d99a596971812733d6fe2dc7b4ba20851d4541467ea493aa
2131	Information Science	SFT-INFO-COARSE-MINIMAL-SUFFICIENT-002	Minimal sufficient record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:db19424a38e4035ad03da8a402288566459711ce4db9c31d7c6522ba5b8c599f
2132	Information Science	SFT-INFO-COARSE-BOTTLENECK-003	Information bottleneck as lawful coarsening	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a8e223eadd37a7481f3b4d7e86c64f73874b5342ef0d7622c97bcf35a1597e
2133	Information Science	SFT-INFO-COARSE-PARTITION-004	Coarse-graining partition structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:53298df77ca5b49459e6f736cce7e341a10de8459dacefe92fd41119c71b16b8
2134	Information Science	SFT-INFO-COARSE-REFINEMENT-ORDER-005	Refinement and abstraction order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:854bcb61f6e1991169d6a533f0d115b7ddf4cb3124f870501a0fb431c2c61877

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2135	Information Science	SFT-INFO-COARSE-LOSS-LEDGER-006	Loss ledger for aggregation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7271f4e4aac9fba5c1d1fb9e9f4f83468773b8de1b2ffd627b9028fbd09d77bb
2136	Information Science	SFT-INFO-COARSE-FEATURE-SELECTION-007	Feature selection by retained distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6f6034b8862bc5167fb3441ba76b71cb16aaac7795d4446fc05209400c4141d2
2137	Information Science	SFT-INFO-COARSE-STATE-LUMPING-008	State aggregation and exact lumping	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c4419bef376017a10c6f4c09e1f6cacc6e2675645cb04a897984bcf4d96e6f83
2138	Information Science	SFT-INFO-COARSE-RECONSTRUCTION-BOUNDARY-009	Summary record reconstruction boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8b53ea67fcfb48054fa3dc7e7cb03681b0cc30535c8f679c69a9fc4c99580f71
2139	Information Science	SFT-INFO-COARSE-COMPOSITION-010	Multi-stage coarse-graining composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:04e613932405f3bd3840ee2ad0012433d18876dd48ceef853726af654c860c7
2140	Information Science	SFT-INFO-COARSE-REVERSIBILITY-011	Coarse-graining reversibility condition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d98b026ead033fe633b15866110646cba156f06f9434e7dc464f2ea64429a71e
2141	Information Science	SFT-INFO-COARSE-COMPLETENESS-012	Sufficiency and bottleneck completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2a994334a9f3743447e8ee7d38a84b77d1217bcb599dff7722f758d63e0f963
2142	Information Science	SFT-INFO-RETR-INDEX-LOCATION-001	Index as a source-bound location relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3094bedd21d986cf9b7f63ba112c136a6f81d95ea86f3bb2b180f7f79af39b46
2143	Information Science	SFT-INFO-RETR-EXACT-MEMBERSHIP-002	Exact retrieval and membership	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9e2ca2e612e6a0462fc5fe76c9e1900cb806254bdb760e3c34b27bfe387dcbd
2144	Information Science	SFT-INFO-RETR-FORWARD-INVERTED-003	Inverted and forward index correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ae60d6a6c7cc6e341bb8f9d9ce9405d94303cf5d57cf4dbf6503ec3b5259d4ce
2145	Information Science	SFT-INFO-RETR-QUERY-OBSERVATION-004	Query as a declared observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1b61a4d68015955f8847b4cf7cd8dbbc819bf41dcd3bfe74499791fa86b56001
2146	Information Science	SFT-INFO-RETR-RANKING-ORDER-005	Ranking as an exact finite order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:69dccac7f3a3c65390d4e04026a4fbf21ed31f3ef1f543b3280b0263b001e2
2147	Information Science	SFT-INFO-RETR-TIE-INCOMPARABILITY-006	Tie and incomparability custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ee4180014b8786d4ecad3cd98ab4e5cd392a1b4a599803c11ac32f3c70b0900b
2148	Information Science	SFT-INFO-RETR-RELEVANCE-RELATION-007	Relevance as a registered relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c550607fac9287a53ec4ced76deff4b98df4b93944267917ad7b4c35a1a65a47
2149	Information Science	SFT-INFO-RETR-PRECISION-RECALL-008	Precision and recall exact-part correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a57b6ba88d5f0fb00942324999058aa7562753f755d1c7ef33c58c5acde62bd5

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2150	Information Science	SFT-INFO-RETR-TAXONOMY-009	Knowledge organization and taxonomy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8150318cc24387a8e2d1bebb8908d558604af3ba4c08dbaa128e50c4e44282ad
2151	Information Science	SFT-INFO-RETR-CROSS-REFERENCE-GRAPH-010	Cross-reference and citation graph	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d2245c1b9469cc2f88abb75687e89ae76f94736b2a9f254372692fd4adb a2394
2152	Information Science	SFT-INFO-RETR-UPDATE-STALE-BOUNDARY-011	Update and stale-index boundaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d6fd5ac22dd4ce19e5594b4f56e332bb946f3a7e1beb405baffea696a369b11d
2153	Information Science	SFT-INFO-RETR-COMPLETENESS-012	Retrieval-family completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:31e9b8db1a4f93fc1b36ea5adc18f7a85c5153ad3f6e81ebeca87da8a4c8955c
2154	Information Science	SFT-INFO-INFER-SUPPORT-RESTRICTION-001	Inference as support restriction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5a05c6e4b4280b01bed0a1dc1abc4fd89a4b184670827653acc73bf724e3a546
2155	Information Science	SFT-INFO-INFER-DETECTION-002	Detection as observation-class decision	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:99b0c51a3c228b310adea6ba2f0acfb43cb5416bab58876bc797ab2219ed9ea1
2156	Information Science	SFT-INFO-INFER-ESTIMATION-003	Estimation as retained representative selection	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cca1bb73b97dbc67d792394b0b0579f60d7c71bf84ff0a6dd32a0011aac46ac7
2157	Information Science	SFT-INFO-INFER-FILTERING-004	Filtering as sequential record update	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8d8f08164eb9cecfa066daf07808217a325464a55a8a6ecd6d666c3917940b28
2158	Information Science	SFT-INFO-INFER-SMOOTHING-005	Smoothing as whole-record reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:51b87b53d8caff87ca4d4743ed0b0af0c9da636c6f778fd36b021491d08f9ae3
2159	Information Science	SFT-INFO-INFER-PREDICTION-006	Prediction as registered transition support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:de5f48db576b5a0f6e31af57fab4fe81b4fbe5022603a58862a466d4b5dba69e
2160	Information Science	SFT-INFO-INFER-EVIDENCE-ACCUMULATION-007	Evidence accumulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3dfafa28b5e96b811e725a521080c4ec557601297fad97f08d5650808c296373
2161	Information Science	SFT-INFO-INFER-LIKELIHOOD-CORRESPONDENCE-008	Likelihood correspondence on exact support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4439c042519652aa49a8c876702cf8815d7140074c716c2acea73d38259e40d4
2162	Information Science	SFT-INFO-INFER-STRUCTURAL-THRESHOLD-009	Decision threshold without fitted scalar	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:72f4c8072030e763982c4166aeb472fb02467bc3d41e178b8595b67b1119a8e5
2163	Information Science	SFT-INFO-INFER-ERROR-CUSTODY-010	False-positive and false-negative custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:29a3735b9dd799d72a1f1513f6de32df493747f163a0a572c97d0f8dbffc0716
2164	Information Science	SFT-INFO-INFER-IDENTIFIABILITY-011	Identifiability and observational equivalence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:85783c389a0eb5a4026e0333b60c7ba90abf6912f7fc20a4fe70c8b74d84467c

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2165	Information Science	SFT-INFO-INFER-SEQUENTIAL-STOPPING-012	Sequential detection and stopping boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2d7d620ce1c627a1bf992164b1cb7b651248b88576dc99fb5b28494ce3232c58
2166	Information Science	SFT-INFO-INFER-MULTISENSOR-FUSION-013	Multi-sensor information fusion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f174110d0bd0aa794981eee40c481bd4bfff6606f74c7730019b4aad2548eb141
2167	Information Science	SFT-INFO-INFER-COMPLETENESS-014	Inference-family completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f2972f07a838cfda163177fd0520b6273828f220ab0d32e26a6713090235aedd
2168	Information Science	SFT-INFO-PRIV-LEAKAGE-001	Information leakage as retained adversary distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d5e1ed5a1b5fe01be6df5bb205feae224c7d37f497e52a7376b7e8a6614ccf65
2169	Information Science	SFT-INFO-PRIV-PERFECT-SECREC-002	Perfect secrecy correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:394da7d7a96dc6428ac5071d9a010365f35e8da0a70e3e5b25d7aa2ddf97039d
2170	Information Science	SFT-INFO-PRIV-SIDE-INFORMATION-003	Side-information and disclosure boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4463e1dc2d066c1746222cdddefdccbeaa42bb0d228ff7e6e1156a345ede63
2171	Information Science	SFT-INFO-PRIV-ANONYMITY-004	Anonymity and indistinguishability classes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1a437ee1b6cc8e81ecfb50465503ac76ea71b02fecbe6a84aab13782b442eb82
2172	Information Science	SFT-INFO-PRIV-LEAST-DISCLOSURE-005	Access observation and least disclosure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3b43fddbe433f91254544278a9b64ed60b222c09a9dd51c94ae3024b15d36e97
2173	Information Science	SFT-INFO-PRIV-COMPOSITION-006	Composition of privacy losses	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b1a70772a665f79302bee0d8b1824f7e0480a91e76fec9af54fbb674fdddee5
2174	Information Science	SFT-INFO-PRIV-QUERY-RECONSTRUCTION-007	Query disclosure and reconstruction risk	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:37d77397646cd879646a1d1c7e8bb548734757510e6a3c082430656463076e98
2175	Information Science	SFT-INFO-PRIV-UTILITY-BOUNDARY-008	Privacy-utility exact trade boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:56e3cbc2c8b0503cbcc009d80f3e36b95875742aed1a5fe90e6fb9c35f66ffad
2176	Information Science	SFT-INFO-PRIV-CRYPTO-HANDOFF-009	Cryptographic ownership handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8d2c7635bdd22d24732063dccb833a9c146aa331fef2ba202dd6eb42a7b88f8d
2177	Information Science	SFT-INFO-PRIV-COMPLETENESS-010	Privacy-interface completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6a7f4ec296e1eb4ac71c80ecde7709c7ed090ee9ef8326130ed6ea5b5b4c4721
2178	Information Science	SFT-INFO-THERM-PHYSICAL-CUSTODY-001	Information record as a physical custody obligation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2d0295b2cf23ff58d8f98e9b1c160d517a1b36b691b686a7d0c88cc816ac76a2
2179	Information Science	SFT-INFO-THERM-IRREVERSIBLE-MERGE-002	Irreversible merge and missing predecessor label	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a99a0ae47e94bee5844a5e063d75dbc92dc9428b94be5ee661b29e6318455463

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2180	Information Science	SFT-INFO-THERM-REVERSIBLE-RETENTION-003	Reversible record retention	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a28cf0219b9c74c8c98b35ff376ebd20c0da17c5993bea2d662a24d50d13ae33
2181	Information Science	SFT-INFO-THERM-ERASURE-BOUNDARY-004	Erasure correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cea4d5f9af27a9f39ae64961bf86cc0b432ef1ef4abff03d4cb08ad51be26976
2182	Information Science	SFT-INFO-THERM-MEASUREMENT-RECORD-005	Measurement record cost	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dcf2b5c1949690047ee8c1ba6acd05b4f1f3fc9f13499925486ab57d98533051
2183	Information Science	SFT-INFO-THERM-MEMORY-RESET-006	Memory reset and provenance custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0d4926053fca5a826d3950cd0c2356836bdd803489a7ebc4e195594e851f82b0
2184	Information Science	SFT-INFO-THERM-MAXWELL-DEMON-007	Maxwell-demon information ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ac479ce2e5d5d8d4ff3e3edf60147a75e60baf9ed7e79dcaba2d303e880733
2185	Information Science	SFT-INFO-THERM-LANDAUER-BENNETT-008	Landauer and Bennett correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:138ca042b0cfac3f6186893aaae10d777cfd53161a559ce274641d8da0b4349c
2186	Information Science	SFT-INFO-THERM-NON-EQUIVALENCE-009	Information-energy non-equivalence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a08b20c3def670d95b04b75bc75d3e53e4dd4a6a6dce9ac61238252da4aa838e
2187	Information Science	SFT-INFO-THERM-COMPLETENESS-010	Information-thermodynamics completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b662bedffa07ff8ade8f3596731a298ca45959b594568080eca0f994b4f4ecf7
2188	Information Science	SFT-INFO-CORR-CLASSICAL-SYMBOL-001	Classical symbol-information correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:118ba8a08ee951156e3c7e9e436b7ce27523ac223143cdf1e804d15e43f0a5ec
2189	Information Science	SFT-INFO-CORR-DETERMINISTIC-PROBABILITY-002	Deterministic-support probability correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:89616d25a9baf24176aa4e95c638f9c4a4359b6d70440a3f179dcf1184f31dd8
2190	Information Science	SFT-INFO-CORR-PROBABILISTIC-MIXTURE-003	Probabilistic mixture as hidden-support description	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2522d7ab8bdd8abb71758e3167a246660634f72b16c1b88a4deaf1644fb638c3
2191	Information Science	SFT-INFO-CORR-QUANTUM-SUPPORT-004	Quantum support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bc3f2bf9ab87ee90bdadfc03a01864f7e07a8559a6956f48d4be341ca27e1bec
2192	Information Science	SFT-INFO-CORR-QUANTUM-UNIT-005	Quantum information unit correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:554e1458658eb5e15d2d128bef721c6539819a0666e5ba44d6b6e16bb20fd09
2193	Information Science	SFT-INFO-CORR-PRODUCT-STATE-006	Composite support and product-state correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f0ff219046370d360d05ac520611db612ba5173cefec635a8e1d19000d352ef6
2194	Information Science	SFT-INFO-CORR-ENTANGLEMENT-BOUNDARY-007	Joint support and entanglement boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5fa979b85852fe123af398b4dc b4d2d49051bfa7e42a8bf6aa9110932fdadeec

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2195	Information Science	SFT-INFO-CORR-PHASE-HANDOFF-008	Phase information ownership handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fe8cd634a75985a8a44fe9418eddf56537e4a3d7fa4783988c7ae32035e04c48
2196	Information Science	SFT-INFO-CORR-MEASUREMENT-SUPPORT-009	Measurement support and record correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b9bf7541d51c25d138e4ac83e062aa5dc4f4c1975b0960cac0d6b985e7b8d75
2197	Information Science	SFT-INFO-CORR-CLASSICAL-QUANTUM-ENCODING-010	Classical-to-quantum encoding boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e8cad4f6644667784e5c6c7cd0e61ca7da360915d55b238f57c0336eccb7d7e5
2198	Information Science	SFT-INFO-CORR-QUANTUM-CLASSICAL-OBSERVATION-011	Quantum-to-classical observation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bfc0c7b211fdbbdfa90769fd16c8923025edc967489bef354b02507b5cd5ce7
2199	Information Science	SFT-INFO-CORR-NO-CLONING-BOUNDARY-012	No-cloning information boundary correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:10224f88e4fb663e8fd3626bbead483ab7b13155f2e2169be9611817245170c5
2200	Information Science	SFT-INFO-CORR-QUANTUM-CHANNEL-013	Quantum channel support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3dc6abbae97fa6b29ca17d100b3604d7d7ba62c62b072a0fbb29f15f629991b2
2201	Information Science	SFT-INFO-CORR-QUANTUM-CODING-HANDOFF-014	Quantum coding ownership handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:824a1118b9f203ef2bfbcb1cb203cc5aae00cac49abada352f3660079aff4eb2
2202	Information Science	SFT-INFO-CORR-NONDUPLICATION-015	Operational-dynamics nonduplication boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:452ba460fdf82a1005400b2764039037fbc6d38b98527a1e8fc554f669036e78
2203	Information Science	SFT-INFO-CORR-COMPLETENESS-016	Classical-probabilistic-quantum completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:776c6b3d98422e38cda9379977c46ef5f93409220e6716024bfffa07200504289
2204	Information Science	SFT-INFO-SEM-SYMBOL-CONTENT-BOUNDARY-001	Symbol count and semantic-content boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b34473d457ce45cd3d8166ab996da33cb3b1a7df8b8eb01b9b892be2e03d7508
2205	Information Science	SFT-INFO-SEM-REFERENCE-INTERPRETATION-002	Reference and interpretation record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7bb7cd84816f3d312986329effe2a9566b7d20e89bb571a394576e18ccaf7a0c
2206	Information Science	SFT-INFO-SEM-CONTEXT-DISTINGUISHABILITY-003	Context-dependent distinguishability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:635490f5a7dec05f2b988b63dc45efc2142717f8146e9512f38af0acffeb12d
2207	Information Science	SFT-INFO-SEM-BIOLOGICAL-HANDOFF-004	Biological information ownership handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f34f12b5b901548bc6b6d5c584b0f82be02aba1113ec310c207c00c1668c05bb
2208	Information Science	SFT-INFO-SEM-GENETIC-CORRESPONDENCE-005	Genetic representation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:43ff998a5845965206a402626cebc2320ab39bdee80a44faed50bde4376699c0
2209	Information Science	SFT-INFO-SEM-NEURAL-COGNITIVE-HANDOFF-006	Neural and cognitive information handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5ade99b0110b049309903db861e798d292250dba1fd0868306f74ebedf265591

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2210	Information Science	SFT-INFO-SEM-CONSCIOUS-REPORT-HANDOFF-007	Conscious access and reportability handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:137fcb0324afef8d1f6ad5a45a1b07f5df3701fcee827086cfdfffa33e0ad5f4
2211	Information Science	SFT-INFO-SEM-SOCIAL-SHARED-RECORD-008	Social communication and shared-record handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:659099aec42117c905586497e38982dc10c861f6e16c2317425326dee8deb57
2212	Information Science	SFT-INFO-SEM-TRANSLATION-PRESERVATION-009	Meaning-preserving translation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cc7e04894916d4ea57454b7306c18f3fb3a5d4ad533a33c577bff3db8e623707
2213	Information Science	SFT-INFO-SEM-PRAGMATIC-ACTION-BOUNDARY-010	Pragmatic action-information boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:28e546ea99c3b74db31ff9da432ccf7377df1f6c643601647ade081acf8af761
2214	Information Science	SFT-INFO-SEM-CROSS-DOMAIN-PROVENANCE-011	Cross-domain provenance and nonduplication	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6cd4e601d33ee9a0519d4c9c783c4c9b08636d6b0d41564d757de57a8e0e248a
2215	Information Science	SFT-INFO-SEM-COMPLETENESS-012	Semantic-handoff completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:996be961ea2a53fba28114d5b99e6d05773712dfd4fdb3a2693d4b9207216658
2216	Information Science	SFT-INFO-VALID-SYMBOL-REPRESENTATION-001	Symbol and representation complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:13905605a50b784959980afaa21acd703491416db290915d3d22d2b8f8fb36d7
2217	Information Science	SFT-INFO-VALID-RECORD-PROVENANCE-002	Record and provenance complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e50c6cb2f1852e84cc550a4d34331d5ef65ac7727ece5b477600c7245e099500
2218	Information Science	SFT-INFO-VALID-SOURCE-MEASURE-003	Source and measure complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e4d2488dcc212f2693e7c48d372c4354bf9fc26aa6ac6dbf7ef93bdf89f542f0
2219	Information Science	SFT-INFO-VALID-SIGNAL-SAMPLING-004	Signal and sampling complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ec09e7739ec73209f79f0998119b960deb7dfbc7056ea62abd7ce98d3f719719
2220	Information Science	SFT-INFO-VALID-COMPRESSION-DISTORTION-005	Compression and distortion complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fa495dfb7aa1f9b9ce3c35a347554c45d2fb3cfc6699a3b78110bea94d78c4a3
2221	Information Science	SFT-INFO-VALID-CHANNEL-CAPACITY-006	Channel and capacity complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:01cb3d3963c79cfd1bba2523d1ac9fb4512f2f7de977d8a66276aa2142961438
2222	Information Science	SFT-INFO-VALID-NOISE-CODING-007	Noise and coding complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5601ea6e1088c2d50f796c28718b70a11a3a95f95fb0d12378daf3c53bb3f289
2223	Information Science	SFT-INFO-VALID-RELATIONAL-COARSE-008	Relational and coarse-graining complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:826d35df20d7ea259d64e873ba161a578b8597dfad41020a5ef13a95b4dfd64d
2224	Information Science	SFT-INFO-VALID-RETRIEVAL-INFERENCE-009	Retrieval and inference complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:26e531f060e701c76464ad95b078892b2c2d507ed1d867e3583bbf41378bb5ae

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2225	Information Science	SFT-INFO-VALID-PRIVACY-THERMAL-CORRESPONDENCE-010	Privacy, thermodynamics and correspondence validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:69afe4173c3da6539dfae6b81f5768559180a87b37c86efb0905278ece5d090f
2226	Information Science	SFT-INFO-VALID-ADVERSE-BOUNDARY-011	Adverse, absent, unresolved and boundary vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:49d52801554c6c2e7504c642f2a72e1519d41728f2cd87e6aec326f1553cbb90
2227	Information Science	SFT-INFO-VALID-GRAND-LOCK-012	Information Science empirical and formal Grand Lock	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7b67ff978891dc8cc9b0e1a956def0b6006b545966bb1d8876d767ce10fda35b
2228	Information Science	SFT-INFO-HAND-ONE-OWNER-DOWNSTREAM-001	Information Science one-owner downstream handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:84fd6deba477535e0957ec15ce74dd798ba082e143eaac8668b3e695a3c91e3a
2229	Information Science	SFT-INFO-HAND-MEASUREMENT-BOUNDARY-002	Information Science measurement-boundary handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0d9849de5d10c9b2274eeda65888ba039347b56ef77a2351e22fc634a3b56ebf
2230	Information Science	SFT-INFO-HAND-FORMAL-EMPIRICAL-003	Information Science formal-to-empirical handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:79f833a305f0f49b5a7db1a6e8f0ef8989f7efff4ed0e488fd9d29cd8b8e03cbf
2231	Information Science	SFT-INFO-HAND-CONVENTIONAL-CORRESPONDENCE-004	Information Science conventional-correspondence handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ce76e7148ac9ea1c6e9ffca7d2ca72072cb35d4167744c4d6eab6f4ce5eafc7e
2232	Information Science	SFT-INFO-HAND-OPEN-EXTENSION-005	Information Science open-extension handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:521d97d351870c06956513f18cd6a77b7221da4b62697213d8a692ca4ee56a1
2233	Information Science	SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006	Information Science cross-branch completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8e464b9679a76b39f142b622489e4be2beda543d18c7de51abd6bd2567dac8d0
2234	Classical Computation	SFT-COMP-FORMX-CONFIGURATION-IDENTITY-001	Configuration identity and canonical state encoding	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:05ef72b504a615bf9f8e04128b538c827f594e0e7f55391e473c57ce8463b86e
2235	Classical Computation	SFT-COMP-FORMX-PARTIAL-TOTAL-TRANSITION-002	Partial and total transition relations	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:922ed7d7b1ff3af08a6442d7d3a0d47a4c9b0c76ea5089c3e6ba56ab3c41fd75
2236	Classical Computation	SFT-COMP-FORMX-TERMINAL-OUTCOME-DISTINCTION-003	Acceptance, rejection and nontermination distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ddfc25a2da658167388daf5077efe788462d3216876e360cbcdcd199258b34e7
2237	Classical Computation	SFT-COMP-FORMX-LANGUAGE-BOOLEAN-CORRESPONDENCE-004	Language union, intersection and complement correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a52baf651920e7ff284f84abe21503b55b70d132d5feb1a6f0954f9264f1f17c
2238	Classical Computation	SFT-COMP-FORMX-CONCATENATION-ITERATION-005	Concatenation, iteration and generated-language closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:308dd8c15433530c5782863b61e213beba2cc3df8538fd053a598284655a58
2239	Classical Computation	SFT-COMP-FORMX-DERIVATION-TREE-AMBIGUITY-006	Grammar derivation trees and ambiguity custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e4b80cdfce27e9bb0379164532369b43b3dd83978a0c0f6ec53a506ba30235886

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2240	Classical Computation	SFT-COMP-FORMX-PARSE-RECOGNIZE-GENERATE-007	Parsing, recognition and generation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3941b75acdf1ab76fcb29f63ca1351cc1aa20b2f9a497b368325c4621e595f28
2241	Classical Computation	SFT-COMP-FORMX-AUTOMATON-PRODUCT-QUOTIENT-008	Automaton product, quotient and minimization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:72a37de3f8ee4eba738462f193e7b0134ede3e14cf36dd962891a8eece3b7422
2242	Classical Computation	SFT-COMP-FORMX-FINITE-TRANSDUCTION-009	Finite-state transduction and output custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:972b0967ec11b9f859dcc1f1d76b24d285d9a0c00dfef003f3b8ed26bed6a29f
2243	Classical Computation	SFT-COMP-FORMX-STORAGE-MACHINE-CORRESPONDENCE-010	Stack, queue and register storage correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:42d1ee0d0b70f6af59c575cc6499f17220e2c1a4d753a5709f71185e3a9d5085
2244	Classical Computation	SFT-COMP-FORMX-REWRITE-NORMAL-FORM-011	Rewriting termination and normal forms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ec6b6dd611fa6125e4580ecd3dcc50e2aba6c1c1bcad91d697787575e1a302b
2245	Classical Computation	SFT-COMP-FORMX-REWRITE-CONFLUENCE-012	Rewriting confluence and critical-pair custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:75151477eacbcf657be6c606b04e8059925e185c6fbc60ee5afc449cdfac047f
2246	Classical Computation	SFT-COMP-FORMX-RECURSIVE-COMPOSITION-013	Recursive composition and finite iteration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:29fbc24de44568a20cd2c2f495e5cfe22f7f15a3ac6d061cf2392226eb8ccf2
2247	Classical Computation	SFT-COMP-FORMX-PRIMITIVE-RECURSION-MINIMIZATION-014	Primitive recursion and minimization correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6f820a230c272c19e12c63a5a3c787ffa2dd423065d162799a908692d62175df
2248	Classical Computation	SFT-COMP-FORMX-LAMBDA-CAPTURE-NORMAL-015	Lambda reduction, capture avoidance and normal forms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6bf29af9584f6bf9beddd7af94026b30bf783ccc3d75161d3be77f3d724c6747
2249	Classical Computation	SFT-COMP-FORMX-MACHINE-SIMULATION-INVARIANT-016	Abstract-machine simulation invariants	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:74b76cc1c1ac9a63fae07f6b56264188473a100804e25e1077bbbd4fadfb6cd1
2250	Classical Computation	SFT-COMP-FORMX-CIRCUIT-ACYCLIC-EVALUATION-017	Circuit fan-in, fan-out and acyclic evaluation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bf7008c723dd9079f6efcb7ff27fea97bdd4b94e1d6f483d6ae158e85e650218
2251	Classical Computation	SFT-COMP-FORMX-SEQUENTIAL-COMBINATIONAL-018	Sequential and combinational process correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6909c6914657670ae1b1930ed093622de2d1fc793a5aa15131d20456cb1b832b
2252	Classical Computation	SFT-COMP-FORMX-PROCESS-ALGEBRA-EQUIVALENCE-019	Process algebra composition and observational equivalence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cd1f9cd0c479e29f67359615d2436e9c7846ce517ecf9cb66d1f67d54515d4d5
2253	Classical Computation	SFT-COMP-FORMX-UNIVERSAL-SELF-INTERPRETATION-020	Universal interpreter and self-interpretation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a02e52a528eced7ce952aed8e063a18fd4ad825c971d29a0036cac34f6787811
2254	Classical Computation	SFT-COMP-FORMX-MODEL-TRANSLATION-OVERHEAD-021	Effective model translation and overhead custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cc88c1b0901080bea719456844010d90fb80ac69a08d521bb8db23813935a933

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2255	Classical Computation	SFT-COMP-FORMX-COMPLETENESS-022	Formal-computation completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4cdcde71060d5990ee806c8c9ea98ae13a031e61e0eef641c695b7b854feece3
2256	Classical Computation	SFT-COMP-CBLX-DECIDABLE-RECOGNIZABLE-CLOSURE-001	Decidable and recognizable language closure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ebb5093c994b260f89fe43d90812062a281979b5491777b7f4384eaf7160afed
2257	Classical Computation	SFT-COMP-CBLX-CO-RECOGNIZABLE-BOUNDARY-002	Recognizable and co-recognizable decision boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4d2ac779b57b7ba851730b976f8c756d98bd79d99a9a2be70690d5b9ff59b4b7
2258	Classical Computation	SFT-COMP-CBLX-DOVETAIL-ENUMERATION-003	Effective enumeration and dovetailing	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:206e232b7ac03edf9a031fa59b6bf1c02b2beec9f2e0a4912e9298449f8d2cc0
2259	Classical Computation	SFT-COMP-CBLX-DIAGONAL-LANGUAGE-004	Diagonal language construction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:85e2e72ac9a0953cad077e8dfb8b5ff44a5f8d42e51bbc15e2988a0498c598c1
2260	Classical Computation	SFT-COMP-CBLX-SELF-REFERENCE-FIXED-POINT-005	Self-reference and fixed-point construction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8bff973be333e45d57e4f8197e626478b13d105eecab15216b4ebaa9bb80ce77
2261	Classical Computation	SFT-COMP-CBLX-RECURSION-THEOREM-006	Recursion-theorem correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b4ffa3ec2dbc99836f568bbb592e0358ffad889d575770044821893ad2e320db
2262	Classical Computation	SFT-COMP-CBLX-SEMANTIC-PROPERTY-UNDECIDABILITY-007	Semantic-property undecidability boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8a7e29f98636fb2ab938f2a2c3e2cec40b4846594d743e1df9d4a955aee2b85e
2263	Classical Computation	SFT-COMP-CBLX-MANY-ONE-REDUCTION-008	Many-one reduction and composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f3a170cd132c5f82e27e3c0f8a65ab640585bc024e8cd8228ac6c6c03ebbf04
2264	Classical Computation	SFT-COMP-CBLX-TURING-REDUCTION-009	Turing-style reduction correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c88f6e8bf37f767383f7a32619430ca628605d38daeb13bbab36a3e89907853f
2265	Classical Computation	SFT-COMP-CBLX-ENUMERATION-REDUCIBILITY-010	Enumeration reducibility correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c8e6aed32024c375f07ad947ed27a740cafffc906239b1ad675ccf91947dbddb1
2266	Classical Computation	SFT-COMP-CBLX-DEGREE-ORDER-011	Computability degree ordering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6fcea0a78e3759e302f40a2b8500b0d22bfb3907c01d044b3f3eed3e960163e6
2267	Classical Computation	SFT-COMP-CBLX-JUMP-RELATIVE-SUCCESSION-012	Jump and relative-computation succession	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1726d2afb081664fda9c8468d6c156e0c0efa030eb635901f804df7c4fb0b247
2268	Classical Computation	SFT-COMP-CBLX-ORACLE-ANSWER-CUSTODY-013	Oracle answer-record custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:08b45db8aa6523cfeee683c6307838e61ade9ff4c294538785a3b787e0f12b72
2269	Classical Computation	SFT-COMP-CBLX-ARITHMETICAL-HIERARCHY-BOUNDARY-014	Arithmetical-hierarchy correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f105b1030b22d9a7ca3016a29975066fb44d6f07510a91c845664a0cf3148d2c

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2270	Classical Computation	SFT-COMP-CBLX-POST-CORRESPONDENCE-WITNESS-015	Post-correspondence finite witness boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0a5997a26e1f1fd9b3d9b1faa90a7fb37a02ad74d4ee65590f090b322072a02b
2271	Classical Computation	SFT-COMP-CBLX-ENTSCHEIDUNGSPROBLEM-BOUNDARY-016	Entscheidungsproblem correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:185ee2b6e356466324cbe70cc81198cbbfdafa4588e6ce4fc49e976ae53cf9936
2272	Classical Computation	SFT-COMP-CBLX-INCOMPLETENESS-CONSISTENCY-BOUNDARY-017	Incompleteness and internal consistency boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d21509d08b502f830693c6d19eb030d8c3e71c0c3762ea15e4b1a45dc070ad7c
2273	Classical Computation	SFT-COMP-CBLX-BUSY-BEAVER-DOMINATION-018	Busy-Beaver domination theorem	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bd7d42e6dcdc3d369578cb0b8bb334b024147eb7fa95557ea28f504c3d620e46
2274	Classical Computation	SFT-COMP-CBLX-BUSY-BEAVER-FINITE-CENSUS-019	Finite Busy-Beaver census protocol	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c119219f70bdd4b936a398e1ba5106f21274910e7fe24d6fe10e7b60b5b1374d
2275	Classical Computation	SFT-COMP-CBLX-HYPERCOMPUTATION-ADMISSIBILITY-020	Hypercomputation admissibility and physical-record boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8fd953b4892eb44c0a1d20601b2121c553b0852b558c490af8f007412444ea6a
2276	Classical Computation	SFT-COMP-CBLX-COMPLETENESS-021	Computability completeness and no-omission certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e96c59bfea8b5a95027908eb23c03f3da2a172e044b6bf3181060f30c39cd028
2277	Classical Computation	SFT-COMP-CPLXX-INPUT-LENGTH-CARRIER-001	Canonical input-length and instance-family carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5ae2c108a4e22c1175fb4feccf1870d4925263cea2921a47c49d05495cd c39c7
2278	Classical Computation	SFT-COMP-CPLXX-DETERMINISTIC-TIME-CLASS-002	Deterministic time-class correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:43c83ffdd9b8eba6895e93c55ee3419c0e4df5d2579780f891183601e72ace07
2279	Classical Computation	SFT-COMP-CPLXX-DETERMINISTIC-SPACE-CLASS-003	Deterministic space-class correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ac351aee769a0e2311437f55a34476ea31426c47e721e4f979e1f05583ac56c2
2280	Classical Computation	SFT-COMP-CPLXX-HIERARCHY-SUCCESSION-004	Time and space hierarchy succession	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3fb3a7123665c18a4b00b674f5e1c043a0ef171d2fee060c77efe4af8df2a191
2281	Classical Computation	SFT-COMP-CPLXX-NONDETERMINISTIC-SUPPORT-005	Nondeterministic support as complete deterministic branch support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5132f99ae3fb750cf58830a8c5466356ac4b0ebfcfe46dde5569a031a32e17f
2282	Classical Computation	SFT-COMP-CPLXX-CERTIFICATE-VERIFICATION-RESOURCE-006	Certificate length and verification resource	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e6b15c6b24274b3854fee82be148d5ba275b206a669b447c5d3806a0d2c217d
2283	Classical Computation	SFT-COMP-CPLXX-FOLD-P-NP-SCOPE-007	Fold-P and Fold-NP scoped equality boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:defaca0bdd112782bf55a8c0ef3c716526c6a9f975da569aa1240237e354aba7
2284	Classical Computation	SFT-COMP-CPLXX-CONVENTIONAL-P-NP-TRANSPORT-008	Conventional P-versus-NP transport boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:399f5778bedbf1152360ec18540b8785170ff1b25e8e195311b4c2389d78660b

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2285	Classical Computation	SFT-COMP-CPLXX-COMPLEMENT-CLASS-009	Co-decision and complement-class correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5f5be91c1b06981bdbcd5df244115e4ee81c0c8f4f69bc918fcfc64379b7a3297
2286	Classical Computation	SFT-COMP-CPLXX-ALTERNATION-SPACE-010	Polynomial-space and alternating-process correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:035ffff1d6218f0c2b5706f5f40cce634be22c2aedd95879b60f6a62ff6860d31
2287	Classical Computation	SFT-COMP-CPLXX-EXPONENTIAL-SUCCESSION-011	Exponential-resource succession	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3f4a2e70c6a97c4e31201af7ff72e59896208a1c817f044476fddde20560550f
2288	Classical Computation	SFT-COMP-CPLXX-CIRCUIT-UNIFORMITY-012	Circuit family uniformity and nonuniformity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:71a76ad6406f38c2c57786270aafae84c2fd99f8599be3617079832aace2ce3d
2289	Classical Computation	SFT-COMP-CPLXX-CIRCUIT-SIZE-DEPTH-WIDTH-013	Circuit size, depth and width trade ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:04a764d2868070278f9b3a0373ad1df827f0e450645b5152c9e5a518687d2907
2290	Classical Computation	SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014	Formula, branching-program and circuit correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:230d2589acde78139b2b2a6e6447467107881a8d69715d4712418379a94955d2
2291	Classical Computation	SFT-COMP-CPLXX-PARALLEL-WORK-DEPTH-015	Parallel depth and processor work	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:00b10208411ce8ad75203482b23b25e53712e75dc77c14c7f6e317fd5a92af50
2292	Classical Computation	SFT-COMP-CPLXX-COMMUNICATION-ROUNDS-DISTINCTIONS-016	Communication rounds and transmitted distinctions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cebe1722bcc156d61c89a210e3b7739a3fb128ebbd5de85689f030d3ba5edfc7
2293	Classical Computation	SFT-COMP-CPLXX-DECISION-TREE-QUERY-LOWER-017	Decision-tree and query lower bounds	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a1d051b66c6793bfafabd4cd98c8b0dba995eef687a39ba84afe1685bb914e99
2294	Classical Computation	SFT-COMP-CPLXX-RANDOMIZED-RESOURCE-018	Randomized resource as unresolved deterministic support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9823096535f3dea363d02f3535e87e7eccb23aab1c16245f30706a7654ea70b6
2295	Classical Computation	SFT-COMP-CPLXX-DERANDOMIZATION-BOUNDARY-019	Derandomization correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5e8a3d58ed0a745b7e330fade666a9ec718b0cea5b04b0fb0823a4de27b9349f
2296	Classical Computation	SFT-COMP-CPLXX-COUNTING-SUPPORT-020	Counting-complexity support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:762a065484a88a128480137c509c3e33043aa846355c018cb7a908ba3b4de072
2297	Classical Computation	SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021	Reduction closure and complete-problem construction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:04bcd8150ae9201d3922ffff124d4e4dea5f83bc08a7d5429afcd1407c660e37eb
2298	Classical Computation	SFT-COMP-CPLXX-UPPER-BOUND-CERTIFICATE-022	Upper-bound witness and exact resource certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d1976900019513aa47be3e617a48201c6ff957e658eab69ed821ecc723725c6
2299	Classical Computation	SFT-COMP-CPLXX-LOWER-BOUND-ADVERSARY-023	Lower-bound adversary and indistinguishability certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:15e9760aaf52bc4208057d4c02b56d46178cc25973c384ebcc0e64cdf1bb2f7a

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2300	Classical Computation	SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024	Arbitrary circuit lower-bound theorem boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7390b05427a5d0b2b977e869671412ea085fb74eb8030be9906b89529d4e11b4
2301	Classical Computation	SFT-COMP-CPLXX-WORST-AVERAGE-DISTRIBUTION-025	Worst-case, average-case and distribution custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dae9c818e1f164594ebc834ddcfac60c53394c27287ca7c0adf723c35f6148c6
2302	Classical Computation	SFT-COMP-CPLXX-APPROXIMATION-RATIO-026	Approximation ratio as exact-part relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:001f2200d972eeb636174b144c096dd66c7dcec9602bae3a596559e3b61c73f4
2303	Classical Computation	SFT-COMP-CPLXX-PARAMETERIZED-SIZE-027	Parameterized size and fixed-parameter resource	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:704b2a3563f7cfe8c1bea6e61f126a9e4ed800e110eff891c6d9411f4cbd7b3b
2304	Classical Computation	SFT-COMP-CPLXX-KERNELIZATION-BOUNDARY-028	Kernelization correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:87638f73e2058c88517b04585491f76378c6294e71582e710f0532b3d368b616
2305	Classical Computation	SFT-COMP-CPLXX-AMORTIZED-RESOURCE-029	Amortized and aggregate resource accounting	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4225a625872ee296077521ee036f4f1f376ce1111528dff9ab57c9f3af12306c
2306	Classical Computation	SFT-COMP-CPLXX-ONLINE-COMPETITIVE-030	Online competitive-resource correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:279968d86a5e21e52f32c08a227c4b5569f59713faf0c0d715687743d51e891e
2307	Classical Computation	SFT-COMP-CPLXX-DESCRIPTION-PROGRAM-SIZE-031	Description and program-size complexity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0c967d21b5fb44f18bcca78966604644c5be20ee266a00611cd12d28f6ea6b08
2308	Classical Computation	SFT-COMP-CPLXX-REVERSIBLE-TRADEOFF-032	Reversible time-space-record tradeoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d26bf5549335ce6b922488c1533e1f67572b1f0e333c00c4f8d683f8e4312fb6
2309	Classical Computation	SFT-COMP-CPLXX-COMPLETENESS-033	Complexity-family completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7f5c0426902fa647237c31c60bd398fb98183edfebb74e11f14a62c3e916ad48
2310	Classical Computation	SFT-COMP-ALGX-SPECIFICATION-INVARIANT-001	Algorithm specification, invariant and termination certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:70eb3cb7e25655fecad3c36d1930fe5102cb6c821dd7dba96c7cbbbbc0740367
2311	Classical Computation	SFT-COMP-ALGX-SEARCH-002	Exact linear and ordered search	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:80f98e73bea6d1180666a9966985a84433df743beb73f2f0ffd687ffe3faf374
2312	Classical Computation	SFT-COMP-ALGX-COMPARISON-SORT-003	Comparison sorting and permutation custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6b19d5a7cb429e8eb756b6995520a73de408a1e0f55aa48709a80bb75e1cb2af
2313	Classical Computation	SFT-COMP-ALGX-NONCOMPARISON-ORDER-004	Noncomparison ordering correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3e8e751e67ac2709bf1ae0719c70012f0eca7b19576a1a07e9d9f8333c02423a9
2314	Classical Computation	SFT-COMP-ALGX-INTEGER-ARITHMETIC-005	Integer addition, multiplication and division algorithms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1986b80f0a0c3f213bb6f96367d88010f554e9c5979caf3346e831b2ce505ddb

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2315	Classical Computation	SFT-COMP-ALGX-GCD-MODULAR-006	Greatest-common-part and modular arithmetic algorithms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a25b98e0c5b5fdef7ac59ce929ae72bab0b879c7b2267d323f469780e0ca1ca1
2316	Classical Computation	SFT-COMP-ALGX-RATIONAL-ARITHMETIC-007	Exact rational-part arithmetic algorithms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3529fbb809c8a267c1c8aa95c98e8a7dc7a1c5c6c250a73585686a98cb8df073a
2317	Classical Computation	SFT-COMP-ALGX-STRING-MATCH-008	String matching and finite-pattern search	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:875c43ecbc48ec320bac5b9772905896bc33dc395fd02ad2747cfd7cd5a41121
2318	Classical Computation	SFT-COMP-ALGX-SEQUENCE-EDIT-009	Sequence alignment and edit structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:907e84359f8e9d7abcfafdfed6694683d4869b3022986be0eb269effdd22416a
2319	Classical Computation	SFT-COMP-ALGX-TREE-ORGANIZATION-010	Tree traversal, balancing and search organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cb77a18fe66ea6934eae0804e816160510d73199c16f178833816fd04314557a
2320	Classical Computation	SFT-COMP-ALGX-GRAPH-REACHABILITY-011	Graph traversal and reachability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:091ce9e2862a174871a807be845136ed822cf7af7ad94d0a9d1bd4ad9e93a6a6
2321	Classical Computation	SFT-COMP-ALGX-SHORTEST-PATH-012	Shortest-path and path-composition algorithms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bf61c594bae20ae2c96e2d397d9a9300b716bf1e8f163c8903cd58414aa98260
2322	Classical Computation	SFT-COMP-ALGX-SPANNING-CONNECTIVITY-013	Spanning-tree and connectivity algorithms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:70706956bf7f0dc9ab038a422c8f117b34019e51033f2387240679e4de1b1030
2323	Classical Computation	SFT-COMP-ALGX-FLOW-CUT-014	Network flow and cut algorithms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:950fb2aad5b407823e69b4870424de88307396fea4001321eefaa069c040ea85
2324	Classical Computation	SFT-COMP-ALGX-MATCHING-ASSIGNMENT-015	Matching and assignment algorithms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f508ce7800234e1a4491a4a3276d4ef851a3df3a9b58b1fb13337321b1b6cfa2
2325	Classical Computation	SFT-COMP-ALGX-ALGEBRAIC-SOLVE-016	Algebraic elimination and exact linear solving	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f960a59398cee625bef0c860554d09baa09d7c1e5e0077072467f753ab1c70ec
2326	Classical Computation	SFT-COMP-ALGX-POLYNOMIAL-SYMBOLIC-017	Polynomial and symbolic algebra algorithms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0af11b5c6ee29033e717f8237d168a0a6d1bf0840b960685207197e6057d9c29
2327	Classical Computation	SFT-COMP-ALGX-GEOMETRIC-ORIENTATION-018	Computational geometry orientation and intersection	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1746097e1dfcee054d9d781b7703c49209c85135683d56487f9a60a7f1ea74d1
2328	Classical Computation	SFT-COMP-ALGX-CONVEX-HULL-019	Convex-hull and spatial-order algorithms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f35c33680cf31cc79c4ff6d89224667c03f3c9d49f95f82cff95528a526c0b5a
2329	Classical Computation	SFT-COMP-ALGX-DYNAMIC-PROGRAMMING-020	Dynamic-programming optimal-substructure law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:83155df4a73403027cf9d4e77931b849978ec7215c40bd8e13c5a5bb0fa2641d

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2330	Classical Computation	SFT-COMP-ALGX-GREEDY-BOUNDARY-021	Greedy-choice admissibility boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b651accbad18cf0e4312bc77e789e9057dd7a0ea2740cd1baa5bdf31683c9411
2331	Classical Computation	SFT-COMP-ALGX-BRANCH-BOUND-022	Combinatorial optimization and branch-and-bound	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5a5d109a8c5ef478bf8b0a1ede52e69c6a5042f4e5d39225908211589eaffd8e
2332	Classical Computation	SFT-COMP-ALGX-RANDOMIZED-SUPPORT-023	Randomized algorithm complete-support execution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:70966a9f35449066307818f149845c0eb686985d4328883283390b80996857d0
2333	Classical Computation	SFT-COMP-ALGX-PARALLEL-WORK-DEPTH-024	Parallel work-depth algorithm law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2926e2fa0084479f138e46819e36fe57faaec786a5001ca610c8f7ff91530213
2334	Classical Computation	SFT-COMP-ALGX-DISTRIBUTED-LOCAL-025	Distributed local-state algorithm law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:176b072112610b2f4d138a97978cab490064602572f1c46e26515e2affd6840c
2335	Classical Computation	SFT-COMP-ALGX-ONLINE-COMPETITIVE-026	Online decision and competitive ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4b79f08cea3b5df1eeef3bd3e01ec15dd28a182b8a4eb92ee4d15834f30044489
2336	Classical Computation	SFT-COMP-ALGX-STREAMING-MEMORY-027	Streaming memory and approximation ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fd34848982a108037ca3be62f1205b7081a01c391a1d075c7ba5f9b60cef754d
2337	Classical Computation	SFT-COMP-ALGX-NUMERICAL-ERROR-028	Numerical iteration with exact error custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9e090d95bf38a8e080707a935673ec0c7edea4c3c95ceef7a0c47ab2d5eec6e8
2338	Classical Computation	SFT-COMP-ALGX-SYMBOLIC-SIMPLIFY-029	Symbolic simplification and equivalence custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4eedc31ec7fa86ec449aa4d61561831d7647943182e18d8ab437f45329f94a78
2339	Classical Computation	SFT-COMP-ALGX-APPROXIMATION-SCHEME-030	Approximation scheme and guarantee certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e837401c4a85055e82fed3fc6ef60c3702d6cdf2856848f2e43da01229125e26
2340	Classical Computation	SFT-COMP-ALGX-COMPLETENESS-031	Algorithm and data-organization completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:183e2469566102a9df47110d8430f40aa28e1917d6650f6f8649da02614c1bdc
2341	Classical Computation	SFT-COMP-SEMX-AST-WELL-FORMED-001	Abstract syntax trees and well-formed program terms	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f9937dfa65924f7a8ea2f1189be92d03892a5554d26f201352894a02b11761d6
2342	Classical Computation	SFT-COMP-SEMX-FREE-BOUND-SCOPE-002	Free, bound and scoped-name distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5b0d956dd758d1728213b2b4e68eb1751d10c3a30c72f4e150359b9532f6a705
2343	Classical Computation	SFT-COMP-SEMX-ALPHA-RENAMING-003	Alpha-equivalence and lawful renaming	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:516639864237c32835cffdf9bb17be08545ed86aeb5fd7573bc622c18976c4dc
2344	Classical Computation	SFT-COMP-SEMX-CAPTURE-AVOIDING-SUBSTITUTION-004	Capture-avoiding substitution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1536ffef25ca51f5f5e4daec28011fbaad98577f6578320f2119256446fc8bf1

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2345	Classical Computation	SFT-COMP-SEMX-SMALL-STEP-005	Small-step operational evaluation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8fa06f93a05cc64a2b93e882b1f177e1edc2082db91afcdfc1587480765623b9
2346	Classical Computation	SFT-COMP-SEMX-BIG-STEP-006	Big-step operational evaluation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:17707d9d4ffe36a0157f4aacbd46f2e07283520bddec0a0e16995d0e82d693d
2347	Classical Computation	SFT-COMP-SEMX-EVALUATION-CONTEXT-007	Evaluation-context composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c5461e1574d94896c0637f5ac18f3603e2f437bb732f81cf6f35a1b219092ec1
2348	Classical Computation	SFT-COMP-SEMX-DENOTATIONAL-COMPOSITION-008	Denotational meaning as compositional Fold map	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:161a065120e382997cd400a6e7ece0719d31ad8c41185c32349db9a11e6d3231
2349	Classical Computation	SFT-COMP-SEMX-OPERATIONAL-DENOTATIONAL-ADEQUACY-009	Operational-denotational adequacy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:16c5aa07ecd2b339aad20d57f2e7c2746b1870afe4ee1053c5b010dd0be3037c
2350	Classical Computation	SFT-COMP-SEMX-FULL-ABSTRACTION-BOUNDARY-010	Full-abstraction correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:486d14064cfb1261819ac89b7f8de2c7d0acbb564fc8024e7345d6169acd0354
2351	Classical Computation	SFT-COMP-SEMX-TYPE-RULES-011	Type formation, introduction and elimination	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:65b9f5ff0aab1c7f63eeb73273e56a2b50b130833e33fbeb0b69f7c3f10e8e77d
2352	Classical Computation	SFT-COMP-SEMX-TYPE-INFERERENCE-012	Type checking and type inference correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8fe747263cf1a16065934e787e315a030bb431bae39d32b3bc2d3b960a520636
2353	Classical Computation	SFT-COMP-SEMX-POLYMORPHISM-PARAMETRICITY-013	Polymorphism and parametricity boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:35f9a63bdc416323e4b251126fb3b9883d38222321e1f892790ef7e23e61a65
2354	Classical Computation	SFT-COMP-SEMX-DEPENDENT-EVIDENCE-014	Dependent evidence and proposition-as-type correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6c4c0457213d3fbc0f5c60c4a7bc54a65a214c0395d37e159509bd2799d9549
2355	Classical Computation	SFT-COMP-SEMX-STATE-EFFECT-EXCEPTION-015	State, effect and exception semantics	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6be8516daeee0e7564b5cc97f2730f07bd9b66d5233bc3a71773f5cc55e90dee
2356	Classical Computation	SFT-COMP-SEMX-CONTEXTUAL-BISIMULATION-016	Contextual equivalence and bisimulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ec7102f78869885bf468157071f968a44265cd0d503a331ba5d9ae5b57cd5a23
2357	Classical Computation	SFT-COMP-SEMX-TERMINATION-DESCENT-017	Termination measure and well-founded descent	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e8a0d80e975829d3f611d06b45ea68622cebe30115c9047aaca1a5c563a9ef71
2358	Classical Computation	SFT-COMP-SEMX-PARTIAL-TOTAL-CORRECTNESS-018	Partial correctness and total correctness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4d14a78649a97b9fc51447970158d5ba0414783efffb003e633f0fc50ede15e0
2359	Classical Computation	SFT-COMP-SEMX-ASSERTION-INVARIANT-019	Assertion logic and invariant preservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2c8b3d8de65efb5076cfc6383c4370d169baf0bef3101bf2f35183ed89712b35

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2360	Classical Computation	SFT-COMP-SEMX-SPECIFICATION-REFINEMENT-020	Formal specification and refinement ordering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bfe321ce3fca7ca7a2b0fb078b9947bfb2cfbb2fd249305a14faedf32ea1d19b
2361	Classical Computation	SFT-COMP-SEMX-PROGRAM-TRANSFORMATION-021	Program transformation and optimization correctness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:365414c6f3faeec0c984d8eb78ce99d23a57a2075629a0f07af8da53d10e1371
2362	Classical Computation	SFT-COMP-SEMX-COMPILER-SIMULATION-022	Compiler pass simulation and semantic preservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eefea6935ebb19346833ee549a4249a0089c2108f42145393e9e7274fac357ba
2363	Classical Computation	SFT-COMP-SEMX-INTERMEDIATE-COMPOSITION-023	Intermediate-language composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e1791bc5734b7e9811dbc79d78284476ec6de29fc3d46b8e7103fb17ca508e81
2364	Classical Computation	SFT-COMP-SEMX-PROOF-CARRYING-024	Proof-carrying program and certificate checking	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d29d272caf8b9e28b5704cae106e38047a590116668c8b145af76e825b062abb
2365	Classical Computation	SFT-COMP-SEMX-COMPLETENESS-025	Semantics and programming-theory completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4b394dd0722839dc2179f6647fc7d4fbbb1acbf1b9e87c15398211141dbaf0a1
2366	Classical Computation	SFT-COMP-DISTX-EVENT-LOCAL-ORDER-001	Event identity and local process order	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:76821b4e759f6be4268ebf825735416f5c56ea3661b07098fc945a054c9bbf9f
2367	Classical Computation	SFT-COMP-DISTX-INTERLEAVING-PARTIAL-ORDER-002	Concurrent interleaving and partial-order equivalence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f5564192b19f9e0afa64ca8318f15064287ff61073b66b6de88d25bdae00522
2368	Classical Computation	SFT-COMP-DISTX-HAPPENS-BEFORE-003	Happens-before causality relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bc4bfbdf5ddd12999046bce7d97cbec501389e934168b445b4bd61e565808337
2369	Classical Computation	SFT-COMP-DISTX-MESSAGE-CUSTODY-004	Message send, receipt and channel custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bebaca914d3d511b36c4991d8bc13c1a3567e0565b4200f13bb77887fba08337
2370	Classical Computation	SFT-COMP-DISTX-SYNCHRONY-BOUNDARY-005	Synchronous and asynchronous execution boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7b9b75463296e04a5c938f15e118920fe44e0e82a2629bec0e207050c7042907
2371	Classical Computation	SFT-COMP-DISTX-MUTUAL-EXCLUSION-006	Mutual exclusion and critical-section safety	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aef98b4628abf0ac53c11479eca1b2cd2f60e55ba3bfcde93274a4322a1268d2
2372	Classical Computation	SFT-COMP-DISTX-PROGRESS-CLASSES-007	Deadlock, livelock and progress distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:93b28be26d36fadbb84da23cb02720beeb12ef3d7cbc379520470b5d32c989ad69
2373	Classical Computation	SFT-COMP-DISTX-COORDINATION-PRIMITIVES-008	Barrier, semaphore and rendezvous correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ee454fbd5f51cc371c0ad9f39e4dd9c2b2e2abb061cb25929175ed1f2aa95675
2374	Classical Computation	SFT-COMP-DISTX-LOGICAL-CLOCK-009	Logical clock and causal timestamp correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d0dabe6250f5edd299efba51b0c34b7cded09c3dca59d4cf714dc6ce77cea2f2

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2375	Classical Computation	SFT-COMP-DISTX-DELIVERY-MODES-010	Broadcast, multicast and point-to-point communication	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b153e59c46ccbcf5f7606d96f5aeffadca07d5598b94d04ff135f191e569f958
2376	Classical Computation	SFT-COMP-DISTX-FAILURE-FREE-CONSENSUS-011	Failure-free consensus construction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b520393f78b6a4187857256f726510749d181ab03a10081a5a8a9a010e80ba3f
2377	Classical Computation	SFT-COMP-DISTX-CRASH-CONSENSUS-BOUNDARY-012	Crash-fault consensus boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:00ad944609968e9b642476482beb8fc9c52e1659ff1476ef053557f40842e589
2378	Classical Computation	SFT-COMP-DISTX-BYZANTINE-AGREEMENT-BOUNDARY-013	Byzantine-fault agreement boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f881eb13ee8addfbae479e2e9aa490fbcd3cb907db78475b936864d860671384
2379	Classical Computation	SFT-COMP-DISTX-HIDDEN-PREDECESSOR-014	Hidden-predecessor agreement impossibility	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:153ace35214c8e9bd4b9e7c4613122f017673b3050e2da32c95b3986e6079571
2380	Classical Computation	SFT-COMP-DISTX-FAILURE-DETECTOR-CUSTODY-015	Failure detector and synchrony-assumption custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5e0d11563372ced650fc8c8be5e5021399eafa4801a3a6db67e363ea2ba9e2c4
2381	Classical Computation	SFT-COMP-DISTX-REPLICATION-STATE-MACHINE-016	Replication and state-machine correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d0f84ac77f0430c92fab8269d9956ef954f2a3b77d910d37d43a2cad9c05637d
2382	Classical Computation	SFT-COMP-DISTX-LINEARIZABLE-SEQUENTIAL-017	Linearizable and sequential consistency boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9d8774d36c1fe34228d6f3254bc4ba189698bf729cdcbba04cbf128c288495a31
2383	Classical Computation	SFT-COMP-DISTX-CAUSAL-EVENTUAL-018	Causal and eventual consistency boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:58eb9e40384fe05b3259221b8d7c63ea92c4d75e6f3de0ce7a7cf6570f693171
2384	Classical Computation	SFT-COMP-DISTX-QUORUM-INTERSECTION-019	Quorum intersection and replicated decision	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b77fa2a83844e81f064af60c6422096ea7fb6d981e05b74d6750decad8f08043
2385	Classical Computation	SFT-COMP-DISTX-TRANSACTION-ATOMICITY-020	Distributed transaction atomicity boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bdb5547fa3b213665530242badc2aa35da2addce3bbc50c82b4f3481ceb f1da0
2386	Classical Computation	SFT-COMP-DISTX-DISTRIBUTED-KNOWLEDGE-021	Local, shared and common knowledge distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5b97b6c393021b68652cacb90e23057bfc5d5957368f8b3437361108f9176f92
2387	Classical Computation	SFT-COMP-DISTX-LOCALITY-RADIUS-022	Locality radius and information-propagation lower bound	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:458a83a0584cbfddae4adfbb898f55f52a346e81a1c6c2311b44753dd8215a6
2388	Classical Computation	SFT-COMP-DISTX-NETWORK-TOPOLOGY-023	Network topology and distributed computability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:803d514dbacbe8457e1d1240c0dc3118c833396f23e26839b8ff2b0f8524c233
2389	Classical Computation	SFT-COMP-DISTX-PARTITION-CUSTODY-024	Dynamic-network and partition custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:21aac73eb668978af9f4749daadd622ae1338e322f1360292fd438914ec c9399

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2390	Classical Computation	SFT-COMP-DISTX-SAFETY-LIVENESS-FAIRNESS-025	Distributed safety, liveness and fairness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ed7ecefdab3d3a01655579c0a95aa84c9c1a9e24c198f7788224933d4d00e6ce
2391	Classical Computation	SFT-COMP-DISTX-COMPLETENESS-026	Concurrent and distributed completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:65e0b337b2ee8af57d45b7377189789b050396602af45f818603f7368ff7b49d
2392	Classical Computation	SFT-COMP-SECX-ADVERSARY-DEFINITION-001	Adversary, view, resource and success-event definition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6957c2467a28c4a498771da9800f20de2e76ba62bc4af7ae2930ca2544cc84af
2393	Classical Computation	SFT-COMP-SECX-INFORMATION-THEORETIC-SECURITY-002	Information-theoretic secrecy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a557544e4ebec032beb64533484d8f25f320ddb2af9b4ebd0f1fba4244b2e625
2394	Classical Computation	SFT-COMP-SECX-INDISTINGUISHABILITY-003	Computational indistinguishability correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e8d08965120d7c93451b16e0358e627b3928dd6b4b6bfe3bcc53e0605ebc5a7f
2395	Classical Computation	SFT-COMP-SECX-ONE-WAY-RESOURCE-004	One-way transformation and inversion resource	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:87d821ee1643ce93d8ecadca3a74dd3c2d6f140724fad609e64d3ab6cf00421
2396	Classical Computation	SFT-COMP-SECX-HARD-CORE-BOUNDARY-005	Hard-core distinction correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:793a3b0d423f60b0a952227cdeb402fc097b7c15cbd05db708e09b8b7d473cca
2397	Classical Computation	SFT-COMP-SECX-PSEUDORANDOM-GENERATOR-006	Pseudorandom generator support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:872f5e886b09ea01822d81f7a77e293855d6ec2af77c105767fe2ed83862d68b
2398	Classical Computation	SFT-COMP-SECX-PSEUDORANDOM-FUNCTION-007	Pseudorandom function correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:98ac85b848c75f8c79c39911e0b42703a007503558740258239259807377fb99
2399	Classical Computation	SFT-COMP-SECX-SYMMETRIC-ENCRYPTION-008	Symmetric encryption correctness and secrecy	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b0926efc08f26bf1e09a141d9624734281fab28b8636ade2aa54d4e915501d0f
2400	Classical Computation	SFT-COMP-SECX-PUBLIC-KEY-BOUNDARY-009	Public-key encryption correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3e6e390f842a737be687676f55f75f62aa41e64e1ac96fed8dd091f57555224b
2401	Classical Computation	SFT-COMP-SECX-MESSAGE-AUTHENTICATION-010	Message authentication and integrity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:69ea57ab41bd6a76e29c1e4e313f8300f57a15955d3e240a36cbbb8152b61541
2402	Classical Computation	SFT-COMP-SECX-ENTITY-FRESHNESS-011	Entity authentication and freshness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b862ae4e306a3daa5dd406e536d746bca83c8baed622b1588314a631be2e1ed5
2403	Classical Computation	SFT-COMP-SECX-HASH-PROPERTIES-012	Hash compression, preimage and collision properties	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dc7cab9f930db38e057aabac09e435d632a748a2dfedc42509ad578f952dc2a
2404	Classical Computation	SFT-COMP-SECX-COMMITMENT-013	Commitment hiding and binding	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:59d06a56168b9c2987a390594bd50b788a193ee0abcf90e34b98e8ada70ee4f2

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2405	Classical Computation	SFT-COMP-SECX-DIGITAL-SIGNATURE-014	Digital signature correctness and unforgeability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:56ce0a88da8b990e4037fd6e10df1bfdc63a80e31ac264e2c85b6f548d221637
2406	Classical Computation	SFT-COMP-SECX-KEY-ESTABLISHMENT-015	Key establishment and authenticated exchange	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2a2c4a0983fbeb18243f9d266346d4dc2bd45ec0329b4e8087e32136c70132003
2407	Classical Computation	SFT-COMP-SECX-SECRET-SHARING-016	Secret sharing and reconstruction threshold	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:275ed3e54bd78277b771634df6f26e4c7018887f3d8758f1f1cbd1e81367cfc6
2408	Classical Computation	SFT-COMP-SECX-PROOF-KNOWLEDGE-017	Proof of knowledge and extractor boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:17692a70e5ab4bb87c83eb3865dc7a1480a811181f6f411fc09d822145fabcb6
2409	Classical Computation	SFT-COMP-SECX-ZERO-KNOWLEDGE-018	Zero-knowledge simulator correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:72ff687bf0660893b61147ef1df53895108c5d13593f9750a525d3a4e1ac2823
2410	Classical Computation	SFT-COMP-SECX-MULTIPARTY-COMPUTATION-019	Secure multiparty computation and view custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:437eeb9f9fe1ad076ed626e291603c03ab8736d01870a13a5b2682cc91ebabbf
2411	Classical Computation	SFT-COMP-SECX-OBVIOUS-TRANSFER-020	Oblivious-transfer correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b98601429101853318921511a3a508edd8f4a45e6d58c90ffa1690f2075125fa
2412	Classical Computation	SFT-COMP-SECX-COMPOSABLE-ADVERSARY-021	Adaptive, concurrent and composable adversaries	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1e5ae793371e0436a70e4d5c5e3587485ce8a5006e0daa7c52e317df53de537f
2413	Classical Computation	SFT-COMP-SECX-SIDE-CHANNEL-HANDOFF-022	Side-channel information ownership handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b3fc57d1251004cff61d2d91ea4a1bd3acc53d7f3bdaa7c53352bd0014db3b88
2414	Classical Computation	SFT-COMP-SECX-POST-QUANTUM-BOUNDARY-023	Post-quantum security reduction boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0e17c394d4e2a6f268636f8a1d2f84f595e45d040bfa265be76f329372427d8b
2415	Classical Computation	SFT-COMP-SECX-QUANTUM-CRYPTO-HANDOFF-024	Quantum cryptography ownership handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8fc26fb47557c5f1d8702eccf9c6248883f3ad5964470c14b623b062b7681879
2416	Classical Computation	SFT-COMP-SECX-COMPLETENESS-025	Cryptographic security completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0aeaaa54bd0851429dd8bcac639036b5ae8eb44720318c61f708172f8cd8fd77
2417	Classical Computation	SFT-COMP-LEARNX-PROBLEM-IDENTITY-001	Learning problem, example and target identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:30941a547292c1e8f50330e38964be8887913739e45194cd982be8c1350260bd
2418	Classical Computation	SFT-COMP-LEARNX-HYPOTHESIS-FAMILY-002	Hypothesis family generation and representation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:63236200da2e56d3f92710237df821370f395e31dde8e57dbc271929b52bfff40
2419	Classical Computation	SFT-COMP-LEARNX-LOSS-RISK-003	Exact loss and risk as parts of a finite whole	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b5dfccf83410a4048f4d2db964d69d3b5037e947fa5d89854ca28a976734c638

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2420	Classical Computation	SFT-COMP-LEARNX-HELD-OUT-CUSTODY-004	Training, validation and held-out observation custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:86f6702af9187302c91d9167566ea3883f74368356094acb54e237bcd81e5f7
2421	Classical Computation	SFT-COMP-LEARNX-EMPIRICAL-RISK-005	Empirical-risk minimization boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:040b6d1498580d30954a204f39c0f494a466455ae6128162db761d1da6bc7b1d
2422	Classical Computation	SFT-COMP-LEARNX-GENERALIZATION-006	Generalization as unseen-support preservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f38a67360079b5bb74f153bc527433fce719d4e9f5ba10ee6005185353503b91
2423	Classical Computation	SFT-COMP-LEARNX-SAMPLE-COMPLEXITY-007	Sample complexity from distinguishability count	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2d12207bb8fe7ee9ea656193d8c07276f66114c6f37ae47e05cc2884ce18c370
2424	Classical Computation	SFT-COMP-LEARNX-CAPACITY-SHATTERING-008	Capacity and shattering correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d94094145ec4c004883e62028162c8ebf8fb252fe79ba1a84893b8c1b7e97022
2425	Classical Computation	SFT-COMP-LEARNX-PAC-CORRESPONDENCE-009	Probably-approximately-correct correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5f357fcfb279edda673850eee251e29431fdbcc347b3d7c103b9ad8552e2c4f8
2426	Classical Computation	SFT-COMP-LEARNX-CLASSIFICATION-010	Classification and decision-boundary computation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9c60131c72dfa432787498f3fa9a2ed2cf9a1bfe8a4812b3fc3ce0d3bebb3d32
2427	Classical Computation	SFT-COMP-LEARNX-REGRESSION-011	Regression as exact representative relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5d56e6c97dd7a707dff1ecdd466cb1fdbbbf9db0fba47306b7366e2a8ca573f4
2428	Classical Computation	SFT-COMP-LEARNX-FEATURE-SELECTION-012	Feature selection and sufficient representation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a77a3b0d20e2c5f1d27b6705513297c345fdd06ff436b7447d345f3a0368dc43
2429	Classical Computation	SFT-COMP-LEARNX-CLUSTERING-013	Unsupervised partition and clustering custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:38c359fe762df8857cf434575c13253c812005e5743f2f71787bb382fd96ab5f
2430	Classical Computation	SFT-COMP-LEARNX-GENERATIVE-SUPPORT-014	Generative-support reconstruction boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:124a2783afd7773c5811dbc7f1d6aa9e3f4d28e41282a04046bf138c4841ce27
2431	Classical Computation	SFT-COMP-LEARNX-BAYESIAN-CORRESPONDENCE-015	Deterministic-support Bayesian correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:734446248f1f30ac3c7982aad99b1ca661aec54b44e4e35618edda3f95a1317f
2432	Classical Computation	SFT-COMP-LEARNX-OPTIMIZATION-CONVERGENCE-016	Learning optimization and convergence certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1292904052c2a8fbc2059f7313cf8970e0b9cd690875d0540c004f2fd09d213c
2433	Classical Computation	SFT-COMP-LEARNX-ONLINE-REGRET-017	Online learning and regret as exact ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:11fb4503637f00d22cf13372a640746d1a4f65f3c3ba28b0f088da202c25a7a7
2434	Classical Computation	SFT-COMP-LEARNX-CONCEPT-DRIFT-018	Concept drift and adaptation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fc6dfbeaf677c3065f6d89f7cc8492d02cee3caba9f1efcdd58b20081e662e2d

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2435	Classical Computation	SFT-COMP-LEARNX-SEARCH-PLANNING-019	Search, planning and heuristic admissibility	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2501fb3af3240a93b1a9efcb75d755aba0c4894e8133b38e3786315faf2f1894
2436	Classical Computation	SFT-COMP-LEARNX-REINFORCEMENT-020	Reinforcement state, action and return custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4576edd5a65dde5aaa6103cf3b03ee6256f1986320bd47e08df66a7a5957c1fd
2437	Classical Computation	SFT-COMP-LEARNX-MULTI-AGENT-021	Multi-agent learning and strategic observation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d21f568f31b6f1df324828ca3fe3984c72b10fad3d23a645793ade7f8dd84107
2438	Classical Computation	SFT-COMP-LEARNX-INTERPRETABILITY-022	Interpretability as reconstructible decision trace	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:998e6fbeda643e6dc34b8196fb58636dca1f9fae64365f5b2c1ccdf55afc857
2439	Classical Computation	SFT-COMP-LEARNX-ROBUSTNESS-SHIFT-023	Robustness, distribution shift and adversarial examples	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3c099fc6e9523688b49828cd95ff8278058f88cae65fad8b4d1cbb79de7d5d6a
2440	Classical Computation	SFT-COMP-LEARNX-LEARNED-VERIFICATION-024	Verification of learned-process behavior	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7a75ae59b3b58abd147e008ff6b88867576b3f49ab7a498a0fd3f7204e74d48f
2441	Classical Computation	SFT-COMP-LEARNX-IDENTIFIABILITY-LIMIT-025	No-free-lunch and identifiability limits	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b803edc846a67c4a5bf8a750e73ba026174dbed38c67c54f4ddd1b2aa62295eb
2442	Classical Computation	SFT-COMP-LEARNX-COMPLETENESS-026	Computational learning completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:91c835ed6de921461bb462de45296841fa7d509444e1ab83d85d4ea66ca74b75
2443	Classical Computation	SFT-COMP-SCIX-EXACT-APPROXIMATE-001	Exact and approximate result distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:84970622b0cf5e5e93dd36403e37249069a6538b6e55a8ed6d31d8f2081afdbd
2444	Classical Computation	SFT-COMP-SCIX-FINITE-PRECISION-002	Finite-precision representation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aecf5eb5e4d25c707b0d79cbbaacd39602cebff9958623fc2498ec4fa7fb309c
2445	Classical Computation	SFT-COMP-SCIX-ROUNDING-ERROR-003	Rounding and truncation error ledger	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d157fdac7fee80f80dedd8cc48a478392527392a70fb91f2ff9a4e44962bfc77
2446	Classical Computation	SFT-COMP-SCIX-FORWARD-BACKWARD-ERROR-004	Forward and backward error	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1c3ddd7a6308e1ae90df969f0866502e89ab06df231c3df724838f4fa6fb0b98
2447	Classical Computation	SFT-COMP-SCIX-CONDITIONING-005	Conditioning and sensitivity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5d9c3952e1472f918727bf655f82a7779a118e79b6ed9e15af68d9210ee707d4
2448	Classical Computation	SFT-COMP-SCIX-STABILITY-COMPOSITION-006	Numerical stability under composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:638203d938543911fa1934c6e2812bf7f9633778432e5212c5cde289feb18aab
2449	Classical Computation	SFT-COMP-SCIX-CONVERGENCE-STOPPING-007	Convergence order and stopping certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3186d61ac593644f8382867ff73dccc9751b34d3ad8e988b4965a40a0c8c37af

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2450	Classical Computation	SFT-COMP-SCIX-DISCRETIZATION-008	Discretization and consistency	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7e2ac4ca4d88351b3194eb9ec9e4e63713811f381a1b381aac4935b29c33a253
2451	Classical Computation	SFT-COMP-SCIX-INTERPOLATION-009	Interpolation and approximation custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:706080777f8f25f8e0b0c0ba13d6e64676bb3d6c323339b294e0e5782d922e7c
2452	Classical Computation	SFT-COMP-SCIX-QUADRATURE-010	Quadrature and exact residual bounds	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f17f245b625c883f6361ef370b156ff9f0e6e0b5824b415391fc03e571bbcb3ab
2453	Classical Computation	SFT-COMP-SCIX-ROOT-INTERVAL-011	Root-finding and interval custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:80eefe844eff96b435c08072e5ff6bd0908a04f0f6268a0b0892507ccb232265
2454	Classical Computation	SFT-COMP-SCIX-LINEAR-SYSTEM-012	Exact and approximate linear-system solving	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4a587a26b1637905f09ca4076c568e13b070f1c4a59588f1a4ef6ea37c474400
2455	Classical Computation	SFT-COMP-SCIX-EIGEN-MODES-013	Eigenvalue and mode computation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:faab21d93c3569d70ab00eb9c910efcc1702b38fa9824b4ceb083072c08a5126
2456	Classical Computation	SFT-COMP-SCIX-ORDINARY-SYSTEM-014	Ordinary differential-system discretization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4093d5dc1741e934fe74530d37058a6b2503c72675ab0264508d4e282c18eb91
2457	Classical Computation	SFT-COMP-SCIX-PARTIAL-SYSTEM-015	Partial differential-system discretization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:49cc1bb64ed0ec8d967548abe8668ce6cffe2780f1b94dfac4f8c8093eda694e
2458	Classical Computation	SFT-COMP-SCIX-STOCHASTIC-SUPPORT-016	Stochastic-simulation deterministic-support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f0aff985707147ecf0c4774347e4c55889b03a67ad65468bce7db9f4c34afd39
2459	Classical Computation	SFT-COMP-SCIX-MONTE-CARLO-017	Monte-Carlo support and sampling correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e9a9aaa784128e934df6eabda95521ecec9f1e144ded58198f6bffa4cfcf2dbf
2460	Classical Computation	SFT-COMP-SCIX-INVERSE-IDENTIFIABILITY-018	Inverse-problem identifiability and regularization boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:49b61f2bdba7a6f227f6f40403032d511b57cec32fc7e3b893dc78f33257c96e
2461	Classical Computation	SFT-COMP-SCIX-COMPUTATIONAL-STATISTICS-019	Computational statistics and estimator custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:47c01bce3a142e2488bc94eb8b8f54c449b81498274d47c131e534f8cffeec45
2462	Classical Computation	SFT-COMP-SCIX-SPARSE-COMPUTATION-020	High-dimensional and sparse computation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f1ed9b4cf0352258f2949baf28ffa8e75ae80ff092eb776589ba891d9602f48
2463	Classical Computation	SFT-COMP-SCIX-MANY-BODY-021	Many-body state-space organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ce3a4d103911ceb9bba216ba594faba7ace61018364ea674e458de6f7dc0312a
2464	Classical Computation	SFT-COMP-SCIX-SYMBOLIC-NUMERIC-022	Symbolic-numeric correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:260edf1246e3aaaae8bd9da08ab57d55079f15eaf82639c7bfff7060efd496431

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2465	Classical Computation	SFT-COMP-SCIX-SIMULATION-VALIDATION-023	Simulation verification and validation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:faa8d98dc849ef1b84b040db3cf804cf34f67132c6f52a8ecd576c8514f867fb
2466	Classical Computation	SFT-COMP-SCIX-MODEL-PROVENANCE-024	Reproducible mathematical-model provenance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a4c13c0e5c6508aeaf516ae056755ea8ef564d38bd7ccf810da767e3f35f7cd8
2467	Classical Computation	SFT-COMP-SCIX-COMPLETENESS-025	Scientific-computation completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e90843fef110814cfe879738e175945b56260eed2950bf4aa9382c209593a515
2468	Classical Computation	SFT-COMP-VALID-FORMAL-VECTOR-001	Formal-computation complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2214e4f5c89ae7ad15ccc694fe5f0e288eea4e2f7c29197a1dc0b7504fb2e511
2469	Classical Computation	SFT-COMP-VALID-COMPUTABILITY-VECTOR-002	Computability complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:169f52f6943eb9820aa3ed78670f24269c81664551b0e692c02a121db4db7e51
2470	Classical Computation	SFT-COMP-VALID-COMPLEXITY-VECTOR-003	Complexity complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8296b2a8bce1d2a4b4a6134a4d87a2558a45fb6b232501d5cc3db6cb931c93c6
2471	Classical Computation	SFT-COMP-VALID-ALGORITHM-VECTOR-004	Algorithms and data-organization complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:69e96ff30bdc4259556241a51ccf3e61ebd2166a10ae860bd94c73e46e0362b2
2472	Classical Computation	SFT-COMP-VALID-SEMANTICS-VECTOR-005	Semantics and programming-theory complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e30690b6087cf1eab319a1420f9bc902159ad30f77cf8e14182205d501dd605f
2473	Classical Computation	SFT-COMP-VALID-DISTRIBUTED-VECTOR-006	Concurrency and distributed complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:833de968a9cb7abd35b2a08a889d570e055976acf6d75d01424b6a10651a1c2e
2474	Classical Computation	SFT-COMP-VALID-SECURITY-VECTOR-007	Cryptography and security complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:199feee7fae46ae0cb3af0dea58d10e1b061b49c76c6979472c5403c78397e30
2475	Classical Computation	SFT-COMP-VALID-LEARNING-VECTOR-008	Learning and intelligence complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8f85c9c55cb5f772ddcb17e7a64f963a52f699a0c604e972db1911ced7d94b4c
2476	Classical Computation	SFT-COMP-VALID-SCIENTIFIC-VECTOR-009	Scientific-computation complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:16de2551ef53859b6dd85b7fa5d2785b7c0830248e156b09a4f196edfaa120bd
2477	Classical Computation	SFT-COMP-VALID-FAMOUS-BOUNDARY-010	Famous-problem theorem and finite-census boundary vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:21cc1ef42523def5e87ac5eee6b194d094ba54cda2ebe2e4dae2191c39e13c5a
2478	Classical Computation	SFT-COMP-VALID-ADVERSE-BOUNDARY-011	Adverse, absent, unresolved and ownership-boundary vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:044dea4ce05cb1dd1c9d2a14a6c2b1bc0a5825e71e8a6e952016e49e5a197948
2479	Classical Computation	SFT-COMP-VALID-GRAND-LOCK-012	Classical Computational Science empirical and formal Grand Lock	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:31489be2c8fb8943fe78df5391ce2b191dcb3d8cc7b5281a88478addb67e8a50

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2480	Classical Computation	SFT-COMP-HAND-ONE-OWNER-001	Classical Computation one-owner downstream handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c66f7306bf18e9aadf4ca9b63c96087ea0db83585443c133b11d9c840a8bcbf8
2481	Classical Computation	SFT-COMP-HAND-MEASUREMENT-BOUNDARY-002	Classical Computation measurement-boundary handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ca36797c3c58ac3c2f9572975fcf54bbcd302d0826471c4c5a3a8863ae205ad3
2482	Classical Computation	SFT-COMP-HAND-FORMAL-EMPIRICAL-003	Classical Computation formal-to-empirical handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e5436a3b978b62a59992e52c92021f48eb9fc768a9e67e9b109806a94f78afdd
2483	Classical Computation	SFT-COMP-HAND-CLASSICAL-QUANTUM-004	Classical-to-quantum operational handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2ce0d9f2abe0141ce4f7283069e50825792dea85fa3904e2ff0c8bf75629c596
2484	Classical Computation	SFT-COMP-HAND-OPEN-EXTENSION-005	Classical Computation open-extension handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:78da65a6b8b65d6d3c3462075f93d3300df2ff87cb6f7925dadf56e254d3e434
2485	Classical Computation	SFT-COMP-HAND-CROSS-BRANCH-COMPLETENESS-006	Classical Computation cross-branch completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c035c034f2623e06068a000dff4a6211a9ace07869fb7fcf48d3350ee119f2ca
2486	Reversible and Quantum Computation	SFT-QUANTUM-REVS-CONFIGURATION-TRANSITION-001	Reversible configuration and transition identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c1a6af8987b1c3ec5ebf02d54926aadb94976b59321379bedea165ff1e677d3
2487	Reversible and Quantum Computation	SFT-QUANTUM-REVS-MAP-CLASSIFICATION-002	Injective, surjective and bijective process distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d57a9265e397e3ce1308c7873bec25654a19faf28d8ff9c0c0609824f0a3d58e
2488	Reversible and Quantum Computation	SFT-QUANTUM-REVS-LANGUAGE-GRAMMAR-003	Reversible language and grammar relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:54259a3fe2986ed640b13e38d61df6b5ba1c284f65af2c9445b0a423cb7782c8
2489	Reversible and Quantum Computation	SFT-QUANTUM-REVS-AUTOMATON-TRANSDUCER-004	Reversible automaton and transducer construction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:25bf5a2ab341a8b3399ea18a694936140752d2ce9f0ee174d8003d34d734fac2
2490	Reversible and Quantum Computation	SFT-QUANTUM-REVS-REWRITING-PREDECESSOR-005	Reversible rewriting and retained predecessor custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:881d51963db1786c680ef872808abf0be3aa51c5253225602e48de746087b40e
2491	Reversible and Quantum Computation	SFT-QUANTUM-REVS-TAPE-MACHINE-006	Reversible tape-machine configuration law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e8e58c21f295a561303c01aef080746002f60b6e468924ed0866027ff08852eb
2492	Reversible and Quantum Computation	SFT-QUANTUM-REVS-UNIVERSAL-INTERPRETER-007	Reversible universal interpreter	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:601a25567e990e88812397ae239c63cdf96c06a45d4ddbe080de44f8bc5c6f9c
2493	Reversible and Quantum Computation	SFT-QUANTUM-REVS-HISTORY-UNCOMPUTATION-008	History tape and uncomputation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dc23c8b9762fd8b728f1b2ab6f9d631afd7540540c75e30cf5277b4f3c001a5a
2494	Reversible and Quantum Computation	SFT-QUANTUM-REVS-ANCILLA-RESTORATION-009	Ancilla preparation and exact restoration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5cc7d3edb3e0dafb19726ebbb3908ee64c2dbbf966eb8e031b58ead36c563c1

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2495	Reversible and Quantum Computation	SFT-QUANTUM-REXX-GARBAGE-CLEANUP-010	Garbage record and cleanup boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:de29a80e561df8683ed002701bdf8ff9f8993886a28e63d13241e3b66ff7211
2496	Reversible and Quantum Computation	SFT-QUANTUM-REXX-ERASURE-HANDOFF-011	Logical reversibility and physical erasure handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:18e94b65d4a134c94b03b17c2b2ed92df203f3ad878bd5dec3287fc0a2b51fe4
2497	Reversible and Quantum Computation	SFT-QUANTUM-REXX-IRREVERSIBLE-SIMULATION-012	Reversible simulation of irreversible computation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4af6000bc8cc6ac4b302c79deab9545af71a5e4c622a5fa1a99b05ee753a456a
2498	Reversible and Quantum Computation	SFT-QUANTUM-REXX-OBSERVATION-SIMULATION-013	Irreversible simulation of reversible computation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a228793a7a7417741da1ecccacda28672dbddf562b38bf7f906f3566fd075399
2499	Reversible and Quantum Computation	SFT-QUANTUM-REXX-RESOURCE-TRADEOFF-014	Reversible time, space and record tradeoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7dad641a707a1137855e03b850b4ef39b1168855c75e7f5b88dd59bc4a1c43e5
2500	Reversible and Quantum Computation	SFT-QUANTUM-REXX-CIRCUIT-SYNTHESIS-015	Reversible circuit synthesis and decomposition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:00b3f31892ee8f8d15dc5cff96ed50443153952bd9b5b75fe5bea34665a491ee
2501	Reversible and Quantum Computation	SFT-QUANTUM-REXX-CONTROL-016	Reversible control and conditional execution	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5ce14dd5b5ee37e3d9fcdcfdc00d9d17d9867a5da9cfb8d1e8ed183e9ec1f448d
2502	Reversible and Quantum Computation	SFT-QUANTUM-REXX-FAULT-RECOVERY-017	Reversible fault and recovery trace	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4d43437bc6ce9aa6ebd6fb865848bc023d82f3c0bf086c78d7e26edf58397cc8
2503	Reversible and Quantum Computation	SFT-QUANTUM-REXX-COMPLETENESS-018	Reversible-computation completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:286e0e1e0213794cb1715babb4f3d239b86df8eff7275082ba352b84529e66df
2504	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-INFORMATION-UNIT-001	Quantum information-unit identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4509e5c5939f5156b1060685cd4b7d1dde127e01f8f7a31dff98ac1e69268c98
2505	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-REGISTER-SUPPORT-002	Finite register and word-support construction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9ca570a51adb715efd5f6f8ab7cb77051f83767b7a94c9f7e03cd04f31c40f4c
2506	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-STATE-IDENTITY-003	Canonical state-description identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2402952ebc75d9763eb55653db0c6d375dbc5314072973eabebdc9d0125061da
2507	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-PREPARATION-004	State preparation and provenance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:70979d98c9cbdaadc5b4cce292ddf8255d0653cfd0582dad870731a779b1c45
2508	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-PRODUCT-STATE-005	Product state and component reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2d85c3e3e84bd556997ecf69a106bc4ff554f17a6751a120e5fd8a5dc4cbcc75
2509	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-JOINT-MARGINAL-006	Joint state and marginal support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2b3c0b6607af9ccc4509b1d83a3c33d5b36cad8271ceeb01abb85175e1540f02

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2510	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-PURE-MIXED-007	Pure-support and mixed-record correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:28e296f15f825dc2c6925f7429132601eb8be317655293c597fcd88dbf2a6755
2511	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-SUPERPOSITION-SUPPORT-008	Superposition-equivalent complete support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4a42e08c5efe05880736cd8ce0198fe6db4dbfd82c3480e3de048240df13bc30
2512	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-RELATIVE-PHASE-009	Relative phase-label identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:60dbdbeaa6e2e2942b88084b2468aa50d46427d845004f91f90f4eb8e94e3ee6
2513	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-GLOBAL-PHASE-010	Global and relative phase distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d2a8f2002a3c22e4db71a7022f70f045177edceee2c17db6ef8286cc4ab24181
2514	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-PHASE-COMPOSITION-011	Phase composition and inversion	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e4e489d082ea1614dc7bec1fc93b7b502168bd7f955afd564619c4c51c8dc610
2515	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-INTERFERENCE-CLASSSES-012	Constructive and destructive interference classes	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:419507ed46703949d71cabbf3cf5b234ef96968a1ca353f22002910ef697ca27
2516	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-PATH-MERGING-013	Path composition and predecessor merging	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9d64a42b48be4ca9f5a3f705acfeb91e0b00c7d94a972d098c9e98fd1a62f5a6
2517	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-WHICH-PATH-014	Which-path record and interference boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9e88fa83b62ba6010900f67dbaade3367cbda6ec56b649da014a64d1ed35d41
2518	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-BIPARTITE-ENTANGLEMENT-015	Bipartite nonfactorable support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:00feac58eb639f6b1ee6b52635ef198ffb08ed4ed536cde5b8c0023026a1cda4
2519	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-MULTIPARTITE-ENTANGLEMENT-016	Multipartite nonfactorable support	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:73551624cfce10acd3164061ed9d02b2ccd8c201b173984cbcd86e7492a810c0
2520	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-ENTANGLEMENT-CUT-017	Entanglement partition and cut structure	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:11f278f78be095cf6e69ad062885f2f1768d30ab562db5c08d028a01d4ae95e3
2521	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-ENTANGLEMENT-SWAP-018	Entanglement swapping correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5e0753f708ad7d648ea6abae55c2f7f642a3ff7e5191ea4d472f8690d1650ad
2522	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-MONOGAMY-019	Monogamy and shareability boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a2d4e6c7faeaa6ac9de03d76fe857bd7498a4b1f1c79f1e1acb63009e477a7dc
2523	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-PURIFICATION-020	State purification correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:34ff5cdf590a29ac0dcc8ef08d29aba42c5ceb495a80c893c5a1c20adebdaa83
2524	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-REDUCED-OBSERVATION-021	Reduced observation and retained environment record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0b3a739b0445173ab3a2276746cc431103a49279235c514f8fa2a1294d1c6633

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2525	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-MEASUREMENT-CLASS-022	Measurement question and outcome-class identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:cfa379963b05bf0e9994669b8486a01235b077ea5ed9fb1f6ec81160ae0ecee2
2526	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-MEASUREMENT-REPEATABILITY-023	Measurement composition and repeatability	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eb3cddc2b5a75d3984308d8fcb37ba1714e194d646da05e6959ac0e01d8ff7ca
2527	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-OBSERVATION-COMPATIBILITY-024	Compatible and incompatible observation relations	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f3b117b3af9056e91d0dbe0fe8539ad1307eb559fae5250a5e438bda6604656
2528	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-DEFERRED-MEASUREMENT-025	Deferred-measurement correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:93e35d078d24df28131ffa88edbf9abb96alde1bd0a83141dc2e366f660a412
2529	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-NO-CLONING-026	No-cloning and exact copying boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a70d2584130a0489c6d1442e137f8cd77dc8a5bfda38d0cc316ba458b6eab556
2530	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-NO-DELETING-027	No-deleting and retained-record boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b6891e68c3fb693be79ac5332d00165c11aa974b310b84d63c48f87f4ed4ea17
2531	Reversible and Quantum Computation	SFT-QUANTUM-QSTATEX-COMPLETENESS-028	Quantum-state structure completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dba29903af1c70b92bf9bb278fceddd10f820809bef4b4a5f534223e130bd2d
2532	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-TRANSFORMATION-001	Reversible transformation identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8b5001ec7f6cb3acb0b6b310de22de61b4eb5083ca6baa4f838d8b97ef273dbd
2533	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-SINGLE-UNIT-002	Single-unit permutation and phase actions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ea3b9e967ed4dbbdf3bf60e2c3d4b9df6a45d2a4c51b19b8b1fda2f5673409ba
2534	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-CONTROLLED-003	Controlled transformation law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3e13e2a5631fd926b66b5a7f0ddebe18c9ea742fa984b28aeb29b63f0de71391
2535	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-ENTANGLING-004	Two-unit entangling transformation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2d5cf54185479899eac16361b268746ab889ea24a98db34c68ff7fa91fd0e701
2536	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-MULTICONTROL-005	Multi-controlled transformation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:66beb7c02e47e1d52ae04e98a0f05df366c44074f999999f98177e7d12c5ea8b
2537	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-COMPOSITION-INVERSE-006	Gate composition and inverse	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b723ce59cdd03a86c6e71f0dde9dcf301bb35aeb96b6002cd417a19f21702bfc
2538	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-COMMUTATION-007	Gate commutation and causal reordering	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e03cfd16be48d36d84bca744439b11df8d28babfd801df29e48ec206b6825f7a
2539	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-DESCRIPTION-GRAMMAR-008	Universal finite gate-description grammar	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b49926c4a573df6af7dfbf6245d2d626bd7c6db330d4848244bacce2ada6ef0c

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2540	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-EXACT-SYNTHESIS-009	Exact gate synthesis	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:91f18eef9502551589f2f1af93744b35ae326f7f6a075246c547ee221a75271a
2541	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-APPROXIMATE-SYNTHESIS-010	Approximate gate synthesis with enclosure custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:666608332674b971e5212e4c8cdf54729da6f85e64fd5383e50ab1ca4bcda3c4
2542	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-CIRCUIT-SYNTAX-011	Circuit wire, register and gate syntax	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9cd4df151f0ed87920f6c24f83df29aa963f37525019fc54b6e51407abc3ce88
2543	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-CIRCUIT-SEMANTICS-012	Circuit branchwise operational semantics	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:83794eb58a4c8b04026d2a321375704fff221cee9ee8264a16c78e5ec41dde19
2544	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-CIRCUIT-OBSERVATION-013	Circuit observation semantics	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8e6b9386b3c4e1a6984bbc0b5e56215a5775c2f2fd9f99bf6190e81a4d301e5c
2545	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-CIRCUIT-INVERSE-014	Circuit inversion and uncomputation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:345c266b83686cf0eba59615abd04386331e054d61397110e1fe5c20946ff714
2546	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-CIRCUIT-EQUIVALENCE-015	Circuit equivalence and normal form	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3fd607c9b9b332cd1a6c7c3dbb213bfecfb4a9b114b1d19d691ca7c9bed2f522
2547	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-LOCAL-DECOMPOSITION-016	Circuit decomposition into local transformations	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:86aaf62f90fd19f92b19f67fbbfc374a9bb1a4109ccd190bac12888930ba2c9e
2548	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-CIRCUIT-RESOURCES-017	Circuit size, depth, width and live-support resources	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:05ca55eae472f5265d17cb719681ede3ab8baa7611c8df02eb7198e40a83e7e9
2549	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-COMPILATION-018	Circuit compilation and semantic preservation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:35e44849027ef13fda2c95f5474fb6d77f83c305219ea4ed5fe1e9415f1ef19e
2550	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-MEASUREMENT-BASED-019	Measurement-based computation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:471565a1baece55943aa9b8f82ba9a1f79180ee216e84635c681ee9cde1a8e25
2551	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-ADIABATIC-BOUNDARY-020	Adiabatic computation correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:75a421b6ff07ed1a747a991322b59212b6bf2d6b507ba562512b298fd06497a8
2552	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-TOPOLOGICAL-BOUNDARY-021	Topological computation correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2a5819d1404b1d485a21f9d1268905df7677faea70e4be6bd3cbe43b7312fa97
2553	Reversible and Quantum Computation	SFT-QUANTUM-GATEX-COMPLETENESS-022	Gate-and-circuit completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:529a40311f05d60d99080f86ba2bbcaa8392a15a8e006c8c7e2c7bf89a49ae8f
2554	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-SPECIFICATION-001	Quantum algorithm specification and correctness trace	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:99f791836aa124c14c32cbeb4ee8ae4063f52f0fe0df9353ebe43a593ec643e7

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2555	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-ORACLE-002	Reversible oracle and query interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c4dcd4b669adf954991cd5fa669eb73bfe980fca4db19f5256c3e0f65feddfdf
2556	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-PHASE-KICKBACK-003	Phase kickback correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8e53cd4b8504ed63af45bd97dceaab40cd60c89858b0f9d421db5b0ed965fc0f
2557	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-FOURIER-004	Quantum Fourier transform correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a350b7f7c0159d1aa7361166777e966c06fc38f32cd979b6833b2c6f9bb77c23
2558	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-PHASE-ESTIMATION-005	Phase estimation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:62e453962fba7cf89d6471b6520fafea927970b3b759623bf50cflb66a315d79
2559	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-PERIOD-ORDER-006	Period finding and order finding	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a27386bfddeb7ccf0ce82c1bb6618f0e04f37625a5080cf9f87ea4a261e20bfb
2560	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-DEUTSCH-PROMISE-007	Deutsch-style promise distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:23c735cd44ecd083f63373633c3ac08bc0b7629502fd994a52cb5bdeb9d570
2561	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-SIMON-HIDDEN-008	Simon-style hidden-structure distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:363db564f687e848d11636c013b2143ad1ba7dcca1af2d216b7a501ab9b22e64
2562	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-FACTORIZATION-009	Factorization reduction and resource custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:57a2dac780fbab05c787ffea3bb8edb902c42f906751602e898e294523d9063b
2563	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-SEARCH-010	Unstructured search and amplitude-amplification correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:58330f107cb92f7460edcbd01f36e92580181d0995720eaf0aba244de49d766a
2564	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-COUNTING-011	Quantum counting correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4c131477b1ac328d4426a2bf4f0d34c6d8e0754765c5dbd8345c09499d4edb8b
2565	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-AMPLITUDE-ESTIMATION-012	Amplitude-estimation correspondence with exact enclosures	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ed56af082d66d5075507a46a54ecc5d6a1f420a4a97f47e437aec0307a6599a5
2566	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-WALK-013	Quantum walk state and transition law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f7810258c596ea6d594fd4dbddfc6f81a049dbaba23aeed8d675404ec1fc82b
2567	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-WALK-SEARCH-014	Quantum walk search boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aa9edc20c205fcde5dfd05deb41c7143947c3470b0fbb5ee23e2c093f644ba8
2568	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-LINEAR-SYSTEM-015	Linear-system algorithm correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c77327afa3678f6c95665c5d3aa98d3c95563a71b848d56467eec659fbf9c00c
2569	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-EIGENMODE-016	Eigenvalue and mode-estimation algorithm	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7881703e0b15e360d809276d83265d1bc3ab18b2165ebecd76010eecdcb b33f6

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2570	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-HAMILTONIAN-INTERFACE-017	Hamiltonian-simulation algorithm interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fa1aeab8485737f751127afb14d71b1a26e963daff7c869e73ea142ead6de278
2571	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-PRODUCT-FORMULA-018	Product-formula simulation and error custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:edff0ad0eaa1280f08480f78331c5be991ee4e6b91e49b13db1f31c152833de2
2572	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-COMBINATORIAL-OPTIMIZATION-019	Combinatorial optimization algorithm boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ecb3da65494f8e3e97a64631d914f7f6bac906efea433bf9fc48030750e98003
2573	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-VARIATIONAL-BOUNDARY-020	Variational algorithm parameter-prohibition boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:155051cedf924f8b67569617fa99546c6fb91c54a9d9c5fb59722dcd12cced54
2574	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-ANNEALING-BOUNDARY-021	Quantum annealing correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:246ec51cdcc2fc1ae0962f9cab40abf60edce266a71a521820b330f5bdb53683
2575	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-SAMPLING-022	Quantum sampling and output-support custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6ef91e7464c4f4369c2564a180da11dc5e444b75e6b18583755157deec3cc382
2576	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-BOSONIC-SAMPLING-023	Bosonic-sampling correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0ec830399029b17ae876e01bf52d27acc07e9a14c5362487626389a21a190243
2577	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-HIDDEN-SUBGROUP-024	Hidden-subgroup algorithm family	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3bb52cf39d5ccd8e08410e32bd8b33164a43b5bad73cac2890cf8905093dd435
2578	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-DYNAMIC-PROGRAMMING-025	Quantum dynamic programming correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e6657124994f2f4af35f0984d4555a2a08b920aa1f0768d8b1d9ca93dc06afa9
2579	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-PARALLELISM-BOUNDARY-026	Quantum parallelism and output-access boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c9f957aa2df8a9c632991d04bdf4f5ad1dfe4f915c15400081fd542229cfd7069
2580	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-CLASSICAL-CUSTODY-027	Classical preprocessing and postprocessing custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:09ca9d6e5a3e5efa4ed3d02f7fd63605a8b625e79b9c7fb64ed264e2a937b9dd
2581	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-SPEEDUP-028	Speedup definition and comparison grammar	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6fb381c39102f62d4e9ac871ba972d2d74e1ac52c4acc43c8226fe343440d92b
2582	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-RESOURCE-BOUNDS-029	Algorithm lower and upper resource witnesses	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:adf146bef541f68b5015d0838d5d652173d444744fbd347e36ec9d8d7fb4a244
2583	Reversible and Quantum Computation	SFT-QUANTUM-QALGX-COMPLETENESS-030	Quantum-algorithm completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be183aeb9c83cd98d8d6a59b0df147aa0f8f334a2cae8c2ff521fba02c33eb0d
2584	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-INPUT-SIZE-001	Canonical quantum input-size carrier	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9ae4fd9294b5e8ab57800c754a037ea8ea0b805366df78c8fdd00b4627326e78

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2585	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-GATE-DEPTH-002	Gate count and circuit-depth resource	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9c3a3b0862561a294c83710b159ad630f0183658c957b3d8601171ac4ad1621b
2586	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-BRANCH-SUPPORT-003	Live branch-support and state-description resource	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:312b3f9ec032529c5caba6067369ba55fc83709380a6c13d5831956ef80638fe
2587	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-RECORD-RESOURCE-004	Ancilla, measurement and retained-record resource	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bbf945918ab9ed9da16b1bf8761208494543a0704e29cc564cf72a0a71c9f6d7
2588	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-QUERY-005	Quantum query complexity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9917d13cf77dfd378f79bf7f2e9a5d38c416e94733667a71288671200fc045d1
2589	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-COMMUNICATION-006	Quantum communication complexity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:553aab1eb1ef19db3570bf61927723037a9219fa63acb87c202d472f9bfc30
2590	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-DECISION-CLASS-007	Quantum decision and recognition class	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c4620b96478b9638b904831a3198b63473ad396f90517378f3687a9cdef77fbb
2591	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-BOUNDED-ERROR-008	Bounded-error deterministic-support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a6a049352785a4bb608475615a833a54f1315371cc67a98108e6ce3dec04573b
2592	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-EXACT-DECISION-009	Exact quantum decision class	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bb305d629813ac758b15e228becacbfa5501e36d0d1409e7c4211513b6a7579
2593	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-ONE-SIDED-ERROR-010	One-sided-error correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:dc3dc960cf5ffcd638c2d09c0bb4eb26592ab2914e0b5f0b7219ae5a639d4eba
2594	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-NONDETERMINISM-011	Quantum nondeterminism correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9ddbf0268eed7e12174b1e5491df76b3c11c93c92839df0b77d4b965f664eee
2595	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-POLYNOMIAL-TIME-012	Quantum polynomial-time correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0cfdda54a20502f672fac23627d8f7f07af6e13526b93a6bacdf6a7a627e3319
2596	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-WITNESS-VERIFY-013	Quantum witness-verification class	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d380def03ff20463a7c5eceb523208bd0de9c14ae81818c8cb4dcd8bcdef439a
2597	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-INTERACTIVE-PROOF-014	Interactive quantum proof correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4d91ba16b6fde9099733e3584629d28e981b72b94658aac4db74a25a8e7809af
2598	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-SPACE-015	Quantum space complexity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:587dc8e1da18658d8b74e093bcac722ba3e4253784697cfff4b5893c96388d279
2599	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-PARALLEL-016	Quantum parallel complexity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f7a0404fc737283bba59728d9b5424ffab40ac748a06dddf5ac746ec074e3413d

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2600	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-UNIFORMITY-017	Quantum circuit-family uniformity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c55fae1e1863fc5820515fb493be82c11bece3a827995f11b9be688693e3f312
2601	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-REDUCTION-018	Quantum reduction and completeness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c56d004bfc36401e94937879b9c9824273d90bc18915c897562268fa948ad413
2602	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-ADVERSARY-LOWER-019	Quantum lower-bound adversary certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bd32fcf800d0653fc4fffd8b443ea1d1394a3a0e9617f7fd96187e3a990029eca
2603	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-POLYNOMIAL-METHOD-020	Polynomial method correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:49b2fee86b80771f99488941875b603c3442ba63bfff63cbcb1912fa6d39c8e4f
2604	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-CLASSICAL-SIMULATION-021	Classical simulation resource boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c2779f9eda27d0f0e8ecf5305ae16d04db236548913ae73ac2fc867e5c670571
2605	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-ADVANTAGE-022	Quantum advantage and separation certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2acc6d3fec6f694907d190d3c82c0eee707fe40bae87ac2f0e74e107cf92b147
2606	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-AVERAGE-CASE-023	Average-case and distribution custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:691f996178c6f59f55ef921c6812d758d49e607cc9d2eedb48498b803449a3be
2607	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-PARAMETERIZED-024	Parameterized quantum complexity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3d4c9d10a5fcea9713c3f225538e9959cac1e73112a5a60408936f4abb4e75c6
2608	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-DESCRIPTIVE-025	Quantum descriptive complexity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ff4c459d85411a61b053743f19306a30f58c7dc01f0c2153e6dc1f61c3ac4799
2609	Reversible and Quantum Computation	SFT-QUANTUM-QCPLXX-COMPLETENESS-026	Quantum-complexity completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d84ae8e7486c5a85ec27e57471c7f9e5a675f5821f0903ef767c5530f68b30a8
2610	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-CHANNEL-001	Quantum channel as exact relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:59b7627bcc3cad71c2500ada3f272b4e86f4fc0bc3e8bc6cb3433690a3da7b55
2611	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-COMPOSITION-002	Channel composition and memory boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:589c4daf64d0143ba453051b5f1e2c9afad93dcf06e578607d2952b99bbf4a42
2612	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-CLASSICAL-OVER-QUANTUM-003	Classical information over a quantum channel	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ca1b04a7ddcac9e59815f58896a1e47f4af6f9f655b5aadf246f449973afb9f1
2613	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-QUANTUM-TRANSFER-004	Quantum information transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8116bc92126a510b919b089914971c78f3f0ecfeffce88b7fbf2c4db43e76d38
2614	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-ENTANGLEMENT-ASSISTED-005	Entanglement-assisted communication	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9cf3bbe8093792283b6ba3719388ac066eaf273ec4bf336f6c317b6faeb5c7a8

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2615	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-TELEPORTATION-006	Teleportation-equivalent state transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a98394422b828be384aaa52f7f173faaf5e3b24339ae66ed4090830ff36737e4
2616	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-DENSE-CODING-007	Dense-coding-equivalent distinction transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bd8e3adb57c298bad71c94498d786f41d417d70ad485f129287dea6cc5d86f06
2617	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-NO-SIGNALLING-008	No-signalling operational boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b7ff278c0ace85e20357d456d100ef48970e3d4ea0cf867929c8e723e79d0bec
2618	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-CAPACITY-009	Channel capacity as generated support rate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9877068a5d841f06bde5bd2f51ae9610016490f02238e3455a9588167cad58b
2619	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-PRIVATE-COHERENT-010	Private and coherent information correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:99193247bc475c647196d2ac7c4a3145777ebaa5694bef3142f77cff778bb03
2620	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-DATA-PROCESSING-011	Quantum data-processing relation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:22ba20947483cd6a836efbde48b11bf8a5c124ff5967d834287c01939c444415
2621	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-NOISE-ENVIRONMEN-012	Channel noise and environment record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4cf3a01e853a41a74645ee334655711a972807106122d45bb55eb507640d90d4
2622	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-ENTANGLEMENT-DISTRIBUTION-013	Entanglement distribution and swapping	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:71a463ad79adbfe6f9937120beeb8c3938b2213905c0f931f949fbfb1c96f7f
2623	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-REPEATER-014	Quantum repeater correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b09de71a3b5c42d89e8ce4353418f5fe5d5e9719fb357fccfe94dee295d62f58
2624	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-NETWORK-IDENTITY-015	Quantum network node and link identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f23149520ad72273eb34621de6e3d5c21279ab9f8b43d640d5e25668b4efb548
2625	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-CAUSALITY-016	Distributed quantum process causality	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:078d540c771aa19eb0b6a43d1150b5a197b1e9f298078a05bbe0e31aa06caf84
2626	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-ROUTING-017	Network entanglement routing	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ff79382ba10ee289ef5ab35e94d4c873a0295304b8fbbf025df0acb36eb3db88
2627	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-QKD-018	Quantum key-distribution correctness boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:97ca19f650b83a7a40e8951a9f56fefae9a8abae12407193b48ed5b1d75e1ce2
2628	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-AUTHENTICATION-019	Quantum authentication interface	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a031faa6c525e870cc7fa104545e67893ebe5f011d806e8501fe89fd2ba4bf8
2629	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-SECRET-SHARING-020	Quantum secret-sharing correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:be797c98156fcb0498b7a7b3b84668299f469c0e32de24425e5a1d820ee2023e

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2630	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-DEVICE-INDEPENDENT-021	Device-independent security handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c2ec2267046c481b786d01c73946c6aa1708fcbafbccf41a8c973c11da081d8b
2631	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-ADVERSARY-TRANSCRIPT-022	Quantum adversary and transcript custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fe11c0c71438066d6618a813ded5fb6dc41ff8c14b4940d80cee9db69f43421
2632	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-POST-QUANTUM-HANDOFF-023	Post-quantum classical-security handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9647836ad4a693e5e1a7aeafa3168dc04c2f87f7471490157f3b205dc8f46af7
2633	Reversible and Quantum Computation	SFT-QUANTUM-QCOMM-COMPLETENESS-024	Quantum communication and security completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:549cdb70cffd760ae9952d5bef571058e2e583ed5e17cf5a0be52f1a608e00ce
2634	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-LOGICAL-PHYSICAL-001	Logical information and physical carrier distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8ca0045d3a6c9bf4a3e49d5401c9425e067efd91acc4cde2ab5b35ef6ff947e
2635	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-ENCODER-DECODER-002	Quantum encoder and reversible decoder	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8ce63850b94782800e2110a923b7f251498401cf2f6a8e82ac926aa29a228152
2636	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-ERROR-FAMILY-003	Error action and error-family registration	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:821de04dfd308d93ad60a96f3764dc9fe3f320549e4d6940e366dc68716a079c
2637	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-SYNDROME-004	Error detection and syndrome distinction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:51c83c233ac90d7f55a3986eee36eec6e40a3b19d615f90812eac18ca61c81fe
2638	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-CORRECTABLE-CONDITION-005	Correctable-error condition correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1daa05a01c33f5fd18bcd8c745915e7b0a4847314fbb9e2eb2d59130ea1db923
2639	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-BIT-REPETITION-006	Bit-label error repetition code	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2c82e56699ed69e38036bf91e65ba8a7237d39f14a3eccc12c653b4718c9dcb3
2640	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-PHASE-REPETITION-007	Phase-label error repetition code	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e4819c0f9bd9086c5bceb58bb0d4cbc3b0411d5de688e407f0c9cfd0b5f142d
2641	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-JOINT-ERROR-008	Joint bit-phase error composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0f5399fe917d08f0c68691ac90af1c49aabffa7ee7ac58efbe048090af698a87
2642	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-ONE-ERROR-009	Single-error exhaustive recovery	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e035f2b790fb4cf453501440e863a18df5c04e67a267e50029c75266f77716b8
2643	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-TWO-ERROR-010	Two-error exhaustive recovery	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3f3a6c8b07f87d0646a6cd3e6e2a59a81152ad6305c63841a41fc4af2e7b1e63
2644	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-THREE-ERROR-011	Three-error exhaustive recovery	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:11c4d8cbf4530006b9a0d1c79d5fc006dba818518b19c9c4516c0a85a0fcd17b

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2645	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-MULTI-ERROR-SUCCESSOR-012	Positive-finite multi-error successor law	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:0cf0fc7fdcd1e25a550860a42df007fa1ca84f6c0ab5b561998a53a23c27d44
2646	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-ERASURE-013	Erasure error and located recovery	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:97dea103cb489fbbaed2e02e16cfdae13f6fc2826c1aebc0de4637d341dd0ee
2647	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-AMPLITUDE-LOSS-014	Amplitude-loss correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:48827434f3ce4b62a448e87aafc5d438e27fdbb3ae1580e0e01d17c68702873
2648	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-DEPHASING-015	Dephasing correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a3d92c7192cd0cd8abecbefaa0d4a025cf2d72e60082e00453908dc2547124c4
2649	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-DEPOLARIZING-016	Depolarizing-support correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ba048dd0d7e964be0acbea310c29e06215ead43bfb0a51ff7b307a0be2d7c150
2650	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-STABILIZER-017	Stabilizer-code correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e57a4d8fc09f8f601080d354bb32ef3f1339e9d7b4cd091c4a388f373632f198
2651	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-CSS-018	CSS-code correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:67cd6ea19e6be988e13351fe43785fecb6fa354457bb877016cc2bdd910d047c
2652	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-SUBSYSTEM-019	Subsystem-code correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:81d92c24a4c93d689f835e2e8e320a604e243c6acc377b644e090d76891ec27c
2653	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-TOPOLOGICAL-020	Topological-code correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a5d177e8ee3e84961d62932036e533027ca8ff582971a4ca72be1d27b9ebc4f7
2654	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-SURFACE-021	Surface-code correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6cb978aae152056c782701e1f9b375c640f819a73fc1b8b767933a6b2040c8b7
2655	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-CONCATENATION-022	Concatenated-code composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:564c2c3e139937f1de6add70e781a91091a37a14c7884604e205d41df577d172
2656	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-LOGICAL-GATE-023	Logical gate and encoded transformation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7a209b712ac6cb872b7b0e76b39f2fd70bed7e992b097d626b3bca6ccbe9c437
2657	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-TRANSVERSAL-024	Transversal-gate containment	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:756cb6fe6ccd6d93519f6bad52ca7ec393678323fd80c587eee4259956b0d61b
2658	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-SYNDROME-FAULT-025	Syndrome-extraction fault custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:38c39f21129fe64fd1f24a01f6668e89a4453c94cb5d2b333faf7617519930cb
2659	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-FAULT-LOCATION-026	Fault-tolerant location composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c6e3f3fd0f7a2f8de9e95ff7cb32213e0830bda9c25be614f3d6414f17c7481

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2660	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-MALIGNANT-SET-027	Malignant-fault-set boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1d8459ef5bc47efe43799e28591cb59060d0a05a99af9a0d0f50a71349f683ea
2661	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-CORRELATED-FAULT-028	Correlated and nonlocal fault boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:654b4c4a3bf993eda8b2fe3635e84644bb1c57d999542407d591a48fc05337fa
2662	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-LEAKAGE-LOSS-029	Leakage and loss fault boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2f3608afbc751872ca474a3f9d663417aba27959b2a82cd691123b69fbc4e56
2663	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-DISTILLATION-030	State and magic-resource distillation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3b3e32f94925d00e48b57c7d04bba7ae09bbe5d9b530d69b2cfb4f8d693988
2664	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-PHYSICAL-THRESHOLD-031	Physical threshold constant measurement handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2b894a90da74f8447dd997e8013d68513a7ca520fd471ab3fb34cb904b27cb01
2665	Reversible and Quantum Computation	SFT-QUANTUM-QCODEX-COMPLETENESS-032	Quantum coding and fault-tolerance completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9b130b5c4fd4046768d3a237915c0ce918376574e692cb6f5743aaf49836bb0c
2666	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-MODEL-SIMULATOR-IDENTITY-001	Quantum model and simulator identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3da80c2fdf954b7f41a48d6ddf329469d9eb131eb1d2f10fa90996509dcf89e
2667	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-TARGET-SUPPORT-ENCODING-002	Finite target support encoding	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9b40db1a1d66ddfffe51c2d4a8ed1bcbe4126c44ff57e621958e4cccc72d381
2668	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-DIGITAL-SIMULATION-003	Digital quantum simulation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c839b40be7727d9073a5f3d67fb1df6b613d1e1a6cdc78454b089417277cd362
2669	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-ANALOG-BOUNDARY-004	Analog quantum simulation correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:feb6969efd7e3b075ccb68941eab9eef5ccc15f2bca51d5107892c481b6e283
2670	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-LOCAL-UPDATE-005	Local interaction and update composition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d61405660a791dd4190b44c3be267d5c370e1c9850b2ec093c7ca864cb447b7e
2671	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-HAMILTONIAN-CORRESPONDENCE-006	Hamiltonian correspondence without imported continuum proof values	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:28a3e3e662b3773c4a140c1b84da25906492432d0b5d7e2eb4aecf29c151120d
2672	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-TIME-EVOLUTION-ENCLOSURE-007	Time-evolution approximation and enclosure custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:a7816fb5812f8ed1da1a7cc70f6143809f3746c6e975dc0acbab31822e71c9d8
2673	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-MANY-BODY-SUPPORT-008	Many-body support organization	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:01ea3b4501528d0fc6145f2497fa68a65a64cdd8d16a208da1acd9402cb4a063
2674	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-FERMION-BOSON-ENCODING-009	Fermionic and bosonic encoding correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bc581e6f325eaf7407cf3f67d7be08f872c86ecb70dbce74c88edbfcc9a7cc

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2675	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-LATTICE-FIELD-HANDOFF-010	Lattice and field discretization handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bdb24be374b9b9a9b27c39210944aa85fecc73631dd8f784fd95958acddcadd
2676	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-OPEN-SYSTEM-011	Open-system simulation and environment record	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9cb61672aa4c64f08de9a372c44c21dae38e644815f642a4130e1b9d1dc60b32
2677	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-NOISE-CUSTODY-012	Noise simulation and source custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3726582b1fc5fb2242ddf579f5551098c55c7564aca774c2bab5b2b26241932b
2678	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-CHEMISTRY-HANDOFF-013	Quantum chemistry simulation handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e575b052b5482a607eba363a0633f57293ddc89d03299470fc904590dbd9208c
2679	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-MATERIALS-HANDOFF-014	Materials simulation handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3d0fc0ab883e09458a9c9c95cbe7fec4b07457bd8b96ee0e1be3a1d4617206f5a
2680	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-COMPUTATION-VERIFICATION-015	Verification of a quantum computation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:99e6893148d1365275022f7600678f0bfff99d43c6bb9a5239ec634b377a361c7
2681	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-INTERACTIVE-VERIFICATION-016	Interactive quantum verification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:25ca3df8eae3b2418d39a33e03aba3183c924377153274ed1d8e26dd25ba3c1e
2682	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-BLIND-DELEGATION-017	Blind delegated-computation correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3c4869bb0c0ee070e7f57a80f650175f88e00c6f076324f45da920e75e6a7005
2683	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-SELF-TESTING-018	Self-testing correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d0dc6ab0e5fdf134a7b9cb9e1333b4cdc45bbbd724289d26f06c1bc6e5c7545b
2684	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-TOMOGRAPHY-019	Tomography and state-reconstruction boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eb9109ef49265eabccdd261b69bb0a18fdee0cb8ea3f62e7ef15a5ed0f49a390d
2685	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-PROCESS-CHANNEL-VERIFICATION-020	Process reconstruction and channel verification	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3f85d6c823346e7fae0bca236969c49bc41402fa60b6d5cb430a3205406a46b1
2686	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-DETERMINISTIC-BENCHMARKING-021	Randomized benchmarking deterministic-support boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1089b279ab1a0170cd091d632796b64da1cd7b678b52c0439f36a06e05da731f
2687	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-OWNING-DOMAIN-VALIDATION-022	Simulation validation against owning-domain data	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b0309a07cd6d8b0a25f54b31835c4a9bad1dca840373c307e89ca0c43f1d128e
2688	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-WORKFLOW-PROVENANCE-023	Reproducible quantum-workflow provenance	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:956fd986d7484e5804c1bef19f21766c9e203106b0d89ff84bad64f1ff5d3de7
2689	Reversible and Quantum Computation	SFT-QUANTUM-QSIMX-COMPLETENESS-024	Quantum simulation and verification completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eeb3915656fe41fb894b3b70cf49b9b1b64bdd75cbb2621ff2f65a1fea9c3ad5

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2690	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-PROBLEM-EXAMPLE-IDENTITY-001	Quantum learning problem and example identity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:65f97c528556510cd79d279c31548e9e08796bb44d0082b6729a91d99bd08e7b
2691	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-CLASSICAL-DATA-BOUNDARY-002	Classical-data quantum-process boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:66466af4b2f4a7312fc81d4d6d29159a8e3c1db2cdc6d989650637d72d8206cb
2692	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-QUANTUM-DATA-CUSTODY-003	Quantum-data support and observation custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:aaf22c744004c2430b18272dcaab6a0b2f4909d6bf81c0fc61225b7e8374a40e
2693	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-HYPOTHESIS-FAMILY-004	Quantum hypothesis family generation	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4054b109c0d66e370772229353c625ccc26b579f278bf988a9f19bdfbfa00139
2694	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-FEATURE-MAP-005	Quantum feature-map correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ead50c2898ab96c41f3902a7920db272982b56fa1015876a181879ea330b0ae3
2695	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-KERNEL-BOUNDARY-006	Quantum kernel correspondence boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:04ef3e26e8db92331ab1f9d2a2faccd904c0cce51384928ca391a8106c4fa269
2696	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-CLASSIFICATION-007	Quantum classification process	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:3033bc11ffc2918c0c3be978de28cb36b2cc2cbeca51493a7e13932427a0b1d6
2697	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-REGRESSION-008	Quantum regression process	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:51c7e80569552b9bfc0a4862d7b4d372afe0827368c3524872ea3dc9385060f6
2698	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-GENERATIVE-SUPPORT-009	Quantum generative-support reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:698359a3853280aa3f6964cf450acf2f7def7b1ab614b671fb515e65dbb6b0d
2699	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-CLUSTERING-010	Quantum clustering correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:059b43ed158be4743b233a277ce5268a617fd9f96c718039d2d0b619b95d5918
2700	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-PRINCIPAL-STRUCTURE-011	Quantum principal-structure correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ef0e0427f11710f30504f8083d70982cc2dc1e0b99439d3a850092faed9f39c7
2701	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-OPTIMIZATION-012	Quantum optimization in learning	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7a3a839aed9cbd6a6717036844844ba79dcadc50972366a3130d3aed53c459e4
2702	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-VARIATIONAL-BOUNDARY-013	Variational quantum learning parameter boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e9981b411ea9c81555ca809e1f97215cdc6f8eab2e74d60b6cb92aefa3befe4b
2703	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-REINFORCEMENT-014	Quantum reinforcement-process correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:32629b84aa53b7b4c407cfd8f46fb1eac517b525fe6c58fc8a46dc0fa370efb7
2704	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-ONLINE-015	Quantum online-learning correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:58d7f9aeb24dd4337260b52e802ce0cdce9c995cfd6328295a044d37a2f4c8e0

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2705	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-SAMPLE-COMPLEXITY-016	Quantum sample complexity	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eeaa42743d74ca8285fee800856ce254fc11b3396e36d904929cc347166e8c03
2706	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-QUERY-COMPLEXITY-017	Quantum query complexity of learning	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:16decc3feb3ceeb9f811959705a7856a395b6ffb03047b24fafaadc837b92889
2707	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-GENERALIZATION-CUSTODY-018	Generalization and held-out target custody	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:5e6b50d613bf5064b9ad756aef50209728f6d72e553fa187da770e5766270973
2708	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-ADVANTAGE-CERTIFICATE-019	Quantum advantage in learning certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bdd0de5541769817f63620be42a4d26cd9e7243a1a1f4f5f90f2f0a7d261d948
2709	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-INTERPRETABILITY-020	Interpretability and branch-trace reconstruction	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8085c9eed22079f9054685025fa0be89b2d63f8a25646a92229b8f356df18389
2710	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-ROBUSTNESS-021	Verification and robustness of quantum learners	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:84b3219b563e427ed92c822984b710d224a6d626fa956a17f0c2387adf435629
2711	Reversible and Quantum Computation	SFT-QUANTUM-QLEARNX-COMPLETENESS-022	Quantum-learning completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eb2d67d692a56175ec70aa9980d6f5e18ee213e36594c1233817cfe87d22c14
2712	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-CLASSICAL-STATE-EMBEDDING-001	Classical-state embedding in the quantum machine	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:97589da6e1d4dad086cc7acdf225734834f51ab8e88533a194a8f668520ece1
2713	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-REVERSIBLE-SUBMODEL-002	Classical reversible submodel correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:860500bec17e76ea67cf6824b7458f8309c27869821cb44a45b16644c8389494
2714	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-PROBABILISTIC-SUPPORT-003	Classical probabilistic support correspondence	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7692326b8a9a1afc54d31347d3e93018d021cc66369e9c8042372de304add83
2715	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-MEASUREMENT-DECODER-004	Quantum-to-classical measurement decoder	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:30be3ef2a45fceaef7154469bc053fff098f74b6c0f2bf9ae2c51062d8ed060a
2716	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-BIDIRECTIONAL-SIMULATION-005	Operational simulation in both directions	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:13724cc428ecbb46dbbd0bf0f86e0993b76c8d9e6f4b322b2b6a3f7d7fb20957
2717	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-EFFICIENT-REGION-006	Classical simulation efficient-region certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:1b166eda6ee3fb25ff456a662ac72c2ae24d4c2a168e6fb106b86100e9c8296a
2718	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-PHASE-SEPARATION-007	Phase-sensitive separation witness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:36ceb1d4b0a69ccb938467ad9096e76316672fcea7594886055de0e454f7f2463
2719	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-ENTANGLEMENT-SEPARATION-008	Entanglement-sensitive separation witness	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:eb35ea4e46230bfd10c53f9af278ade18e0f375b2f9f31a6356be78975a2c627

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2720	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-NO-CLONING-009	No-cloning computational limit	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:e24b56950bfc68daa0e65624757b9180cbb50ceb276410acba3e7eb0c4f6527e
2721	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-MEASUREMENT-DISTURBANCE-010	Measurement disturbance computational boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d1ee1ea629331f0dccc805e5890f20b1aaf25fe8b6a32266e9ef0e371f47b1de
2722	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-HALTING-SELF-REFERENCE-011	Quantum halting and self-reference transfer	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f47b1c54b9109ad48e8fa13cf7f209bfff60e49cff3d49bc8bd714644c2fe9483
2723	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-UNDECIDABILITY-012	Quantum undecidability boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:29d7ca0b7be87c7945b4e4b0605e7d29a4535bf2eb8b554495bec5d5892047
2724	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-INCOMPLETENESS-013	Quantum incompleteness boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:c158dc4afca40053535c31975bcba6c0970b4a536cd12e52e54cc0690e08f42
2725	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-NO-HYPERCOMPUTATION-014	No-hypercomputation boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:2db3d79829f3ee699197e369c9968d1c4ff376c96be97905d33849b333dbc030
2726	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-FINITE-SUPPORT-015	Finite support and completed-infinity prohibition	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f0433854b807b74562a3c6dc2d9ea09ccf7e9484cd87c29dc3f4377f62420487
2727	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-NO-UNRESTRICTED-ADVANTAGE-016	No unrestricted advantage from bounded examples	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8b69498005cf382b784c448e4c7718a1a6c022d3f1d8c3c332e54899d7cd9b5b
2728	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-NO-UNMEASURED-SPEEDUP-017	No physical speedup without device measurement	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:54cf1ce61fef584862d1941b57662cb8fff20527a0eb849033ad6cdbc28987e8
2729	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-NO-FORMAL-HARDWARE-THRESHOLD-018	No hardware threshold from a formal code theorem	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8bf2ab0b75e1b7b7febb67b936aca8287f252acf62e711cceb1814cb94c31ddb
2730	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-IMPLEMENTATION-HANDOFF-019	Energy, timing and implementation handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:349bfa13c73b0182eb1454536efa2b1fa582f43ecb30341fbb3bda2bc882799b
2731	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-PHYSICS-MEASUREMENT-BOUNDARY-020	Quantum-to-Physics measurement boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bd7fb477471ca1b208e034d7b05723782ea84110187bf53a88f0281fca4b74a2
2732	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-OPEN-FALSIFICATION-021	Open extension and falsification boundary	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b2fb7db464e8881aa24e1b22b3d79852ae7ac258a9a042d71cd4287c642a55e5
2733	Reversible and Quantum Computation	SFT-QUANTUM-QLIMITX-COMPLETENESS-022	Quantum computation limits completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b16d42d36e8e0925408519a05d95c13308a5b24e41ce2f8cfbcdbced800132a0
2734	Reversible and Quantum Computation	SFT-QUANTUM-VALID-REVERSIBLE-001	Reversible-computation complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:48792da96d5f5b8ce8fa11b5363ce2cf8a3c548bf3465c304f43ad43f699cb47

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2735	Reversible and Quantum Computation	SFT-QUANTUM-VALID-STATE-002	Quantum-state-structure complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:b51f0b544cad8490da7ab2b3bbf691b2e4cad802ee5f99b6e4497e7261092154
2736	Reversible and Quantum Computation	SFT-QUANTUM-VALID-GATE-CIRCUIT-003	Gate-and-circuit complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:9b72cec0cdf572064df960865cd370007c6b516537971cfac8bf85f62ee97865
2737	Reversible and Quantum Computation	SFT-QUANTUM-VALID-ALGORITHM-004	Quantum-algorithm complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:81c4b69f3ff4896eb168b883d002c69e654db101b805b52e01097608338a6178
2738	Reversible and Quantum Computation	SFT-QUANTUM-VALID-COMPLEXITY-005	Quantum-complexity complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:165a3b9abd922134c3d6522ccc2a15205078207b1980594f7fd16d5fc964046
2739	Reversible and Quantum Computation	SFT-QUANTUM-VALID-COMMUNICATION-006	Quantum-communication and security complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:6952f6ac6b8d2cf3cf220baaeaf0e8be0d9bd4179f8927881e2351a22a658bfb
2740	Reversible and Quantum Computation	SFT-QUANTUM-VALID-CODING-007	Quantum-coding and fault-tolerance complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:d57b2f167d86ffff531433017aa2233cdf5586539a679277aa80d6fdb69cc1e62
2741	Reversible and Quantum Computation	SFT-QUANTUM-VALID-SIMULATION-008	Quantum-simulation and verification complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:902817b43c056bcd5a32aa52a5b32311283db1d396ce77ba8b6b801ee01ac09
2742	Reversible and Quantum Computation	SFT-QUANTUM-VALID-LEARNING-009	Quantum-learning complete validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:bfe8847f8595aaef4dac29e4cd82f42c0b1f9511c14bab612b995650a67b331d
2743	Reversible and Quantum Computation	SFT-QUANTUM-VALID-LIMITS-010	Classical-quantum correspondence and limits validation vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:fd44600f7c180ac9e7e816cb8588d49f939c6003db44f911ba0f83b1240a6821
2744	Reversible and Quantum Computation	SFT-QUANTUM-VALID-ADVERSE-OWNERSHIP-011	Adverse, absent, unresolved and ownership-boundary vector	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:add981310a5a650816faf2bb9f22471dece1e8ab95e6ecf575e2ee3504077626
2745	Reversible and Quantum Computation	SFT-QUANTUM-VALID-GRAND-LOCK-012	Reversible and Quantum Computation empirical and formal Grand Lock	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:f6ad2363ac3de7e83a60b977ed24642bc1cc68d5f9b535838f8e6aab2a087f37
2746	Reversible and Quantum Computation	SFT-QUANTUM-HAND-DOWNSTREAM-OWNERSHIP-001	Quantum Computation one-owner downstream handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:4cb88c6a40b9b7958701d787327752b3d6bd6b5e11abe04d68ca38108afec28f
2747	Reversible and Quantum Computation	SFT-QUANTUM-HAND-PHYSICS-MEASUREMENT-002	Quantum physical-measurement handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:98289f64312bfb3bc2dec26aec199d4766d193723018ff50a0353f76a228ccc2
2748	Reversible and Quantum Computation	SFT-QUANTUM-HAND-CHEMISTRY-MATERIALS-003	Quantum chemistry and materials handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:10444146c044f15ef45a1cce089151fc0ec6963602b277b8d3d14fd4258c8c38
2749	Reversible and Quantum Computation	SFT-QUANTUM-HAND-SOFTWARE-HARDWARE-004	Quantum software and hardware engineering handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:8cf3f413acff88600c32dc5c26639b0e888da2349158a1aeed746377d15b0258

#	Branch	Claim ID	Title	Formal closure	External status	Candidates	Controls	Receipt ID
2750	Reversible and Quantum Computation	SFT-QUANTUM-HAND-OPEN-EXTENSION-005	Quantum Computation open-extension handoff	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:7617d7cd2287fbe3eb9e0e724b6a3eb70934e995878357a0a14792524ea52b7c
2751	Reversible and Quantum Computation	SFT-QUANTUM-HAND-CROSS-BRANCH-COMPLETENESS-006	Quantum Computation cross-branch completeness certificate	depth_independent	empirically_tested_and_independently_replicated	256	4	sha256:ef8df1b6bb47eea3ca46acd3458598c247270d8628de0d964b43b104af03c8d6

7. Publication boundary

Maria Smith has authorised this conceptual monograph, audit layer and machine inventory for first publication as standalone V3 version 0.1.0 under reserved DOI 10.5281/zenodo.21717584. This publication creates a new Zenodo record rather than updating the deprecated V2 ToE lineage. After the new V3 concept exists, the later complete version must update the V3 lineage, preserve this chronology and re-run the final publication and verification gates after the computational programmes and global programme requirements close.